

GE Healthcare

Optima MR450w System Installation



OPERATING DOCUMENTATION

5670002
Revision 1.0

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ

(BG)

Това упътване за работа е налично само на английски език с изключението на случаите, когато се изисква изрично по друг начин от местното законодателство, или от споразумение на местно ниво.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告

(ZH-CN)

本维修手册仅提供英文版本，除非当地法律或本地供应协议另有明确要求。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

Tento provozní návod existuje pouze v anglickém jazyce, pokud není jinak výslovně vyžadováno místními zákony nebo odsouhlaseno na místní úrovni.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

Denne servicemanual findes kun på engelsk, medmindre andet kræves i henhold til lokal lovgivning eller lokal aftale.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar, tenzij expliciet vereist door plaatselijke regelgeving of overeengekomen op lokaal niveau.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

This service manual is available in english only except as otherwise expressly required by local law or agreed to at a local level.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS

(ET)

See teenindusjuhend on saadaval ainult inglise keeles, kui kohalikud seadused ei ütle teisiti või kui kohalikes õigusaktides ei ole otseselt teisiti ette nähtud.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

Tämä huolto-ohje on saatavilla vain englanniksi, ellei paikallinen laki nimenomaan toisin vaadi tai jos toisin on sovittu paikallisella tasolla.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

Sauf exigence contraire des lois locales ou accord contraire au niveau local, ce manuel d'installation et de maintenance n'est disponible qu'en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

Diese Serviceanleitung existiert nur in englischer Sprache, sofern nichts anderes gesetzlich vorgeschrieben oder auf lokaler Ebene vereinbart wurde.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά, εκτός αν η τοπική νομοθεσία απαιτεί κάτι άλλο ή υπάρχει διαφορετική συμφωνία σε τοπικό επίπεδο.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el, kivéve ha a helyi rendelkezések ezt kifejezetten elő nem írják, illetve ha helyi szinten erről külön megállapodás nem születik.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

Þessi þjónustuhandbók er eingöngu fánleg á ensku, nema annað sé sérstaklega krafist, löglega eða samþykkt á landsgrundvelli.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

Il presente manuale di manutenzione è disponibile soltanto in inglese, eccetto quando espressamente richiesto dalle normative locali o convenuto a livello locale.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告

(JA)

このサービスマニュアルには英語版しかありません。ただし使用国の法令に別異の定めがある、あるいは現地で別段の合意がある場合を除きます。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

현지 법률에 따라 명시적으로 요구하거나 현지 수준에서 합의한 경우를 제외하고 본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS

(LV)

Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā, izņemot gadījumus, kad vietējie likumi nepārprotami nosaka citādi vai panākta vienošanās vietējā līmenī.

- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

Šis eksploatavimo vadovas yra tik anglų kalba, išskyrus tuos atvejus, kai vietiniai įstatymai tiesiogiai numato kitokius reikalavimus arba vietinių lygiu sutarta kitaip.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL

(NO)

Denne servicehåndboken finnes bare på engelsk, bortsett fra dersom det motsatte uttrykkelig er fastsatt av lokal lovgivning eller det er inngått annen avtale lokalt.

- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim, chyba że lokalne przepisy lub umowy wyraźnie stanowią inaczej.

- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

ATENȚIE

(RO)

Acest manual de service este disponibil numai în limba engleză, cu excepția cazului în care este o cerință obligatorie stipulată de legislația națională sau convenită la nivel local.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

Данное руководство по техническому обслуживанию предлагается только на английском языке, за исключением тех случаев, когда наличие руководства на национальном языке является требованием местного законодательства или когда выпуск такого руководства согласован с местным представительством.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

Tento návod na obsluhu je k dispozícii len v angličtine, okrem prípadov, kedy tak výslovne vyžadujú miestne zákony alebo je dohodnuté na miestnej úrovni.

- Ak zákaznikov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION

(ES)

Este manual de servicio sólo existe en inglés, salvo que la legislación local exija de forma expresa lo contrario, o así se haya acordado a nivel local.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

Den här servicehandboken finns bara tillgänglig på engelska om inte annat uttryckligen krävs av lokal lag eller har överenskommit på lokal nivå.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT

(TR)

Aksi, yerel bir yasa tarafından açıkça gerekli görülmediği veya yerel bir seviyede kabul edilmediği takdirde, bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

AVISO

(PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês, salvo disposições em contrário previstas pela legislação local ou acordadas no âmbito local.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês, salvo requisição expressa pela legislação local ou acordo efectuado a nível local.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

UPOZORENJE
(SR)

Ovo servisno uputstvo je dostupno samo na engleskom jeziku, sem ako lokalni zakon to izričito zahteva ili je dogovoreno na lokalnom nivou.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORENJE
(HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

警告
(ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

警告
(ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

Revision History

Revision	Date	Description
1.0	February 2, 2010	For review approval and validation of this document. See information below. • DOC 0666573, rev 1, review and approval. • DOC 0666573, rev 2, for validation • DOC 0666573, rev 3, for re-validation based on SPR's: Cable map updates (MRIhc46698). E-Off update (MRIhc47230). Update illustrations for new Teal MDP (MRIhc47445). Revise connection point of gradient cable ground wires at rear of magnet (MRIhc47442). I Provide instructions for installation of dongle to SRPS J50 (MRIhc47443). Rear pedestal installation procedure updates (MRIhc46657). Update installation instructions for enclosure covers shipping separately (ECO2097330).

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Chapter 1 Getting Started

1 Overview Information

1.1 System Configuration

This manual provides instructions to complete the installation of an MR450w 1.5T system.

Additional options are installed according to the installation manual delivered with the option.

1.2 Site Ready Check for Mechanical Installation

Before equipment is delivered, the following must be complete to avoid delays and confusion.

- Pre-installation work must be complete. Refer to: *Direction 5670001, Optima MR450w Preinstallation Manual*
- Magnet Delivery and Installation complete. Refer to: *Direction 5500108, GE 1.5T and 1.0T LCC Magnet Delivery and Installation Manual*
- Magnet Subsystem should be installed. Refer to: *Direction 5670004, Optima MR450w Magnet & Cryogen Subsystem*

1.3 Product Delivery Instructions

The “Shipping Document” lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

NOTE: Two or more people are required to accept delivery of equipment due to weight and size of cabinets and other components.

Labels that are attached to the outside of packing boxes summarize box contents.

NOTE: Project Manager Install (PMI): Make sure the right combination of short and/or long cables have been delivered to site.

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI - M7000BL is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1.4 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with:

- Call 1-800-548-3366. Select “Install Support Services for FOA and MIS”.
- Contact your local service coordinator for more information on this process.

1.5 MR System Shipping and Storage Requirements

1. The customer must provide a shipping dock or other staging area for unloading system components.
2. The customer is responsible for unloading and moving all system components from the delivery truck or van to final locations in the MR suite. This includes:
 - Door and other openings are sufficiently wide to allow passage
 - Sufficient room is provided for any required dollies or rigging
3. The MR system must be shipped and stored within the environmental requirements listed below in [Table 1-1](#).

NOTE: Some equipment is liquid cooled. After coolant is added, the equipment must be kept from freezing.

Table 1-1: Transportation and Storage Environmental Conditions for System Components

System Equipment	Temperature Range	Temperature Change	Relative Humidity	Humidity Change	Atmospheric Pressure
Cabinets and System	-22 to 158° F (-30 to 70° C)	68° F/Hr (20° C/Hr)	5-95% Non-condensing	30%/Hr	1060 to 525 hPa
Host Computer1	-22 to 140° F (-30 to 60° C)		8-90% Non-condensing		1060 to 500 hPa
LCD Monitor	-22 to 158° F (-30 to 70° C)		10-80% Non-condensing		1060 to 525 hPa
8-Port Ethernet Hub	-22 to 158° F (-30 to 70° C)		10-90% Non-condensing		1060 to 525 hPa
Magnet	-22 to 158° F (-30 to 70° C)		5-95% Non-condensing		1012 to 525 hPa

1.6 Non-English Label Installation

Components and accessories delivered to this site may have English labels attached to them. For non-English sites, optional multi-language labels have also been supplied.

If required, non-English labels should be attached over the English labels.

- For Surface Coils and Coil Cables - Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi-language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1.7 Product Locator

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for U.S. ONLY, the preferred method of submitting information is the FE Site Verification Web Site at: <http://supportcentral.ge.com/15563>. The FE Site Verification consists of three components that are available on the web from the main menu. They are:
 - Install/deinstall product locator model and serial numbers
 - Add/modify ship to address information
 - Update CARES FE data for primary/secondary FE's

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page - <http://supportcentral.ge.com/15563>

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1.8 Required Tools



FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

Tools to install the MR450w system are listed in below:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1	---	N/A
2	---	Wrecking bar
3	---	Claw hammer, 3/4 lb
4	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5		Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
6		Gradient Cable Crimper, Thomas & Betts TBM5

Item	GE Part Number	Description
		Gradient Cable Cutter, Greenlee 45207
7	46-320273G3 or G4	Non-Magnetic Tool Kit - Universal (See Note 2)
8	46-301450G1	Fiber optic connector repair kit (See Note 2)
9	46-198094P1	Wrist grounding strap
10	---	Volt Meter
11	---	Extension cords, with ground conductor
12	---	Power strip, grounded type, with minimum of five outlets
13	---	Plastic or aluminum flashlight
14	---	Assorted crimp tools.
15	---	Non-magnetic level, 2 foot length minimum
16	---	Non-magnetic tape rule, 12 ft
17	---	Assorted drill bits
18	---	Inspection mirror
19	---	Hobby and utility knives
20		8 Inch Pipe Wrench (2)
Note 1: Supplied as part of Optima.		
Note 2: Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

2 Lock Out Tag Out (LOTO)

2.1 LOTO for the Main Disconnect Panel (MDP)

The power to the entire MR450w system can be removed by locking out/tagging out (LOTO) the Main Disconnect Panel (MDP). Prior to installing any equipment, the steps in this Lock Out / Tag Out (LOTO) procedure must be performed by a LOTO-authorized GE Field Engineer. Completing all the steps in this procedure ensures a safe environment and avoids equipment damage when working on these parts.

Table 1-3: Affected by LOTO

Name of Equipment	Number of Locks	Titles of Employees Authorized to perform LOTO	Titles of Affected Employees	How to Notify
Heat Exchanger Cabinet (HEC)	2 per GE Field Engineer (FE)	GE Field Engineers	Hospital Personnel	Verbal, Posted signs
Power, Gradient, RF Cabinet Power Distribution Unit (PGR)				

2.1.1 Preliminary Requirements

Table 1-4: Tools and Test Equipment

Item	Part #
Personal Lock	46-194427P320
Multi-locking Device (If multiple service people)	46-194427P313
Red Warning LOTO Tag	46-194427P322
Digital Voltmeter	46-194427P284

2.1.2 Safety



⚠ DANGER

**ELECTROCUTION HAZARD!
HIGH VOLTAGE PRESENT.**

**USE PROPER LOCK OUT / TAG OUT PROCEDURES BEFORE INSTALLING,
SERVICING AND/OR MAINTAINING.**

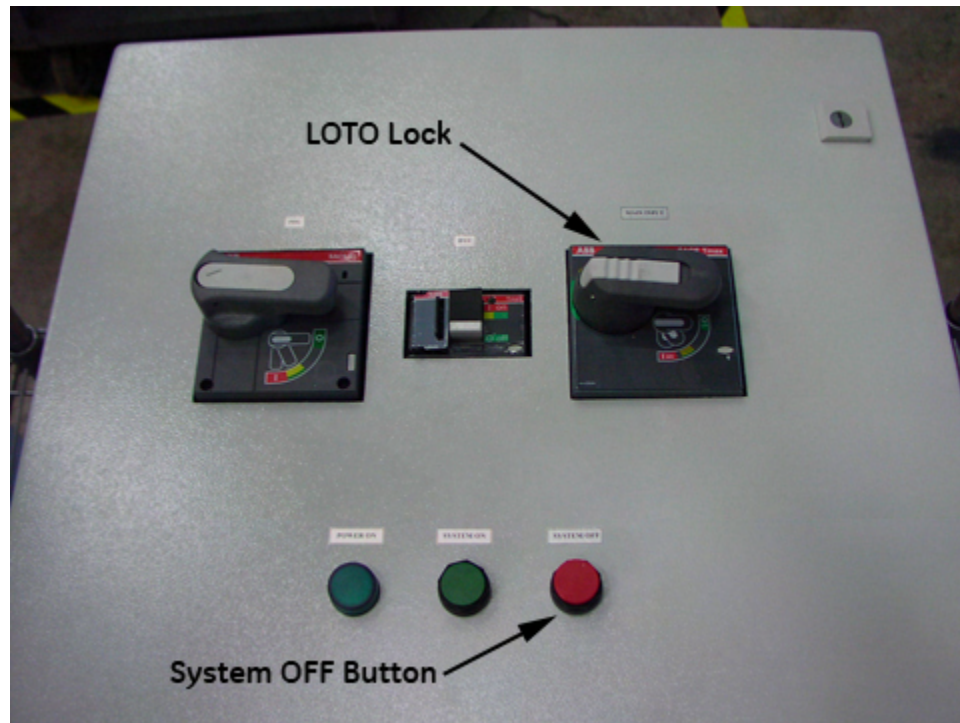
2.1.3 Applying LOTO

1. To prepare for shutdown of equipment, notify affected personnel working in the area the LOTO is being performed.

NOTE: Observe *Power On* light on the MDP. If the light is on, it signifies the main breaker is closed and control power is on.

2. Locate and depress the System Off button on the MDP which feeds the PGR Cabinet and the HEC Cabinet.

Illustration 1-1: Main Incoming Switch on MDP with LOTO lock

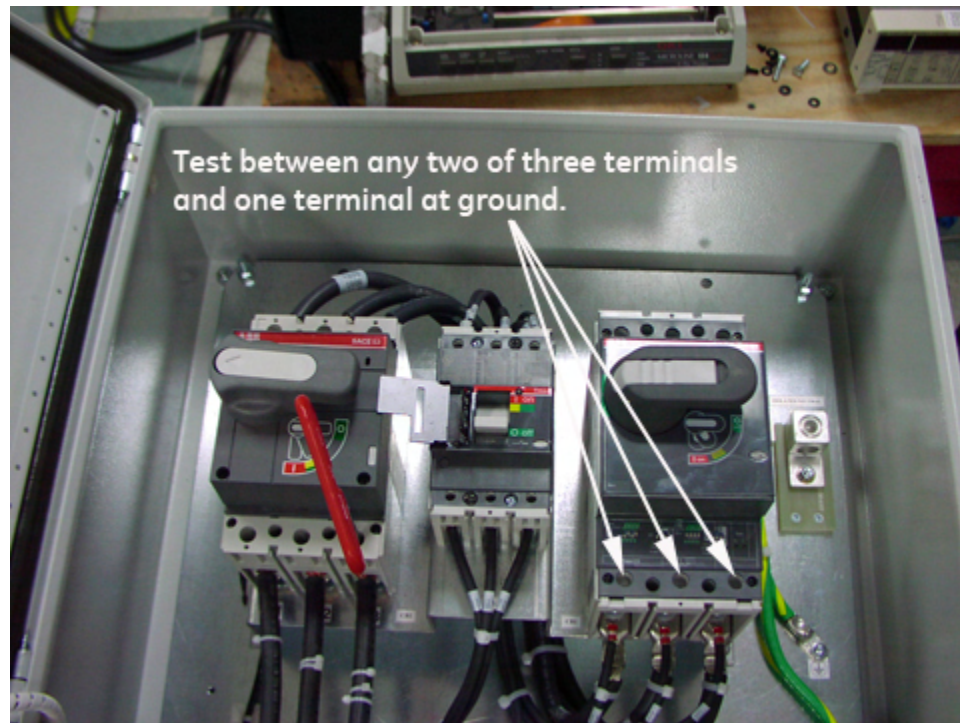


3. Turn the Main Input Switch to the OFF position.

NOTE: Observe *Power On* light on the MDP. The light should now be off.

4. Open MDP door and check that all energy has dissipated. Use a voltmeter to measure the voltage between each **INPUT** phase voltage test point and **GND** (Ground). Reading should be zero.

Illustration 1-2: MDP Test Points



5. Lock Out/Tag Out the main incoming switch on the MDP.

**NOTICE**

Powering off the HEC circuit breaker will cause the Cryogen Cooler Compressor to also lose power. Make sure that the HEC downtime is minimized to avoid boiling off Cryogen.

2.2 Removing LOTO

Refer to MR450w Service Methods DVD for procedures to remove LOTO.

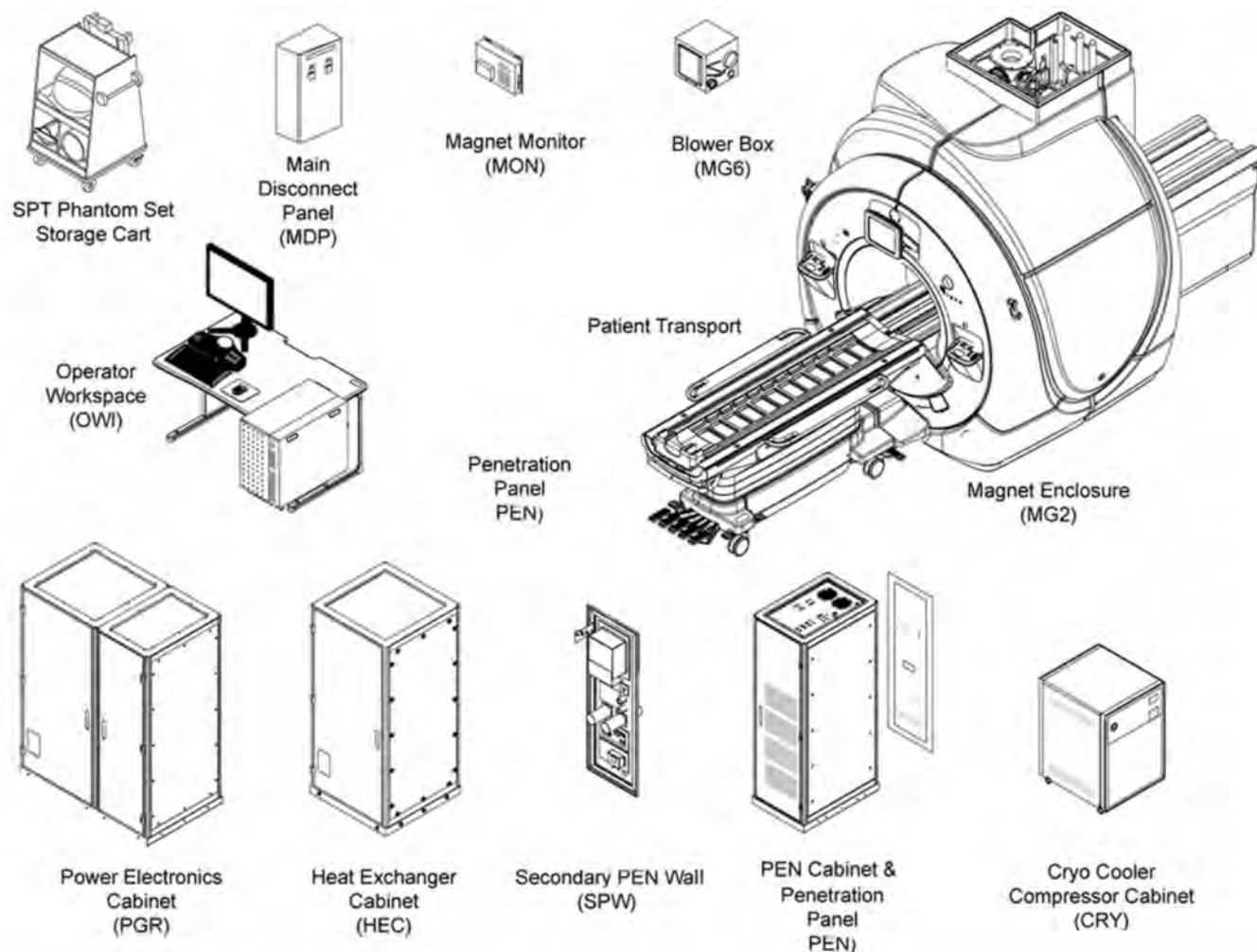
3 System Components

3.1 Basic System

The basic system, Illustration 1-1, consists of the following major equipment:

- The 1.5T (15 kilogauss) Magnet with Magnet Enclosure and Magnet Accessories
- Shield/Cryo Cooler Compressor Cabinet
- Gradient (XRMW) and RF body coils and Head Coil
- Main Disconnect Panel
- System electronics cabinets:
 - PGR Cabinet containing XGD Gradient drivers and Power Distribution Unit module with unregulated transformer 380/400/415/480 Volt, 50/60 Hz with power filter, CAM Chassis with boards for either 16 or 32 channels, components for the Volume Reconstruction Engine (VRE), Narrow Band RF Amplifier
 - HEC Cabinet containing the Cryo Coolers supply, Cooling Loops, and Gradient Blower.
 - PEN Cabinet containing the PHPS2, Scan Room power supply, CAN Fiber I/F, T/R DD Driver module, and Penetration Panel
 - SPW (Second Penetration Wall) containing Gradient Filter, Scan Room Filters & I/F, and Ramp/Shim access
- Operator Workspace equipment: GOC Computer Cabinet (containing Linux PC and GOCAA), LCD Widescreen monitor, keyboard, SCIM, mouse and mouse pad, and chair
- Pneumatic Patient Alert System
- Patient Transport Table and cradle
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows

Illustration 1-3: Optima MR450w System



3.2 System Options

This section lists options for the Optima MR450w system that have site preparation impact.

- Advantage Workstation (AW). NOTE: LCD monitor is recommended for AW to be located near the system Operator Workspace due to gauss field proximity limits.
- Bar Code Reader for use in conjunction with HIS/RIS software option.
- Oxygen Monitor
- BrainWave option
- Additional Patient Transport table with cradle.
- Other hard copy devices and patient accessories.

4 Installation Process

4.1 Installation Procedure

The following flow chart is a guide for an orderly and efficient installation of the Optima MR450w system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

NOTE: All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

4.1.1 Installation Flowchart Explanation

This installation flowchart has been developed assuming that all MR450w equipment has been delivered complete. Make sure that every part required is available before starting.

NOTE: These procedures are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The Procedures must be returned to the binder for future reference upon completion of the installation.

The flow chart references the section in this manual for completion. The columns are defined as follows:

- **Step** - The recommended order of installation procedures
- **Completion Check** - This box **MUST** be checked after completion of procedure per ISO 9001 requirements.
 - System - This box spans the entire column. Mark with a check (✓) when completion of a **System** procedure
 - Eq - This box spans half the column. Mark with a check (✓) when completion of a **Equipment Room** procedure
 - Mag - This box spans half the column. Mark with a check (✓) when completion of a **Magnet Room** procedure
- **System** - Installation procedures that are not specific to the Equipment or Magnet rooms but usually have to be completed before proceeding to the next step
- **Equipment Room** - Installation procedure to be performed in the equipment room or operator workspace area
- **Magnet Room** - Installation procedure to be completed in the magnet room

4.1.2 Installation Flowchart

Table 1-5: Installation Flowchart

Step	Completion Check		System	
	System		Equipment Room	Magnet Room
	Eq	Mag		
PREREQUISITES FOR SYSTEM INSTALLATION				
1			Direction 5500108, <i>Magnet Delivery and Installation</i> completed	

Step	Completion Check		System		
	System		Equipment Room	Magnet Room	
	Eq	Mag			
2			Direction 5670004, <i>Optima MR450w 1.5T Magnet and Cryogen Subsystem</i> , Section 1, completed		
3			Customer supplied water cooling installed		
4			Magnet room exhaust fan installed and operational		
5			Magnet room cryogen vent to building installed		
6			Main Disconnect Panel (MDP) installed		
7			Site electrical wiring and HVAC installed and operational		
8			SPW installed		
9			HEC cabinet installed with customer supplied power and cooling connected		
GETTING STARTED					
10				Section 1.3, Product Delivery Instructions	
11				Section 1.4, Damage in Transportation	
12				Section 1.5, MR System Shipping and Storage Requirements	
13				Non-English Label Installation	
14				Product Locator	
15				Section 1.8, Required Tools	
16				Before starting installation, complete Lock Out Tag Out (LOTO)	
POSITIONING OF EQUIPMENT					
17			In Chapter 2, Cabinet Positioning , complete Cabinet Installation Requirements through Install Penetration Panel in RF Screen Room Opening	Complete Chapter 4, Install Blower Box on Penetration Panel	
18			In Chapter 6, Operator Workspace , complete Operator Workspace Table Anchor Installation and Chapter 6, Section 1.2, GOC Installation		
CABLE AND SYSTEM INSTALLATION					
19			Move lean cable carts into designated rooms Sorting Cables		
20				Chapter 4, Rear Magnet Enclosure Cable Cover Test-Fit	
21				Chapter 4, Cryogen Line Positioning on Magnet	
22			Route gradient cables in Equipment room DO NOT CONNECT at this time.	Route gradient cables in Magnet room DO NOT CONNECT at this time.	
23			Route ground cables DO NOT CONNECT at this time. Chapter 4,	Route and connect magnet ground cable Chapter 4,	
24			Route power cables DO NOT CONNECT at this time. Chapter 4,	Position rear pedestal at back of magnet Chapter 4,	
25			Route all cables to designated end points between cabinets, SPW, and OW. DO NOT CONNECT at this time. Chapter 4, Overhead Cables, RF Cables, Chapter 4, Data Cables, OW Cables	Route all cables to designated end points PEN Panel, SPW, and magnet & rear pedestal components. DO NOT CONNECT at this time. Chapter 4, Overhead Cables, RF Cables, Data Cables	

Step	Completion Check		System	
	System		Equipment Room	Magnet Room
	Eq	Mag		
26			Move cabinets to final position PEN cabinet and PGR cabinet	
27			Complete PDU set-up in PGR cabinet Chapter 3 ,	
28			Connect ONLY the ground and power cables. Refer to Chapter 4, Route and Connect Ground Cables , Chapter 4, Route and Connect Equipment Room Power Cables and Power and Ground Connections	
29			Perform <i>Ground Resistance Checks</i> . Instructions found on applicable MR450w Service Methods DVD at Functional Checks>Grounding & Leakage Current>Ground Resistance Checks	
30			In Chapter 4, Gradient Cable Installation , complete Equipment Room Gradient Cables for connection of cables between PGR cabinet and SPW.	In Chapter 4, Gradient Cable Installation , complete Magnet Room Gradient Cables for connection of cables between magnet and SPW.
31			In Chapter 4, RF Transmit Cable Installation , complete for routing and connecting RF cables in Equipment room	In Chapter 4, RF Transmit Cable Installation , complete for routing and connecting RF cables in Magnet room
32			Complete connection of cables in Equipment room. Refer to Chapter 4, Route and Connect Equipment Room Data Cables	Complete connection of remaining cables in Magnet room. Refer to Chapter 4, Route and Connect Cables in Magnet Room
33			Complete connection of cables to Operator Workspace. Refer to Section 5 in Operator Workspace	
34			Complete the procedures in Chapter 4, Install Air and Water Hoses for both the Equipment and Magnet rooms	
35			Complete the procedures in Chapter 4, Install Fiber Optic Cables for both the Operator Workspace, Equipment room and Magnet room	
36			Complete Chapter 4, Pneumatic Patient Alert Installation	
37			Complete remaining installation procedures in Chapter 6, Operator Workspace	
38			Complete Chapter 4, Install Temperature/Humidity Sensor Components on PEN Cabinet	Complete either Chapter 4, In Room Display Installation OR Chapter 4, LED Display Installation
39				Complete Chapter 4, Dock Installation
40			Complete Chapter 4, Install RF Door Switch Run E3014	Complete Chapter 4, Bridge & Rear Pedestal Installation
41				Complete Chapter 4, LPCA Installation
42			Complete Chapter 4, Emergency Off Connections	Complete Chapter 4, SRI Connections
43				Complete Chapter 4, Route and Connect SPW Cables
44				Complete Chapter 4, Route and Connect Penetration Panel Cables

Step	Completion Check		System	
	System		Equipment Room	Magnet Room
	Eq	Mag		
45			Complete <i>MRU Installation</i> . Refer to Direction 5670004, MR450w Magnet & Cryogen Subsystem manual for installation procedures.	
46				Complete Chapter 4, Installation of Magnet Enclosure Rear Covers
47				Complete Chapter 4, Coil and Dress Cable in PEN Panel Closet
48			In Chapter 7 , complete Mechanical Installation Completion Checklist	
49				Complete Chapter 5, Install Enclosure Covers
50				Complete Chapter 5, Install Laser Alignment Light Warning Label
51			Complete Chapter 5, Install Hearing Loss Warning Sign	Complete Chapter 5, Install Patient Transport
52			Complete Returning Shipping Material	
53			Complete Final Mechanical Installation Steps	
54			Complete Chapter 7, GE Field Engineer Responsibilities	
Mechanical Installation Complete				

4.2 System Start Up

Refer to the applicable MR450w Service Methods DVD for the following procedures:

1. Power On
2. LCD Monitor Screen Adjustments
3. Host Operating System Software Load
4. Either of the following:
 - Class M procedures:
 - Software Option Install with eLicense
 - Installation Calibration Wizard
 - OR
 - Class A and C procedures: Install Flow - MR450w System
5. Turn site over to Applications who will instruct the users

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Chapter 2 Cabinet Installation

1 Cabinet Positioning

1.1 Cabinet Installation Requirements

Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)
- For crane rigging, contact your local field service engineer or Project Manager Installation, PMI.



Damage to Cabinets

Equipment room cabinets must not be installed in an area with magnetic fields higher than 50 gauss.

Refer to site layout plans for positioning of cabinets.



POTENTIAL INJURY HAZARD CABINET WEIGHT

**DUE TO WEIGHT OF CABINETS, AT LEAST TWO PEOPLE ARE REQUIRED TO
MOVE ANY CABINET**

1.2 Cabinet Positioning Preparation

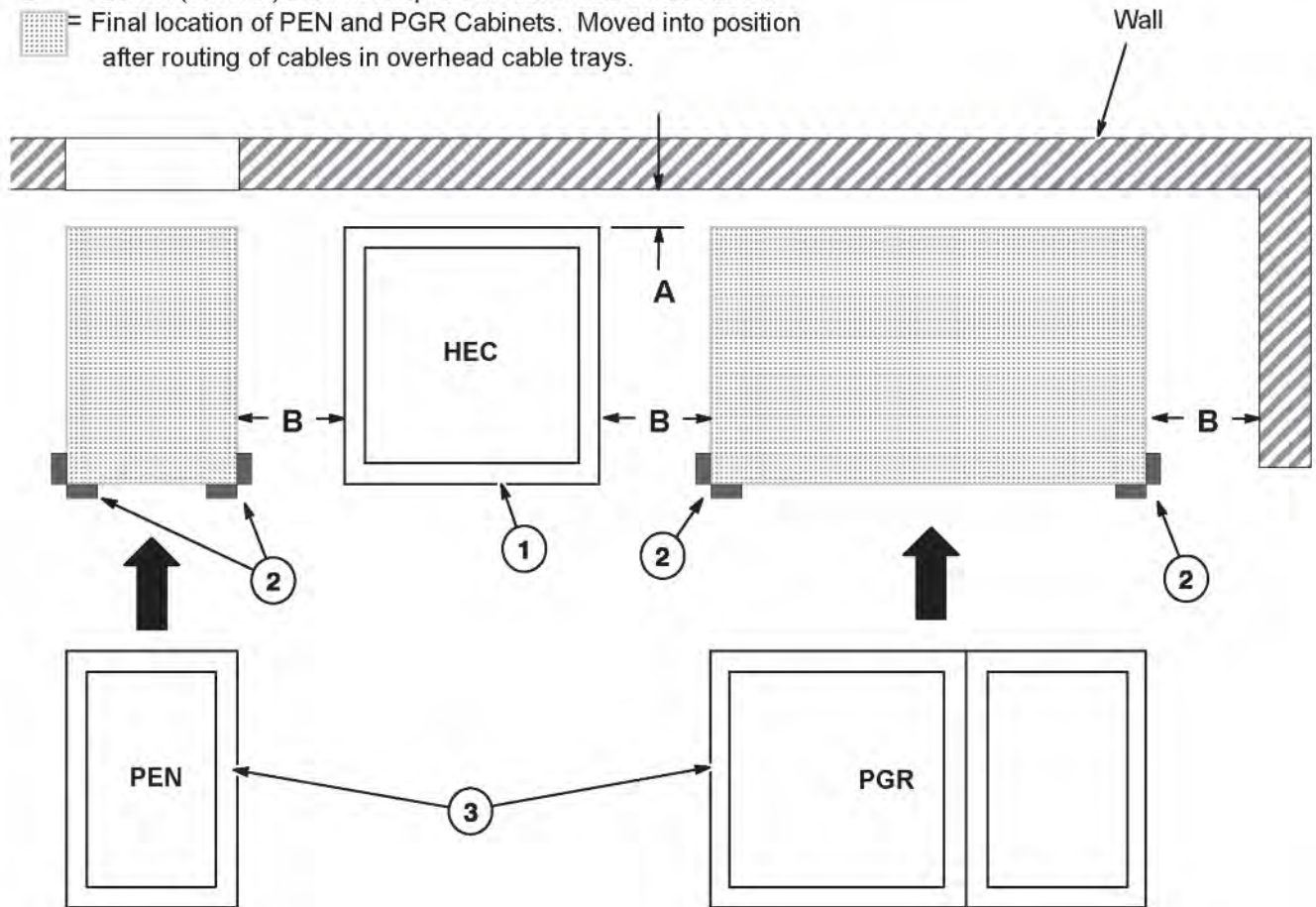
The following illustration shows a typical layout for the equipment room. For ease of cable installation in overhead cable trays, it is recommended that cabinets NOT be positioned in final location until after cables have been routed.

1. The HEC cabinet has already been positioned and installed prior to delivery of PEN and PGR cabinets.
2. Refer to site layout drawings to determine location of PEN and PGR cabinets. Use the minimum spacing distances as noted below, apply tape on floor to mark final location of front corners of cabinets.

NOTE: The procedures for removing the dollies from the HEC cabinet and attaching them to the PEN and PGR cabinets are provided in [Section 1.4](#) through [Section 1.9](#).

3. Using the dollies, move PEN and PGR cabinets into temporary position in front of final location.

Illustration 2-1: Initial Positioning of Cabinets

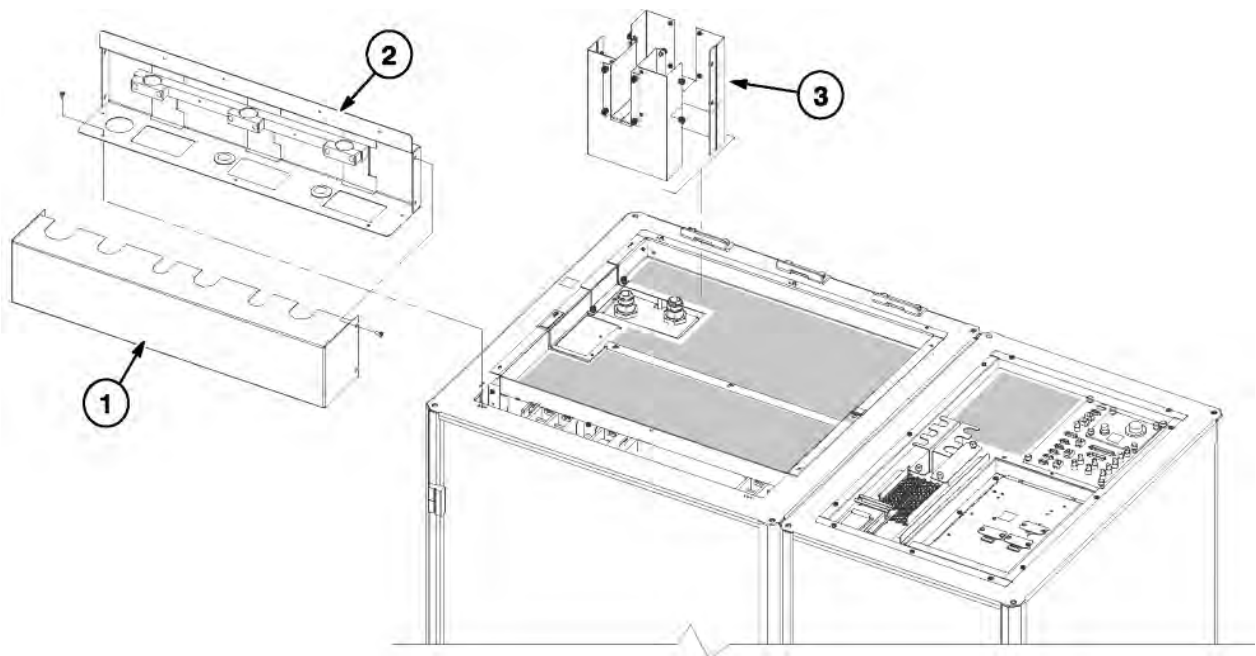
Legend:**A** = 4 inch (102mm) minimum space distance from wall**B** = 15 inch (380mm) minimum space distance at side of cabinet = Final location of PEN and PGR Cabinets. Moved into position after routing of cables in overhead cable trays.

1.3 Removing Components from Top of PGR Cabinet

If the PGR cabinet, with the brackets and covers attached to the top of cabinet, is too tall to go through doorways, then these components will have to be removed. They will be reinstalled after cabinet is placed in final position.

1. Unfasten the four screws that attach the front cover to the cable bracket and remove cover.
2. Unfasten the five screws that attach the cable bracket to the top of the cabinet and remove bracket.
3. If provided on cabinet, lift and remove the protective coolant hose covers. These will be reinstalled after the coolant hoses are connected.

Illustration 2-2: PGR Cabinet Bracket Removal



1.4 Cabinet Moving Dollies

The HEC cabinet has already been delivered to site and installed. The moving dollies that are required for positioning the cabinets should have remained attached to the HEC cabinet.

- If dollies are not on site, they can be ordered online at www.umi-dollyshop.com
- For technical questions, contact: UMI, 1-800-638-7457

The dollies have the following features to aid in their use:

- Wheel Locking Pin - Keeps the wheels locked into a certain direction
- Parking Brakes - For holding the cabinet in position
- Slide and Lock - The dollies can be removed in limited space (15 inches or 380mm)
- Twinning Hardware - hardware to connect each dolly to the other dolly during transport without a cabinet

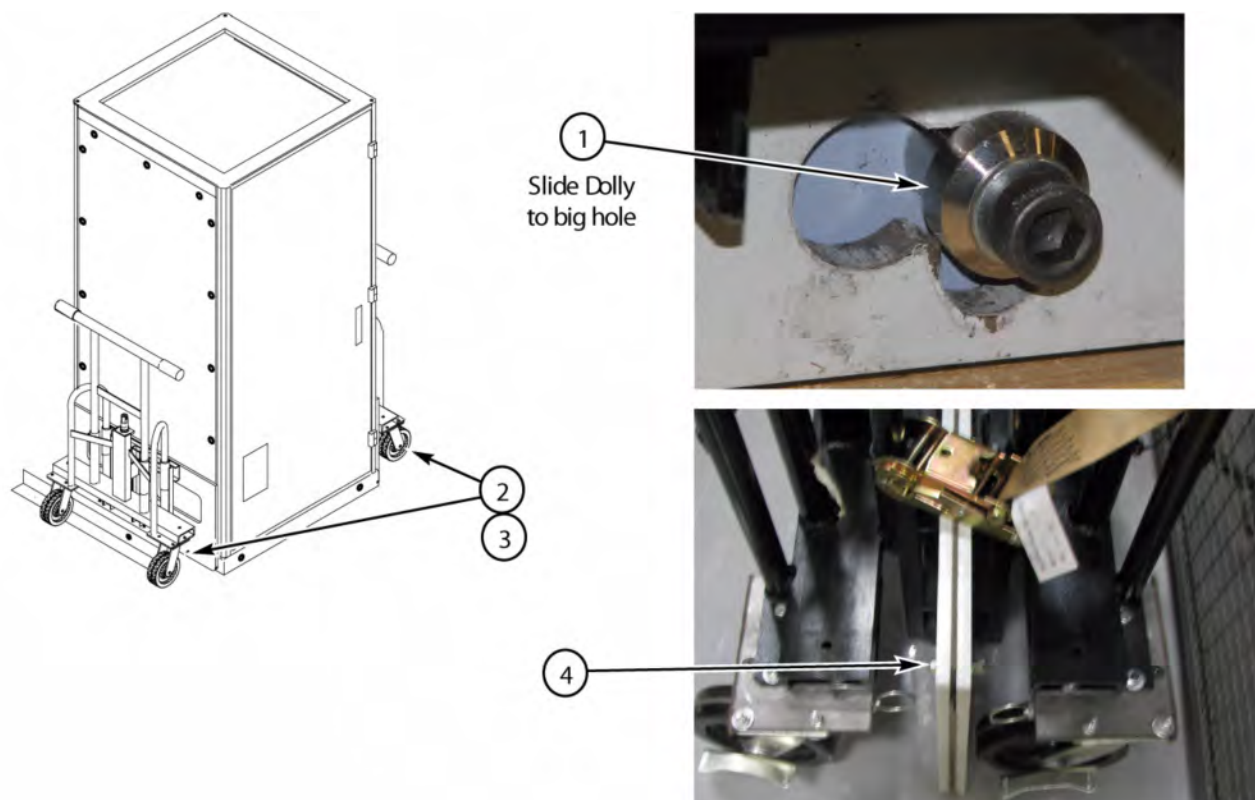
1. The dollies are designed with a slide and lock attachment plate. Slide the dolly until the cabinet bushing lines up with the large opening on the attachment plate.

2. Remove each dolly.

NOTE: Dolly height may have to be adjusted to aid in the removal process.

3. Remove each bushing and place in attached dolly bag.
4. Attach the two dollies together using the twinning hardware, if desired.
5. Move dollies to next cabinet to be positioned.

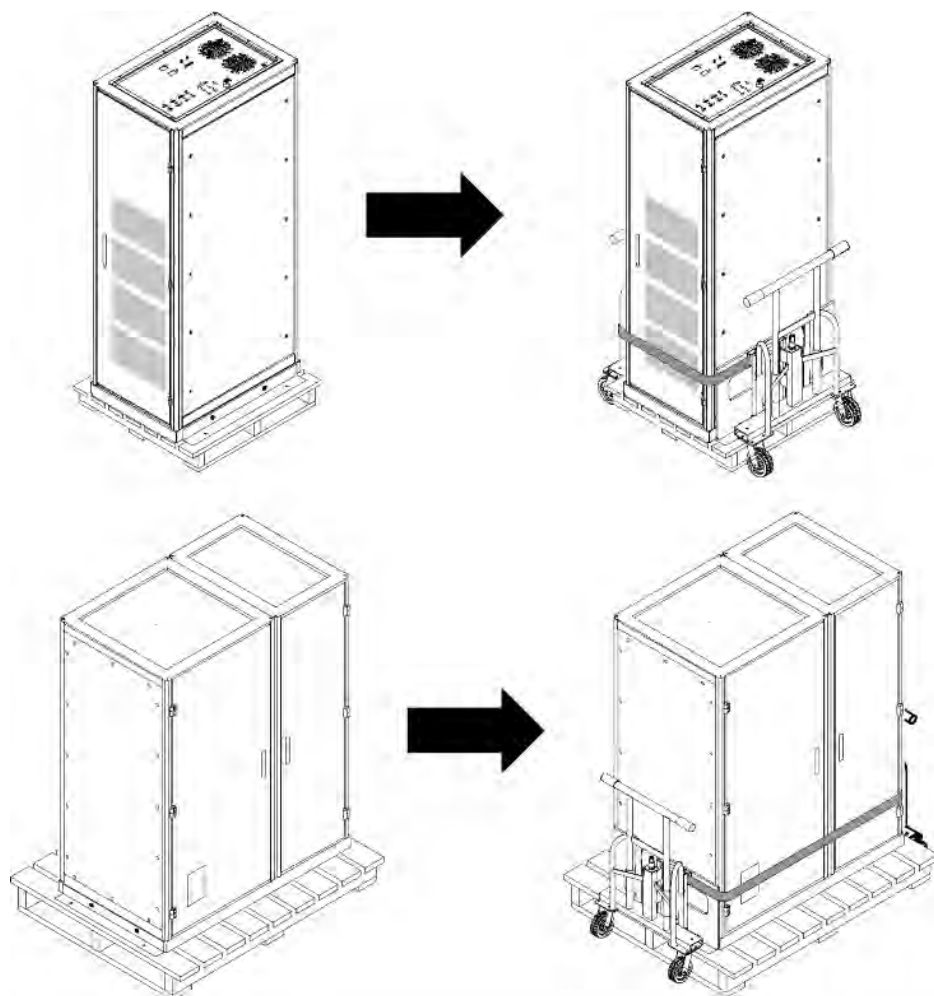
Illustration 2-3: Cabinet Dollies



1.5 Remove Cabinets from Shipping Pallets

The PEN and PGR cabinets are delivered to site secured to pallets. The cabinets must be removed from the pallets and positioned in Equipment Room according to site layout plans. The cabinet moving dollies that were attached to the HEC cabinet, which was delivered and installed prior to magnet delivery, are required to move the PEN and PGR cabinets off the pallets.

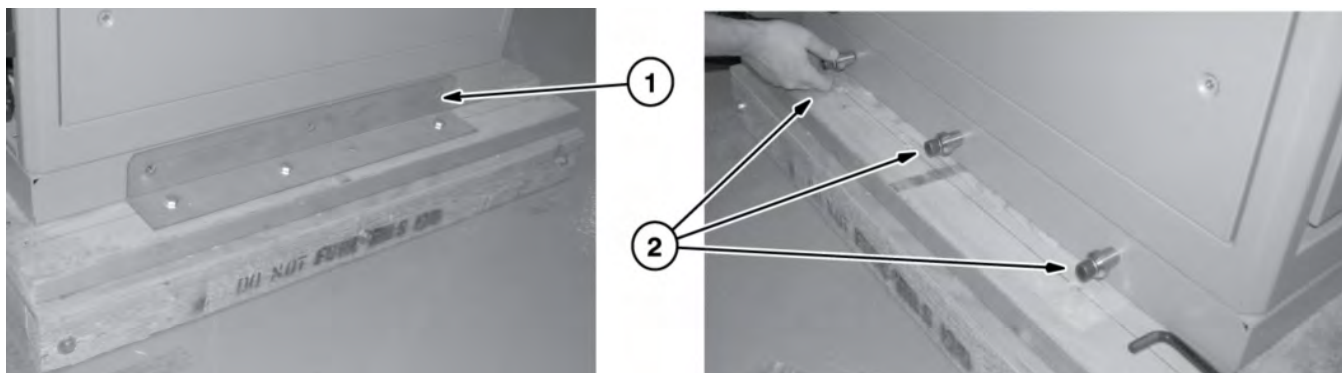
Illustration 2-4: Securing Dollies to Cabinets



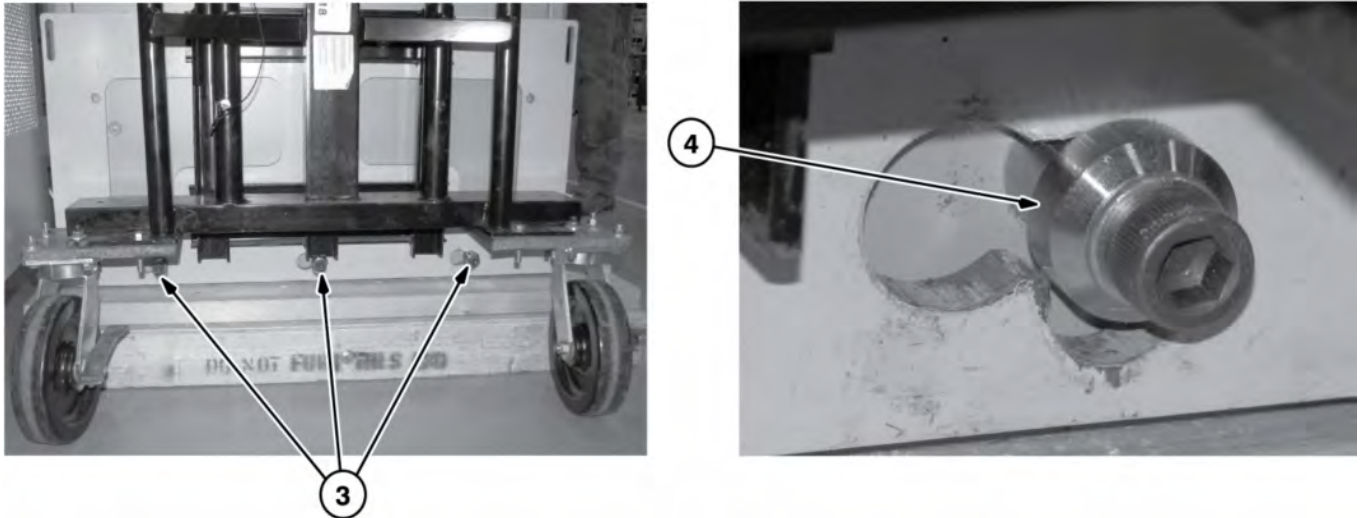
1.6 Move PGR Cabinet into Equipment Room

1. Unfasten screws for each angle shipping bracket and remove bracket from each side of cabinet.
2. Install the six bushings, three on each side, to the bottom of the cabinet.

Illustration 2-5: Installation of Bushings

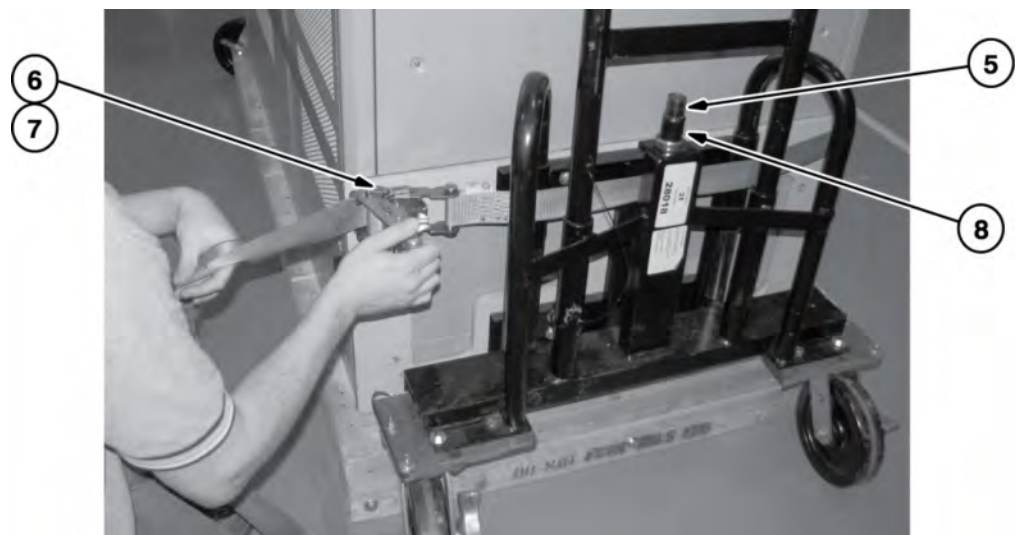


3. Maneuver the dolly so that all three bushings slip into the slide lock on the attachment plate.
4. Slide the dolly sideways to position the bushings in the lifting part of the attachment plate.

Illustration 2-6: Position Dollies on Bushings

5. Raise the dolly by turning the jack screw clockwise. This will lock the dolly into position.
6. Secure to dollies with strap. Make sure the strap goes through the slot of the attachment plate and the clamp is on flat surface of dolly.
7. Tighten strap and close clamp.
8. Lift cabinet off of pallet by turning the jack screw clockwise until the cabinet surface is free from the pallet surface.

NOTE: Alternate raising each dolly in 1/4 inch (6mm) increments until the cabinet is above the pallet.

Illustration 2-7: Secure Dollies to Cabinet

9. Slide the pallet out from under the cabinet.

10. Lower the cabinet by turning the jack screw counter-clockwise until the cabinet is about 1 inch (25mm) above the floor.

NOTE: Alternate lowering each side of cabinet.

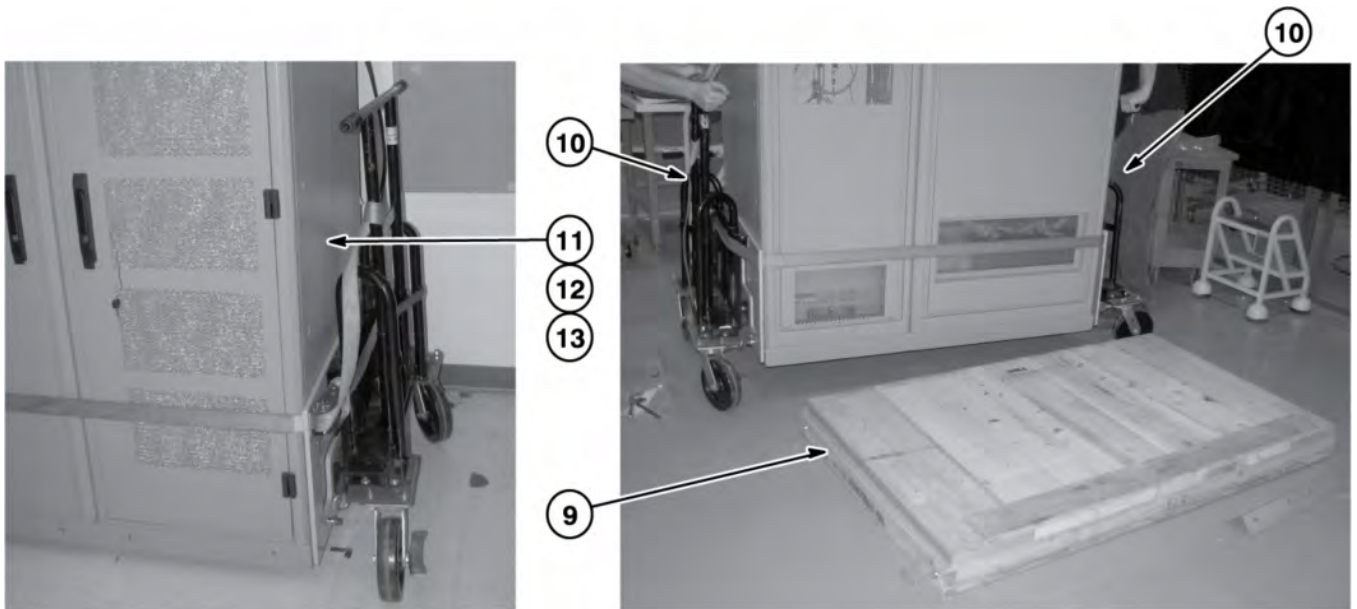
11. Move the cabinet into the equipment room.

12. Refer to [Section 1.2](#) for preliminary placement of cabinet and lower cabinet into position.

NOTE: Cabinet will be moved to its final position after cables have been routed overhead.

13. Remove dollies for attachment to PEN cabinet.

Illustration 2-8: Final Placement of Cabinet



1.7 Move PEN Cabinet into Equipment Room

Repeat the steps in [Section 1.6](#) for removing the PEN cabinet from the shipping pallet and moving it to the temporary position in the Equipment Room.

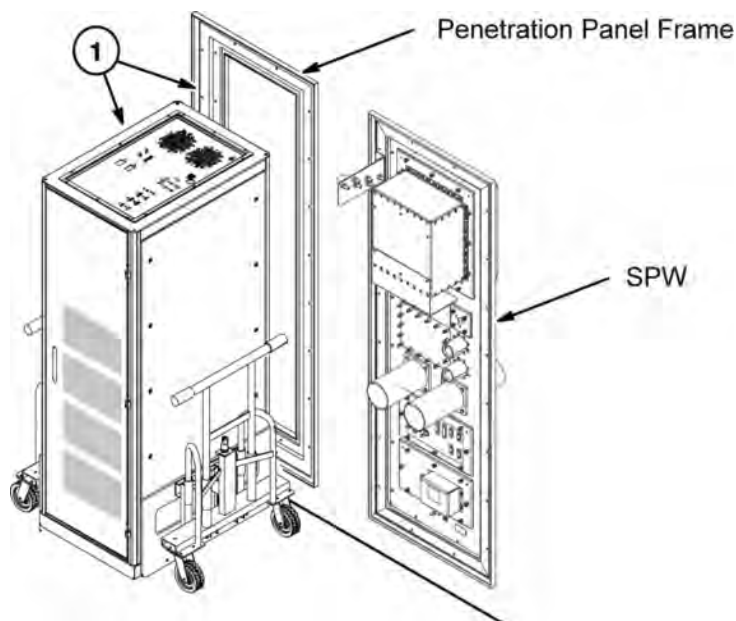
1.8 Install Penetration Panel in RF Screen Room Opening

NOTE: It is recommended that the following steps be completed after routing of cables in overhead cable trays.

1. Move PEN cabinet to within 2 inches (50mm, +/- 25mm) of framed opening in RF room wall.

NOTE: The SPW was previously installed.

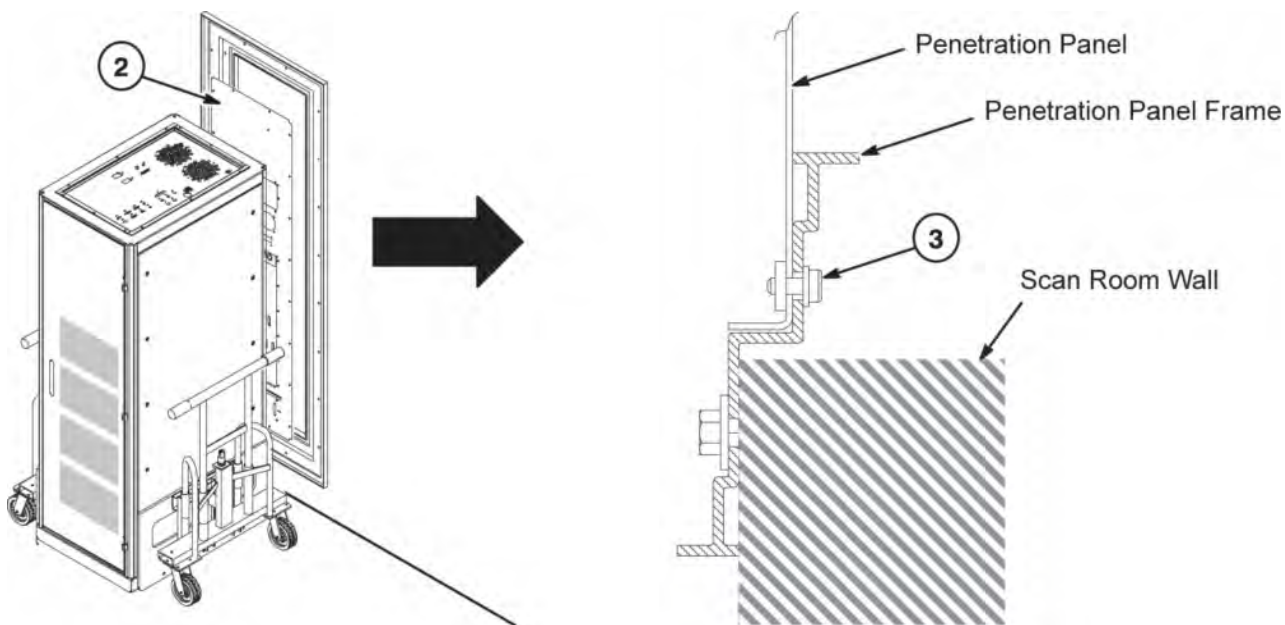
Illustration 2-9: Initial Position of PEN Cabinet



2. Unfasten Penetration Panel from rear of cabinet.
3. Install and secure Penetration Panel into open frame using 20 M5 x 12mm SS Hex screws and 20 M5 SS Flat Washers.

NOTE: The fasteners were delivered to site earlier with the SPW installation.

Illustration 2-10: Install PEN Panel



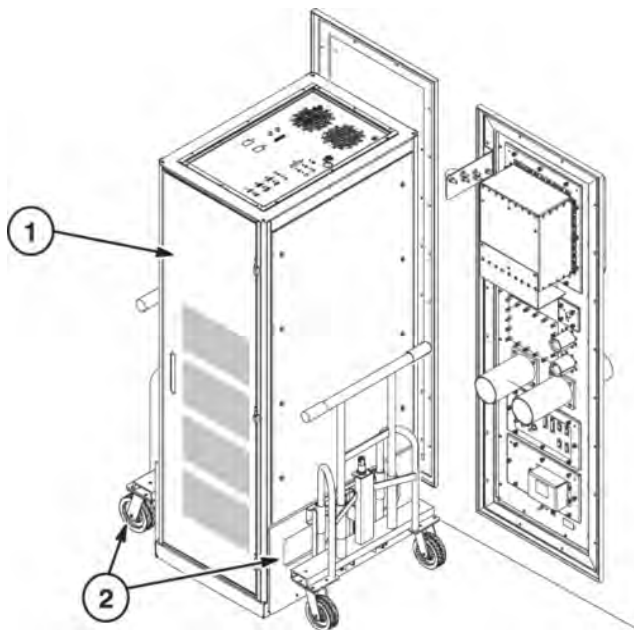
1.9 Final Position of PEN Cabinet

1. Move PEN Cabinet into position and lower cabinet into place.

NOTE: Rear of cabinet must be positioned as close as possible to frame of Pen Panel.

2. Remove cabinet moving dollies. Refer to dolly removal steps in [Section 1.4](#).

Illustration 2-11: Final Position of PEN Cabinet

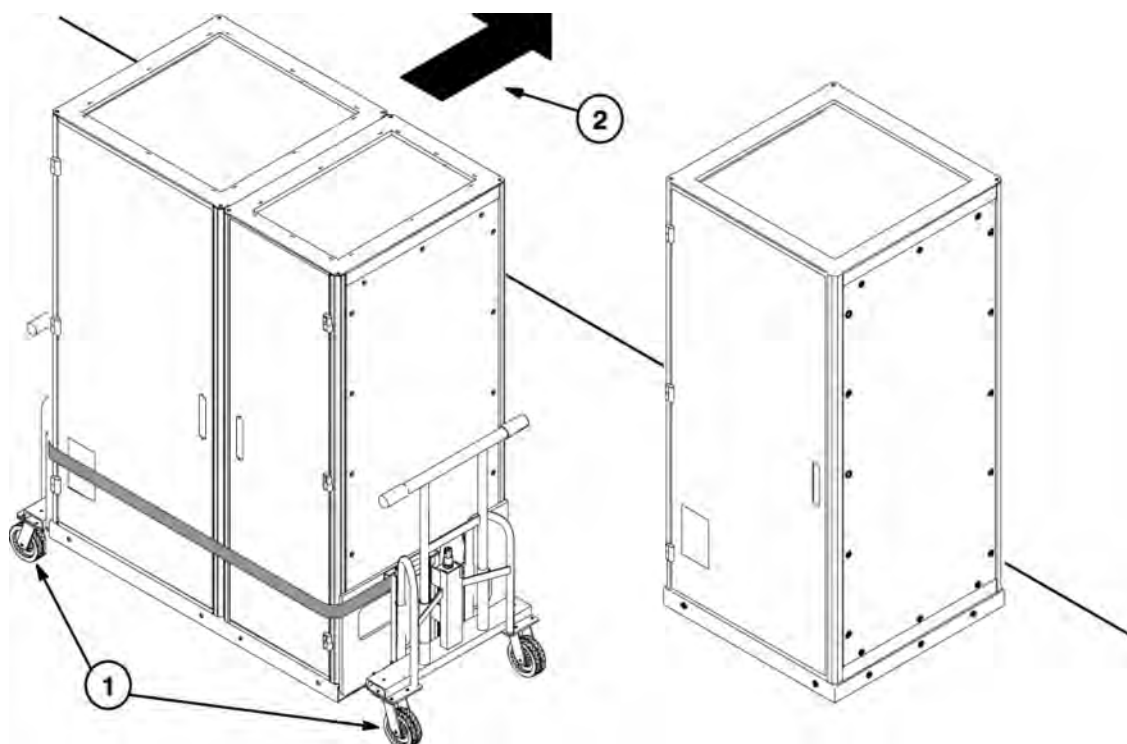


1.10 Final Position of PGR Cabinet

The final positioning of the cabinet is recommended after all the cables have been routed in the overhead cable trays.

1. If dollies are not attached to cabinet, re-attach moving dollies to each side of cabinet
2. Lift and move cabinet to final position as indicated by site layout drawings.

Illustration 2-12: Positioning PGR Cabinet



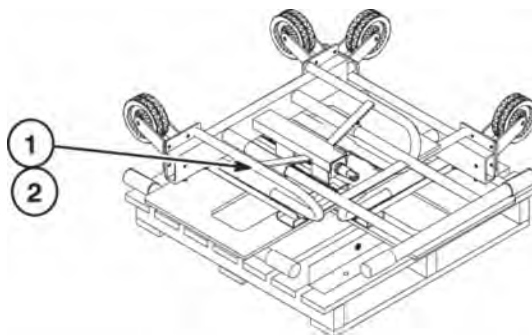
1.11 Return Moving Dollies

Moving dollies must be returned for use on future shipments.

1. Secure dollies to one of the pallets used for shipping cabinets to site.
2. Call Tech-Trans at 1-800-851-9929 to make arrangements for dollies to be returned for next shipment.

NOTE: For sites outside of the United States or Canada, re-use or dispose of dollies per the local requirements.

Illustration 2-13: Secure Dollies to Pallet



Chapter 3 PGR Cabinet PDU Setup

1 PGR/PDU setup

1.1 Input Voltage Selection



⚠ DANGER

ELECTRIC SHOCK HAZARD!

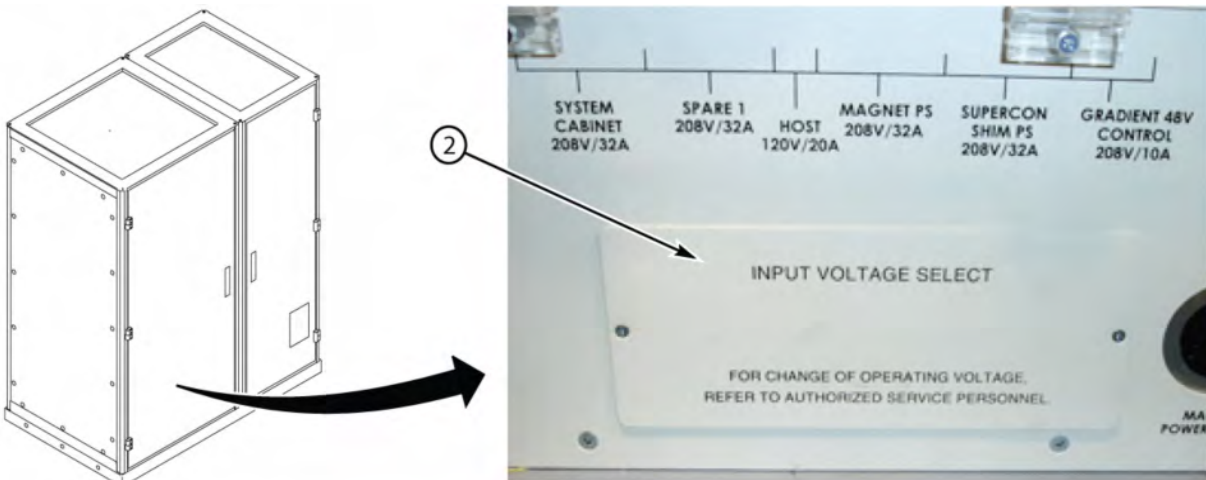
LETHAL VOLTAGES ARE PRESENT IN THE PGR PDU/GRADIENT CABINET

LOCK OUT AND TAG OUT THE POWER AT THE MAIN DISCONNECT BEFORE PROCEEDING. REFER TO LOTO FOR THE MAIN DISCONNECT PANEL (MDP).

The PDU is shipped with the default voltage setting of 480 VAC.

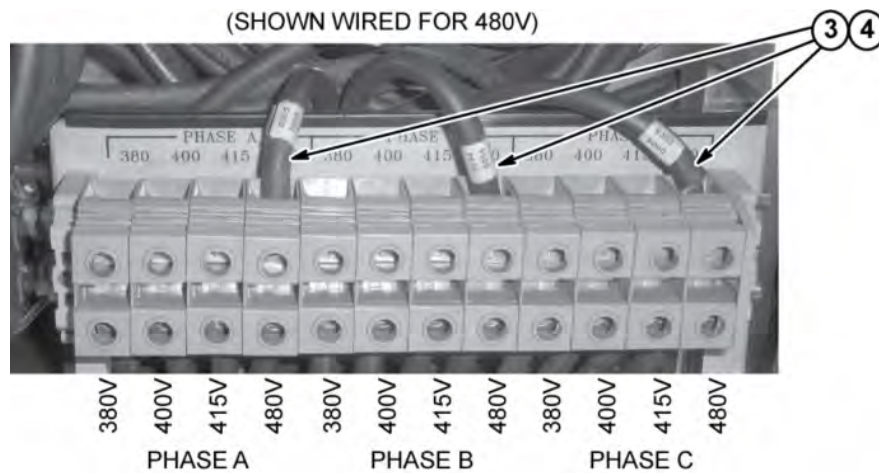
1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT VOLTAGE SELECT.

Illustration 3-1: Remove Cover Plate



3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.
4. If the power connections are not configured properly, make sure that the proper personnel correct the wiring connection. See Teal PDU Installation and Maintenance Manual for additional details.

Illustration 3-2: Power Wire Connections to Terminal



1.2 Circuit Breaker DIP Switch Settings

The overload and short circuit trip settings for the Input Circuit Breaker must be set to the correct values corresponding to the input voltage.

1. Remove the small PDU cover plate marked CURRENT ADJUST under the main circuit breaker which is held in place by a single screw on each side.
2. Lift open the transparent plastic cover on the lower portion of the circuit breaker to give access to the DIP switches controlling the circuit breaker operation.

Illustration 3-3: Access to DIP Switches



The following illustration shows the position of the DIP switches for each input voltage selection. If the dip switches are not configured correctly, make sure that proper personnel correct the dip switches settings after following instructions in the *PDU Operation and Maintenance Manual*.

NOTE: The white portion of the DIP switch indicates the position of the switch.

3. Set the “L” switches according to the input voltage to the left.
4. Leave the “t1”, “S1” and “t2” switches in the DOWN position.

5. Leave the "I3" switches in the UP position.

Illustration 3-4: DIP Switch Setup



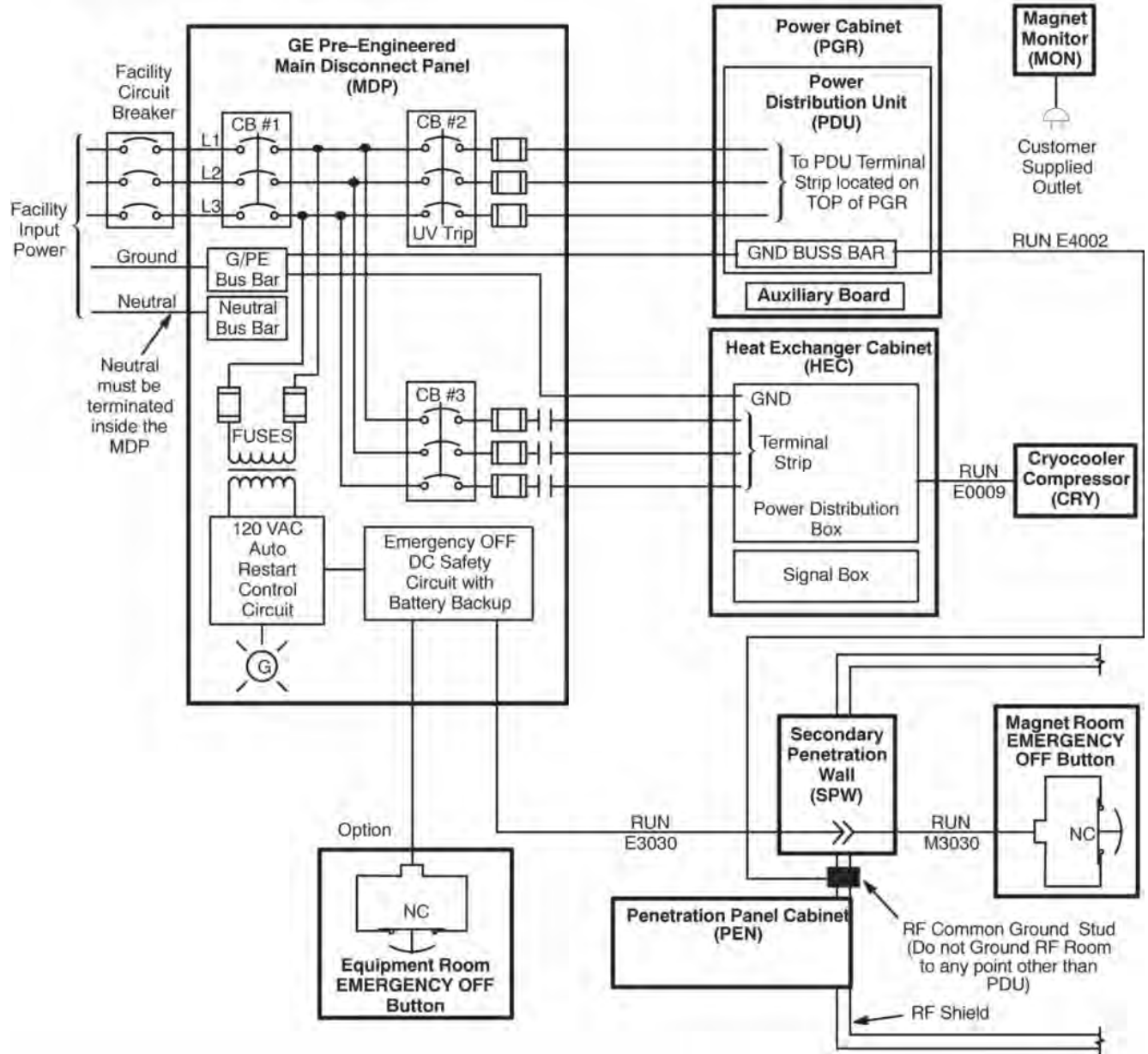
1.3 System Power Connections

All power connections to the Main Disconnect Panel (MDP) and other components should be completed by customer supplied licensed electrician.

- Runs E3030, E0009, M3030, and E4002 are GE supplied. **All other wiring is customer supplied.**
- All circuits require ground wires.

Illustration 3-5: Power Wiring Diagram

- Note:**
- Runs E3030, E0009, M3030 and E4002 are GE supplied cables.
 - **All other wiring is customer supplied.**
 - All circuits require Ground wires.

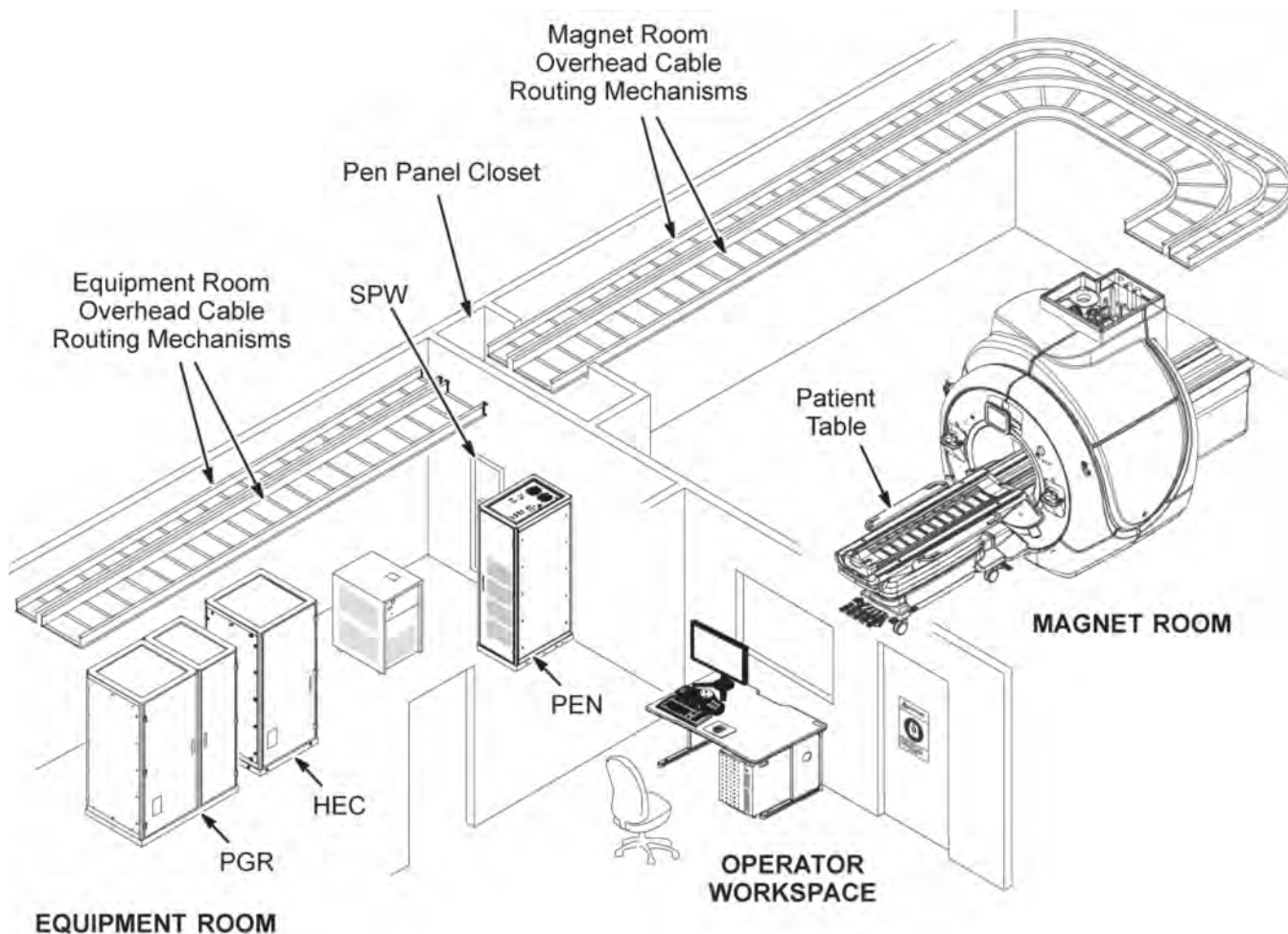


Chapter 4 Cable and System Installation

1 Overview of System Components

The illustration below represents a typical layout for a MR450w system. The procedures on the following pages of this section will provide information on installing cables and completing the attachment of components to the magnet.

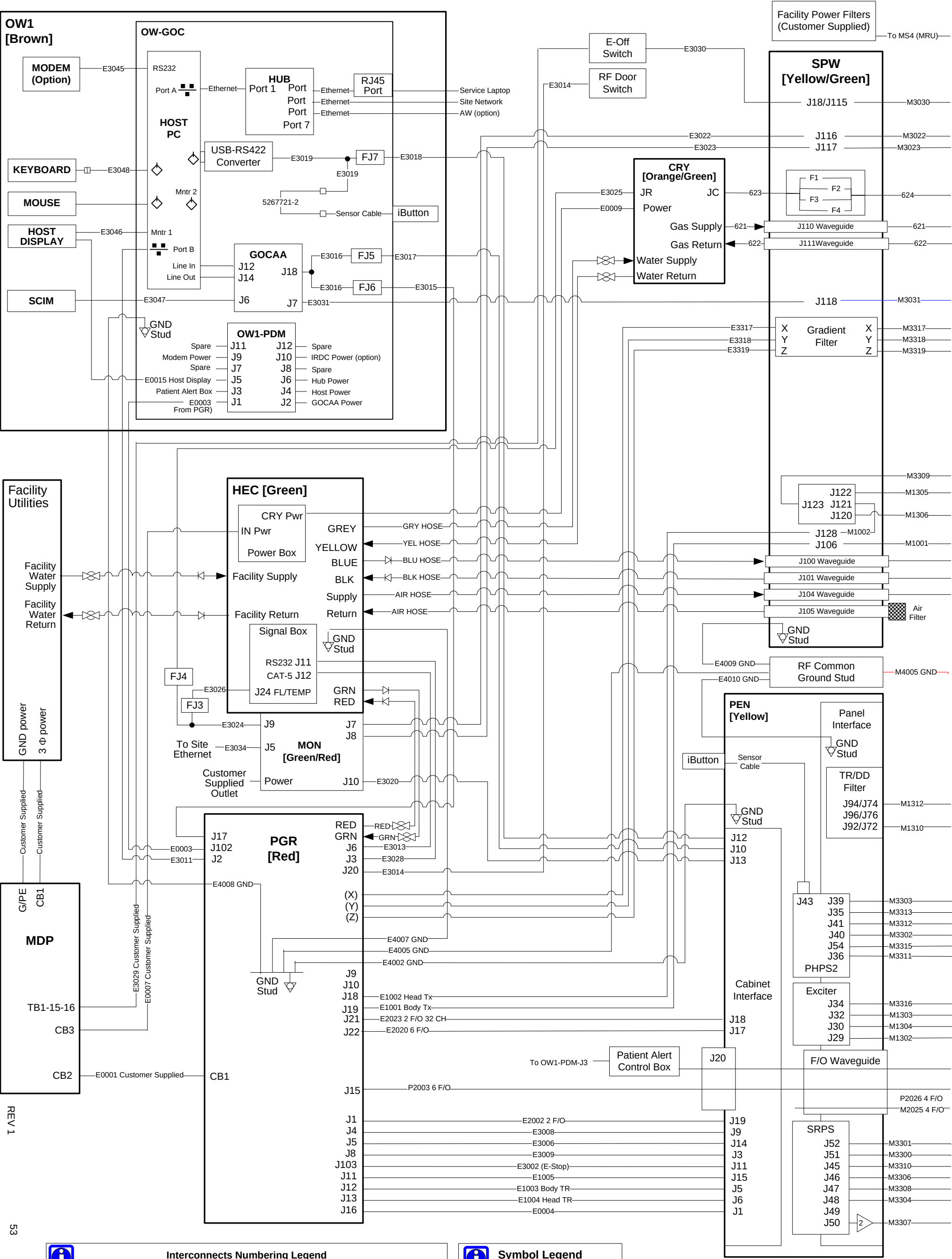
Illustration 4-1: Optima MR450w Room Layout



2 Cable Map Interconnects

Select the applicable Equipment Room and Magnet Room cable map from the following pages for the system installed at the site.

16/0 Channel MR450w 1.5T Interconnect Map with LED Display– Equipment & Control Rooms



REV 1

53

DIRECTION 5670002

Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

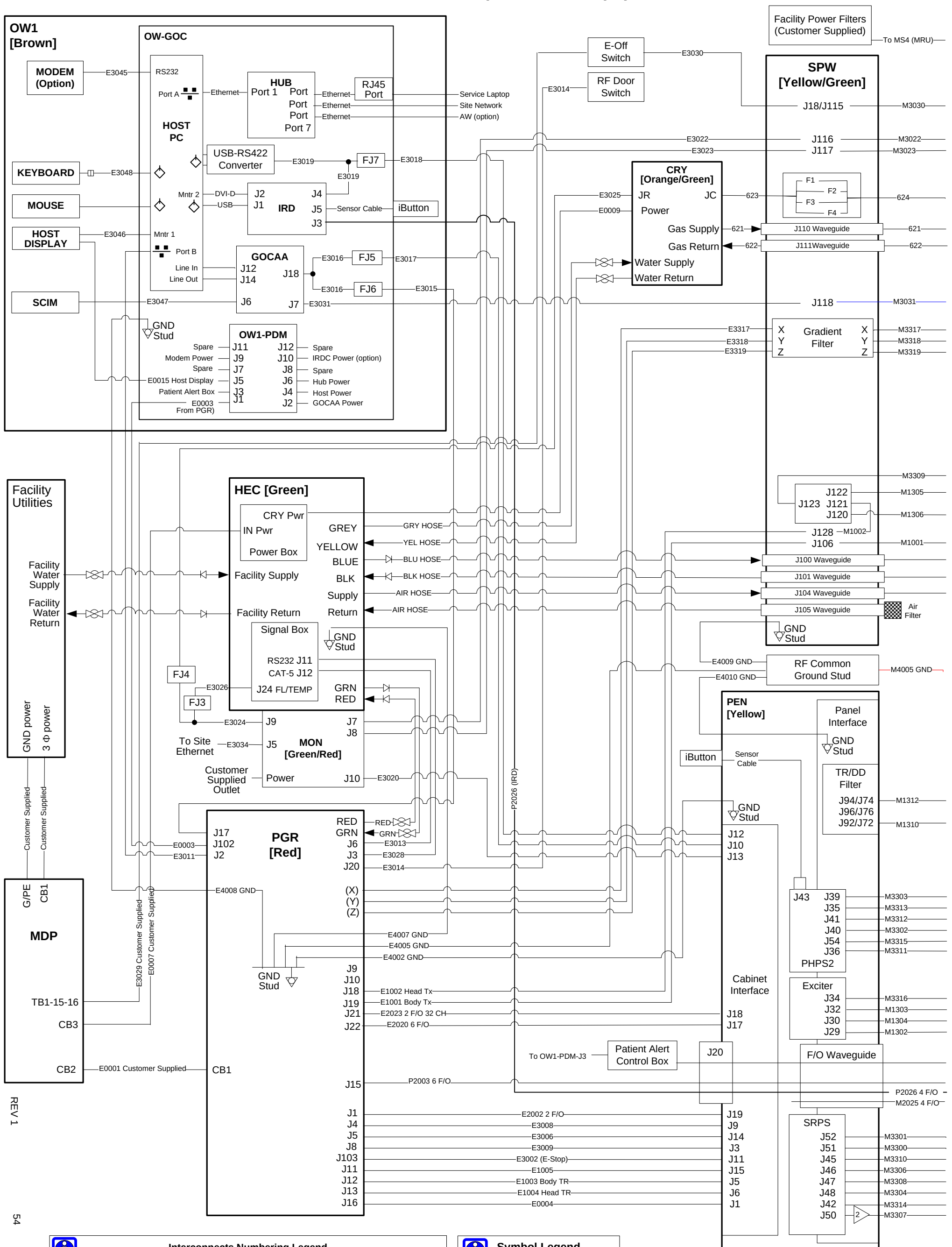
--Dotted lines for module outlines or cables indicate an option

Symbol Legend

- Ball Valve
- Check Valve

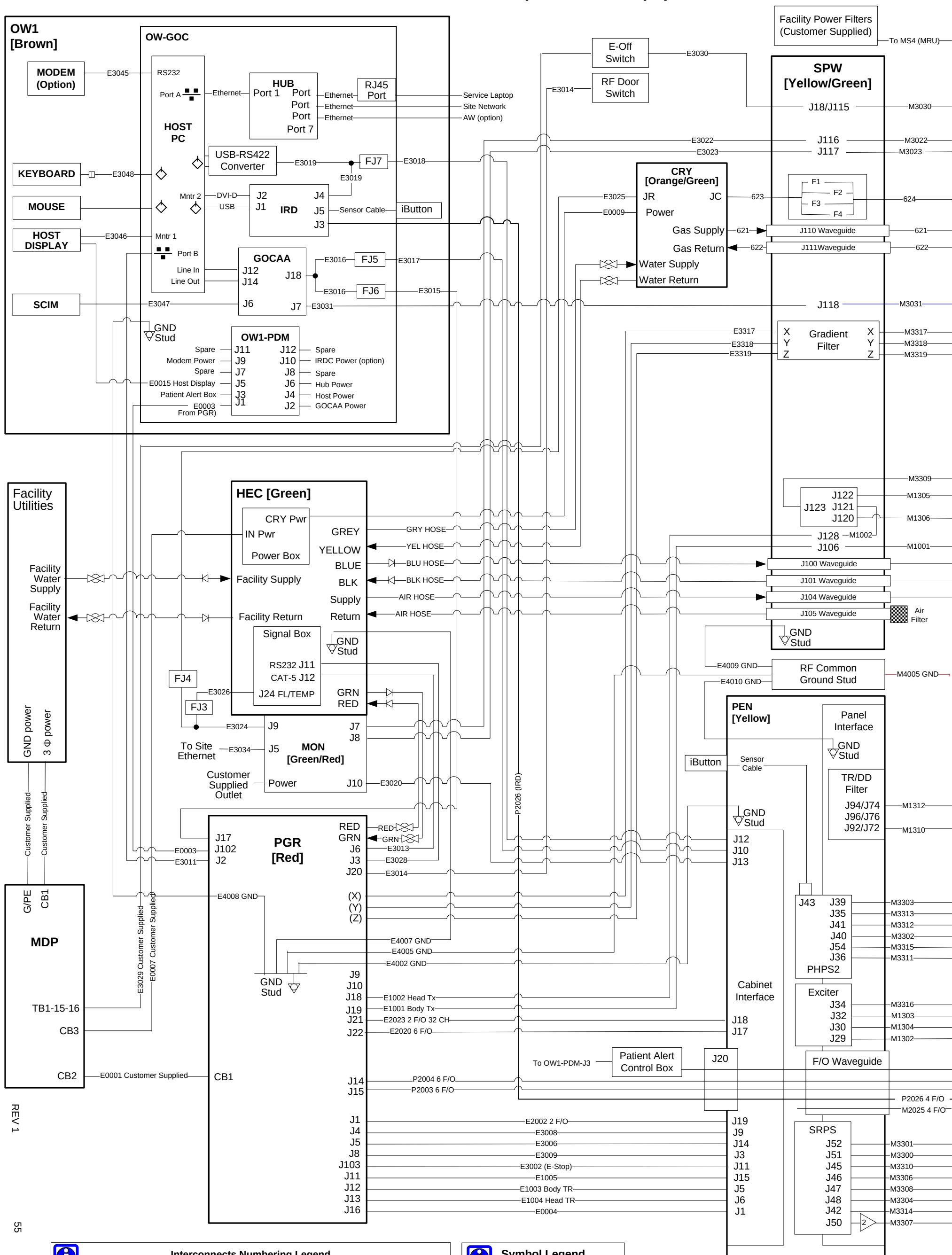
In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

16/0 Channel MR450w 1.5T Interconnect Map with IRD- Equipment & Control Rooms



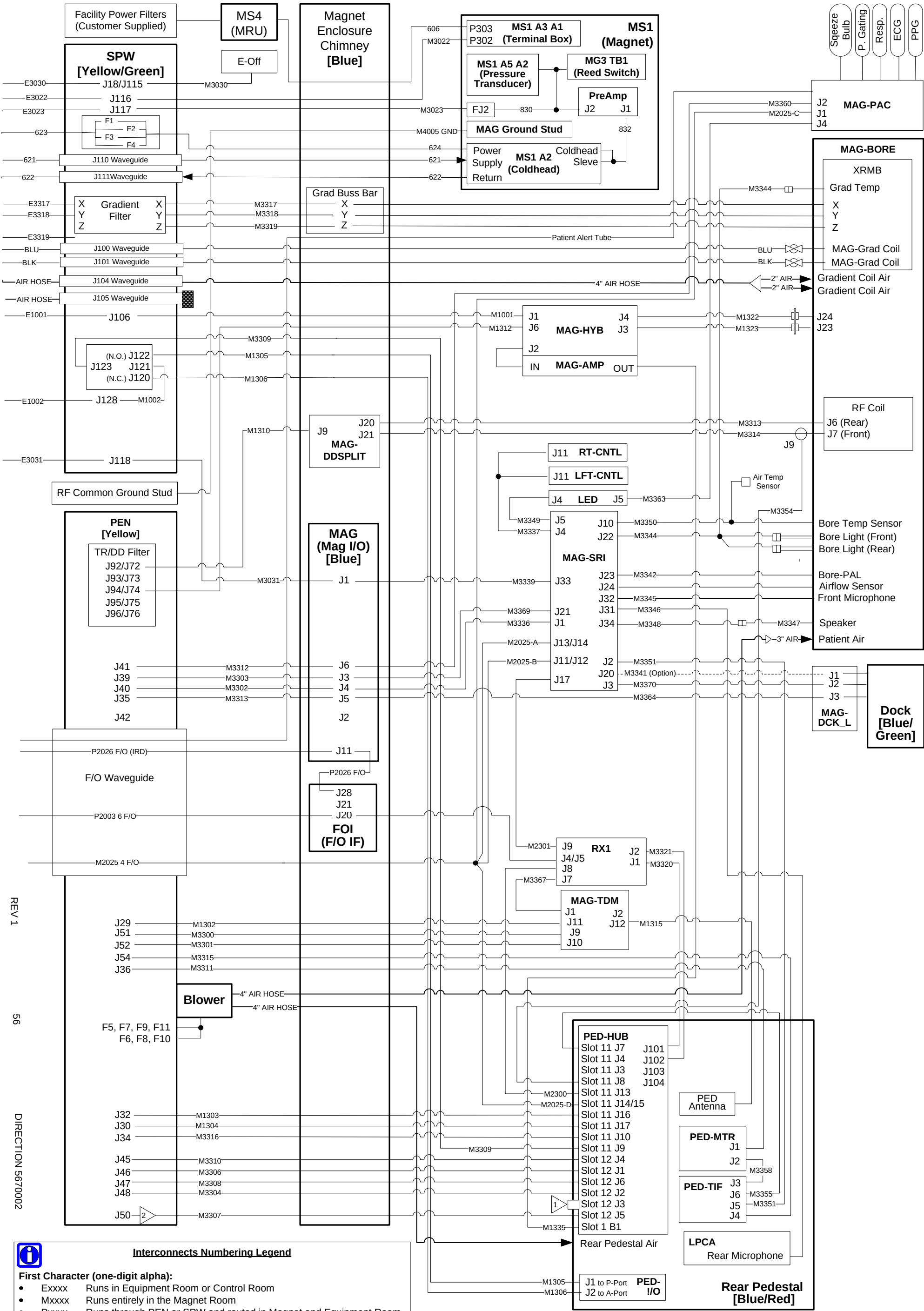
2 In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

32/0 or 32/32 Channel MR450w 1.5T Interconnect Map with IRD- Equipment & Control Rooms



2 In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

16/0 Channel MR450w 1.5T Interconnect Map with LED Display – Magnet Room



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option



Symbol Legend

- Ball Valve
- Check Valve

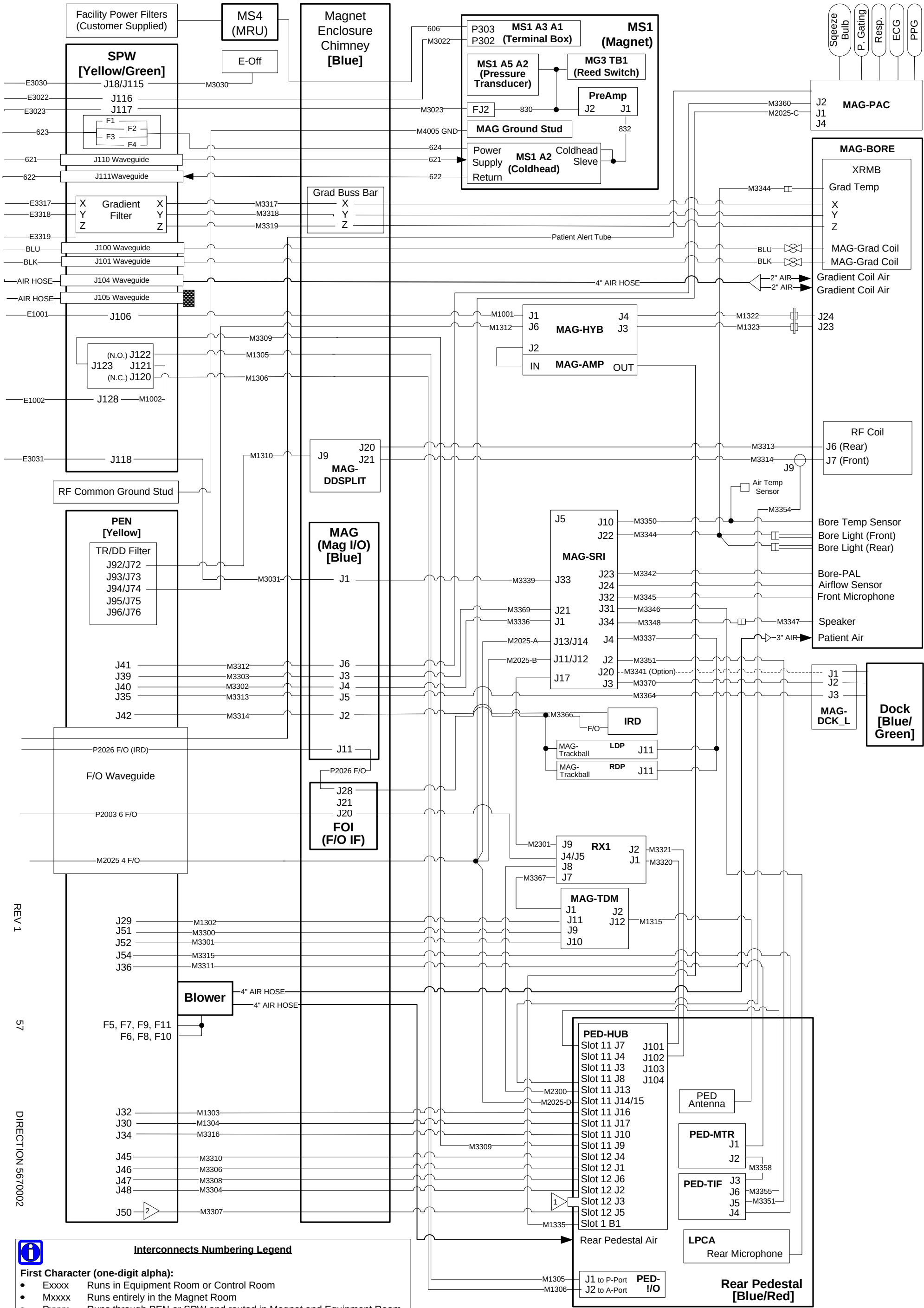


Shunt is installed for 16 & 32 Channel systems.



In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

16/0 Channel MR450w 1.5T Interconnect Map with IRD – Magnet Room



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option

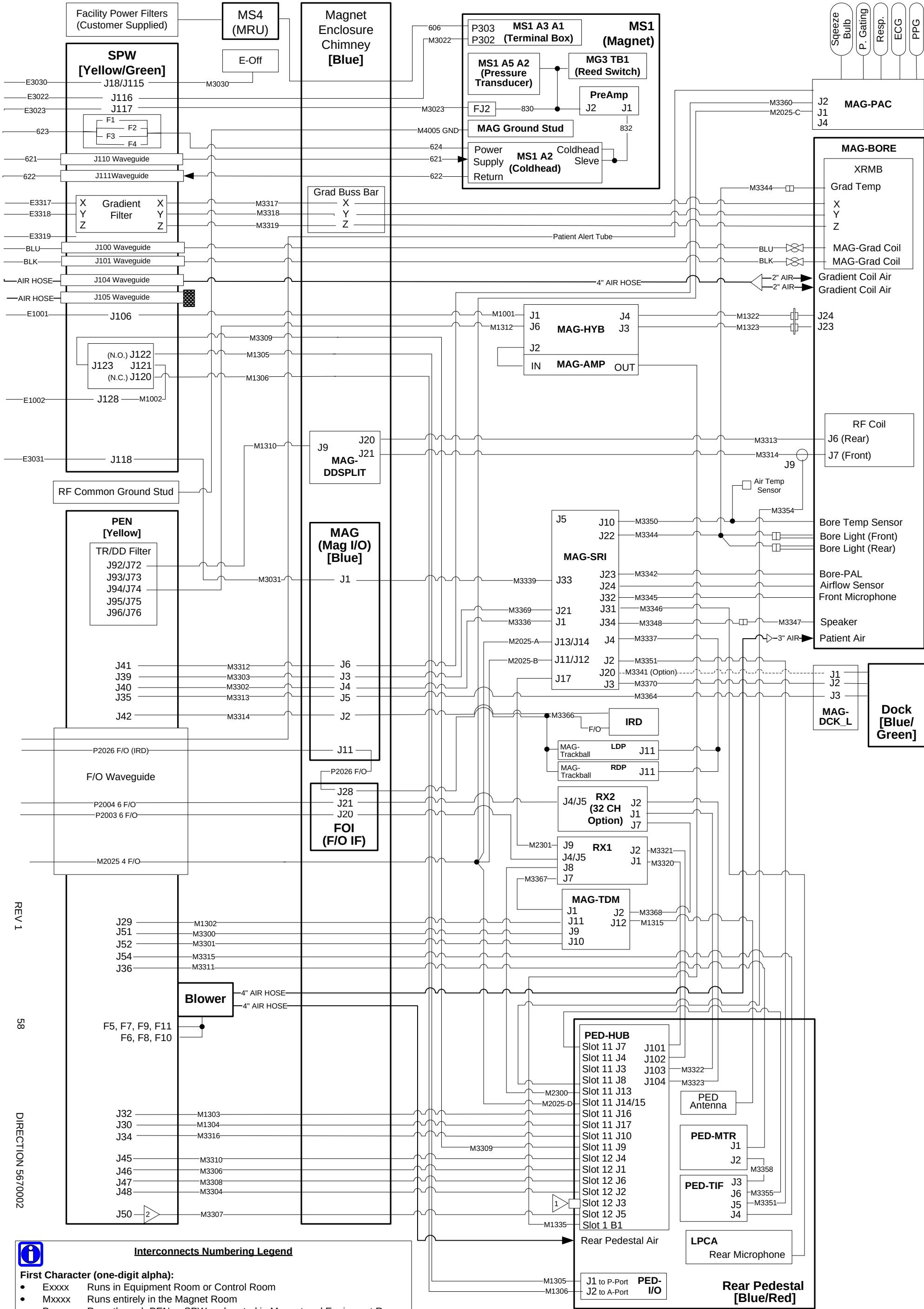
Symbol Legend

Ball Valve Check Valve

Shunt is installed for 16 & 32 Channel systems.

In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

32/0 Channel MR450w 1.5T Interconnect Map with IRD – Magnet Room



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option



Symbol Legend



Ball Valve



Check Valve

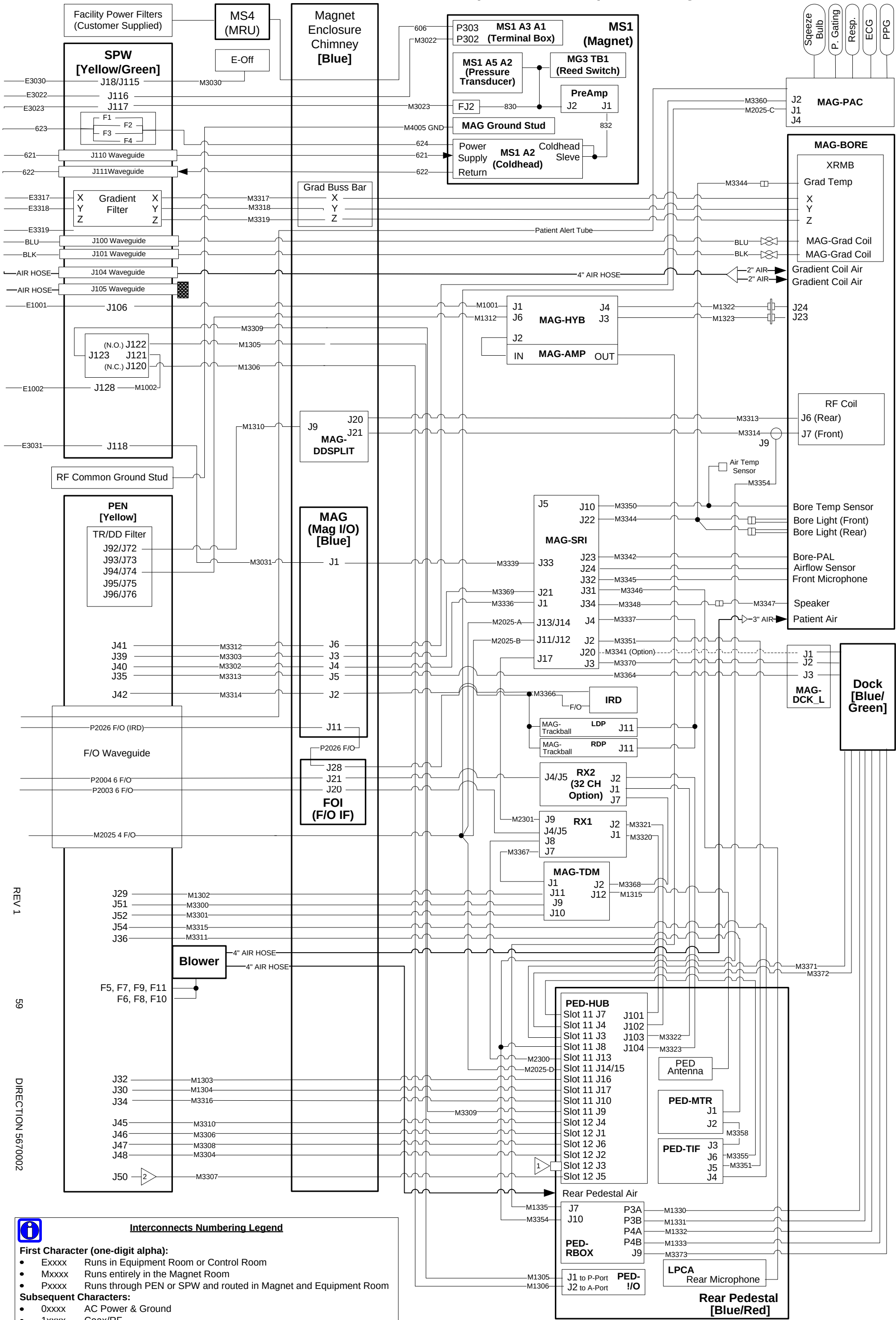


Shunt is installed for 16 & 32 Channel systems.



In certain systems, Run M3307 has been replaced by a terminator located at the back of the PEN Panel at J50 or attached directly to the back of the SRPS module in the PEN Cabinet.

32/32 Channel MR450w 1.5T Interconnect Map with IRD & P ports – Magnet Room



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3 Sorting and Routing Cables

3.1 Sorting Cables



Possible Lacerations

Failure to trim tie-wraps can result in numerous laceration hazards when servicing the equipment.

Tie-wraps must be cut flush with no protruding sharp edges or points.

Move lean cable carts into assigned room. System power, ground, and data cables are color coded at each end by destination per colors shown on cable map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the overhead cable trays.

NOTE: Carefully review architectural drawings to make sure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables to insure that they will route freely without twists or kinks. Route cables in accordance with the applicable Installation Interconnect Diagram.

3.2 Routing Cables

The following cables are grouped together in the overhead cable trays, as indicated in [Cable Locations in Equipment Room Overhead Cable Trays](#) and [Cable Locations in Magnet Room Overhead Cable Trays](#).

EQUIPMENT ROOM Cable Runs

- PGR to PEN:
 - E3002, E3006, E3008, E3009
 - E1005, E0004, E4002, E4005 (GND wall stud)
 - E1003, E1004
- PGR to SPW: E1001, E1002
- PGR to OW: E0003, E3011, E3015, E4008
- PGR to HEC: E3013, E3028, E4007
- OW to PEN: E3017, E3018
- MagMon to SPW & PEN: E3020, E3022, E3023

MAGNET ROOM Cable Runs

- PEN & SPW to MAG I/O M3031, M3302, M3303, M3312, M3313, M3314 (IRD option)
- PEN to REAR PED: M1303, M1304,
- PEN to REAR PED: M3304, M3306, M3308, M3310, M3311, M3315, M3316
- SPW to REAR PED: M1305, M1306
- SPW to REAR PED: M3309
- SPW & PEN to MAG HYB: M1001, M1312
- PEN to MAG-TDM: M3300, M3301
- PEN to MAG-TDM: M1302
- PEN to MAG components: M1310

Any cables not listed above should be routed through the overhead cable trays separately.

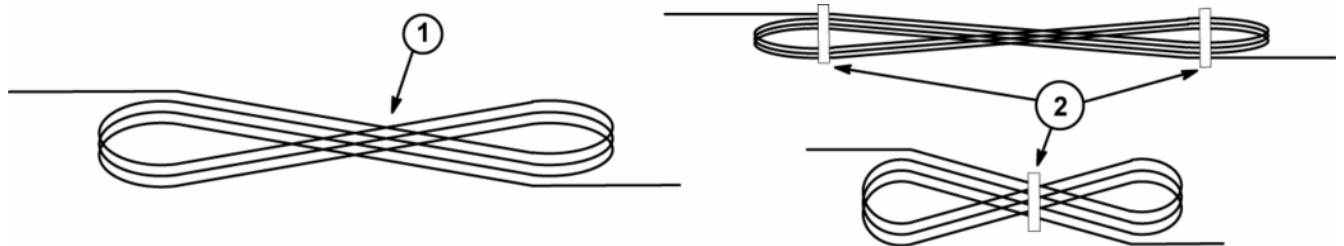
4 Storage of Excess Cable Length

4.1 Coiling Excess Cable Length

Excess length of cables must be coiled in a “Figure Eight” pattern. Only the gradient cable will be cut to length.

1. Excess cable to be stored in overhead cable trays or Penetration Panel closet must be routed in a “Figure Eight” pattern as shown. **DO NOT COIL IN CIRCLES!**
2. Dress and tie-wrap ends of looped cables OR dress and tie-wrap middle of looped cables.

Illustration 4-2: Proper Storage of Excess Cables



3. Repeat as necessary for cables in Magnet or Equipment room.

4.2 Equipment Room Storage

1. Route cables in provided overhead cable trays, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Excess cable length will be stored in overhead cable trays.

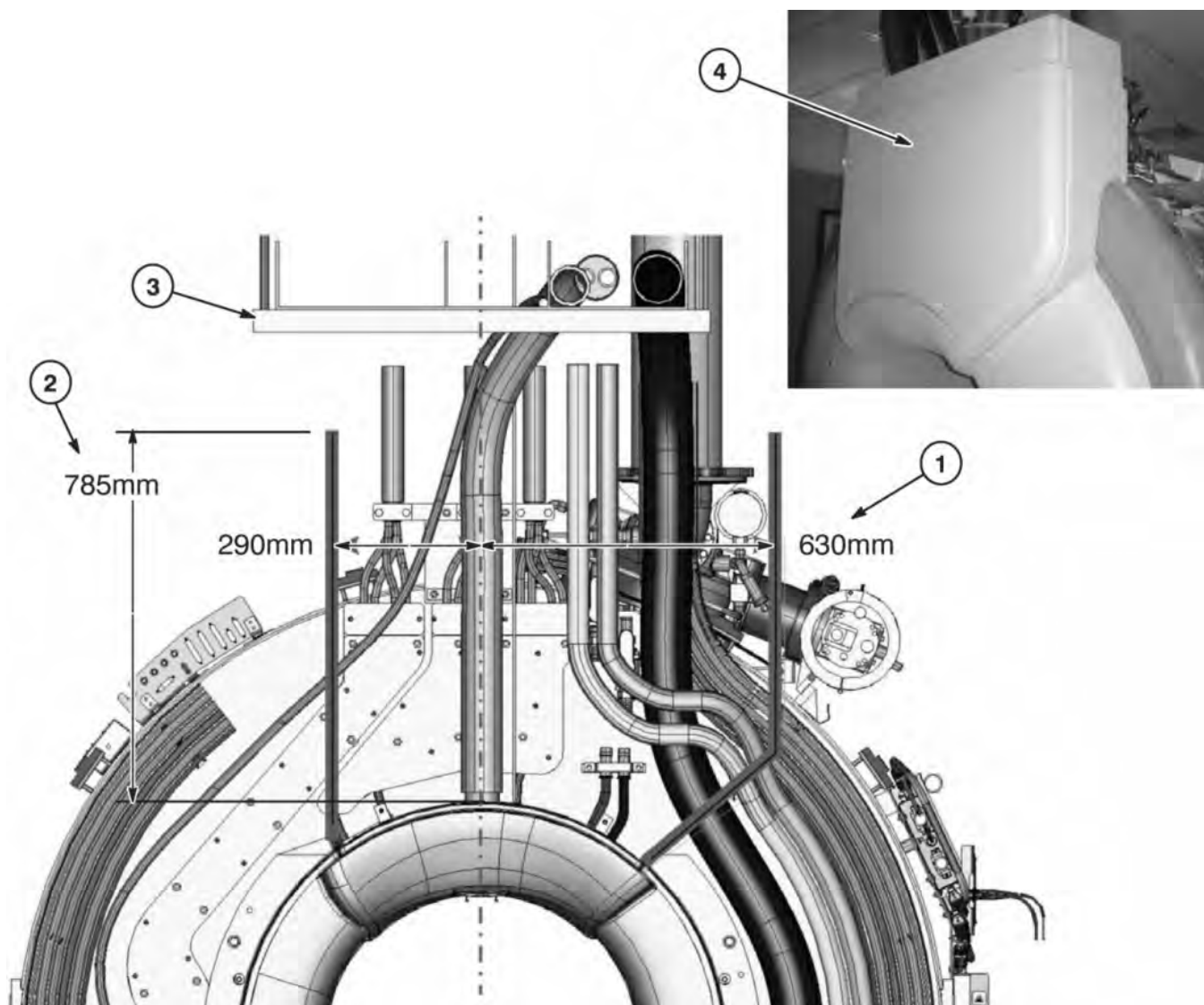
4.3 Magnet Room Storage

1. Route cables inside magnet room thru overhead cable trays, conduit, or ducts to rear of magnet area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
2. Excess cable length will be stored in Penetration Panel Closet.

5 Rear Magnet Enclosure Cable Cover Test-Fit

Before routing any cables in the magnet room, refer to illustration below. Routed cables can not touch the Rear Cable Cover (5328953) when cover is installed at the back of the magnet.

Illustration 4-3: Cable Routing Area at Rear Cable Cover



1. The two dimensions provided, 290mm and 630mm, are measured from the center of the magnet.
2. The top of the Rear Magnet Cover measures approximately 785mm from the top of the rear end bell.
3. Observe the relationship between the end of the overhead cable tray and the area where the cables are to be routed through. **IMPORTANT!** Make sure the cables will not touch the top of the covers when routing of cables is complete.
4. View of rear magnet covers and cable cover after installation.

6 Cable Locations in Equipment Room Overhead Cable Trays

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.



⚠ DANGER

FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.



⚠ DANGER

ELECTRIC SHOCK HAZARD!

LETHAL VOLTAGES ARE PRESENT IN THE PGR PDU/GRADIENT CABINET

LOCK OUT AND TAG OUT THE POWER AT THE MAIN DISCONNECT BEFORE PROCEEDING.

All cables routed between cabinets and the penetration panels must use the overhead cable routing mechanisms or cable trays. For simplicity, in this manual, overhead cable routing mechanisms are referred to as overhead cable trays.



⚠ CAUTION

Muscle strain

Awkward body positioning and / or working from a ladder may cause strain while installing cables.

Take breaks as necessary to prevent personal harm.



NOTICE

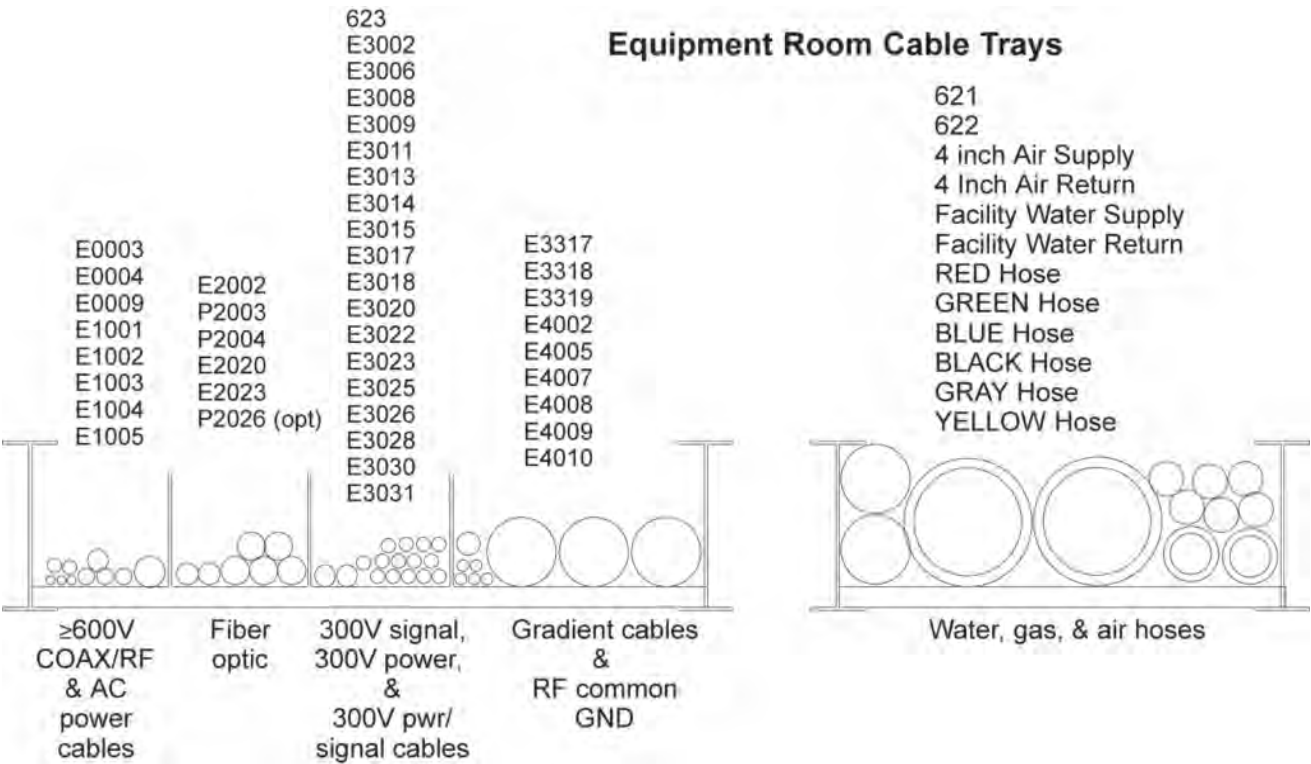
When working in the ceiling area, confirm with the site that no unforeseen hazards exist.

NOTE: The cables must be routed in their designated areas. There are two options for these areas:

- The designated areas are partitioned within one large cable tray (As shown below in the left tray)
- The designated areas are made up of four individual cable trays or other routing mechanisms

The following illustration indicates where the cables, per their Run numbers, are to be routed in a partitioned cable tray. Refer to site architectural drawings and local codes for additional specific guidelines.

Illustration 4-4: Equipment Room Overhead Cable Assignments



7 Cable Locations in Magnet Room Overhead Cable Trays

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.



⚠ DANGER

FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.



⚠ DANGER

ELECTRIC SHOCK HAZARD!

LETHAL VOLTAGES ARE PRESENT.

LOCK OUT AND TAG OUT THE POWER AT THE MAIN DISCONNECT BEFORE PROCEEDING.

The Magnet Room may have a ceiling constructed of tiled panels. These panels will have to be carefully removed for access to overhead cable trays.

- Carefully remove and store ceiling tiles for re-installation after cables are installed.
- All excess cable will be coiled and stored in the Pen Panel Closet.
- All cables routed between magnet and the penetration panels must use the overhead cable trays.



NOTICE

When working in the ceiling area, confirm with the site that no unforeseen hazards exist.



⚠ CAUTION

Muscle strain

Awkward body positioning and / or working from a ladder may cause strain while installing cables.

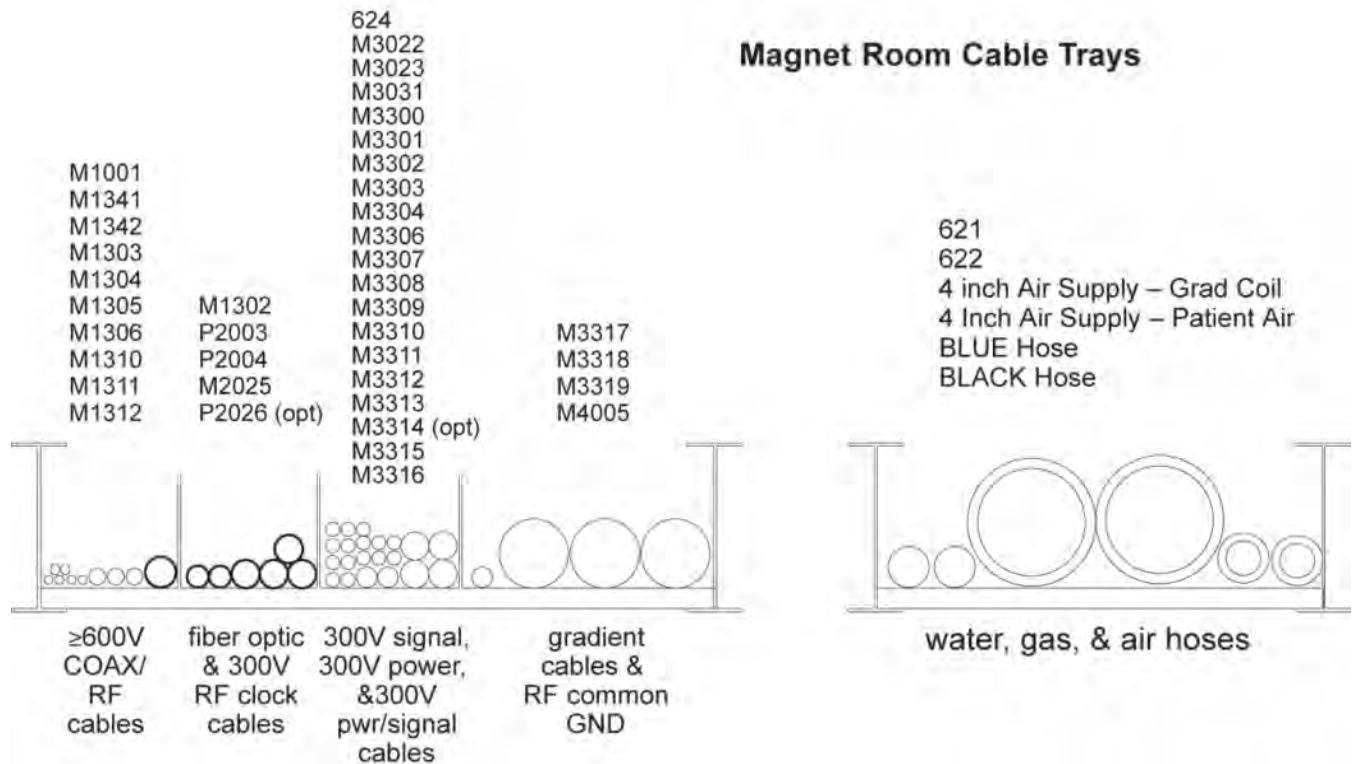
Take breaks as necessary to prevent personal harm.

NOTE: The cables in the cable trays must be routed in their designated areas. There are two options for these areas:

- The designated areas are partitioned within one large cable tray (As shown below in the left tray)
- The designated areas are made up of four individual cable trays or other routing mechanisms

The following illustration (typical view) indicates where the cables, per their Run numbers, are to be routed in the partitioned cable trays, as seen from the magnet end of trays. Refer to site architectural drawings and local codes for additional specific guidelines.

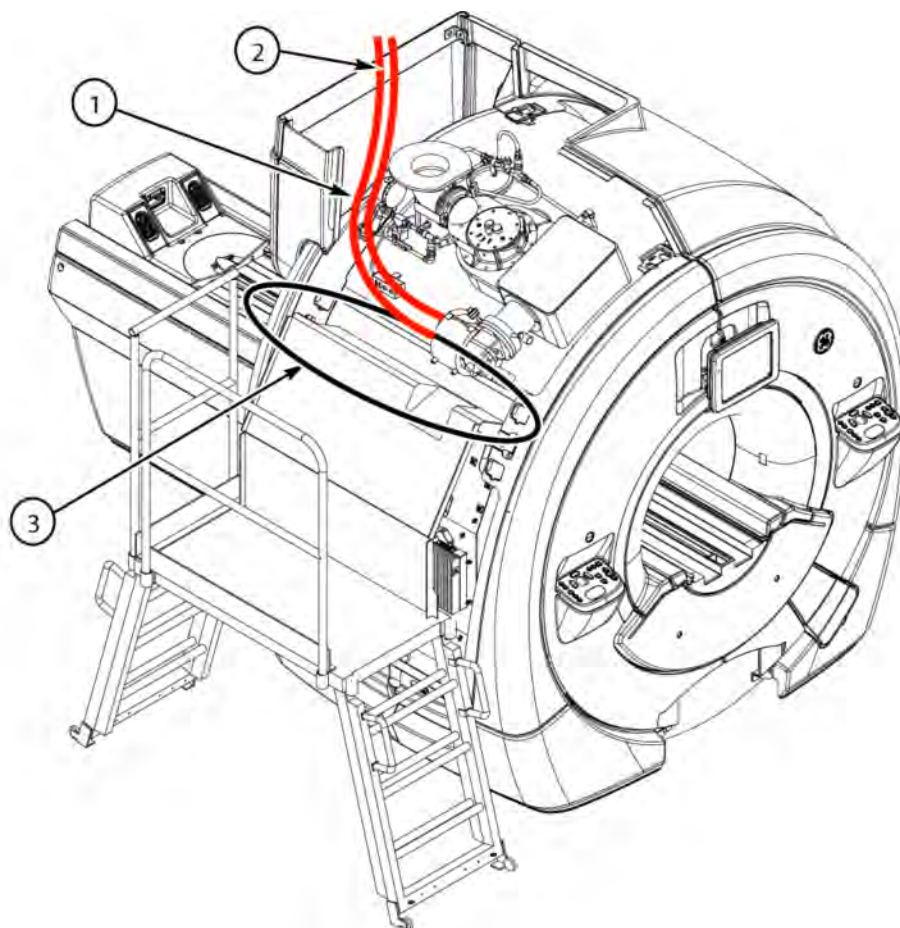
Illustration 4-5: Magnet Room Overhead Cable Assignments



8 Cryogen Line Positioning on Magnet

Cryogen lines should have been routed in the correct overhead tray and connected to the shield cooler during magnet installation.

Illustration 4-6: Routing of Cryogen Lines



1. Make sure that cryogen lines have been routed around vent stack as shown in illustration below.
2. Make sure that there is enough slack in the lines to avoid interference with final installation of all magnet enclosure covers.
3. **NOTE:** When routing remainder of cable, make sure to not route any cables in indicated area. This area on the magnet needs to be clear so that service ladder cover can be placed over modules attached to side of magnet.

9 Gradient Cable Installation

9.1 Equipment Room Gradient Cables

The Equipment Room gradient cables, Runs E3317, E3318, and E3319 are delivered to site pre-terminated at one end for connection to the PGR cabinet. The other ends of the gradient cables are routed to the SPW (Secondary Pen Wall), cut to length, and terminated.



NOTICE

Gradient Cables are heavy, use two or more people when lifting cables.



CAUTION

Muscle strain

Awkward body positioning and / or working from a ladder may cause strain while installing cables.

Take breaks as necessary to prevent personal harm.

9.1.1 Route Gradient Cables in Overhead Cable Tray

It is recommended that the three gradient cables be routed through the overhead cable trough before any of the other cables are routed.



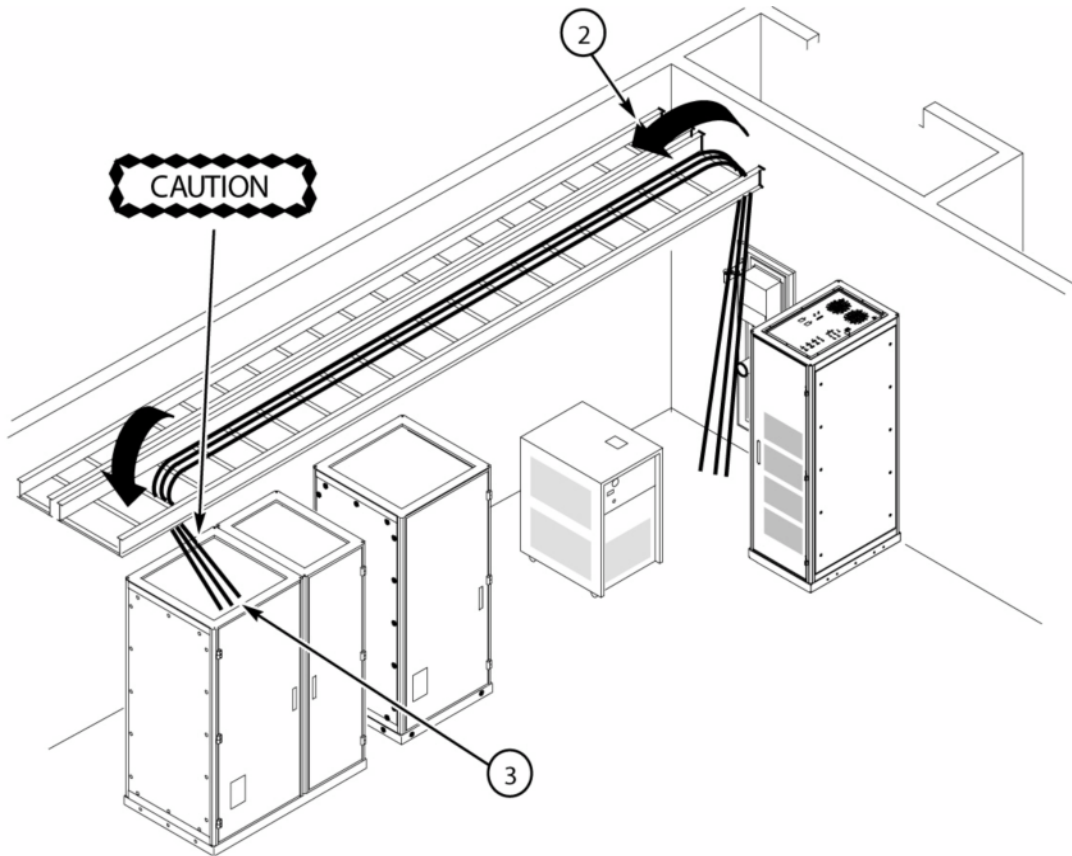
CAUTION

Damage to Overhead Cable Routing Mechanisms/Trays

Horizontal movement of cable trays while routing cables can damage cable tray rods attached to ceiling.

Do not pull on cables at end of cable trays. Route cables in trays by lifting and moving cables to connection points in overhead cable trays.

Illustration 4-7: Routing of Gradient Cables in Equipment Room



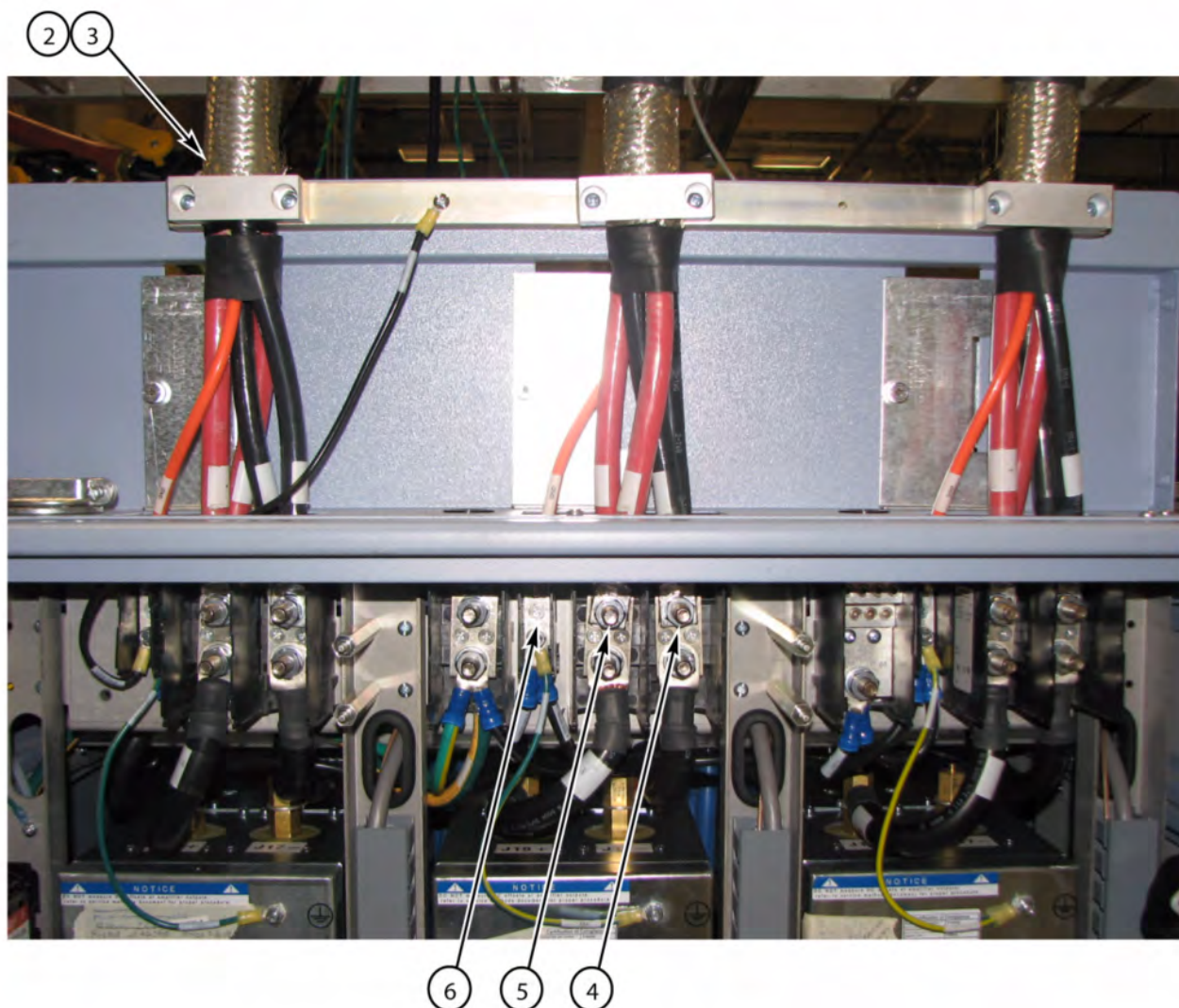
1. It is recommended that the three gradient cables be routed through the overhead cable trough before any of the other cables are routed.
2. Starting at the SPW end of the cable tray, route the pre-terminated end of each cable through the designated overhead cable tray towards the PGR Cabinet. Position in the cable tray as shown in [Cable Locations in Equipment Room Overhead Cable Trays](#).
3. Route the pre-terminated ends of each gradient cable to the top/front of the PGR cabinet for connection to terminals.

9.1.2 Connect Gradient Cable to PGR Cabinet

The steps for this procedure install just one of the gradient cables. Repeat the steps below for installation and connection of the remaining gradient cables and the ground wires.

NOTE: If the gradient cable clamp bracket was removed in the earlier cabinet positioning procedures, reattach using the removed screws.

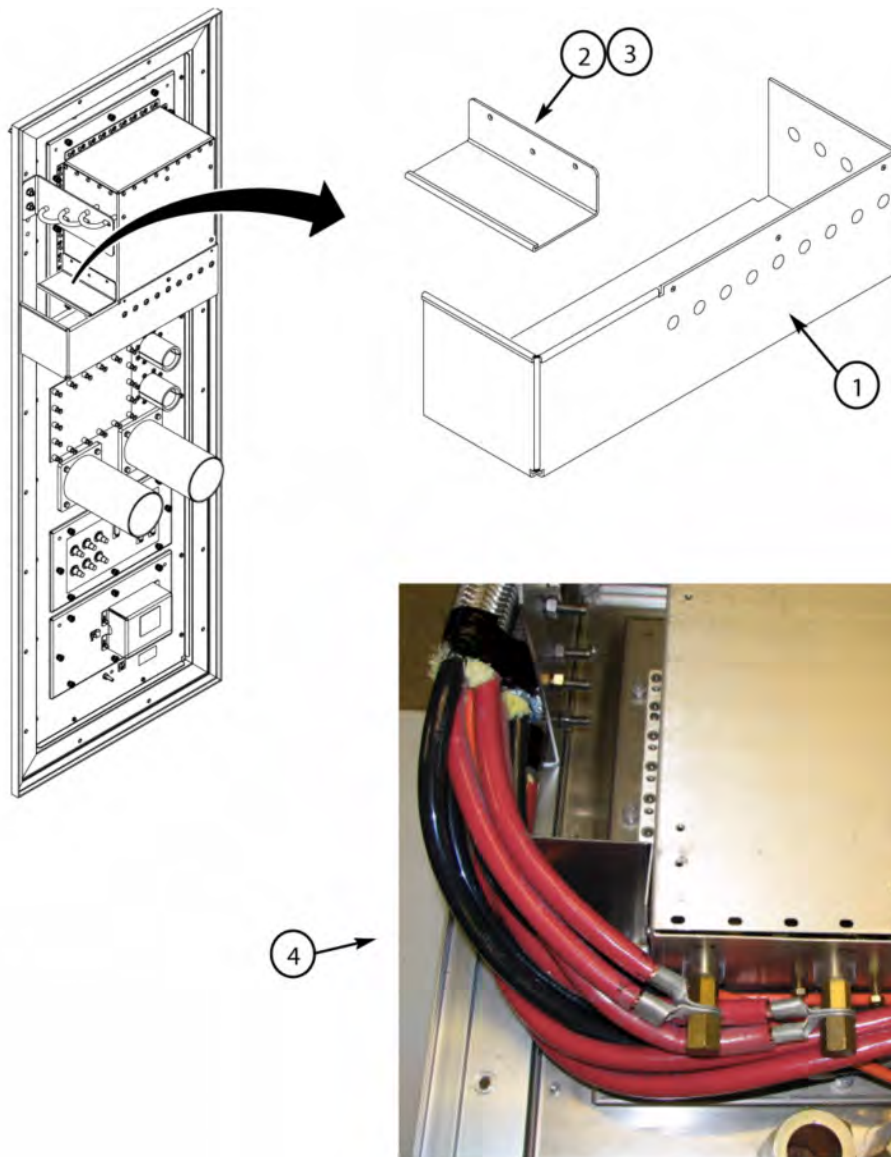
Illustration 4-8: PGR Cabinet Gradient Cables



1. Remove cover at top of cabinet to access gradient cable clamps.
2. Position the braided area of the cable in the clamp at the top of the cabinet.
3. Close clamp around cable, but do not tighten completely. Adjustment may be necessary as the terminated ends of wires are attached.
4. Position the two black wires on the stud of the terminal block and fasten with nuts.
5. Position the two red wires on right stud of the terminal block and fasten with nuts.
6. Attach the ground wire on the ground stud and fasten with nut.

9.1.3 Remove Gradient Filter Safety Shields

Illustration 4-9: Remove Safety Shields

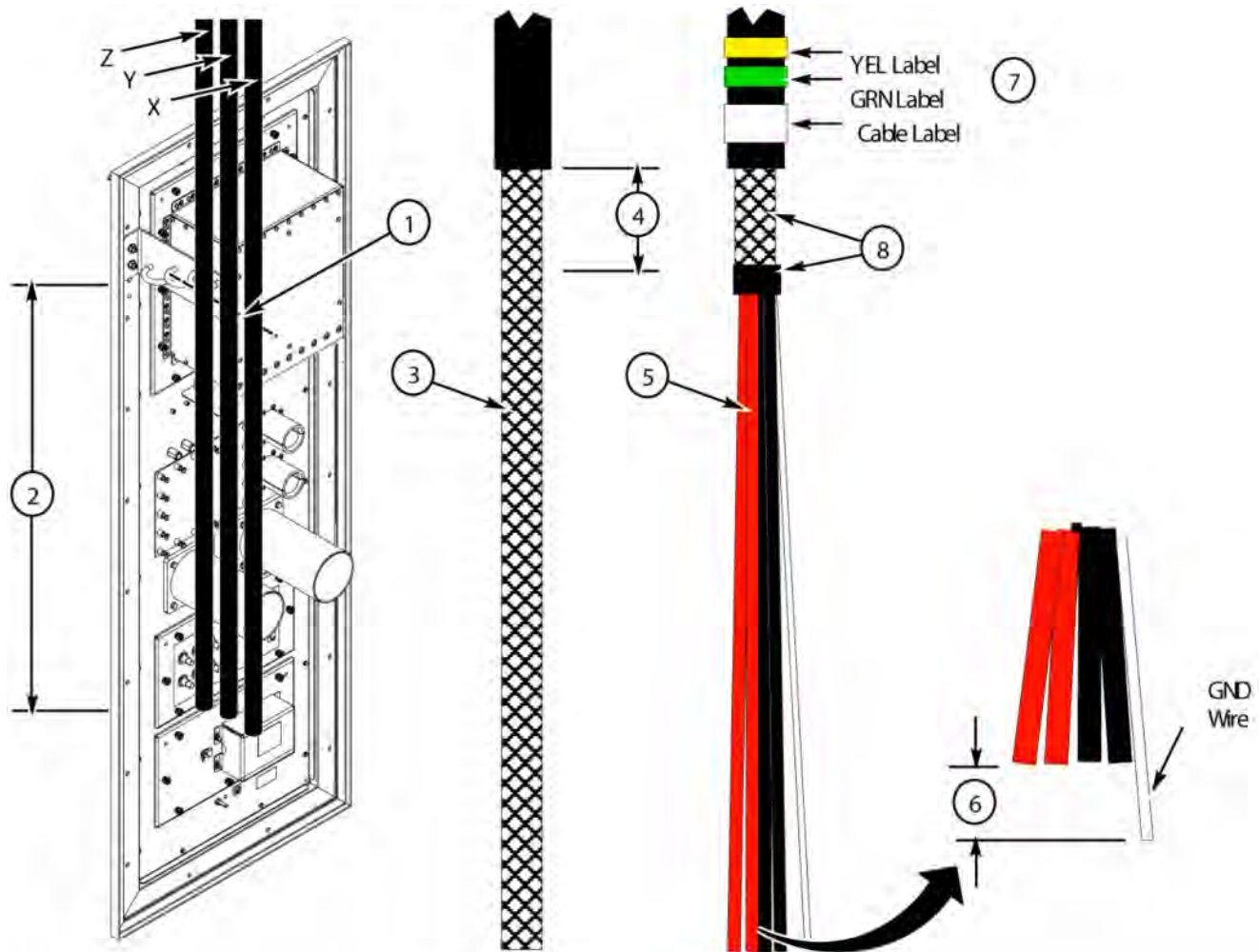


1. Unfasten three screws and remove the front safety shield. Save for re-installation after cables are attached.
2. NOTE: The side safety shield may or may not be removed. It is required to be in place after completion of gradient cable installation. Keeping this shield attached to gradient filter during gradient cable installation will make sure that the cables will have enough length to attach to the studs at the bottom of the gradient filter.
3. If it is necessary to remove the safety shield, unfasten three screws and remove the side safety shield. Save for re-installation after cables are attached.
4. This photograph offers a view of what the gradient cable installation will look like after connection of the cables and prior to re-installation of the front safety cover.

9.1.4 Cut Gradient Cable to Length at SPW

The following steps will provide instructions for exposing the cable braid, cutting to length and terminating the red and black wires for connection to the studs on the bottom of the Gradient Filter.

Illustration 4-10: Trimming Gradient Cables



1. Position cables at U-bolt clamp at side of SPW (Secondary Pen Wall) and mark each cable.
2. From the mark on each cable, measure and cut each cable according to chart below:

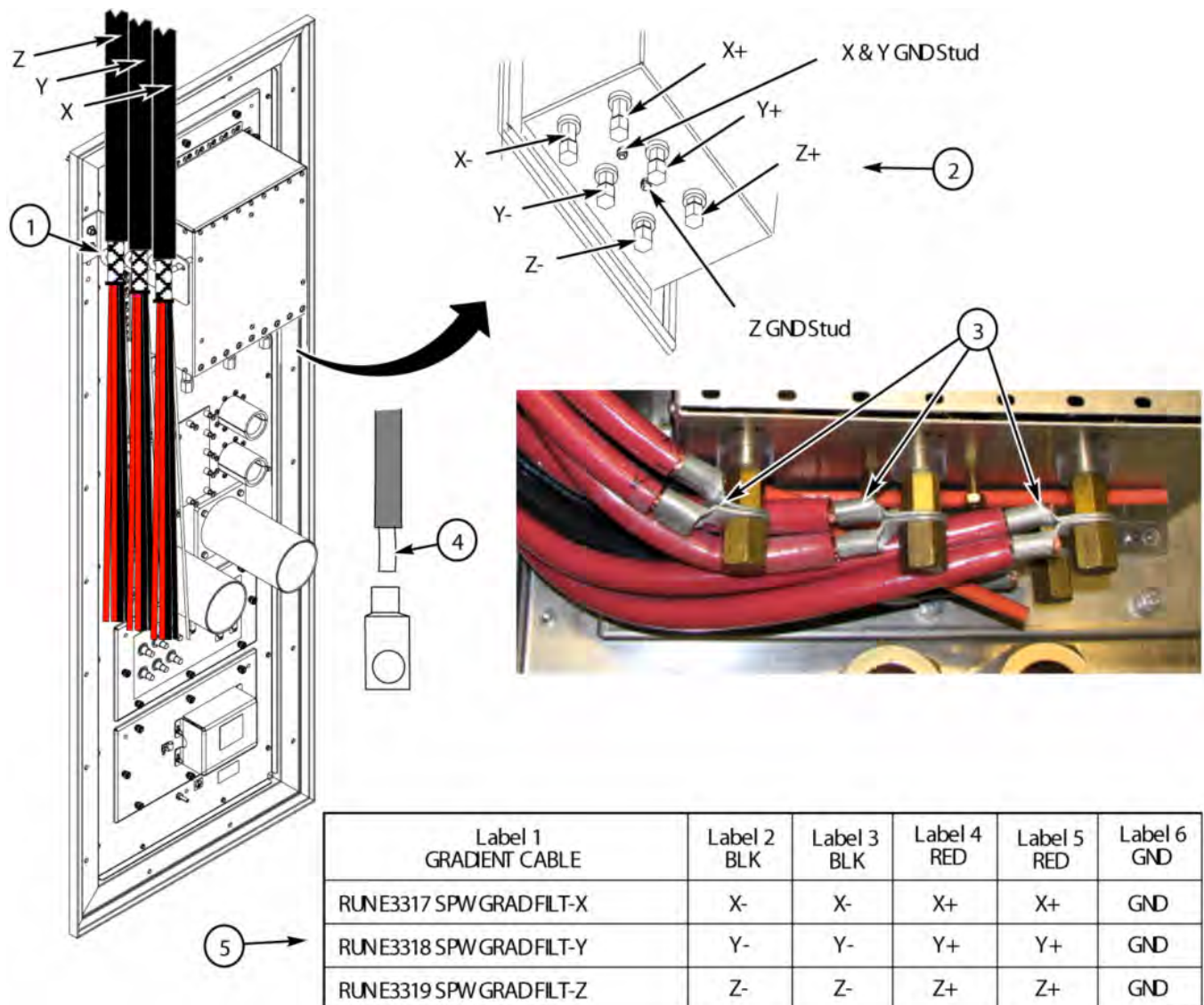
	X	Y	Z
Cut to Length from Mark	20 Inches (508mm)	22 Inches (559mm)	25 Inches (635mm)

3. To expose braid, remove outer cable jacket to mark created in Step 2.
4. Measure and mark 3.5 inches (90mm) from edge of outer cable jacket.
5. Remove remaining braid from the mark to end of each cable exposing the two red wires, the two black wires, and the ground wire.

6. The red and black X wires can be cut back 2 inches (51mm). DO NOT cut back the ground wire.
7. On the end of each cable jacket, apply the provided yellow and green tape as shown below. Also apply the applicable cable label for each Run: Run E3317 = FILT X, RUN E3318 = FILT Y, RUN E3319 = FILT Z
8. Remove plastic covering on braid and apply electrical tape around edge of braid to prevent unraveling of braid.
9. Discard remaining cable.

9.1.5 Connect Gradient Cable Wires to Gradient Filter

Illustration 4-11: Gradient Cables Connection Information



1. Position the X, Y, and Z gradient cables on bracket and tighten U-bolts around braided section of each cable. See following illustration.

**CAUTION****Damage to Gradient Cables**

Over tightening of U-bolts in mounting bracket can damage gradient cables.

Do not over tighten U-bolts.

2. Terminal attachment locations: Black wires = X-, Y-, Z-. Red wires = X+, Y+, Z+.
3. Note the back-to-back position of the terminals attached to studs. Before crimping terminals onto wires, position and mark cables as to alignment of terminals for ease of fastening crimped terminals to studs.

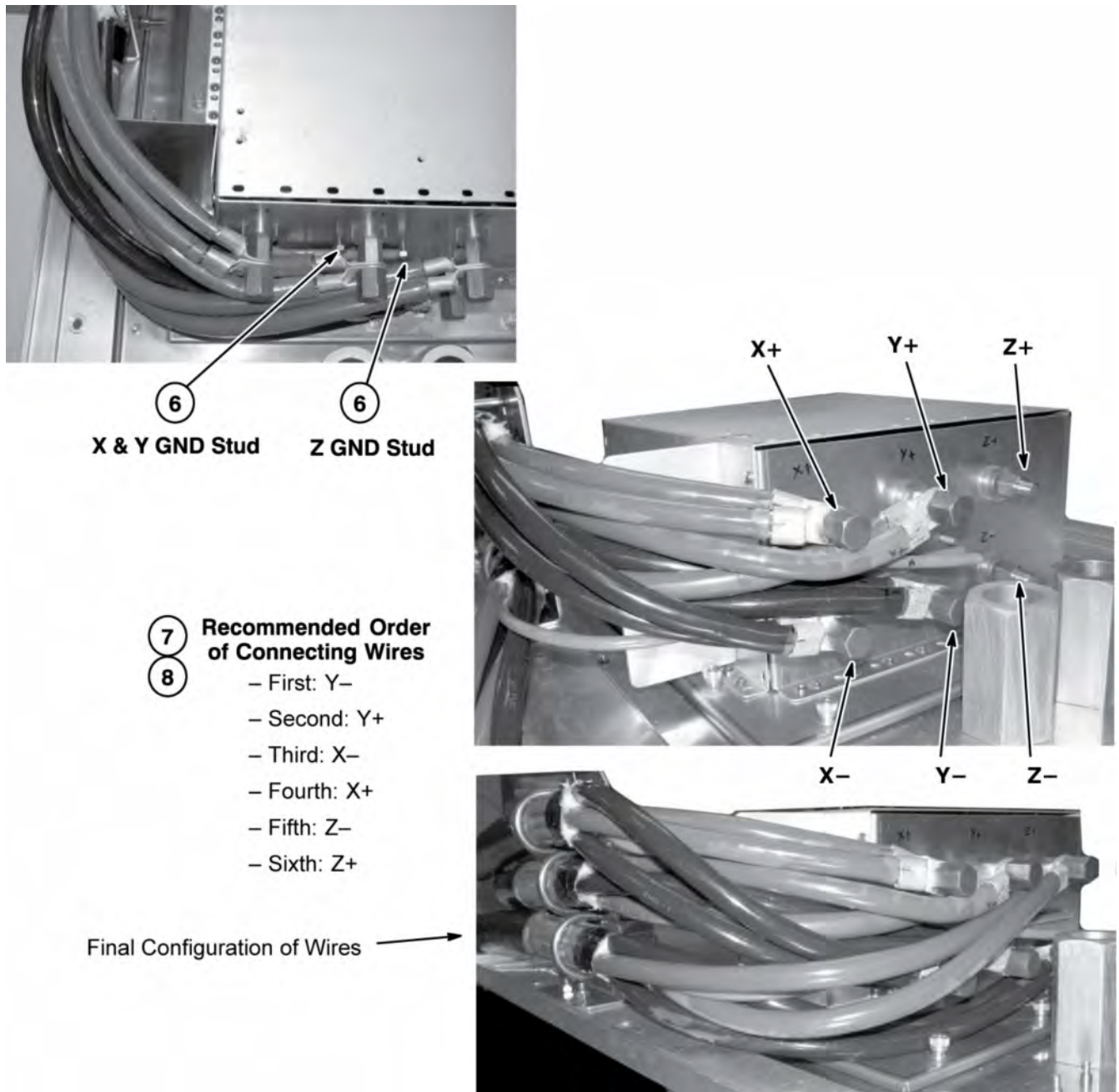
NOTE: Refer to step 7 in next section as to the order of installation of cables on studs. Adjustment to lengths of any of the red or black wires may be necessary to accommodate connections to studs.

4. Strip 0.75 in. (19mm) of insulation from end of gradient black and red wires.
5. Label each of the five wires in each gradient cable per table below. Red wires are "+", Black wires are "-".

**NOTICE**

Make sure terminals are positioned correctly as shown in step 3.

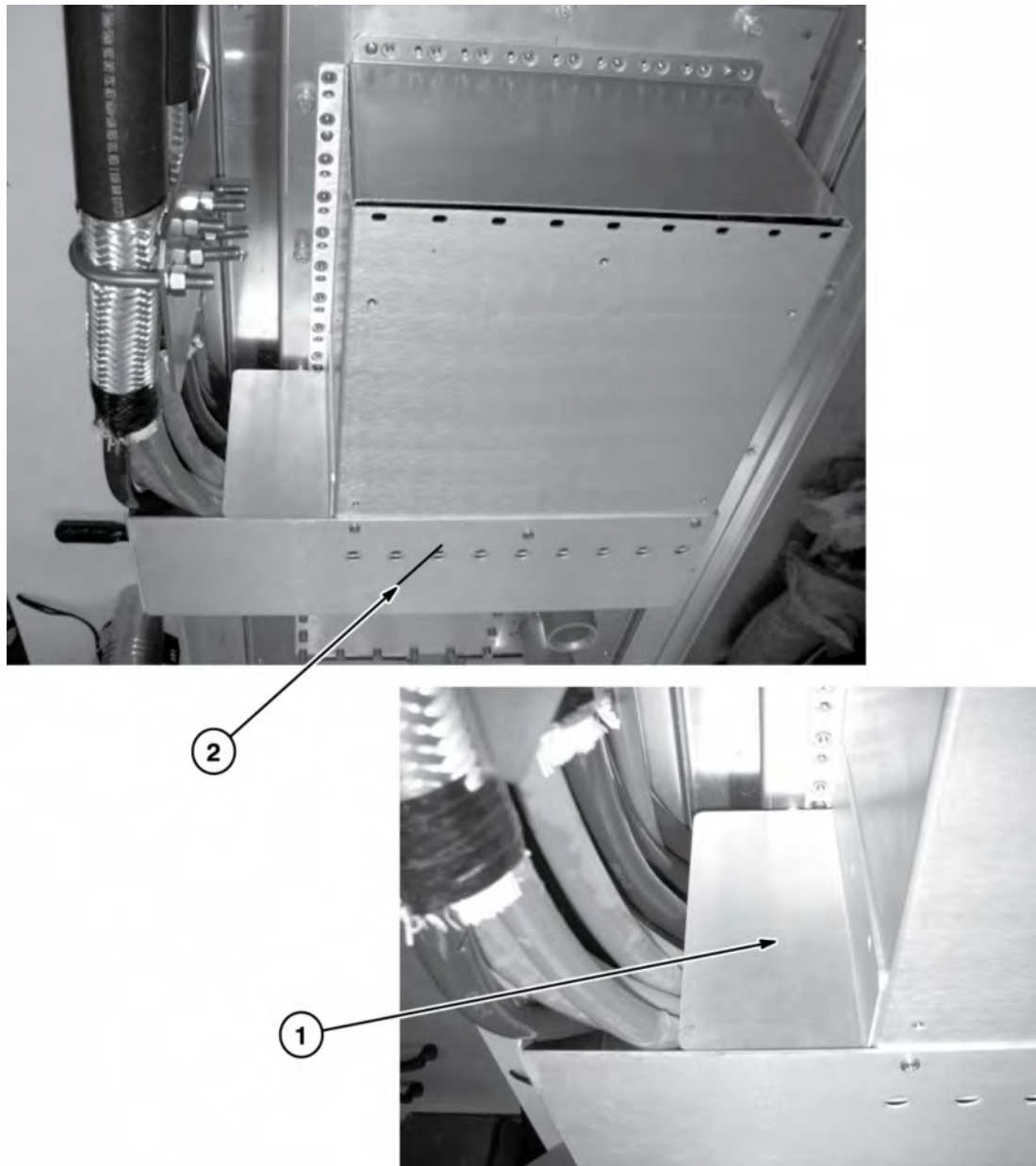
Illustration 4-12: Final Connection of Equipment Room Gradient Cables



6. Ground Wires - Route, mark and cut to length, strip 0.75 in. (19mm) of insulation from end of GND wires, and crimp a furnished T & B 54105 terminal on each ground wire. Connect wires as indicated. **Do not over tighten GND studs.**
7. Gradient cable wires - On end of each end, place (Do not crimp at this time) a furnished T & B 54165 terminal. Route and connect to studs using recommended order below. If too long, cut and re-strip end of individual wires or adjust braided area of cable position in bracket U-bolt.
8. After attaining correct length for each wire, crimp terminals on wires.

9.1.6 Install Safety Cover(s)

Illustration 4-13: Gradient Filter Safety Covers



1. If removed, re-attach the Side Safety Shield using the three previously removed screws.
2. Re-attach the Front Safety Shield with the three previously removed screws.

9.2 Magnet Room Gradient Cables

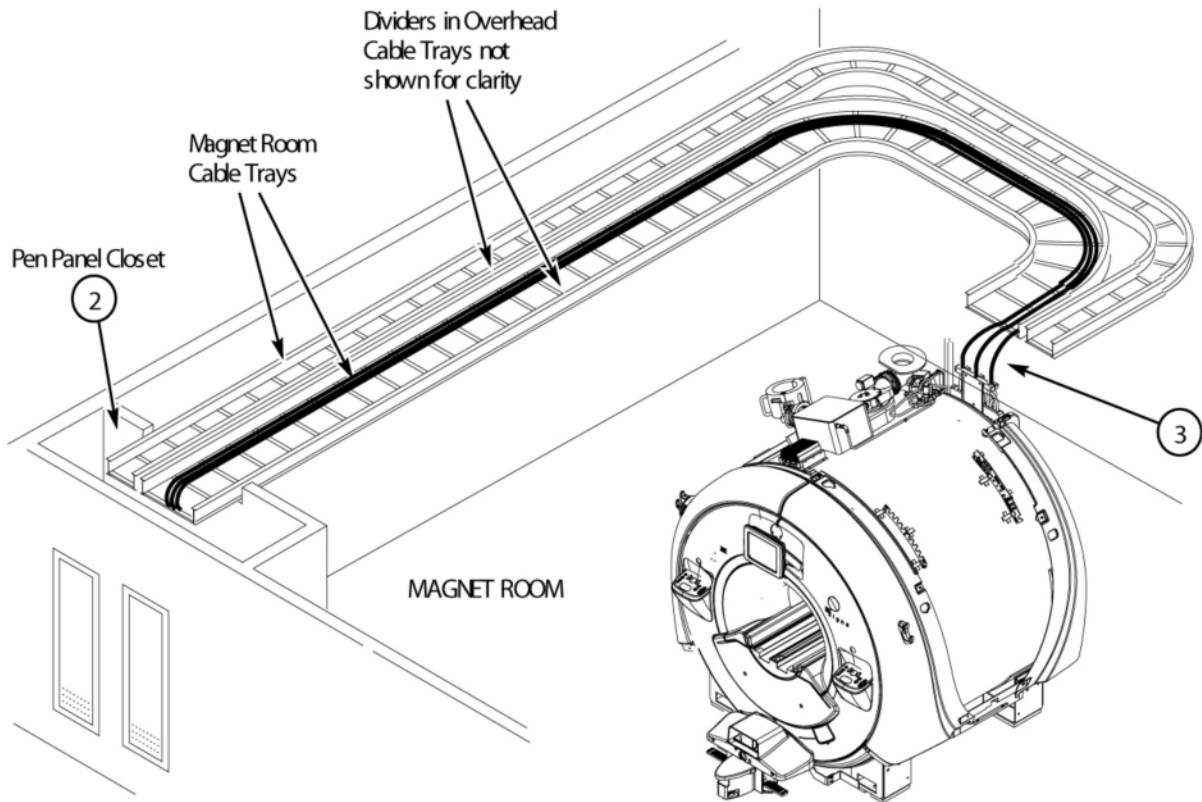
The Magnet Room gradient cables, Runs M3317, M3318, and M3319 are delivered to site pre-terminated at one end for connection to the rear of the Magnet. The other ends of the gradient cables are routed to the SPW (Secondary Pen Wall), cut to length, and terminated.

9.2.1 Remove Magnet Enclosure Covers

Before routing any cables, remove and store in a safe place the magnet enclosure side covers, arc covers, and rear covers at top of magnet.

9.2.2 Route Gradient Cables in Overhead Cable Tray

Illustration 4-14: Magnet Room Gradient Cables



1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray in the Pen Panel Closet, route the pre-terminated end of each cable through the designated overhead cable tray towards the rear of the Magnet. Make sure they are positioned in the cable tray as shown in [Cable Locations in Magnet Room Overhead Cable Trays](#).
3. Route the pre-terminated ends of each gradient cable to the rear of the magnet for connection to terminals as shown in [Section 9.2.3](#).

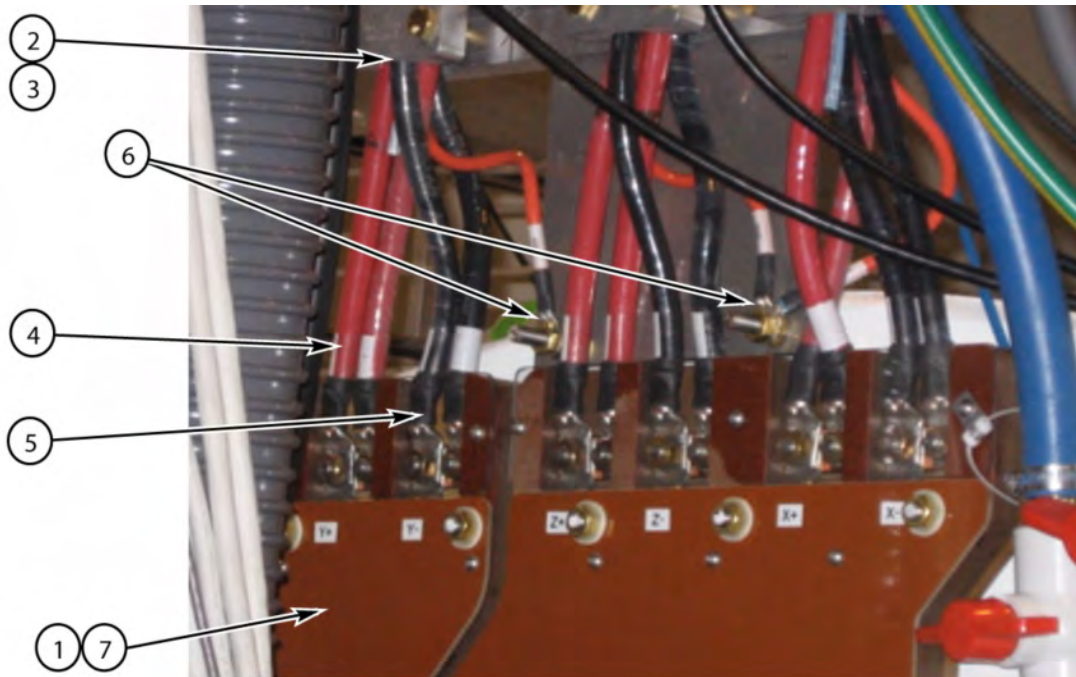
9.2.3 Connect Gradient Cables at Back of Magnet

The steps for this procedure install just one of the gradient cables. Repeat the steps below for installation and connection of the remaining gradient cables and the ground wires.

**NOTICE**

Make sure that the cables are installed in the correct order as indicated in the illustration below. As viewed from the rear of the magnet, “Y” is on the left, “Z” is in the middle, and “X” is on the right side

Illustration 4-15: Gradient Cable Connections

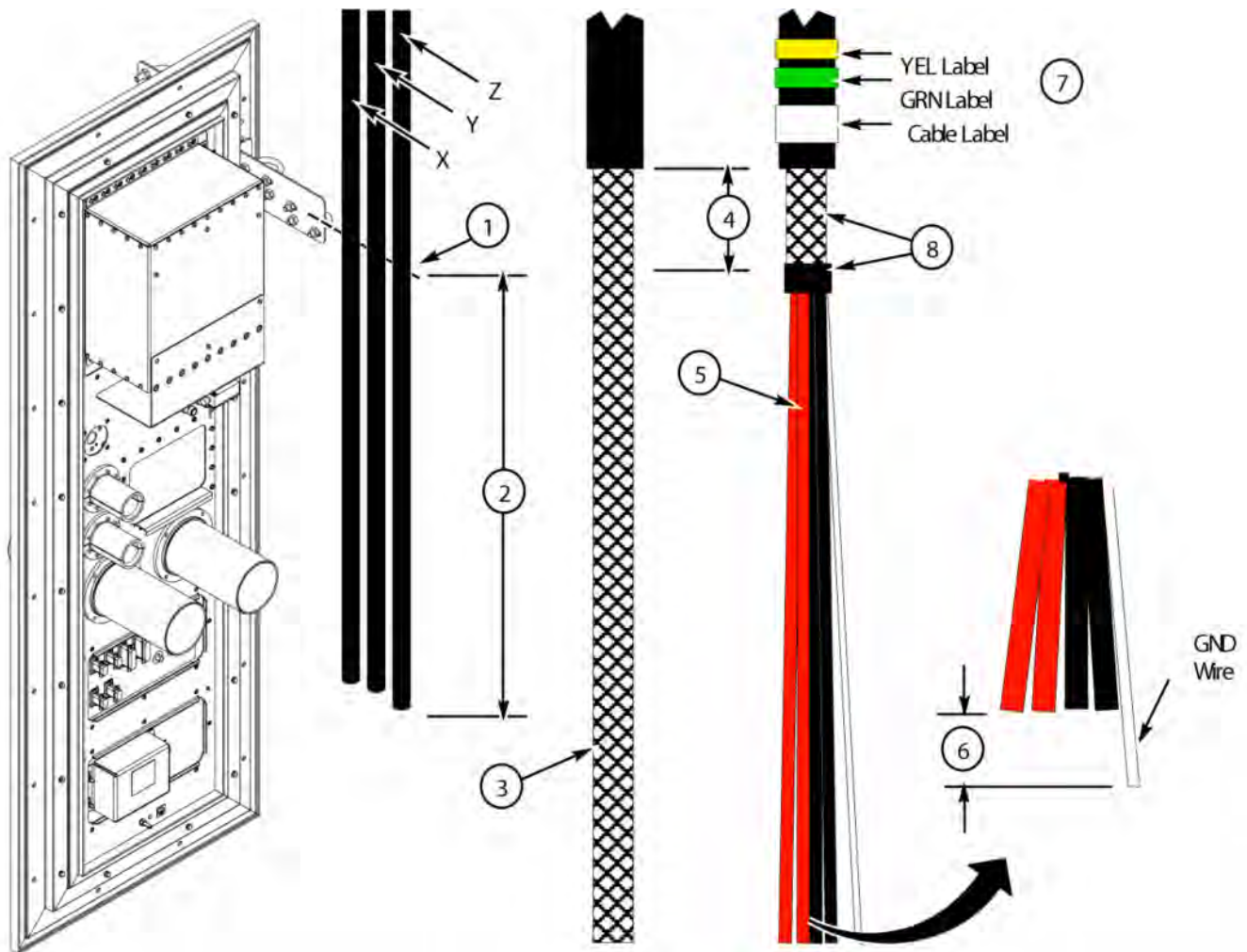


1. Remove safety shield.
2. Position the braided area of the cable in the clamp.
3. Close clamp around cable, but do not tighten completely. Adjustment may be necessary as the terminated ends of wires are attached.
4. Position the two red wires on the two “+” studs and fasten with provided nuts. Repeat procedure for Z and X wires.
5. Position the two black wires on the two “-” studs of the terminal block and fasten with provided nuts. Repeat procedure for Z and X wires.
6. Attach the ground wire on the ground stud and fasten with provided nut. (Z and X are connected together on one GND stud and the Y ground wire is connected to the other GND stud.)
7. Re-attach safety shield.

9.2.4 Cut Gradient Cable to Length at SPW

The following steps will provide instructions for exposing the cable braid, cutting to length and terminating the red and black wires for connection to the studs on the bottom of the Gradient Filter. See following illustration.

Illustration 4-16: Trimming Gradient Cables



1. Position cables at U-bolt clamp at side of SPW (Secondary Pen Wall) and mark each cable
2. From the mark on each cable, measure and cut each cable according to chart below:

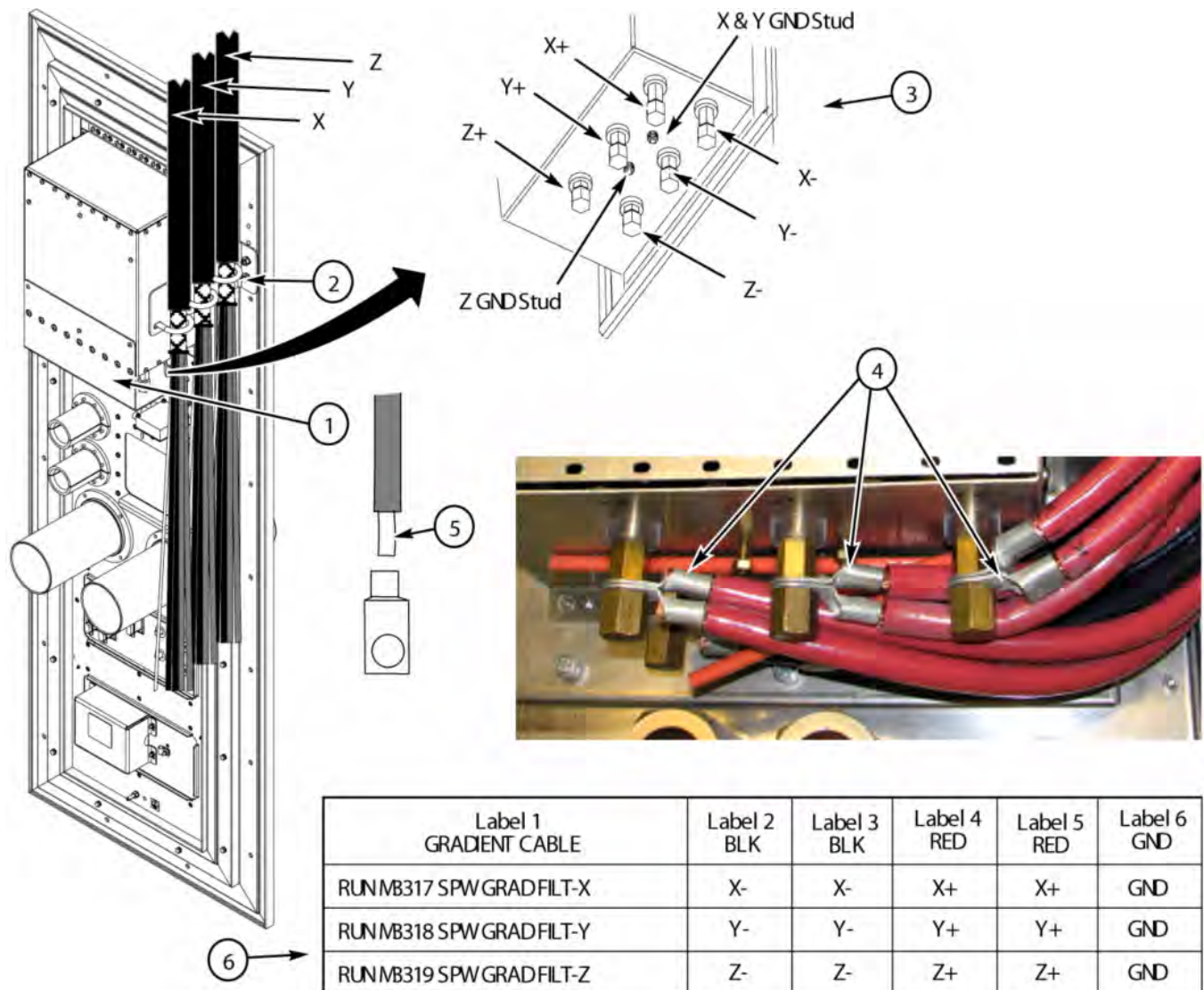
	X	Y	Z
Cut to length from Mark	20 Inches (508mm)	22 Inches (559mm)	25 Inches (635mm)

3. To expose braid, remove outer cable jacket to mark created in Step 2.
4. Measure and mark 3.5 inches (90mm) from edge of outer cable jacket.
5. Remove remaining braid from the mark to end of each cable exposing the two red wires, the two black wires, and the ground wire.
6. The red and black X wires can be cut back 2 inches (51mm). DO NOT cut back the ground wire.

7. On the end of each cable jacket, apply the provided yellow and green tape as shown below. Also apply the applicable cable label for each Run: Run M3317 = FILT X, RUN M3318 = FILT Y, RUN M3319 = FILT Z
8. Remove plastic covering on braid and apply electrical tape around edge of braid to prevent unraveling of braid.
9. Discard remaining cable.

9.2.5 Connect Gradient Cable Wires to Gradient Filter

Illustration 4-17: Gradient Cable Connections



1. Remove plastic shield.
2. Position the X, Y, and Z gradient cables on bracket and tighten U-bolts around braided section of each cable.

NOTE: U-bolts are located on side of bracket away from the gradient filter.

**Damage to Gradient Cables**

Over tightening of U-bolts in mounting bracket can damage gradient cables.

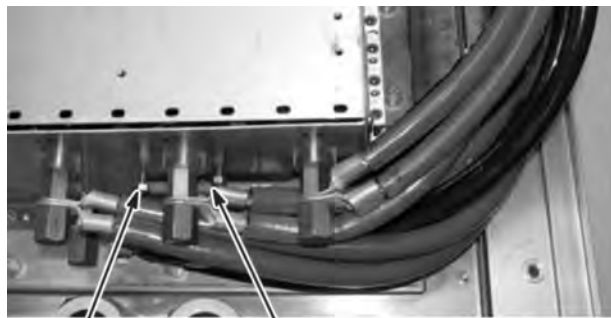
Do not over tighten U-bolts.

3. Terminal attachment locations: Black wires = X-, Y-, Z-. Red wires = X+, Y+, Z+.
4. Note the back-to-back position of the terminals attached to studs. Before crimping terminals onto wires, position and mark cables as to alignment of terminals for ease of fastening crimped terminals to studs.

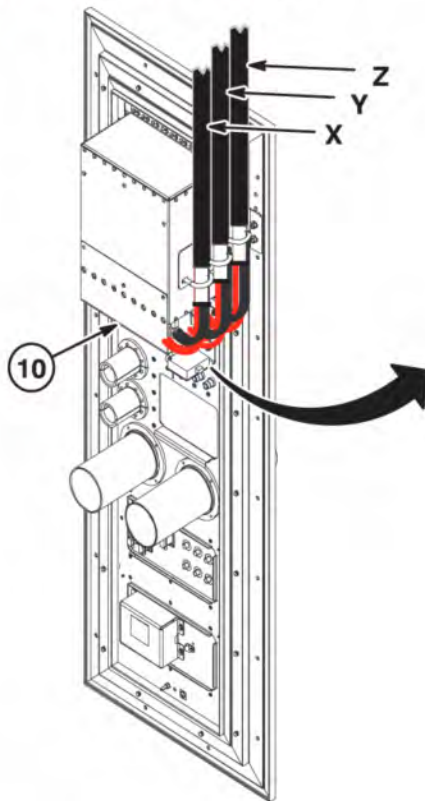
NOTE: Refer to step 8 for the order of installation of cables on studs. Adjustment to lengths of any of the red or black wires may be necessary to accommodate connections to studs.

5. Strip 0.75 in. (19mm) of insulation from end of gradient black and red wires.
6. Label each of the five wires in each gradient cable wire per table in [Illustration 4-17](#). Red wires are "+", Black wires are "-"

Illustration 4-18: Final Connection of Magnet Room Gradient Cables

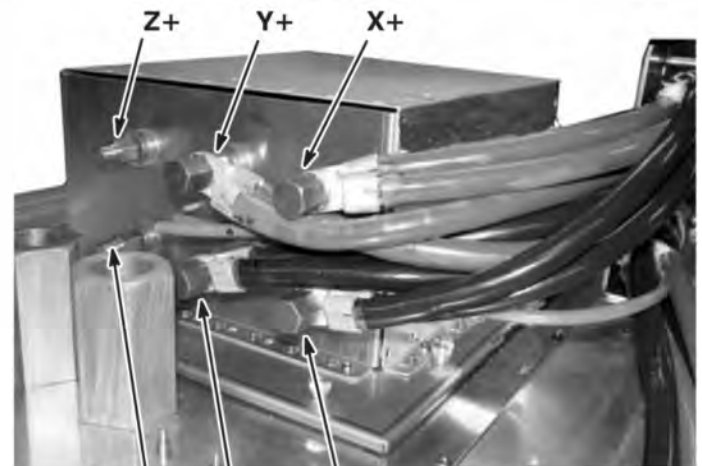


7 Z GND Stud X & Y GND Stud



8 Recommended Order of Connecting Wires

- 9
- First: Y-
 - Second: Y+
 - Third: X-
 - Fourth: X+
 - Fifth: Z-
 - Sixth: Z+



Final Configuration of Wires



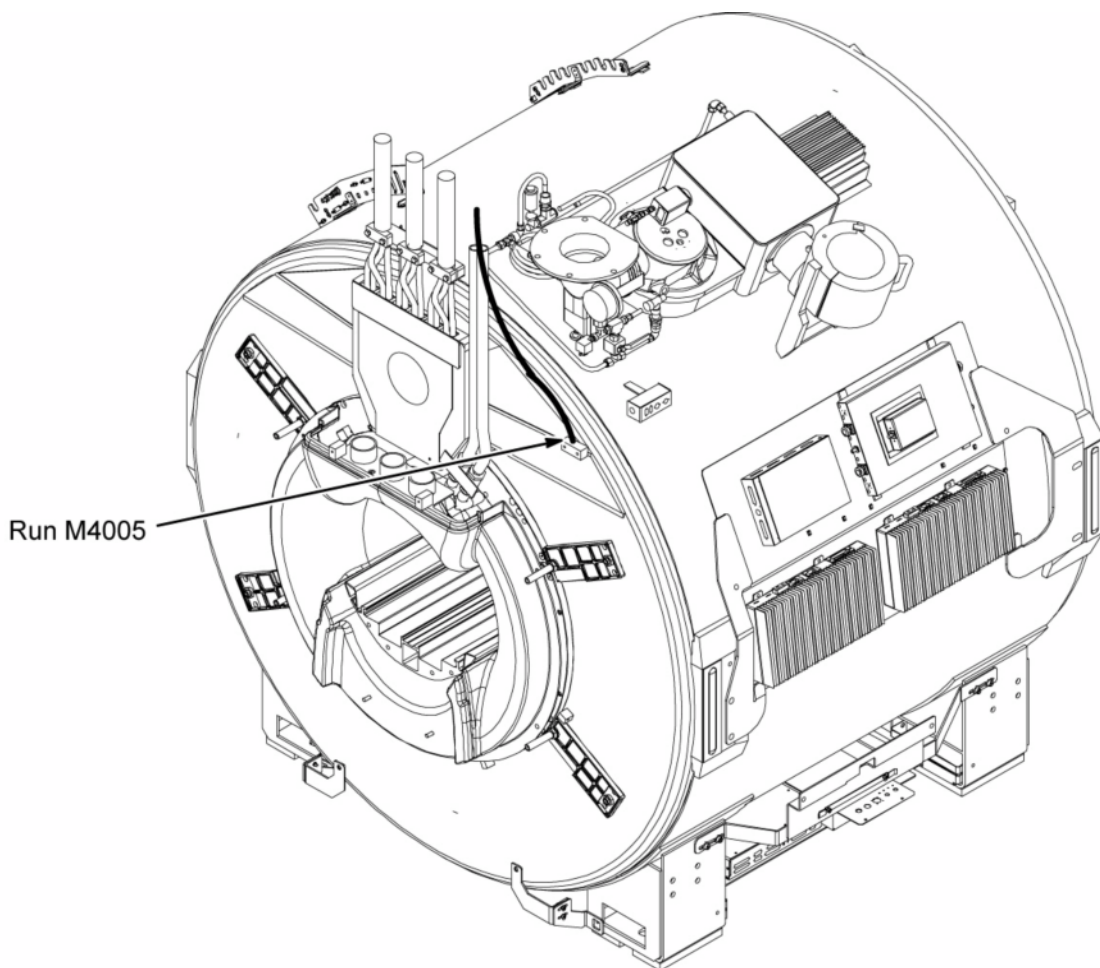
7. Ground Wires - Route, cut to length, strip 0.75 in. (19mm) of insulation from end of GND wires, and crimp a furnished T & B 54105 terminal on each ground wire. Connect wires as indicated.
8. On end of each gradient cable wire, place (Do not crimp at this time) a furnished T & B 54165 terminal. Route to studs using recommended order above. If too long, cut and re-strip end of individual wires or adjust braided area of cable position in bracket U-bolt.
9. After attaining correct length for each wire, crimp terminals on wires. IMPORTANT: Make sure terminals are positioned correctly as shown in step 3.
10. Install plastic safety shield.

10 Connect Magnet Ground, Run M4005

Make sure magnet ground cable, Run M4005, is connected to GND stud at back of magnet. Other end of ground cable is routed through overhead cable trays and connected to RF Screen Room common ground. Refer to cable map.

NOTE: If ground stud on back magnet flange is not available, route Run M4005 to either rear magnet leg. Connect to ground stud on inside of leg.

Illustration 4-19: Connect Magnet Ground

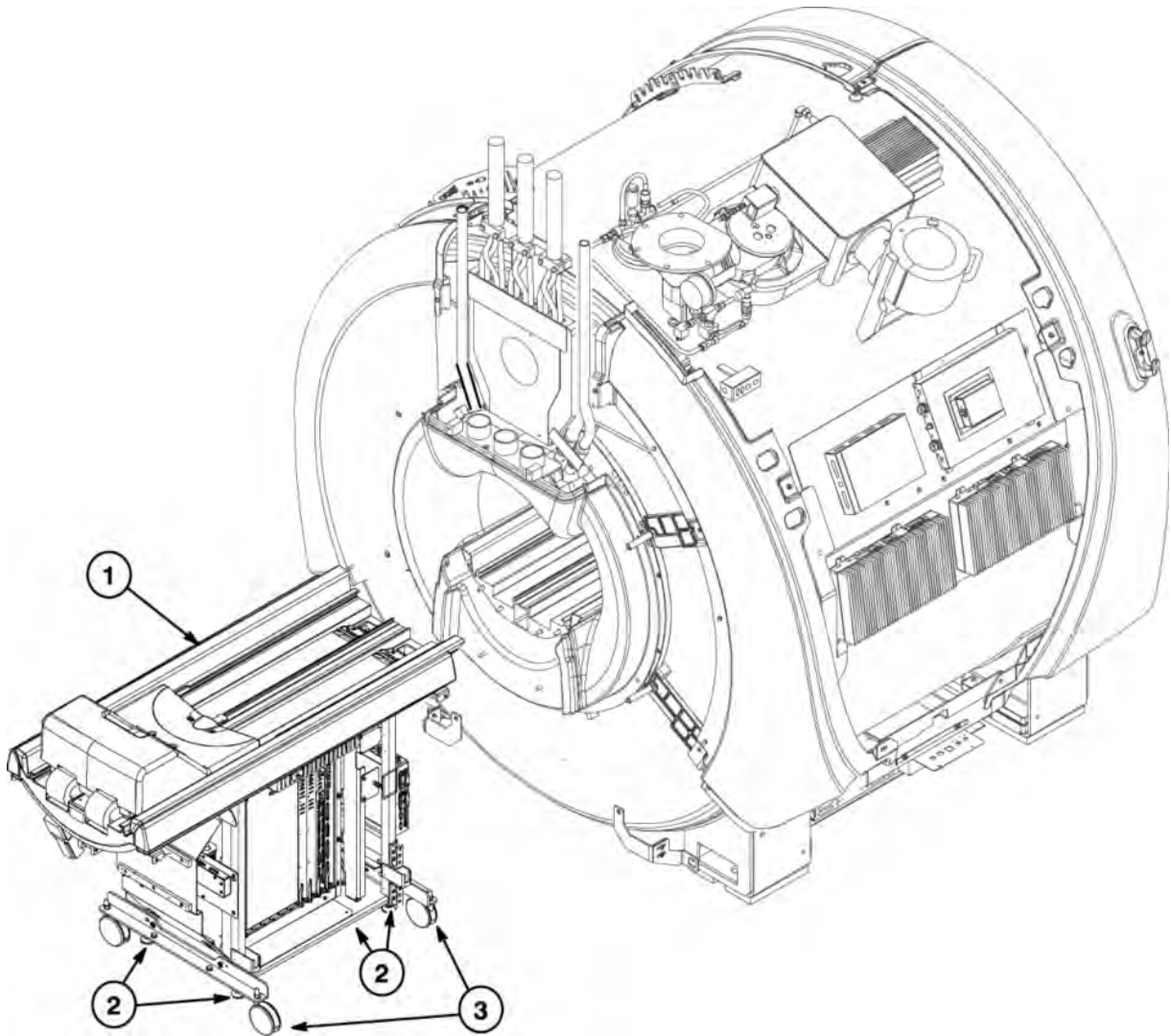


11 Position Rear Pedestal

The rear pedestal has been delivered to site with instructions on how to unload it from the shipping pallet. After unloading the rear pedestal, move it to rear of magnet.

The rear pedestal should initially positioned about 18 inches (458mm) from the installed position when connected to magnet. All cables and hoses will be connected to rear pedestal in this position. This will provide a service loop for any future maintenance or other service procedures.

Illustration 4-20: Initial Position of Rear Pedestal



1. Position the front surface of the bridge on the Rear Pedestal about 18 inches (458mm) from the normally installed position.
2. Lower the four leveling feet until wheels of front and rear shipping brackets are not touching the floor.
3. Remove front and rear wheeled shipping brackets.

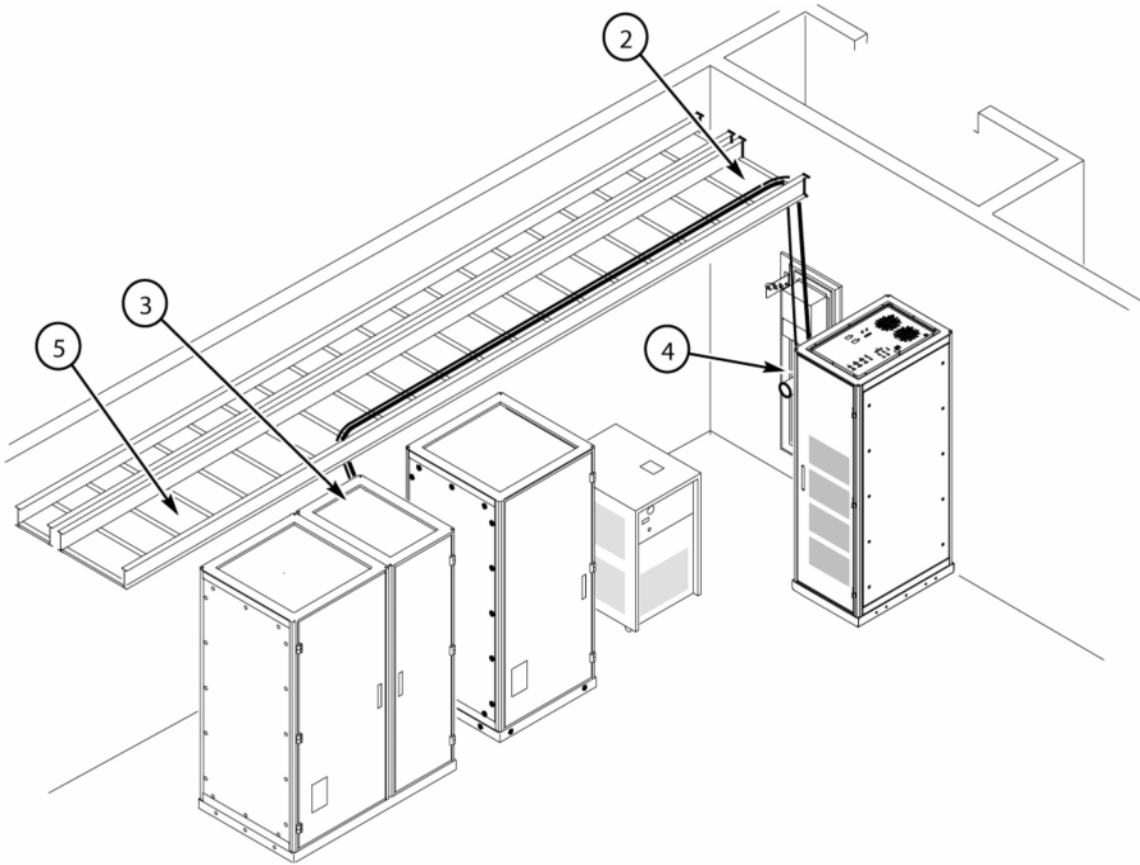
12 RF Transmit Cable Installation

12.1 Route And Connect In Equipment Room

It is recommended that the RF Transmit cables be routed through the overhead cable tray after the gradient cables have been routed and before any of the other cables are installed.

NOTE: The Equipment Room RF Transmit cables, Runs E1001 and E1002 are delivered to site pre-terminated and must not be cut to length.

Illustration 4-21: Equipment Room RF Cable Routing



1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray, route the Head and Body cables marked for connection to the PGR cabinet through the designated overhead cable tray towards the PGR Cabinet. Position in the cable tray as shown in [Cable Locations in Equipment Room Overhead Cable Trays](#).
3. Route the ends of the cable to the top of the PGR cabinet and connect as marked to J18 and J19.
4. Connect other ends of RF Transmit cables as marked to J106 and J128 on SPW.
5. Coil and store excess cable length in overhead cable tray.

12.2 Route and Connect in Magnet Room

It is recommended that the RF Transmit cables be routed thru the overhead cable tray after the gradient cables have been routed and before any of the other cables are installed.

NOTE: The Magnet Room RF Transmit cables, Runs M1001, M1305, and M1306 are delivered to site pre-terminated and **MUST NOT BE CUT TO LENGTH**.

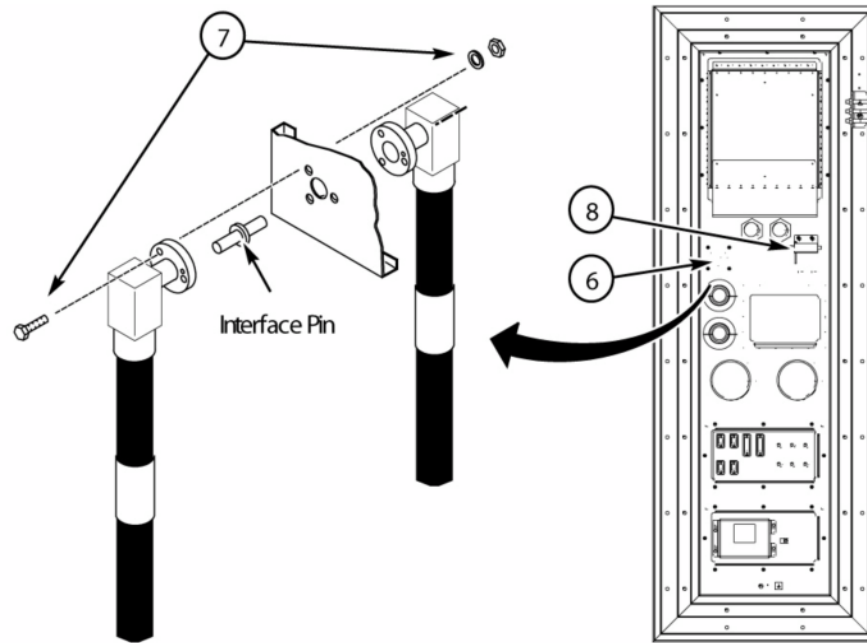
1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray, route Body cable M1001 and Head cables M1305 and M1306 through the designated overhead cable tray towards the rear of the magnet. Position in the cable tray as shown in [Cable Locations in Magnet Room Overhead Cable Trays](#).

Illustration 4-22: Magnet Room RF Cable Connections



3. Connect as marked M1001 to J1 on MAG-HYB.
4. Connect as marked M1305 to J1 on rear pedestal I/O panel.
5. Connect as marked M1306 to J2 on rear pedestal I/O panel.

Illustration 4-23: Connect RF Cables to SPW



6. Connect Run M1001 to J106 on SPW. Two people, one on each side of SPW, are needed to connect M1001 and E1001 to SPW.



NOTICE

Do NOT install O-Ring or Grease that may be provided with Cable Termination Kit. Installation of these parts will prevent the ground from making contact resulting in RF leaks.

7. Insert Interface Pin into one of the connectors. At J106, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Lock Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)
8. Connect Run M1305 to J122 (N.O.) and Run M1306 to J120 (N.C.) on the Transmit Switch
9. Coil and store excess cable length of M1001, M1305, and M1306 on cable hooks in Penetration Panel closet.

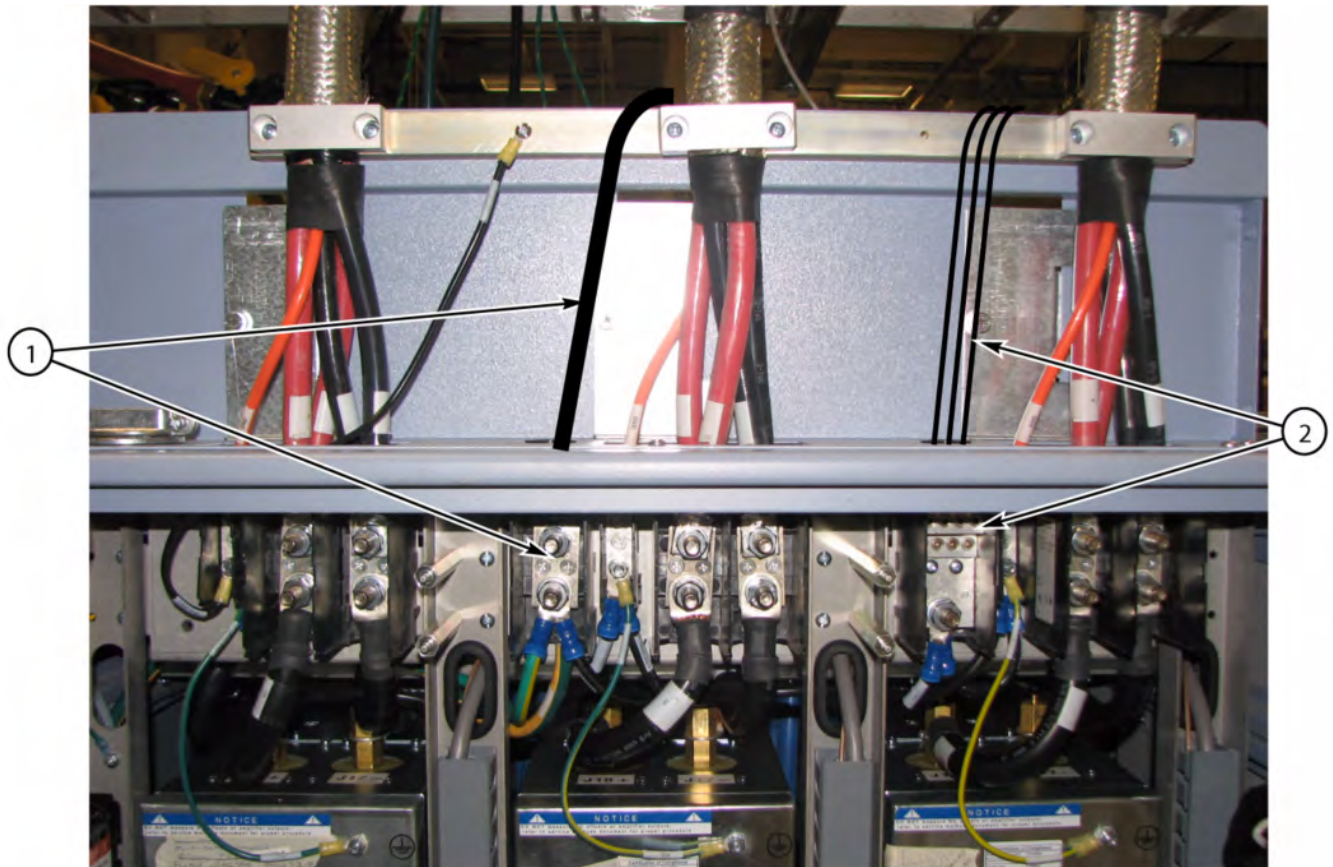
NOTE: The coil radius must be greater than the minimum bend radius for these cables.

13 Route and Connect Ground Cables

13.1 PGR Cabinet Ground Cables

The following ground cables are routed from the Operator Workspace, HEC Cabinet, PEN Cabinet, and RF common ground stud.

Illustration 4-24: PGR Cabinet Ground Connections

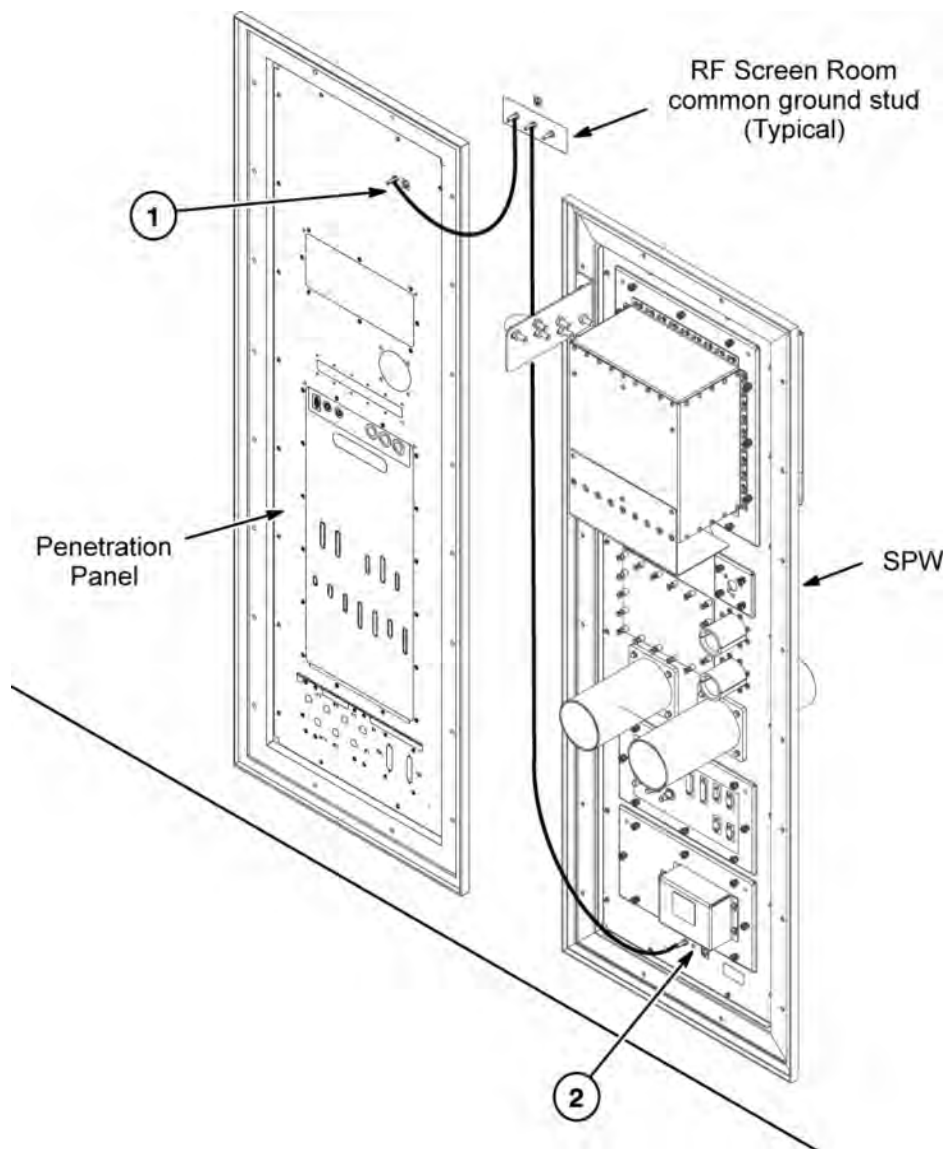


1. Route E4005 from RF common ground stud on screen room and connect to ground stud in PGR cabinet.
2. Route and connect the following to the PGR ground terminal as shown below:
 - E4002 from PEN cabinet.
 - E4007 from HEC cabinet.
 - E4008 from OW.

13.2 Penetration Panel And SPW Ground Cables

The ground cables in the following two steps are installed on the Equipment Room side of the panels.

Illustration 4-25: Pen Panel & SPW Ground Connections



1. Route and connect E4010 from RF common ground stud on screen room to ground stud in Penetration Panel.
2. Route and connect E4009 from RF common ground stud on screen room to ground stud at bottom of SPW.

14 Route and Connect Equipment Room Power Cables

To perform the Grounding and Leakage Current Checks procedure, only the power and ground cables can be connected. The test must be completed before any of the data cables can be connected.

NOTE: For efficiency purposes, all the cables can be routed through the overhead cable trays but only the power and ground can be connected for this test.



NOTICE

Do not connect data cables until the grounding and leakage test has been completed.

Ground cables were routed and connected in Section titled [Route and Connect Ground Cables](#). Connect the following power cables.

- E0003, PGR J102 to OW-PDM J1
- E0004, PGR J16 to PEN J1
- E0009, HEC CRY power to CRY power
- MagMon3 power cable to HEC

Make sure the following has been completed.

- Customer supplied power from MDP CB2 to PGR CB1
- Customer supplied power from MDP CB3 to HEC power

NOTE: The section titled [Route and Connect Equipment Room Data Cables](#) provides instructions on routing and connecting all cables in the Equipment Room.

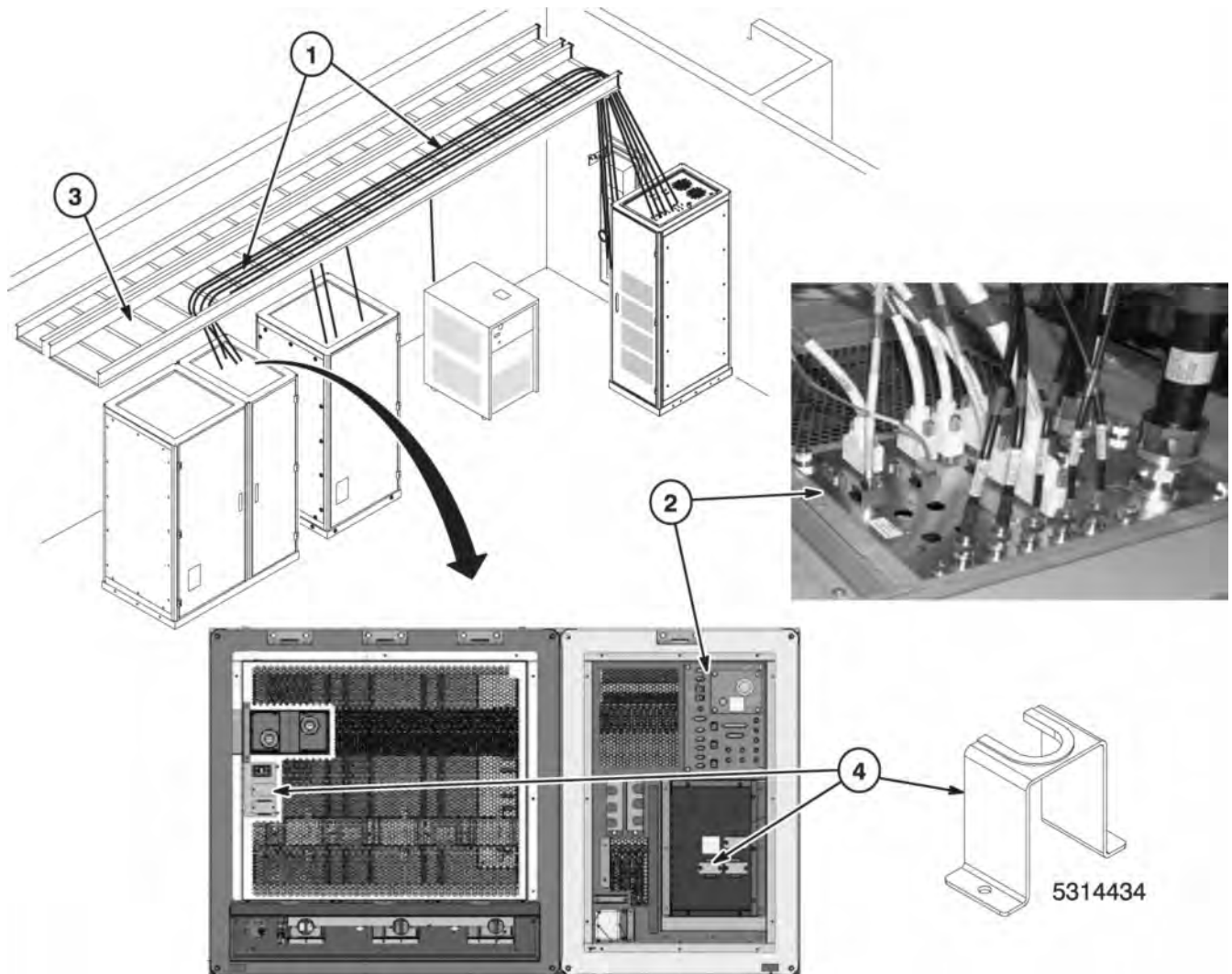
The section titled [Route and Connect Cables to Operator Workspace](#) provides instructions on routing and connecting all cables in the Operator Workspace.

15 Route and Connect Equipment Room Data Cables

Refer to [Cable Locations in Equipment Room Overhead Cable Trays](#) for positioning of cables and hoses routed in Equipment Room in the overhead cable routing mechanisms. Make sure cables are in their designated areas.

15.1 PGR Cabinet Cables

Illustration 4-26: PGR Cable Connections



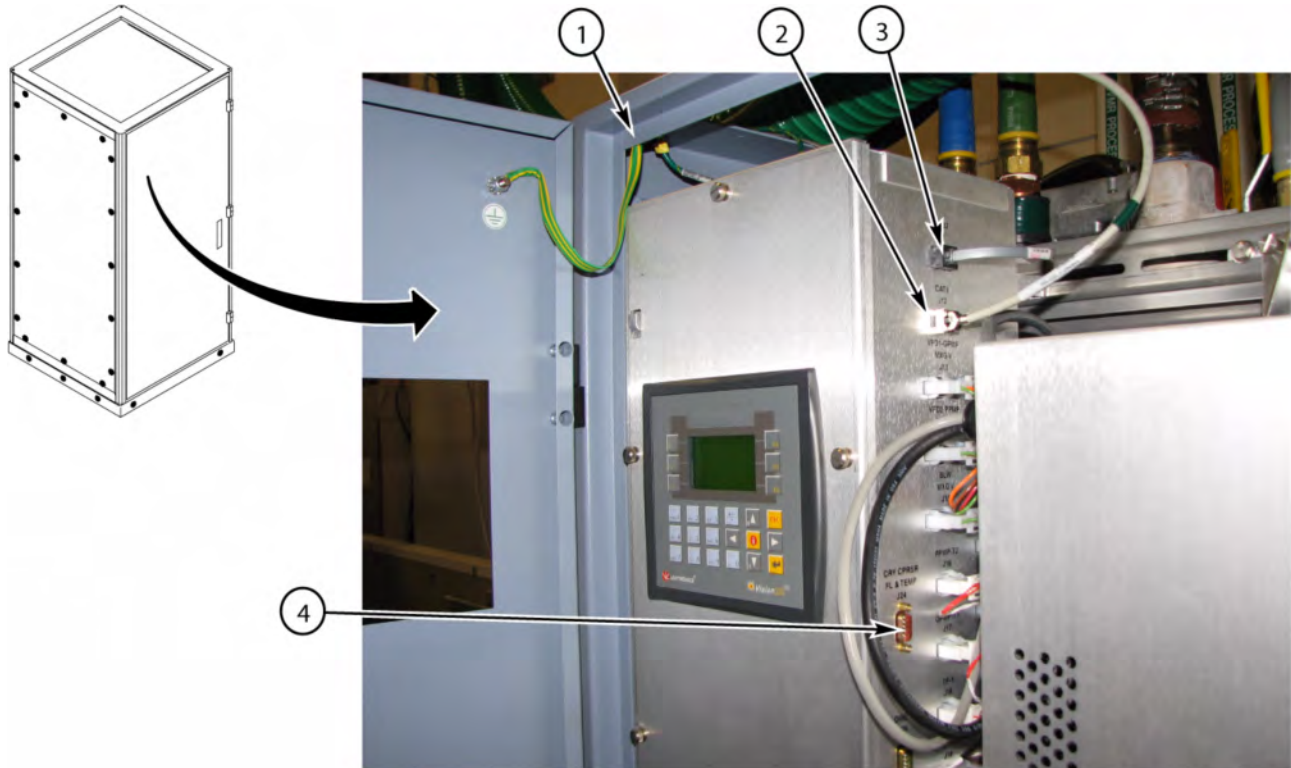
1. Route the cables thru the assigned channel in the 24 inch overhead cable routing mechanism (cable tray) to the cables final connection point.
2. Refer to cable map and connect remainder of data cables to top of PGR Cabinet after Grounding and Leakage test has been completed.
3. Refer to [Storage of Excess Cable Length](#) for coiling of any excess cable in the overhead cable trays.

- At J16 and J102, install Connector Bracket (5314434) on mounting standoffs to hold power cable Runs E0004 and E0003 in place. Bracket is part of Desktop Collector, 2321156-26.

15.2 HEC Cabinet Cables

Refer to cable map and route remainder of cables to HEC Cabinet. Connect data cables after Grounding and Leakage test has been completed.

Illustration 4-27: HEC Cable Connections

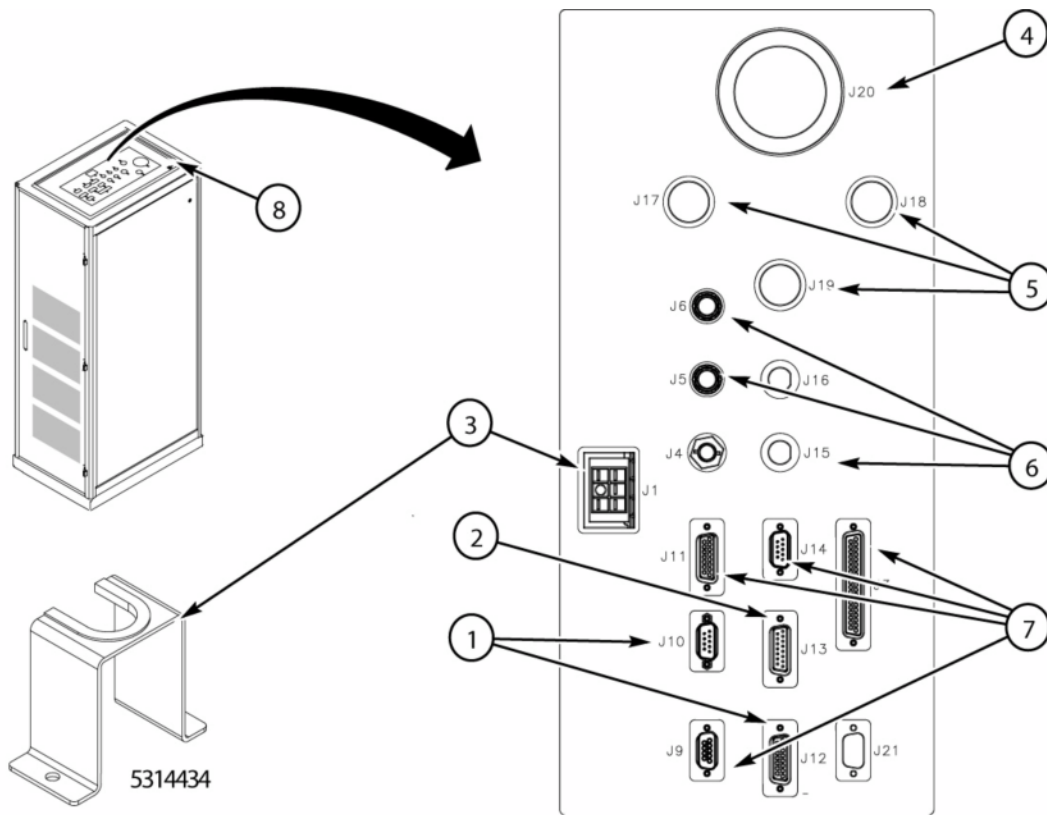


- Connect the ground wire E4007 from the PGR cabinet to ground stud at top of cabinet.
- Connect Run E3013 to J12 CAT5.
- Connect Run E3028 to J11 RS232.
- Run E3026 should have already been connected to J24 CRY CPRSR FL & TEMP.
- Magnet Monitor power cable, Run E3006, should have already been connected to the customer supplied power outlet.

15.3 PEN Cabinet Cables

Refer to cable map and route cables to PEN Cabinet. Connect after Grounding and Leakage test has been completed.

Illustration 4-28: PEN Cable Connections



1. Route Runs E3017 and E3018 from OW and connect to J10 and J12 as marked.
2. Route Runs E3020 from MagMon and connect to J13 as marked.
3. Route Power cable Run E0004 from PGR Cabinet and connect to J1. Install Connector Bracket (5314434) on mounting standoffs to hold E0004 in place. Bracket is part of Desktop Collector, 2321156-26.
4. Route fiber optic cable Runs P2003 and P2004 thru J20 and thru wave guide in Pen Panel to magnet.
5. Route Fiber Optic cable Runs E2002, E2020, and E2023 thru J19, J17, and J18 as marked. Refer to section titled [Install Fiber Optic Cables](#) for connection of fiber optic wires inside of PEN Cabinet.
6. Route Coax/RF Runs E1003, E1004, and E1005 from PGR Cabinet and connect to J5, J6, and J15.
7. Route Signal cable Runs E3002, E3006, E3008 and E3009 from PGR Cabinet and connect to J11, J14, J9, & J3.
8. Route and connect GND cable Run E4002 to ground stud.

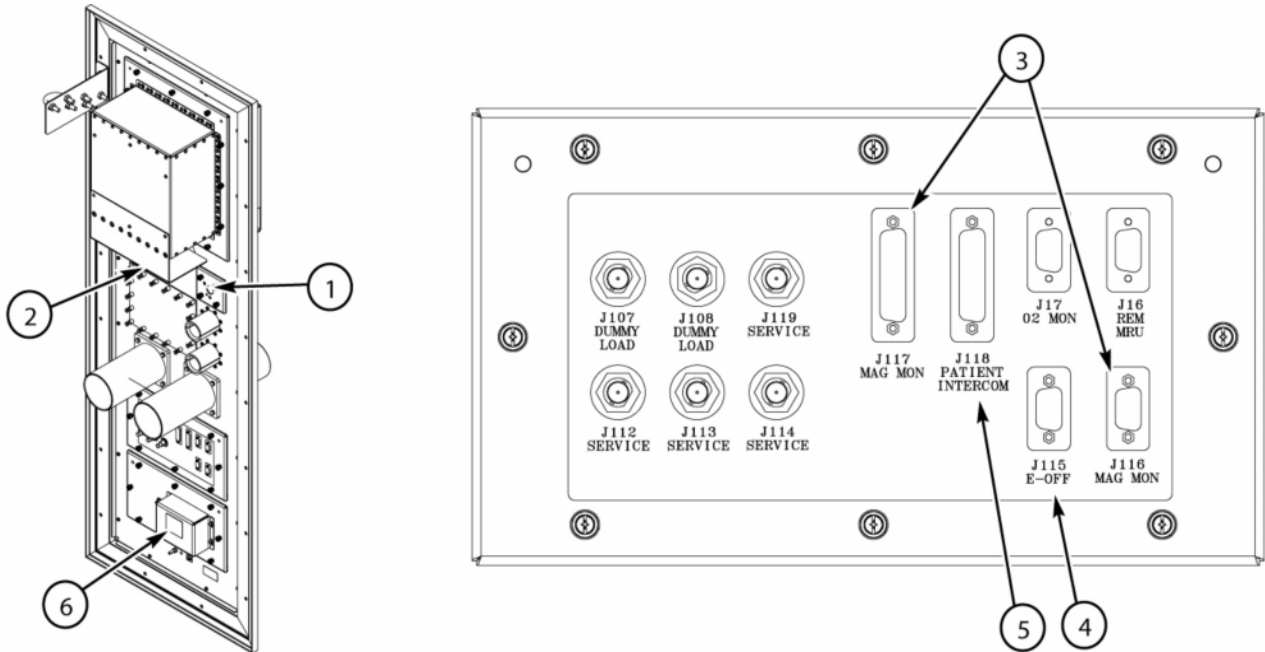
15.4 SPW Cables

The SPW (Secondary Pen Wall) was delivered to site and installed prior to magnet delivery.

- The gas lines, Runs 621 and 622, were routed thru the J110 and J111 waveguides when the magnet was installed.
- The gradient cables were routed and connected to the gradient filter in a previous procedure in the section.
- The coolant lines have been routed thru J100 and J101 waveguides.
- The 4 inch air hose from the PGR cabinet has been routed to J104 waveguide.

Refer to cable map and connect remaining cables to SPW after Grounding and Leakage test has been completed.

Illustration 4-29: SPW Cable Connections

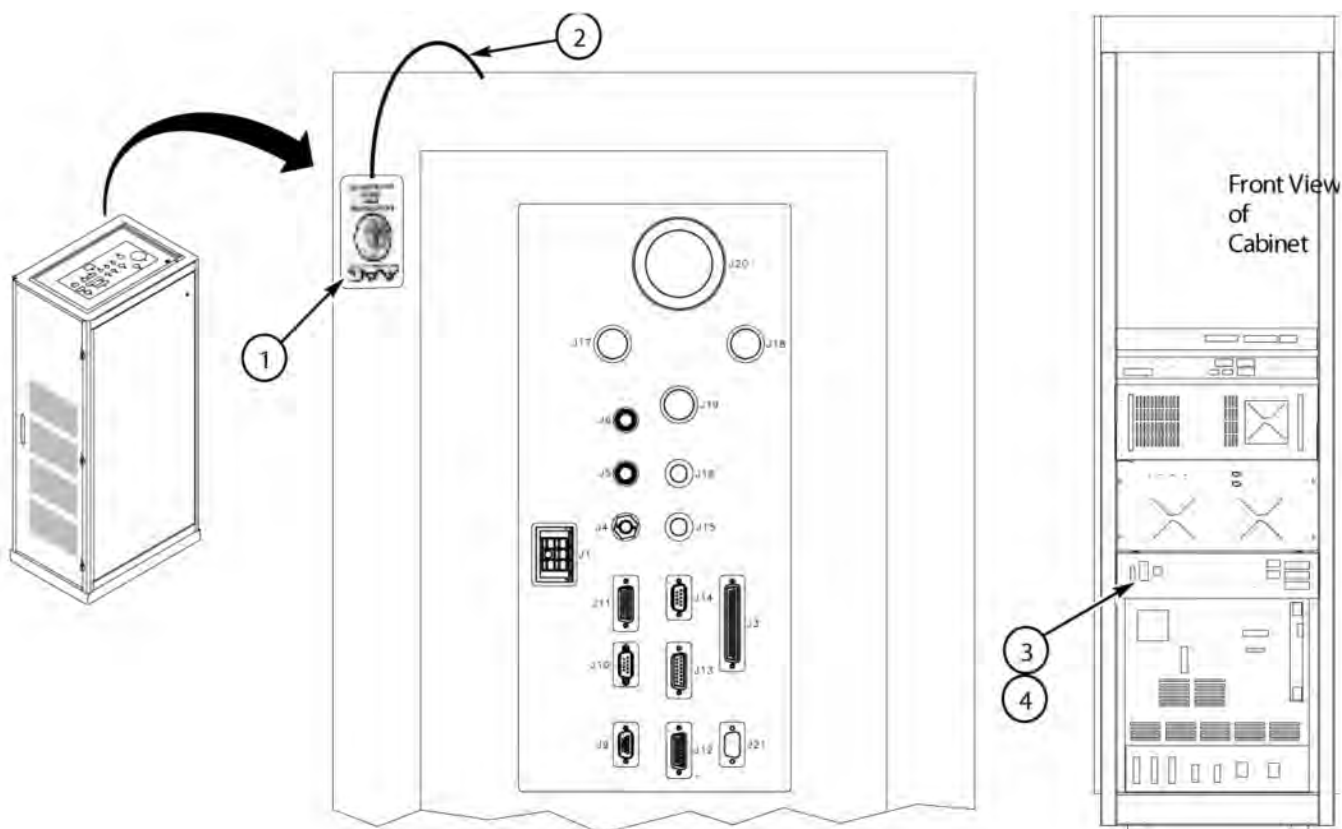


1. Route and connect Run E1001, Body TX, to J106.
2. Route and connect Run E1002, Head TX, to J128
3. Route and connect Runs E3022 and E3023 from MagMon to J116 and J117.
4. Route and connect Run E3030 from E-Off Switch to J115.
5. Route and connect Run E3031 from GOC to J118.
6. Remove safety cover. Route and connect Run 623 to F1 and F3. Replace safety cover.

16 Install Temperature/Humidity Sensor Components on PEN Cabinet

1. Retrieve the iButton sensor (5317298) from the Desktop Collector kit (2321156-26) and attach to top of PEN cabinet using dual locking tape.

Illustration 4-30: Component Installation



2. Connect RJ11 sensor cable (5317299) to iButton and route to rear of PHPS2 module in PEN Cabinet.
3. At J43 on rear of PHPS2 module, install DB9 to RJ11 Adapter (5318097).
4. Connect end of RJ11 sensor cable to RJ11 Adapter.

17 Install RF Door Switch Run E3014

Run E3014 is a 100 ft cable with a 9-pin subminiature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.

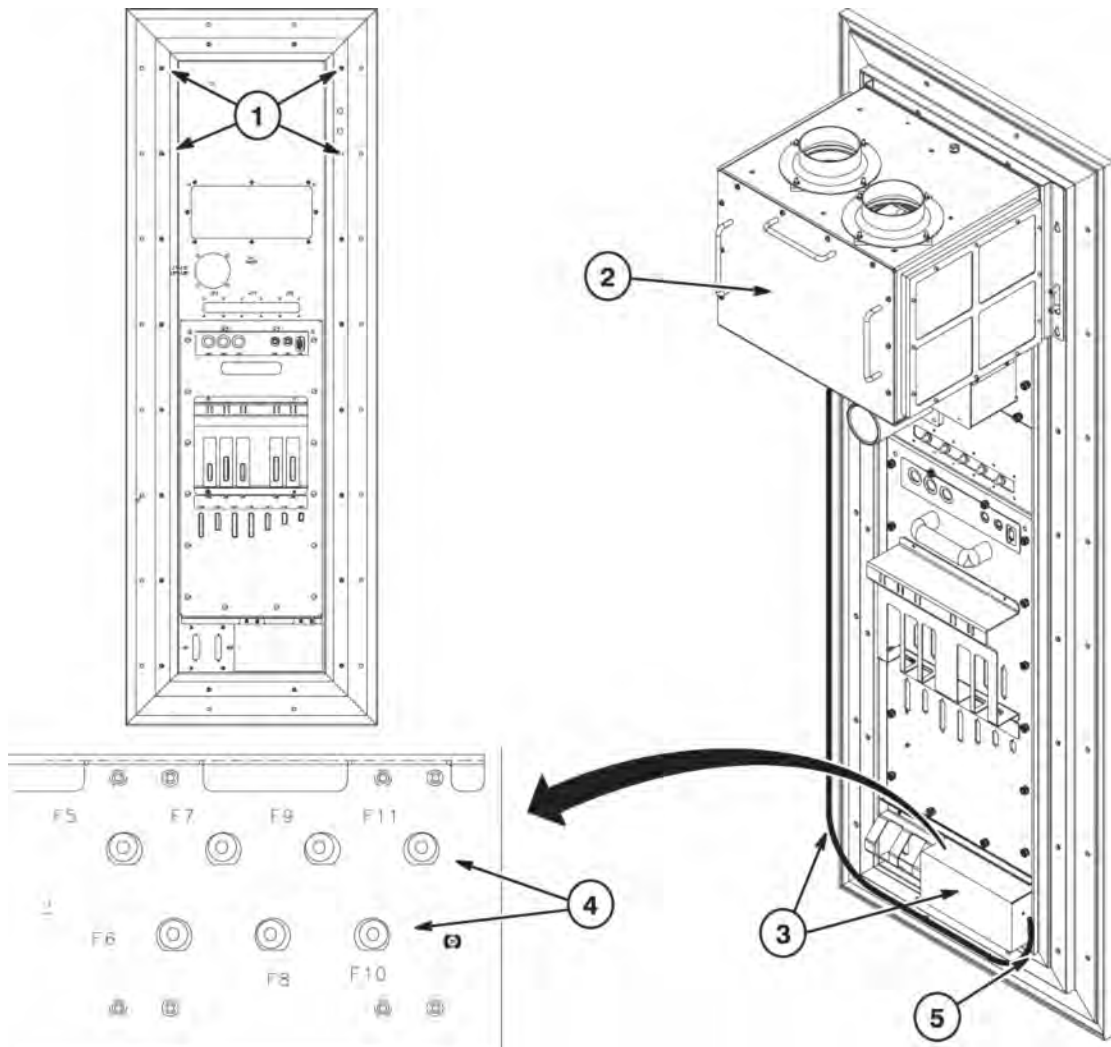
1. Connect Run E3014 to J20 on PGR Cabinet and route to RF Door Switch.
2. Connect black lead on Run E3014 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run E3014 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run E3014 to RF Door Switch N.C. (normally closed) contact.

18 Install Blower Box on Penetration Panel

NOTE: The Blower Box can be installed later in the installation process. Make sure that the area where the Blower Box is to be installed will stay clear of any routed cables and hoses.

1. Loosen, but do not remove, four screws securing top of Pen Panel.

Illustration 4-31: Blower Box Installation



2. Lift and slide Blower Box (5213397) mounting holes over loosened screws. Tighten the four screws to secure Blower Box to Pen Panel frame.
3. Route power cord to bottom of Pen Panel and remove safety power cover.
4. Connect power cord wires as marked to the seven connection points.
5. Re-attach safety power cover making sure that power cord is routed thru access hole.

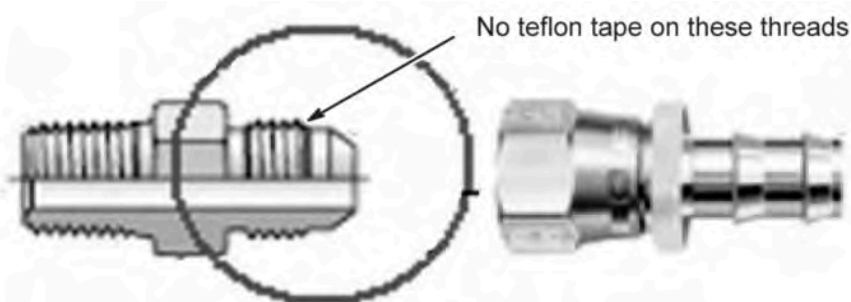
19 Install Air and Water Hoses

Refer to Section [Cable Locations in Equipment Room Overhead Cable Trays](#) for the position of cables and hoses routed in Equipment Room in the overhead cable trays. Make sure cables are in their designated areas of the trays.

Hose and Fitting Installation Notes:

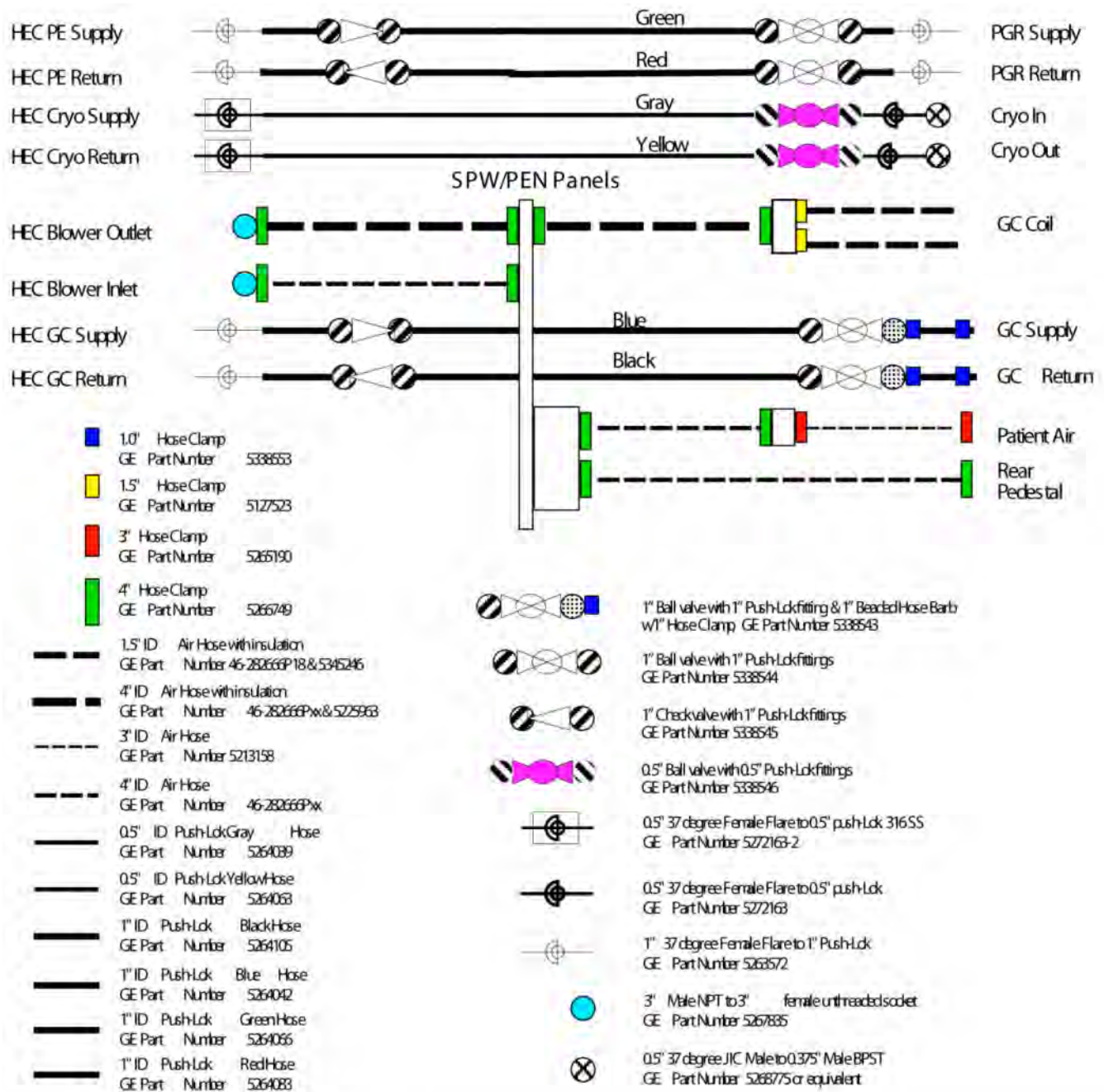
- To prevent cracking of fittings, hold fittings in the HEC stationary with wrench while flare connections are being made.
- Trim extra length off of hose clamps after applying clamp to the hoses. Smooth rough edges.
- DO NOT use teflon tape on threads that have a flare fitting (female and flare).

Illustration 4-32: Flare Fitting Assembly



- A small amount of water may be applied to push-lok barbs to facilitate insertion into hoses.

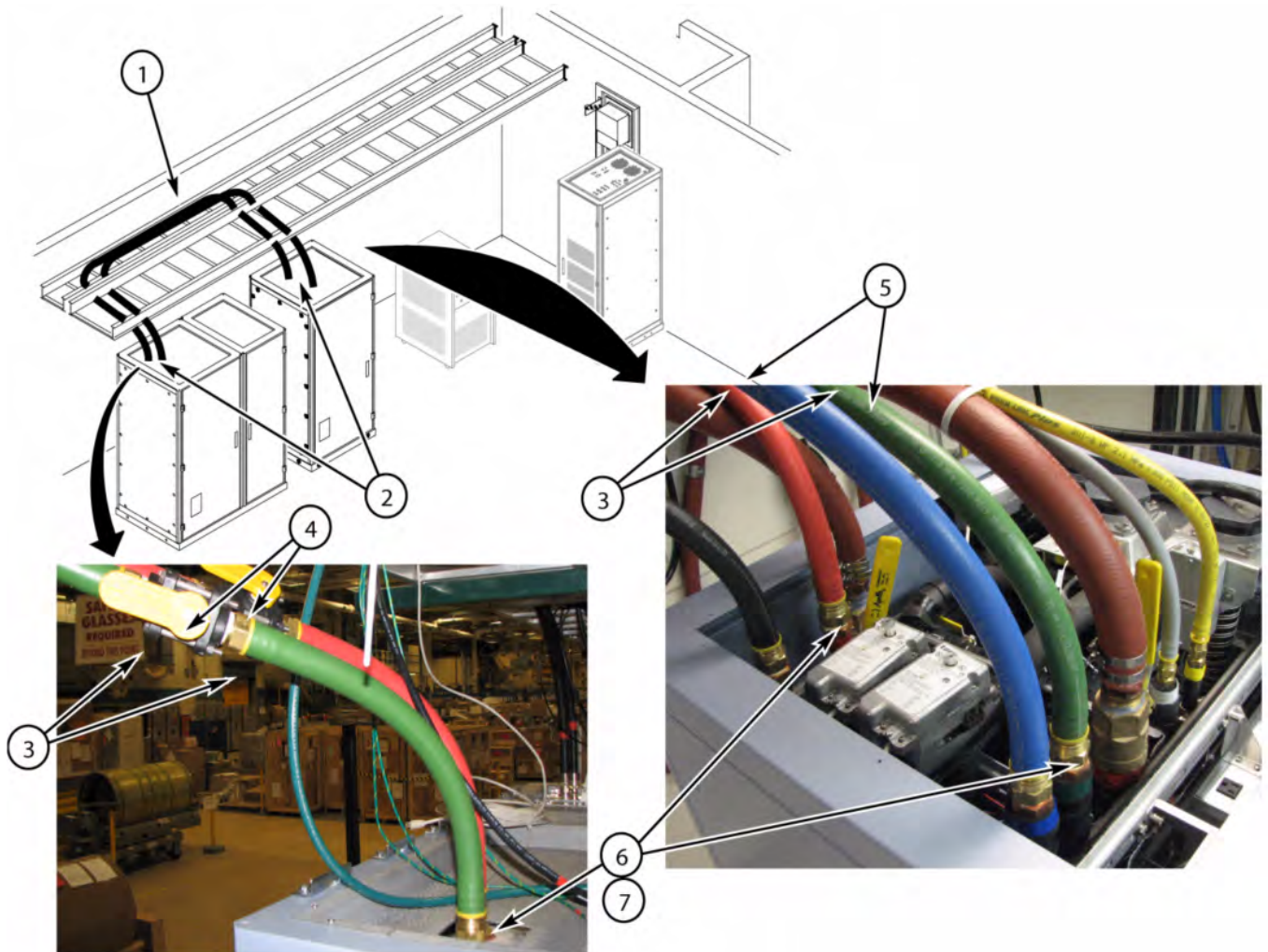
Illustration 4-33: Air and Water Hose Installation Schematic



19.1 Route and Connect Coolant Hoses Between PGR and HEC

Refer to [Illustration 4-33](#) for part numbers and description of hose hardware.

Illustration 4-34: Hoses Between PGR and HEC



1. Route RED (5264033) and GREEN (5264066) coolant hoses from HEC Cabinet thru the designated overhead cable tray to PGR Cabinet.

2. Cut to length.

NOTE: At each cabinet, ball and check valves are to be installed in each hose at a location of 1 to 3 feet (300 to 900mm) outside the cabinet. Make sure the distance measured is outside the perimeter of the top of the cabinet.

3. Cut each hose at the appropriate 1 to 3 foot mark outside the cabinet, as mentioned above.
4. At PGR, install 1" ball valve assembly (5338544).
5. At HEC cabinet, install 1" check valve assembly (5338545).
6. Insert 1" Flare to Push-Lok connector (5263572) at ends of each hose. DO NOT use teflon tape on threads.

NOTE: To prevent cracking of fittings when attaching flare connectors to cabinet connection points, hold fittings in the HEC stationary with wrench while flare connections are being made.

7. Connect RED and GREEN hoses to top of PGR and HEC cabinets as marked.

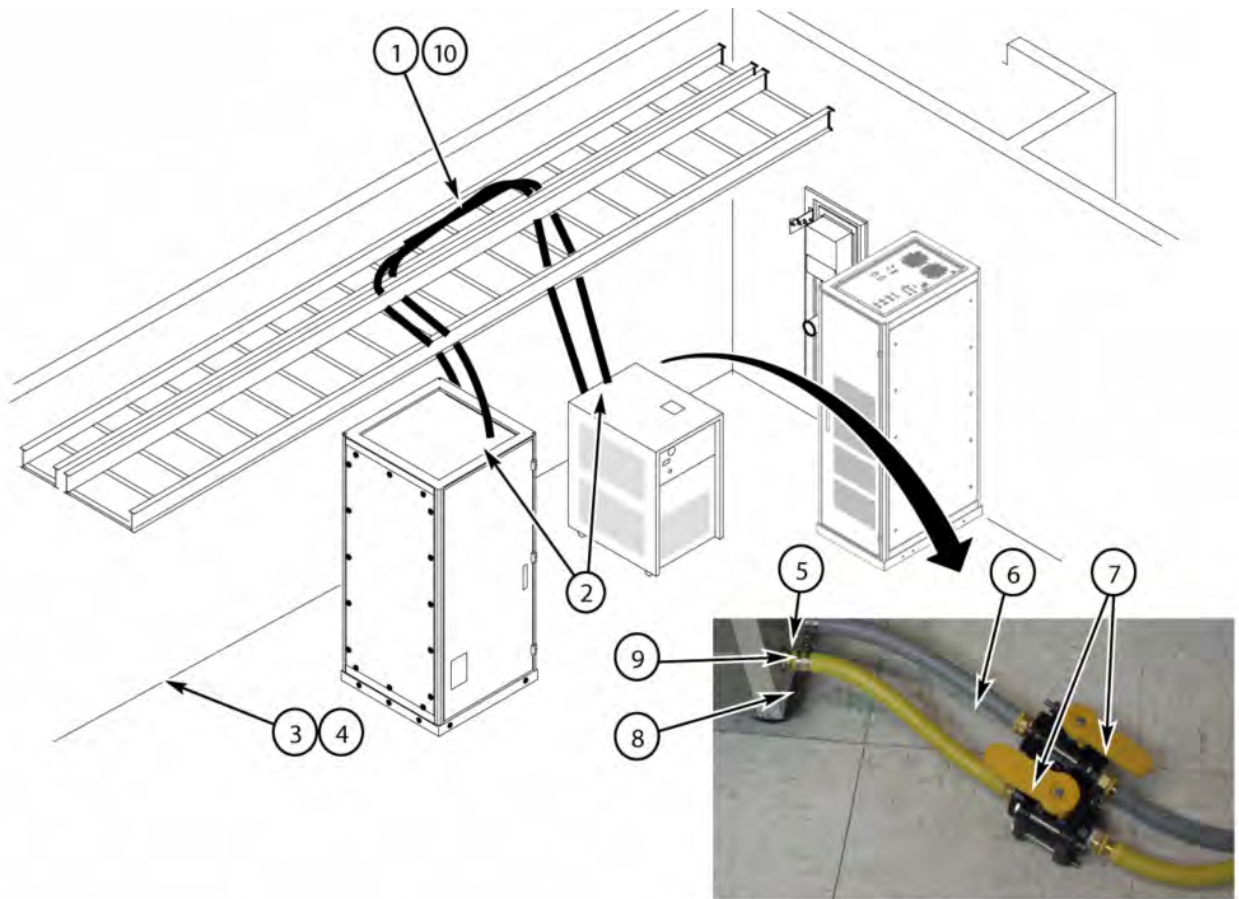
19.2 Route and Connect Coolant Hoses Between Cryogen Compressor and HEC



NOTICE

Use hose barbs and connectors mentioned in the following steps and not the hose barbs sent with F-50H compressor. This is for ease of disconnecting during service and to ensure proper hose barb connections to the hose. No hose clamps should be used with the Push-Lok hose barbs.

Illustration 4-35: Hoses Between Cryogen Compressor and HEC



1. Route GRAY and YELLOW coolant hoses from HEC through the designated overhead cable tray to Cryogen Compressor.
2. Cut coolant hoses to length. Make sure there is enough extra hose for a service loop at the Compressor end.
3. At the HEC Cabinet end of each hose, install the Stainless Steel 0.5 inch 37 degree Female Push-Lok Swivel fittings (5272163-2). DO NOT use teflon tape on threads.

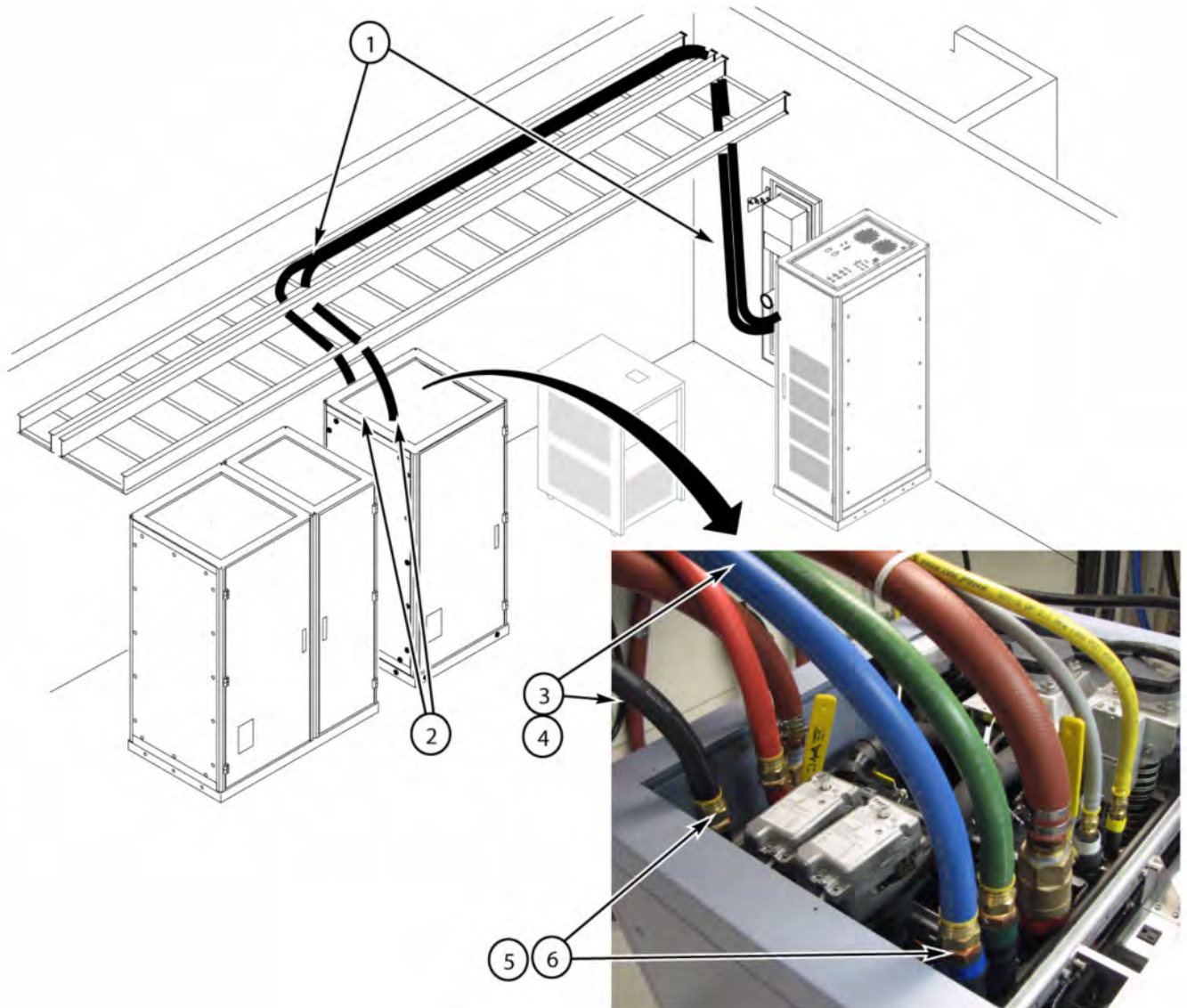
NOTE: To prevent cracking of fittings when attaching flare connectors to cabinet connection points, hold fittings in the HEC stationary with wrench while flare connections are being made.

4. Connect each hose to designated RETURN inlet (YELLOW) and SUPPLY outlet (GRAY) at top of HEC Cabinet.
5. At the Compressor, in both IN and OUT ports, install a 0.5 inch 37 degree JIC Male to 0.375 inch Male BPST fitting (5268775) and a brass 0.5 inch Female Flare to 0.5 inch Push-Lok fitting (5272163).
6. At the Compressor, for both the GRAY and YELLOW hoses, measure 18 inches (457mm) from end and cut hose.
7. Install Ball Valve assembly (5338546) for each hose between cut ends.
8. Attach YELLOW hose to OUT on Compressor.
9. Attach GRAY hose to IN on Compressor.
10. Attach insulation sleeves (5266250) to supply and return hoses.
11. Check for leaks.

19.3 Route and Connect Gradient Coil Water Coolant Hoses in Equipment Room

NOTE: To prevent cracking of fittings when attaching flare connectors to cabinet connection points, hold fittings in the HEC stationary with wrench while flare connections are being made.

Illustration 4-36: Gradient Coil Coolant Hoses in Equipment Room



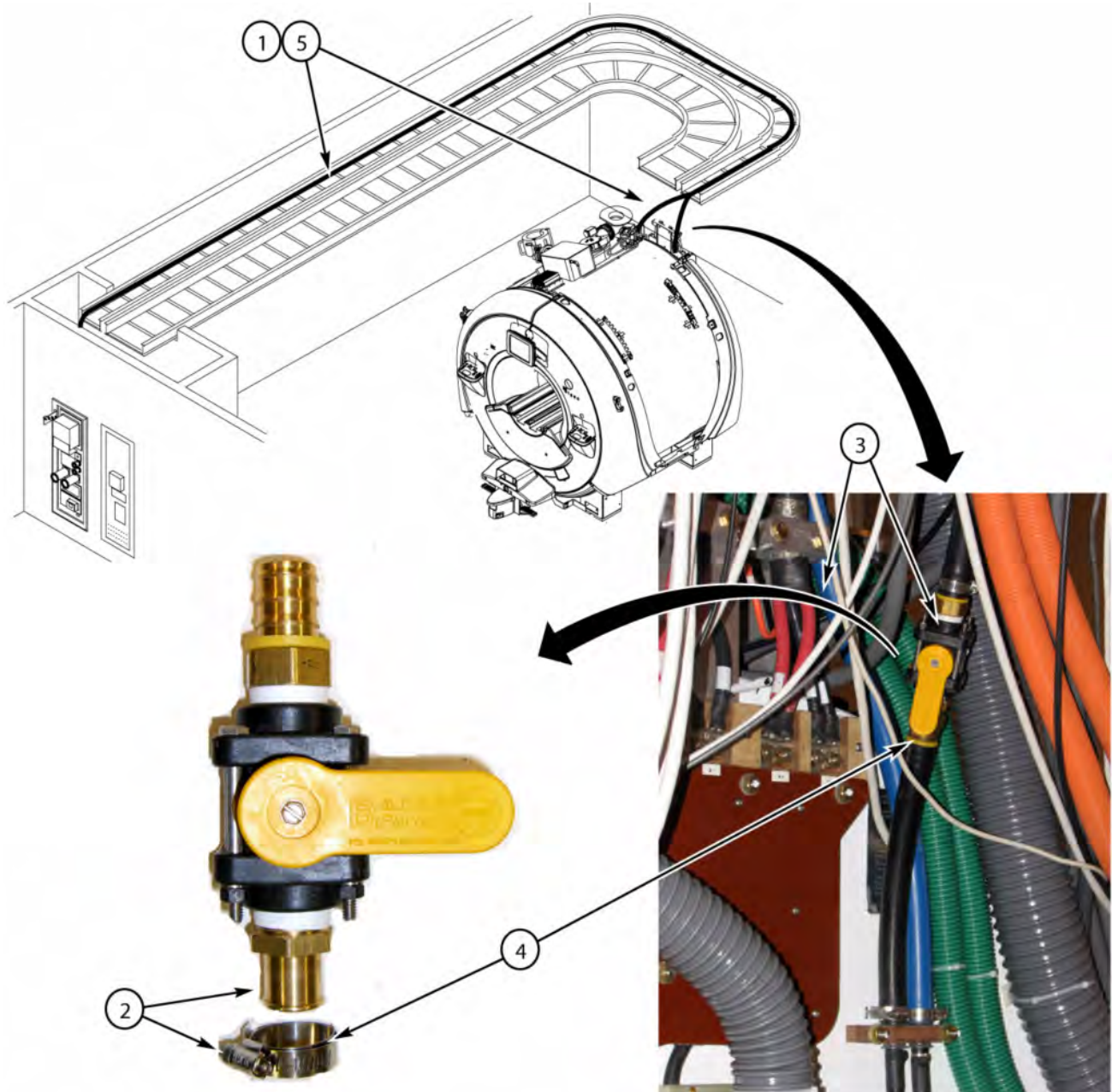
1. Route BLUE and BLACK coolant hoses from HEC Cabinet the designated overhead cable tray to wave guides J100 and J101 in SPW. Route thru wave guides to rear of magnet. Refer to Section 4-20-5 for routing and connection of Gradient Coil cooling water hoses in magnet room.
2. Cut coolant hoses to length at HEC Cabinet.
3. At the HEC cabinet, for both the BLUE and BLACK hoses, check valves are to be installed in each hose at a location of 1 to 3 feet (300 to 900mm) outside the cabinet. Make sure the distance measured is outside the perimeter of the top of the cabinet. Cut each hose.
4. Install Check Valve assembly (5338545) for each hose between cut ends. Make sure valve is installed in the correct direction.
5. At the end of each hose, install a 1 inch 37 degree Female Push-Lok Swivel fitting (5263572). DO NOT use teflon tape on threads.

6. Connect each hose to designated RETURN inlet (BLACK) and SUPPLY outlet (BLUE) at top of HEC Cabinet.

19.4 Route and Connect Gradient Coil Water Coolant Hoses in Magnet Room

One Inch Ball Valve assemblies (5338543) will be installed in each hose.

Illustration 4-37: Gradient Coil Coolant Hoses in Magnet Room



1. Hoses were routed in [Section 19.3](#) from HEC through SPW to rear of magnet and cut to length.
2. NOTE: It is important that the ball valve assembly with beaded hose barb and clamp (5338543) be inserted and clamped to hose routed from the gradient coil. Orientation shown below.

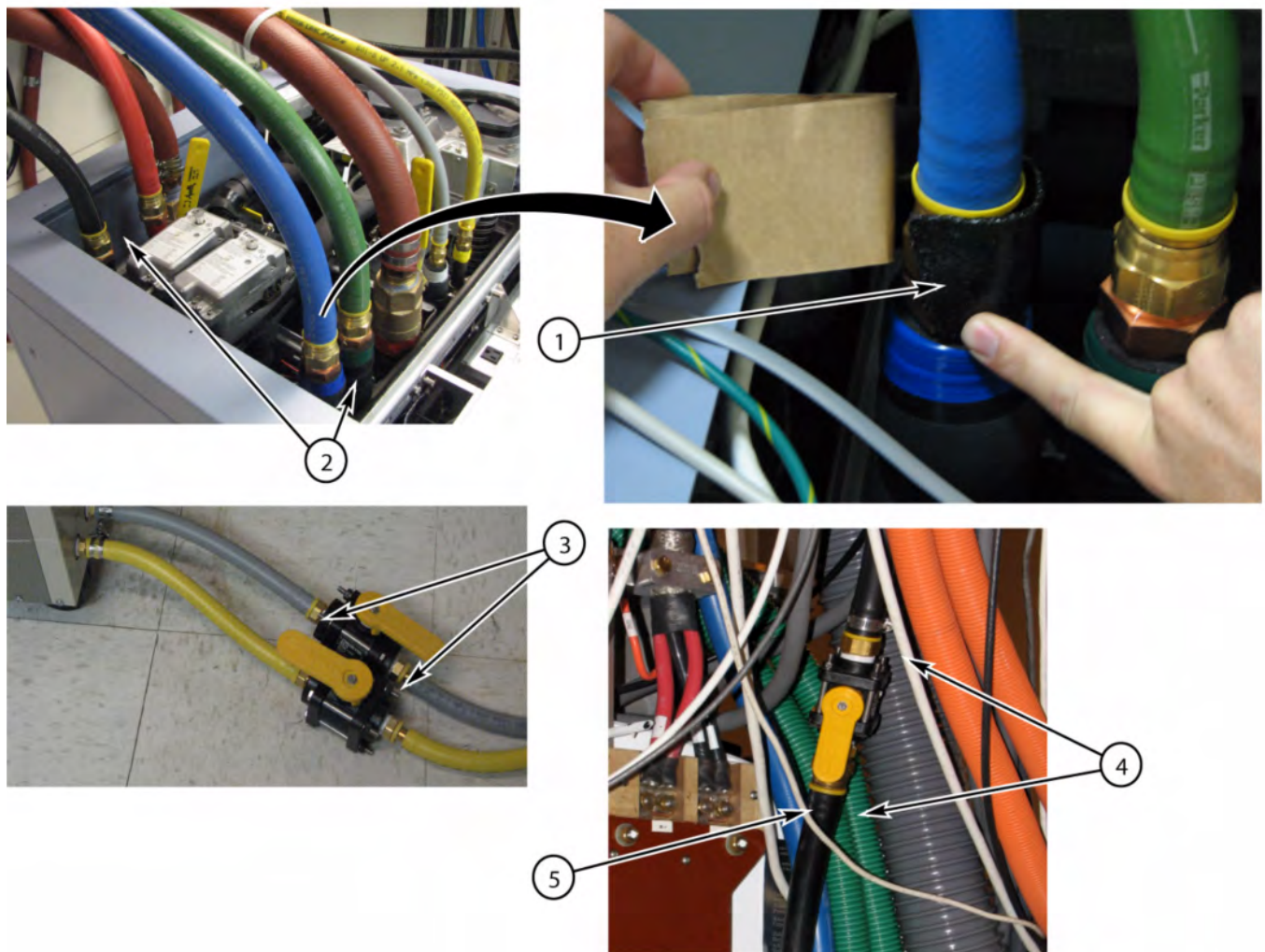
3. Attach each ball valve and fitting assembly to BLACK and BLUE hoses as shown.
4. Install hose clamp on gradient coil hose below the ball valve on the BLACK and BLUE hoses.
5. Install insulation sleeves (5225968) around 1 inch water coolant hoses in both the Magnet and Equipment rooms.

NOTE: When handle is perpendicular to hose, ball valve is closed.

19.5 Apply Insulation on Fittings

The Cork/Rubber Insulation Wrap (5268212) must be applied to all gradient coil and cryogen compressor hose connection points.

Illustration 4-38: Applying Insulation



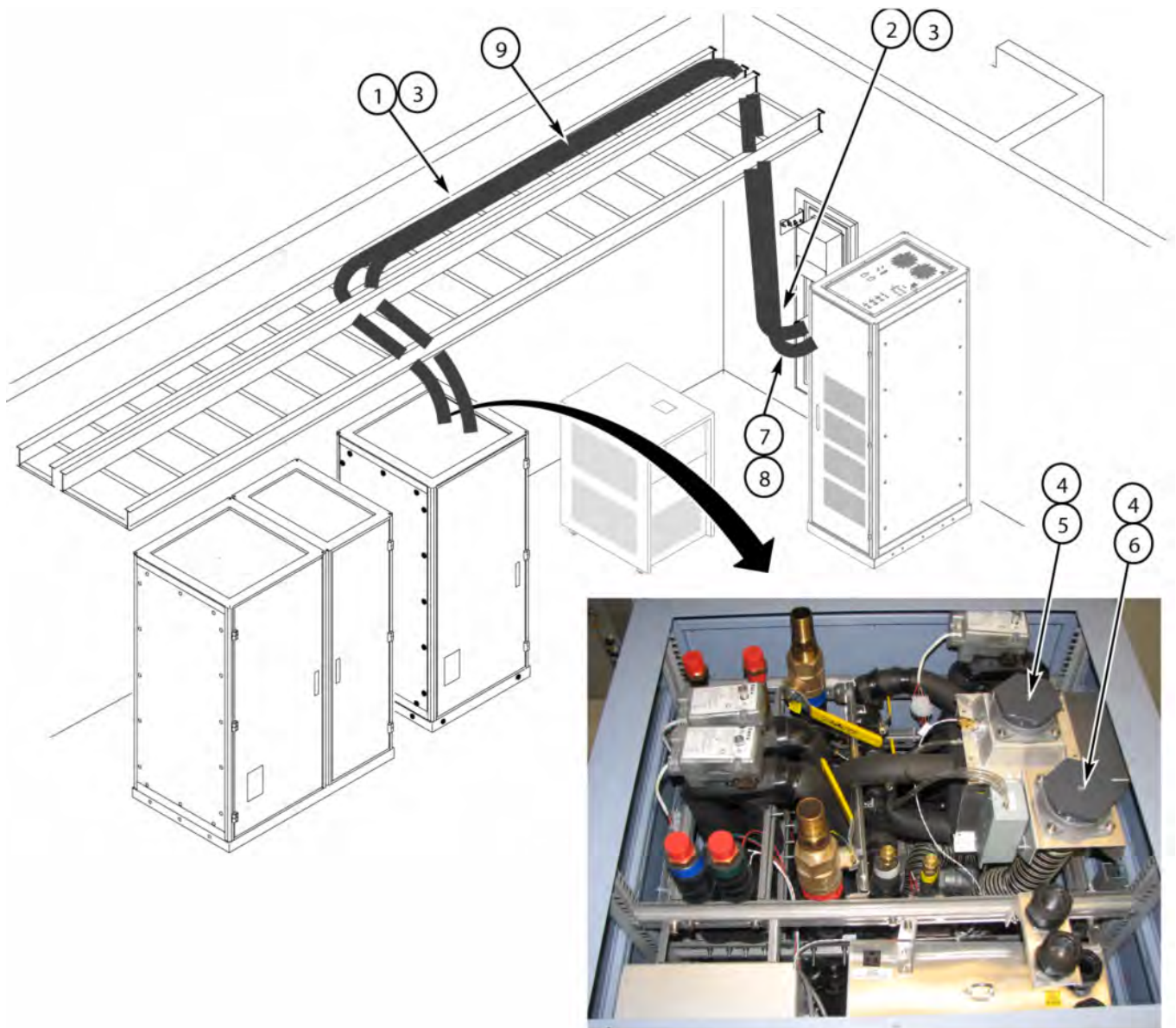
1. Apply and secure a length of Insulation Wrap to cover the BLUE hose connection.
2. Repeat process in Step 1 for the GREEN, BLACK, RED, GREY, and YELLOW hoses at the top of the HEC cabinet.
3. Repeat process in Step 1 for the GREY and YELLOW hoses at the Cryogen Compressor.

4. Repeat process in Step 1 for the BLACK hose at the rear of the magnet connecting to the Gradient Coil.
5. Repeat process in Step 1 for the BLUE hose at the rear of the magnet connecting to the Gradient Coil.

19.6 Route and Connect Air Coolant Hoses Between HEC and SPW

NOTE: Trim extra length off of hose clamps after applying clamp to the hoses. Smooth rough edges.

Illustration 4-39: Air Hoses Between HEC and SPW



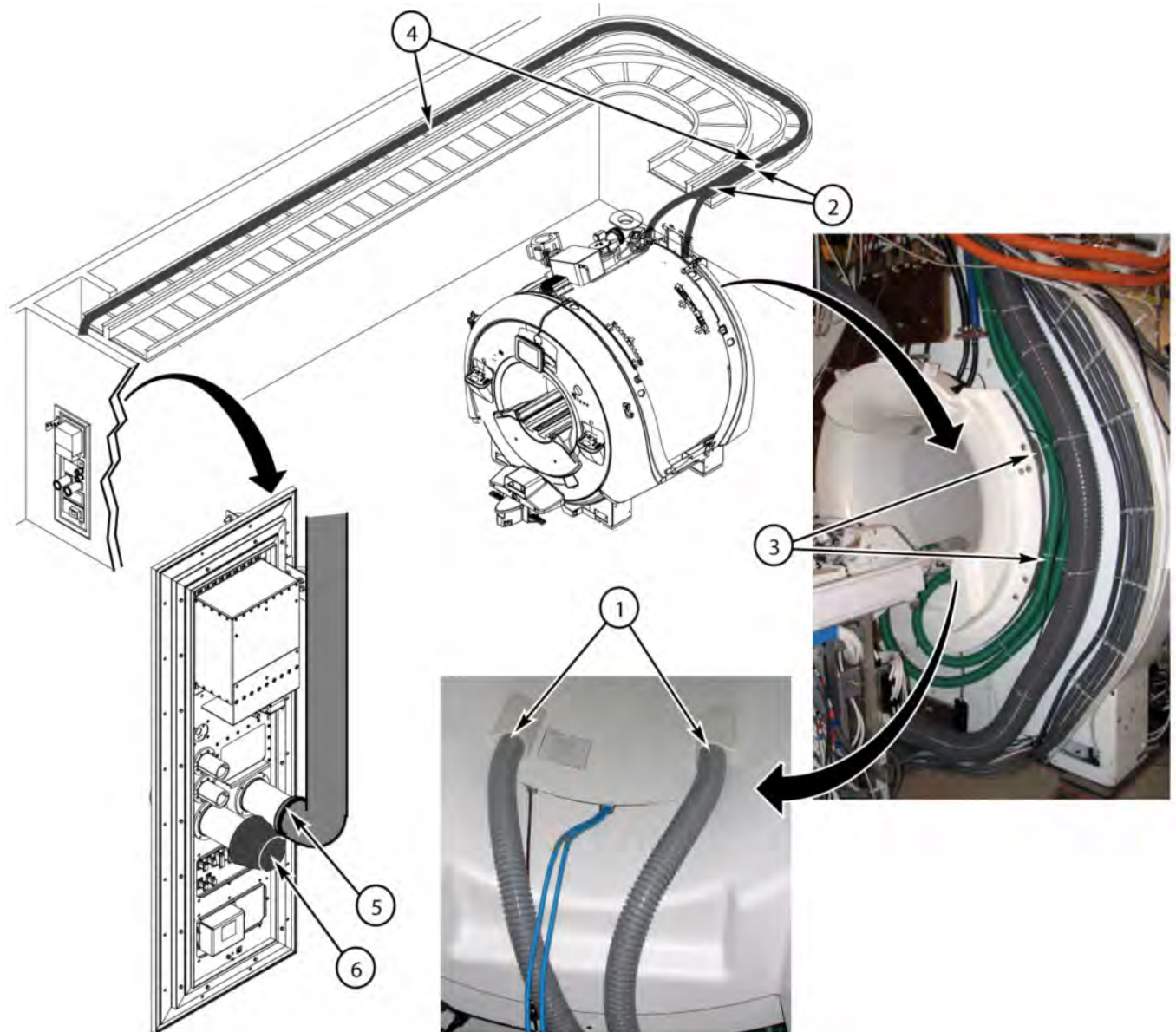
1. Route 4 inch SUPPLY air hose (46-282666Pxx) in the designated overhead cable tray from SPW to the HEC.

2. Make sure there is enough length for insulation that will be applied around the hose in Step 9. Cut to length.
3. Repeat steps 1 and 2 for routing and cutting 4 inch RETURN air hose, with the exception of insulation installation.
4. Remove and discard cap and install 3inch PVC Socket Connection (5267835).
5. Connect 4 inch SUPPLY air hose to top of HEC cabinet marked SUPPLY with 4 inch hose clamp (5266749).
6. Connect 4 inch RETURN air hose to top of HEC cabinet marked RETURN with 4 inch hose clamp (5266749).
7. Connect other end of 4 inch SUPPLY air hose to J104 on SPW and secure with 4 inch hose clamp (5266749).
8. Connect other end of 4 inch RETURN air hose to J105 on SPW and secure with 4 inch hose clamp (5266749).
9. Attach insulation sleeves (5225963) to SUPPLY 4 inch hose.

19.7 Route and Connect Air Coolant Hoses Between SPW and Body Coil

NOTE: Trim extra length off of hose clamps after applying clamp to the hoses. Smooth rough edges.

Illustration 4-40: Air Hoses Between SPW and Gradient Coil



1. At rear of magnet, install two 1.5 inch, 6 meter long, Air Hoses (46-282666P18) on the two sockets.
2. Route entire length of each air hose into overhead cable trays and attach to Air Y-Fitting (5344836) with 1.5 inch hose clamp (5127523) on each hose.
3. Secure each air hose to rear flange of magnet using provided tie-wraps.
4. Route 4 inch Air Hose (46-282666Pxx) in overhead cable tray and attach to Air Y-Fitting with 4 inch hose clamp (5266749).
5. At J104 waveguide on SPW, cut air hose to length and attach to waveguide with 4 inch hose clamp (5266749).

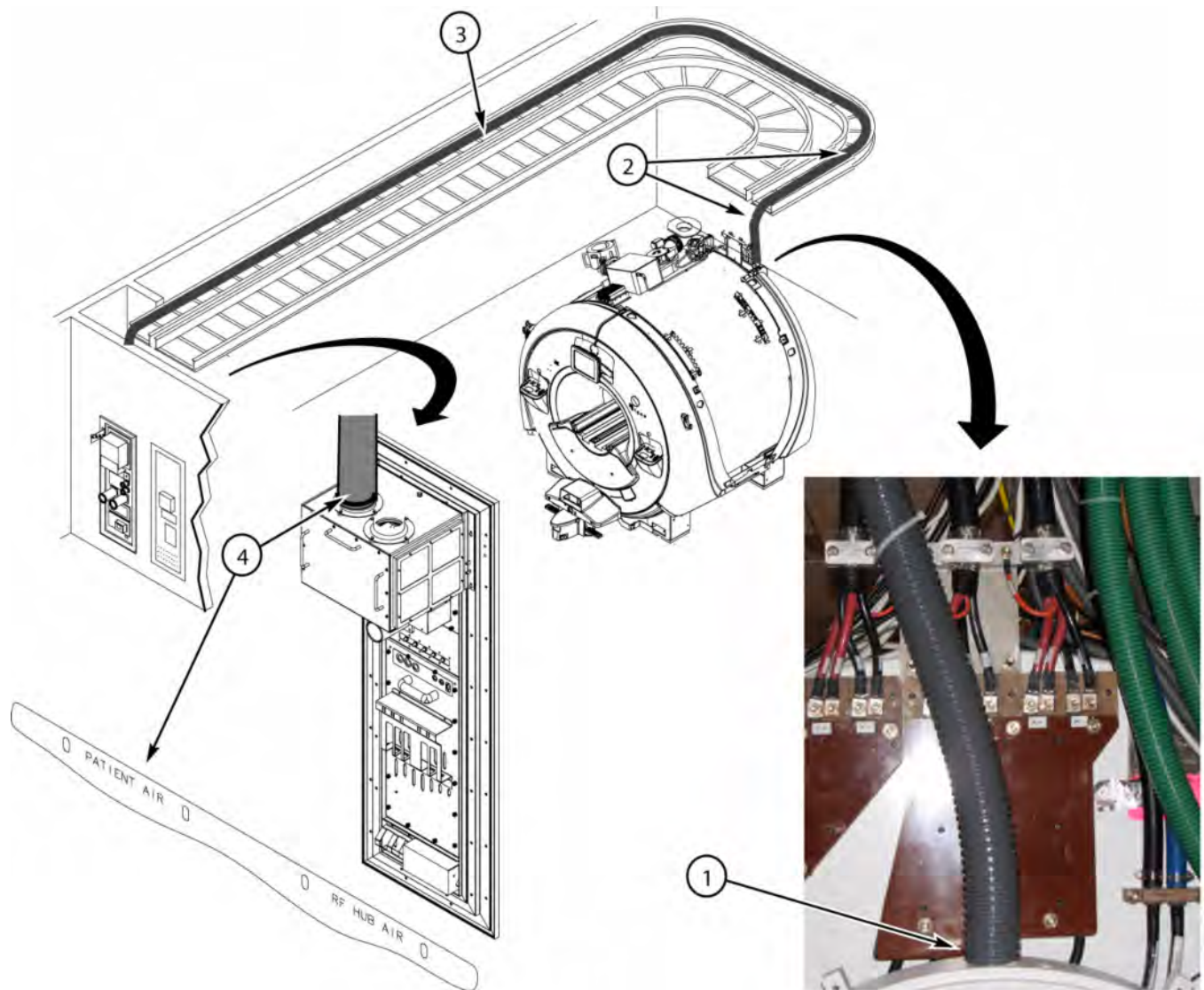
6. Air Filter (5290917) should have been installed with SPW and attached with Hose Clamp (5263772-3).

19.8 Route and Connect Air Hose Between Blower Box and Rear End Bell

This hose is installed to provide air for patient comfort.

NOTE: Trim extra length off of hose clamps after applying clamp to the hoses. Smooth rough edges.

Illustration 4-41: Installing Patient Comfort Air Hose

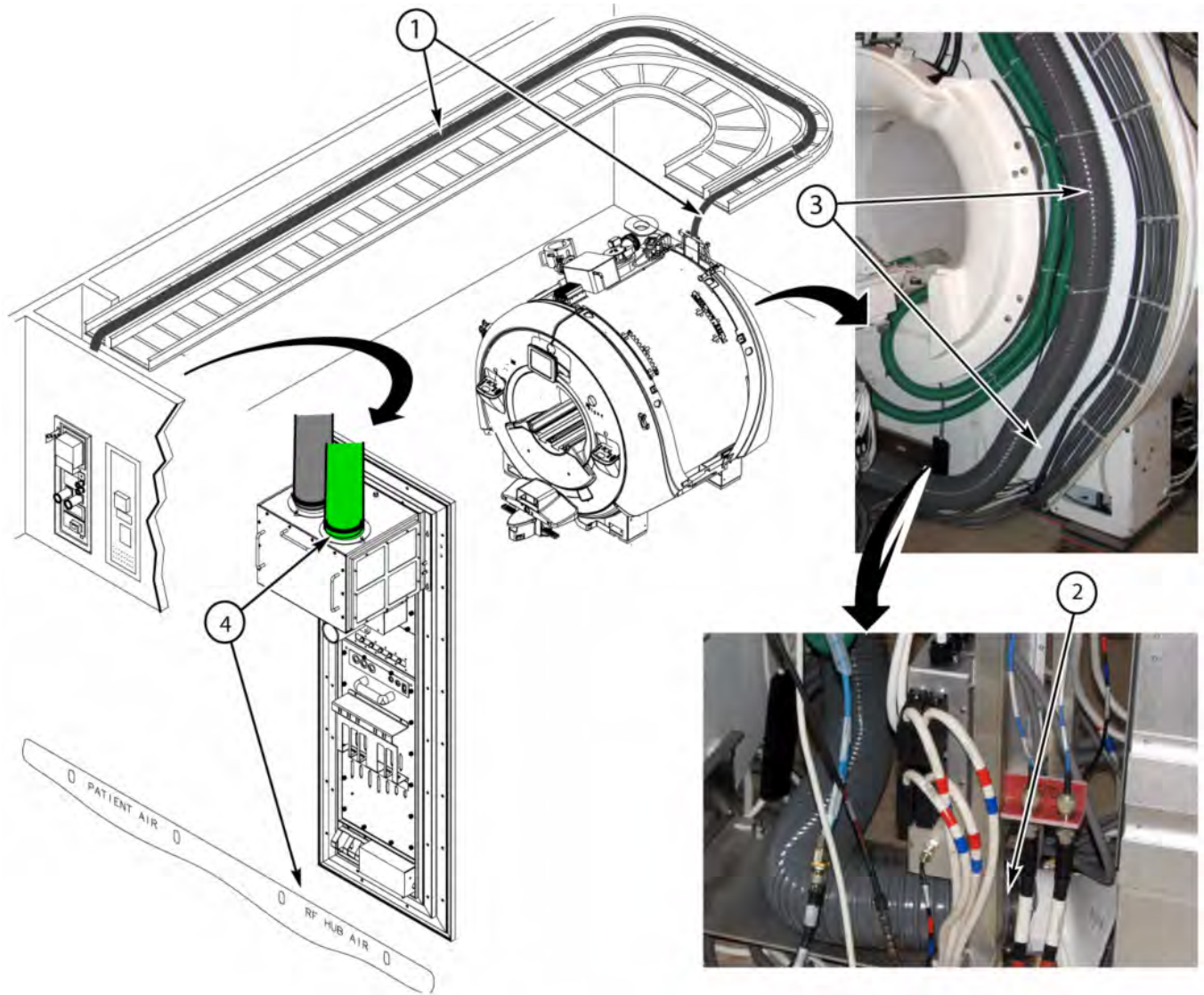


1. At rear of magnet, install 3 inch, 3 meter long, PATIENT AIR Hose (5213158) on the top of the rear end bell. Secure with 3 inch hose clamp (5265190).
2. Route entire length of 3 inch air hose into overhead cable trays and attach to PVC Reducer (5198521) with 3 inch hose clamp (5265190).

3. Route 4 inch, 11 meter long, Air Hose (5199506-2) in overhead cable tray and attach to PVC Reducer with 4 inch hose clamp (5266749).
4. At the Blower Box, cut patient comfort air hose to length and attach to the left side flange labeled PATIENT AIR with 4 inch hose clamp (5266749).

19.9 Route and Connect Air Hose Between Blower Box and Rear Pedestal

Illustration 4-42: Installing Rear Pedestal Air Hose



1. Route 4 inch, 14 meter long, RF HUB AIR Hose (5199506-3) in overhead cable tray to rear pedestal.
2. Attach hose to socket on rear pedestal with 4 inch hose clamp (5266749).
3. Secure air hose to rear flange of magnet using provided tie-wraps.
4. At the Blower Box, cut rear pedestal air hose to length and attach to the right side flange labeled RF HUB AIR with 4 inch hose clamp (5266749).

20 In Room Display Installation

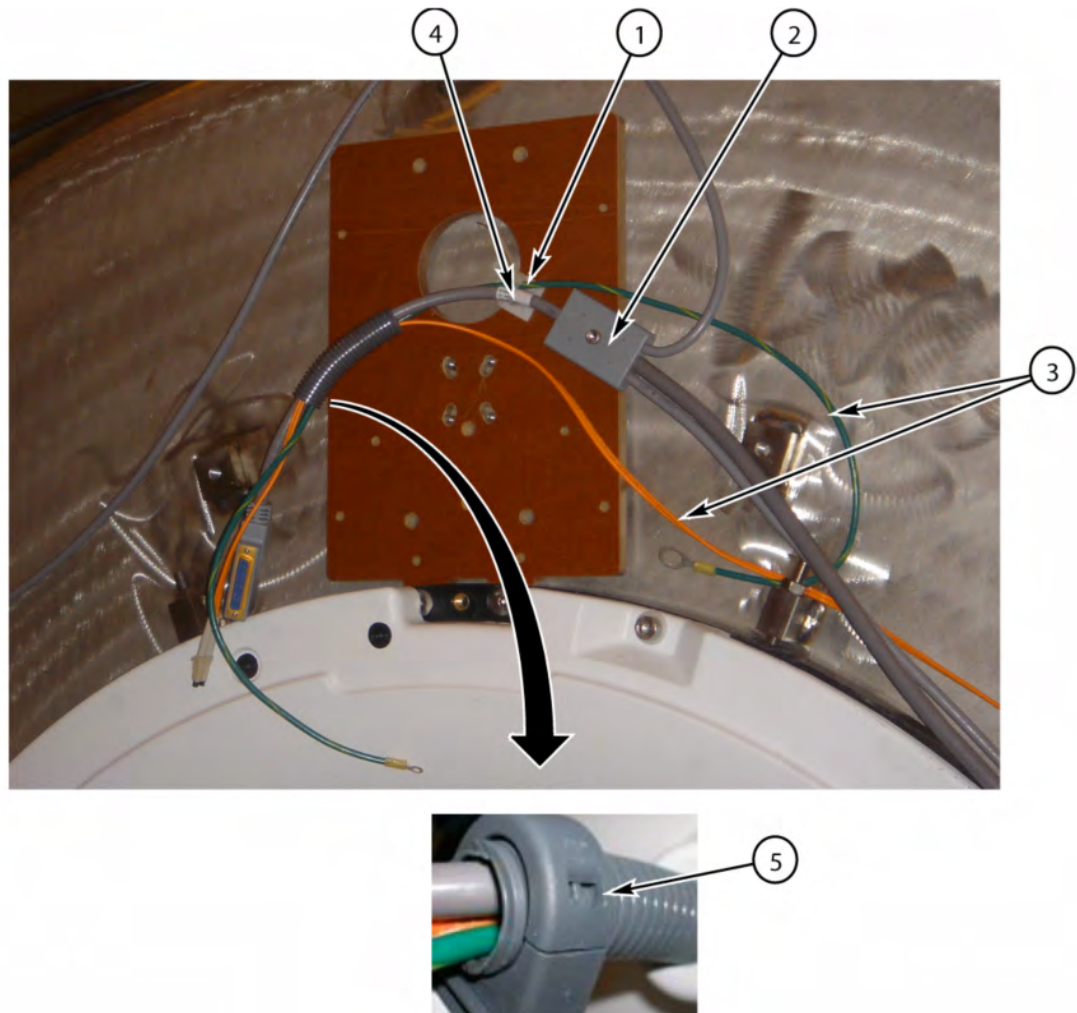
For systems installed with In-Room Display (IRD), the IRD Monitor, Operator Control Panels (OCP), and Run M3366 need to be installed and connected to the pre-routed cables.

NOTE: Two people are required to perform this procedure.

20.1 Install IRD Components

If magnet does not have ground stud available for connecting IRD ground wire, make sure to follow alternate ground wire attachment step below.

Illustration 4-43: Attach Run M3366



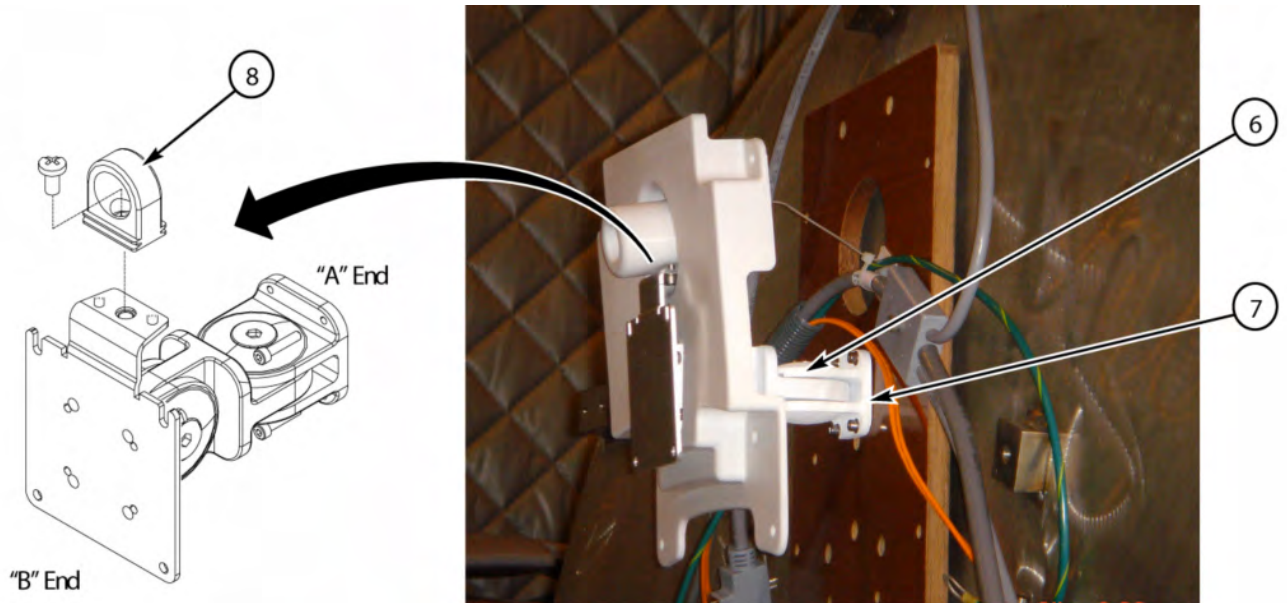
1. Attach Cable Tie Mount base (46-208746P9) to plate with M4 X 10 screw (2109873-14).

NOTE: The Mount should be angled at about 45 degrees as shown.

2. Attach Run M3366 to Arm with (1) M6 X 25 screw (2109867-15) Do not tighten at this time.
3. Route the ground wire and fiber optic cable adjacent to main cable, one on each side. Do not tie wrap either one to the tie mount. Both are to remain free sliding.

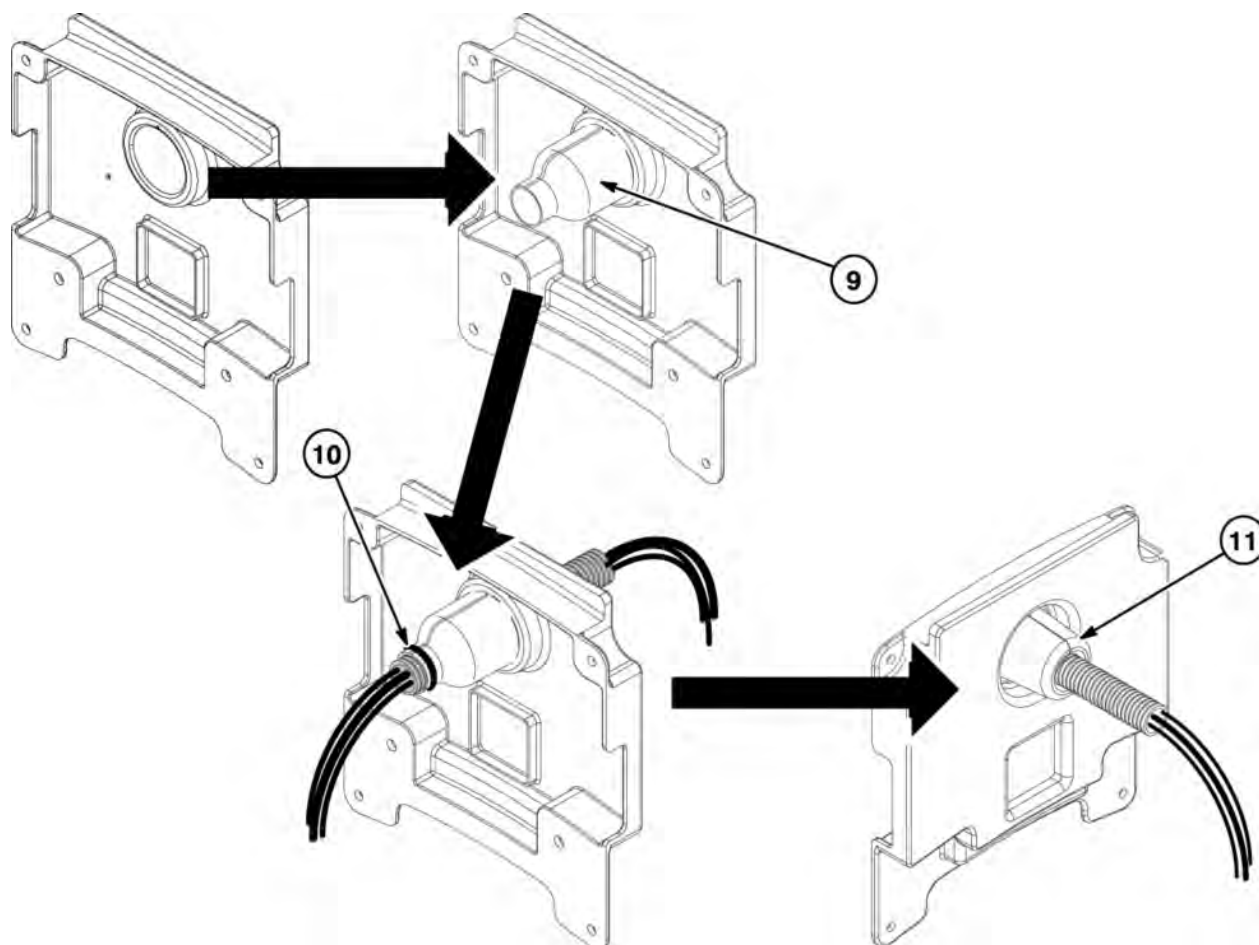
4. Slide tie-wrap through the mount to hold cable in place. Do not tighten.
5. Remove Cable Clamp from Run M3366 cable assembly. Insert a small screwdriver into the side of the clamp that says "Open" until clamp releases.

Illustration 4-44: Attach IRD Mounting Brackets



6. Pass the "A" end of the Pivot Bracket through Interface Cover (5182762). (CAUTION: Do not pass the "B" end, it will tear)
7. Fasten with (4) M4 x 20mm screws (2109867-6) and (4) M4 washers (2109867-6). Add Loctite 242. Using a Number 3 Allen, tighten screws
8. Attach the Cable Clamp to the In Room Display mount (5133768-3), which is attached to the Pivot Bracket, with M6 X 12 screw. (2109873-24)

Illustration 4-45: Secure Run M3366 to Flexible Housing

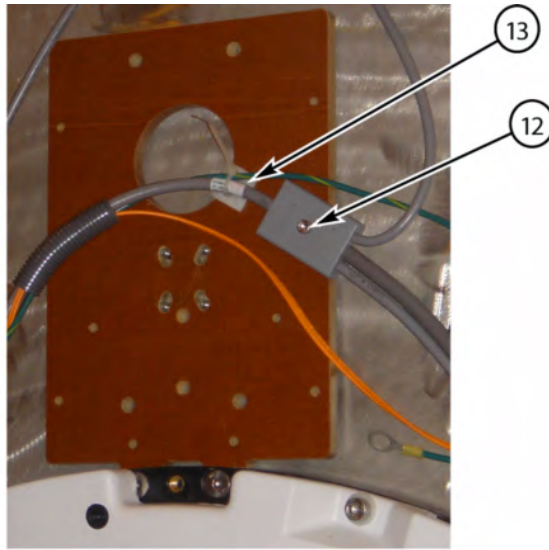


9. On the Monitor Interface Cover, carefully turn inside-out the flexible tube housing to the other side of the cover.

NOTE: Take special care, the flexible tube housing material can tear easily.

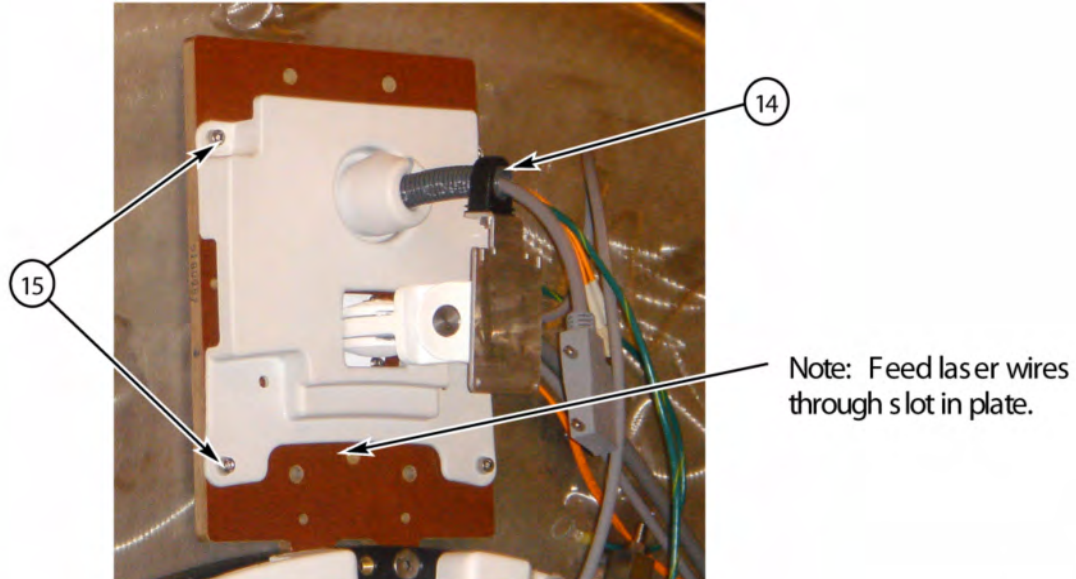
10. Pass connectors and tubing of Run M3366, the Scan Room Monitor Control cable (5149629), through the Monitor Interface Cover.. Fasten together with tie-wrap to secure the cable assembly to the interface cover. Trim excess from tie-wrap
11. After securing cables to the monitor interface cover, carefully return the flexible tube housing to its original position,

Illustration 4-46: Secure Run M3366



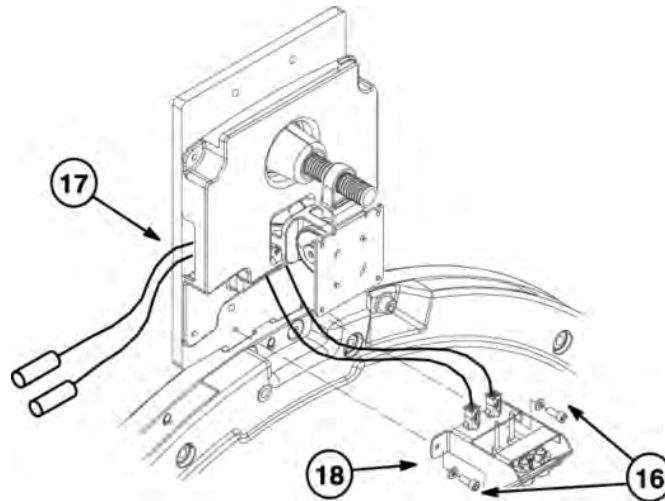
12. Tighten screw holding cable block to arm assembly.
13. Secure the tie-wrap around the gray cable only (through cable tie mount) of the Scan Room Monitor Control cable assembly, Run M3366.

Illustration 4-47: Attach Bracket



14. Snap cable assembly into Cable Clamp on top of bracket and close clamp to secure.
15. Fasten the interface cover to the monitor mount board with (4) M6 x 16 screws (2109867-13) and 4 nylon washers (2229794).

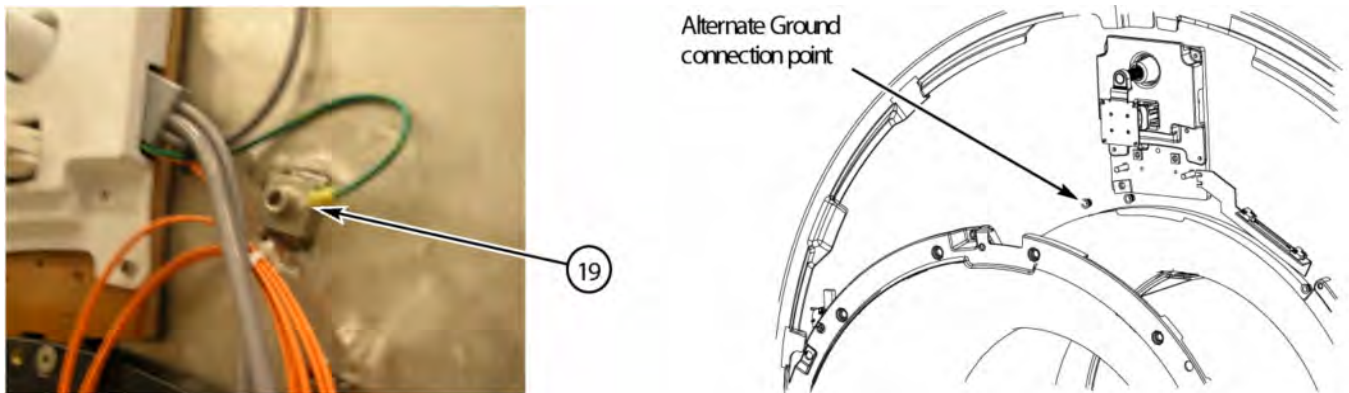
Illustration 4-48: Install Laser Mounting Bracket



NOTE: If previously removed, reinstall the laser-mounting bracket (5178328) to the ring arm using (2) M6 nylon washers (2229794) and (2) M6 x 16 screws (2109867-13).

16. Start right screw & washer. Slide bracket over slot. Add left screw. Be careful of wire position when adding screws so they aren't crimped behind bracket.
17. Route cable under IRD interface cover and out through the slot in the interface cover.
18. Tighten bracket with #5 Allen wrench.

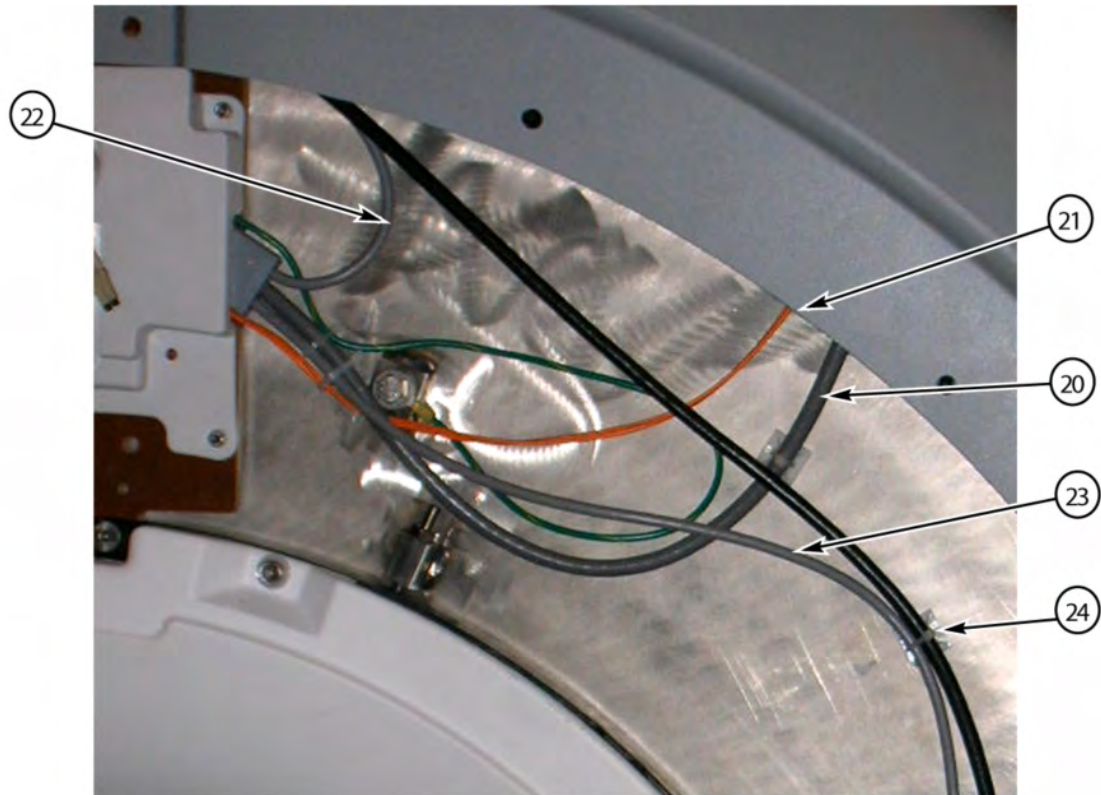
Illustration 4-49: Connect Ground Wire



19. Connect the Monitor Cable assembly ground wire to the front ground stud using (1) Brass M10 x 16 screw nut & (1) M10 SS washer (2109878-4). Note: An alternate ground stud is located on other side of IRD mounting brackets.

NOTE: If above ground block is not available, the ground wire can be attached to the alternate ground connection point on the magnet mounting ring using a M10 x 12mm Socket Head Cap Screw, as shown above, at time of passive shimming when the front end bell is removed.

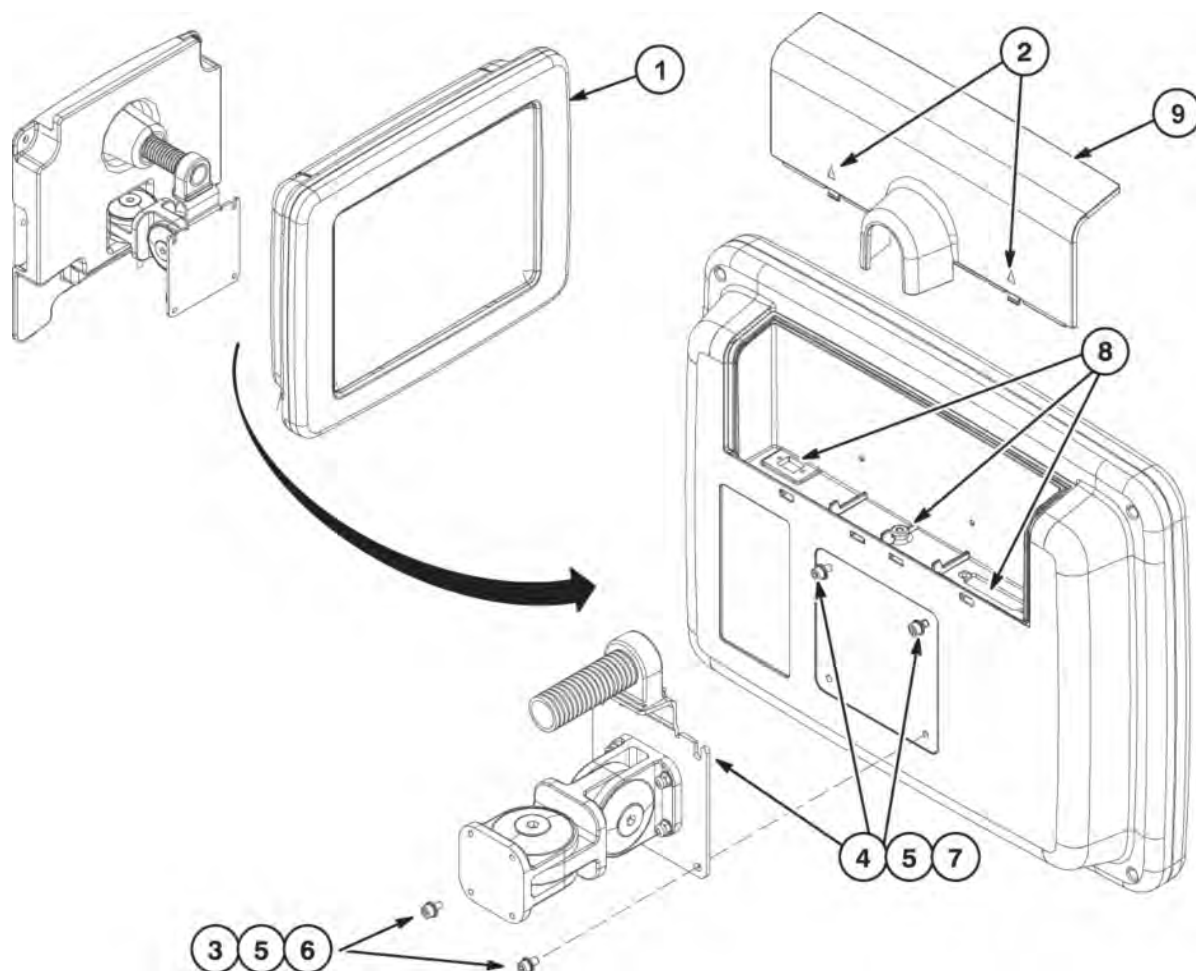
Illustration 4-50: Run M3366 Cable Connections



- 20. Route cable marked MAG-J2 to the top of the magnet and connect to J2 on the magnet I/O board
- 21. Route fiber optic cable to top of magnet and connect to J28 on the fiber optic I/O board
- 22. Route cable marked MAG-TRACK BALL 1 to the left side of the front of magnet
- 23. Route cable marked MAG-TRACK BALL 2 to the right side of the front of magnet
- 24. Attach cables to the magnet using cable clamps to ensure proper placement on the magnet.

20.2 Install In Room Display Monitor

Illustration 4-51: Monitor Components



1. Locate the IRD Monitor (5133768).
2. Remove the top cover on the back of the monitor by pushing in at the arrows and sliding upwards.
3. Remove the two bottom screws on the back of the Monitor.
4. Loosen, but do not remove, the top two screws on back of Monitor and position Monitor on IRD Mount plate.
5. Apply Loctite 242 to mounting screws.
6. Fasten the two bottom screws and hand tighten the two top screws to secure Monitor to mounting plate. Tighten with hex wrench #3.
7. Hand tighten the two top screws, then tighten with hex wrench #3.
8. Connect the three Run M3366 cables: ground, fiber optic, and power, to the back of the Monitor.
9. Replace top cover on to monitor.

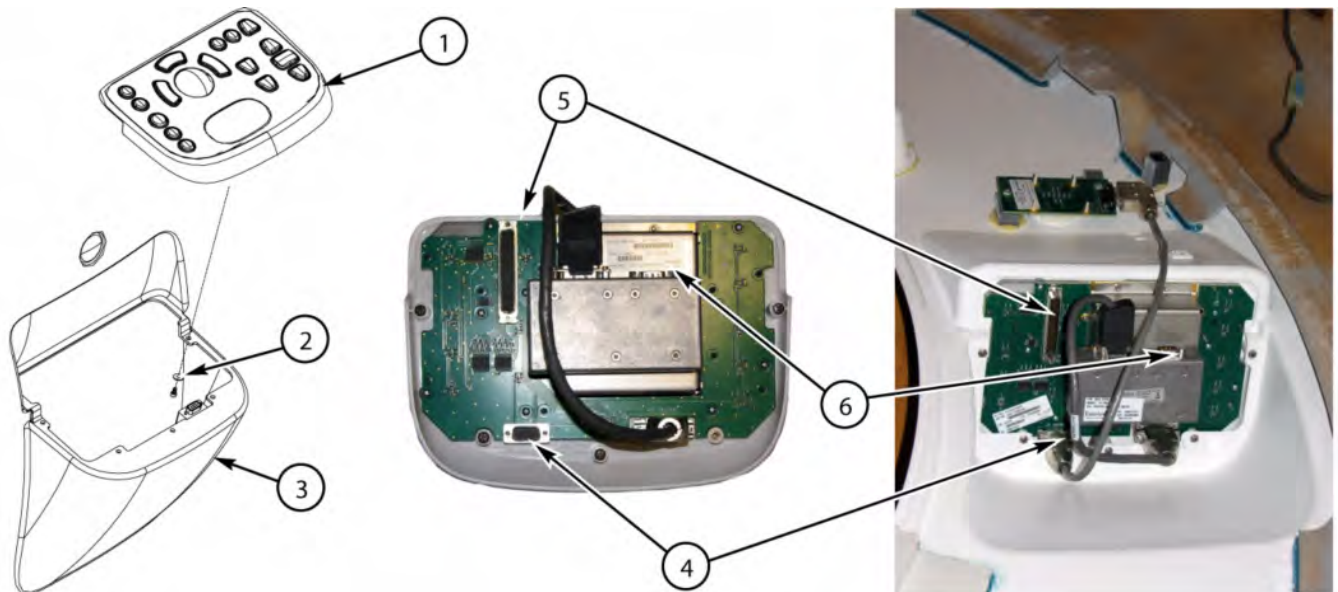
20.3 Install Left and Right Operator Control Panels



NOTICE

Take special care in connecting Runs M3366 and Estop cable. Any reverse connection of these 9-pin Sub-D connectors will mean at system power-on it will not be possible to recover the system from E-Stop condition.

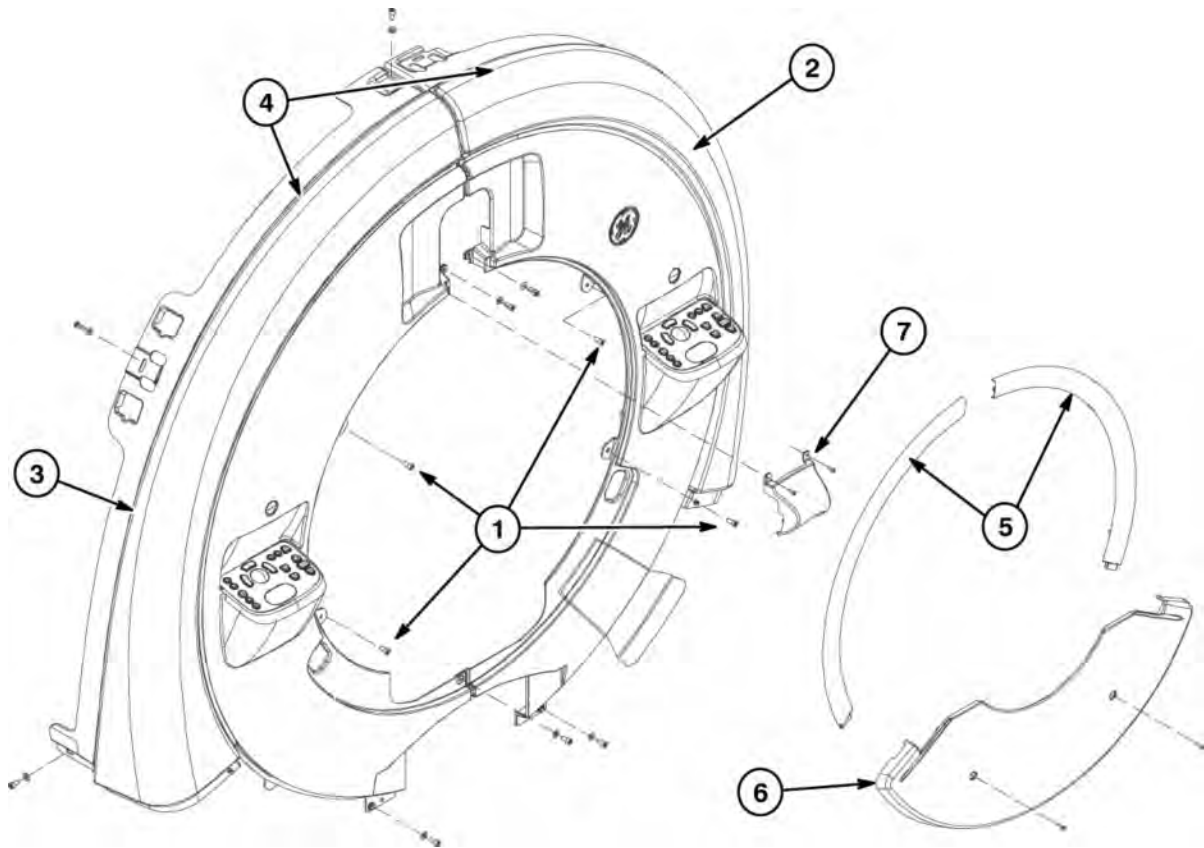
Illustration 4-52: Control Panel Installation



1. Locate the first Operator Control Panel (OCP).
2. Fasten to right Front Cover using the provided five M4 screws and M4 washers.
3. Repeat steps 1 and 2 for installing the second OCP to the left Front Cover.
4. Connect Emergency Stop cables to Estop connector on left and right front Covers.
5. Run M3337 will be connected to left and right OCP marked J1.
6. Run M3366 will be connected to left and right OCP marked Trackball.

20.4 Install Front Covers

Illustration 4-53: Front Covers Components



1. Do not install washers when installing the four M10 x 25mm hex socket screws (2109867-13) that fasten the two inside tabs of each front cover to the magnet.
2. Move Right Front Cover (5328960) to front of magnet and attach Front Cover to magnet using M10 x 25mm hex socket screws and M10 washers (46-294167P73) on all screws except the two that attach the inside tabs as noted in Step 1.
3. Move Left Front Cover (5328961) to front of magnet and attach Front Cover to magnet using M10 x 25mm hex socket screws and M10 washers (46-294167P73) on all screws except the two that attach the inside tabs as noted in Step 1.
4. Secure together left and right Front Covers at top and bottom of covers using M10 washers and M10 x 25mm screws..
5. Attach left and right Front Trim Rings (5177586 & 5177586-2).
6. Install Front Bridge Cover (5328959) using provided fasteners.
7. Install Laser Cover (5182894) using provided fasteners.

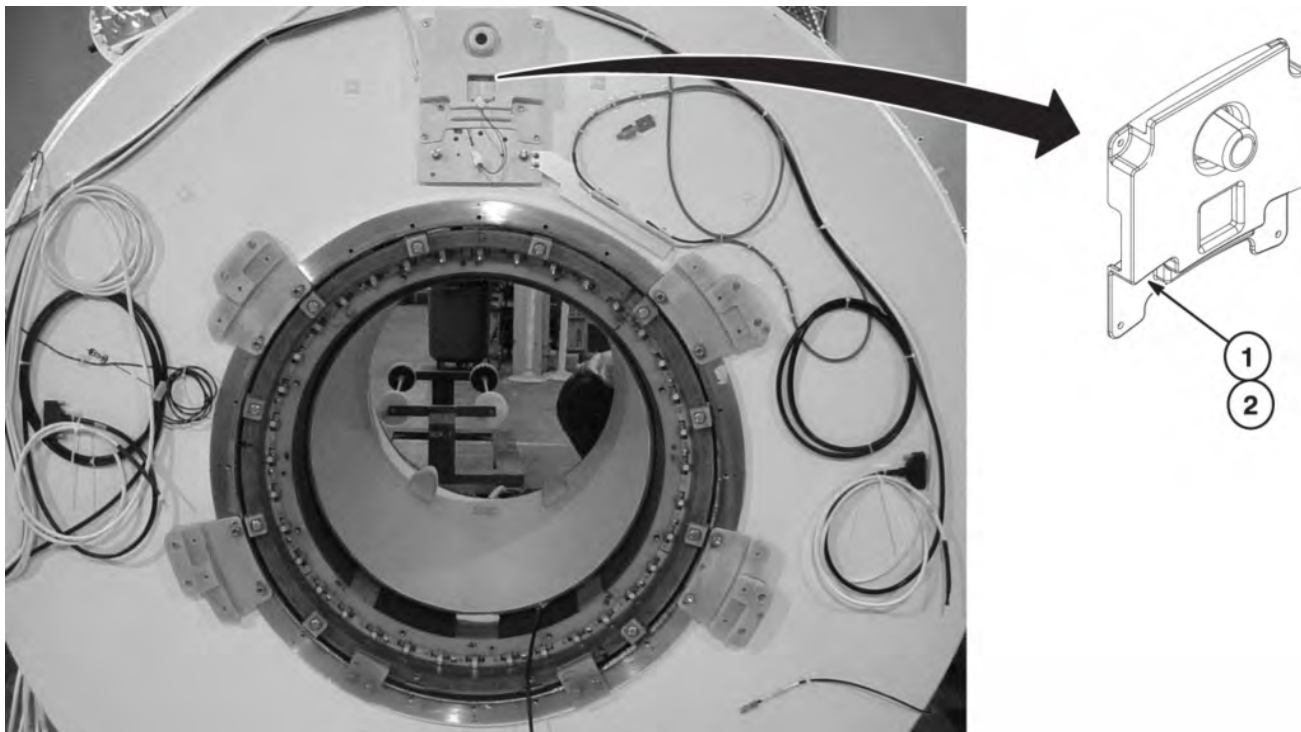
21 LED Display Installation

For systems installed with LED Display, the LED Monitor and Operator Control Panels (OCP) need to be installed and connected to the pre-routed cables.

NOTE: Two people are required to perform this procedure.

21.1 Remove Monitor Interface Cover

Illustration 4-54: Monitor Interface Cover

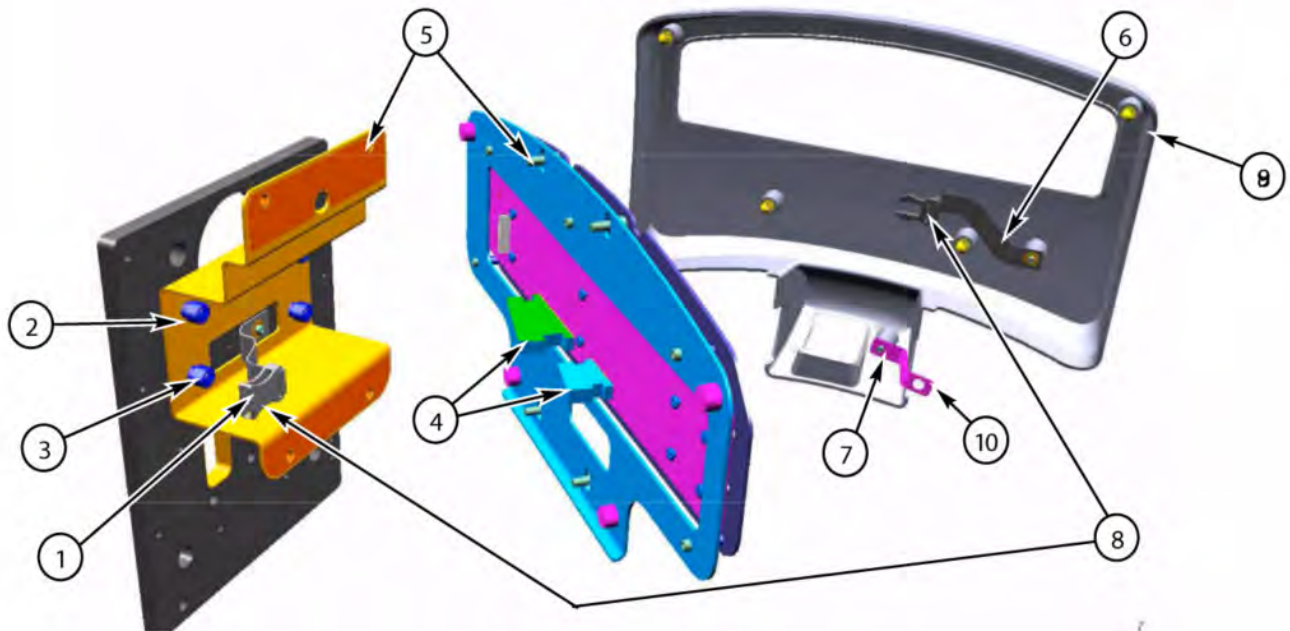


1. Unfasten the four screws attaching rubber interface cover to mounting plate.
2. Discard Monitor Interface Cover and fasteners.

21.2 Install LED Display

NOTE: Do NOT install any washers that may be supplied with the LED Display kit of components.

Illustration 4-55: Display Installation



1. Attach Magnet Side LED Display Strap (5346326-2) to mounting plate.
2. Install, but do not tighten, the upper two socket head cap screws. Slide the LED Display Bracket (5343349) over the two screws.
3. Install the lower two socket head cap screws. Tighten all four socket head cap screws.
4. Route and connect Run M3349 to J4 and Run M3363 to J5 on back of LED Display Panel (5343350).
5. Attach LED Display Panel to bracket using four screws.
6. Attach Display Side LED Strap (5346326) to bezel using one screw.
7. Attach Display Bezel Mounting Brace (5344837) to bezel using one screw.
8. Route Display Side Strap (5346326), that is attached to bezel, through hole in LED Display Panel (5343350) and connect to Magnet Side Strap

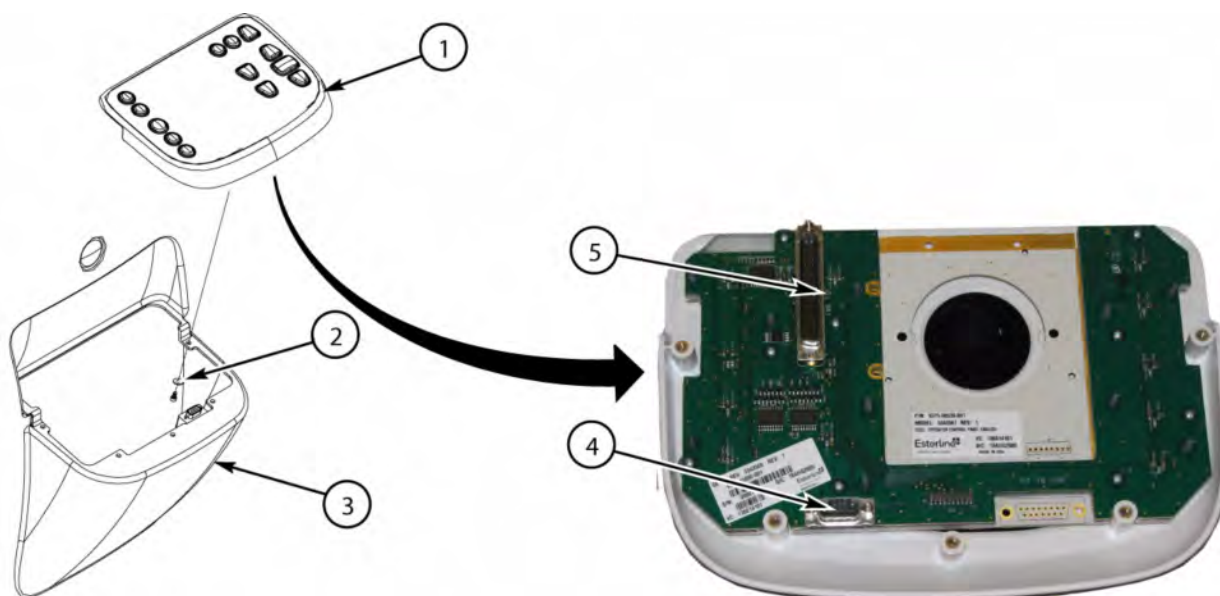
NOTE: Complete the following two steps for fit and alignment. When complete, the bezel has to be removed. It will be reinstalled after the front covers are installed later in this procedure.

9. Attach Bezel (5346325) to LED Display Panel by carefully pressing and engaging the four poppers.
10. Fasten the Display Bezel Mounting Brace (5344837) to the magnet ring.

21.3 Install Left and Right Operator Control Panels

1. Locate the first Operator Control Panel (OCP).

Illustration 4-56: Control Panel Installation



2. Fasten to right Front Cover using the provided five M4 screws and M4 washers.
3. Repeat steps 1 and 2 for installing the second OCP to the left Front Cover.
4. Connect Emergency Stop cables to Estop connector on left and right front Covers.
5. Run M3337 will be connected to left and right OCP marked J1.

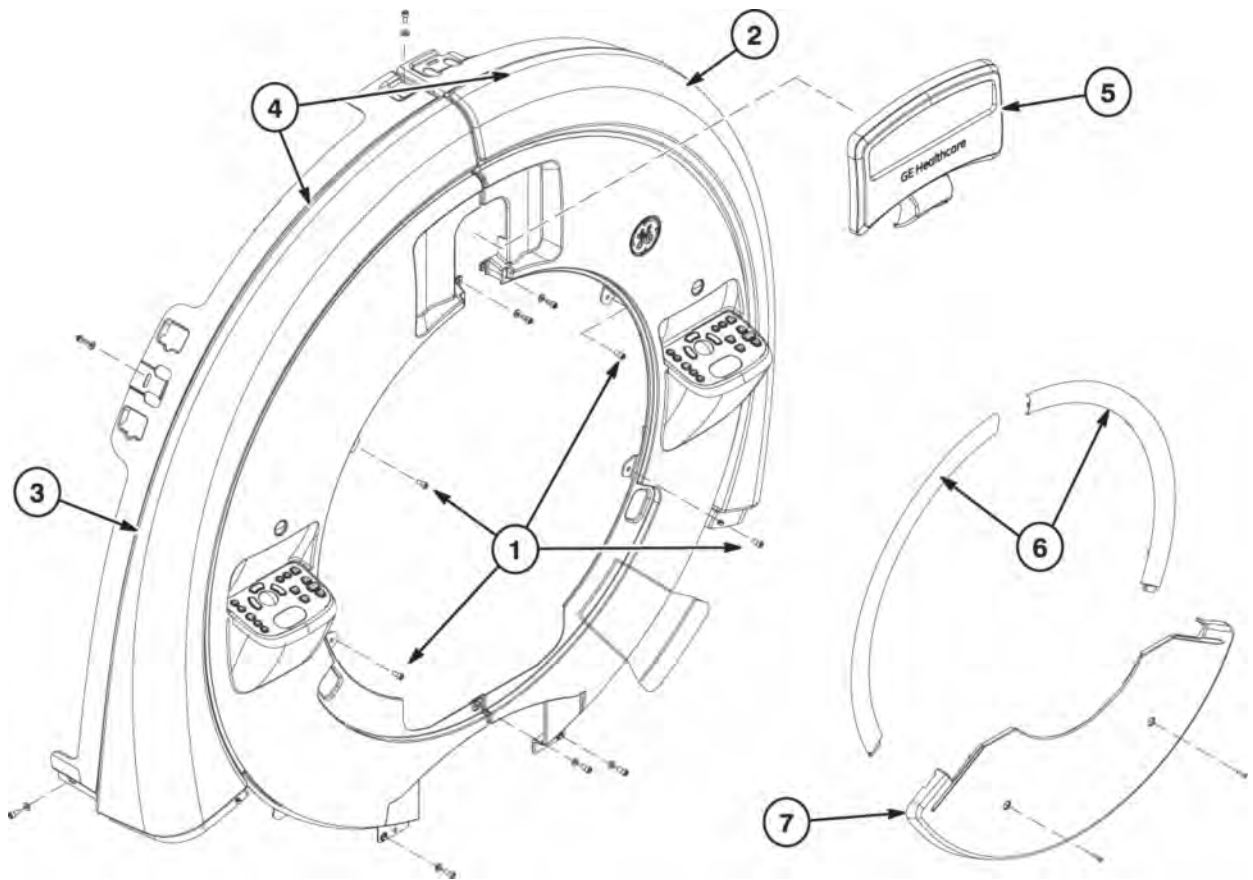
21.4 Install Front Covers



NOTICE

The front covers are heavy. Avoid awkward body positioning. Use proper lifting techniques when handling covers.

Illustration 4-57: Front Cover Components



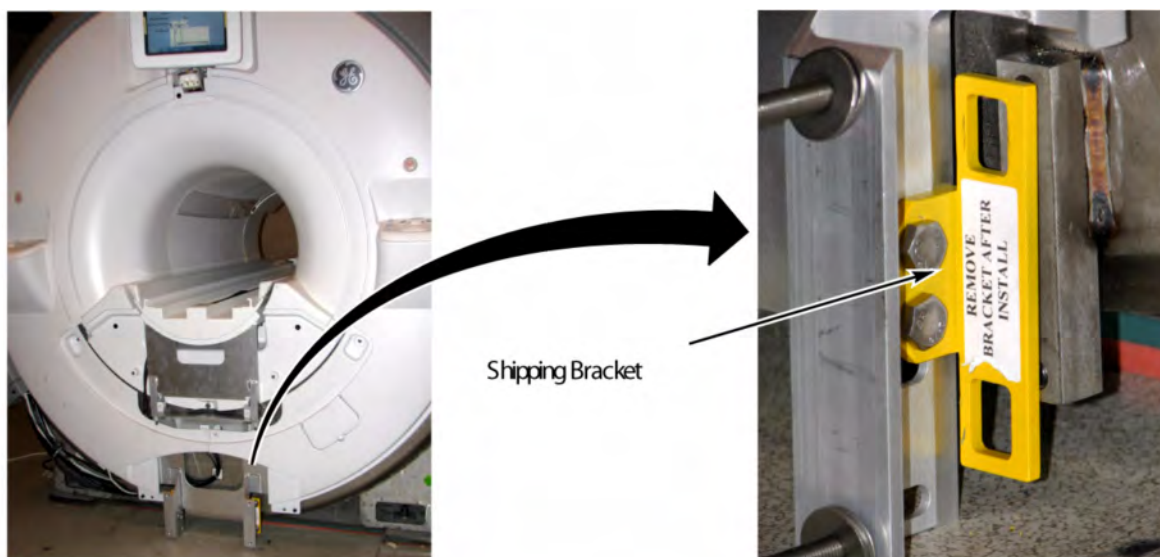
1. Do not install washers when installing the four M10 x 25mm hex socket screws (2109867-13) that fasten the two inside tabs of each front cover to the magnet.
2. Move Right Front Cover (5328960) to front of magnet and attach Front Cover to magnet using M10 x 25mm hex socket screws and M10 washers (46-294167P73) on all screws except the two that attach the inside tabs as noted in Step 1.
3. Move Left Front Cover (5328961) to front of magnet and attach Front Cover to magnet using M10 x 25mm hex socket screws and M10 washers (46-294167P73) on all screws except the two that attach the inside tabs as noted in Step 1.
4. Secure together left and right Front Covers at top and bottom of covers using M10 washers and M10 x 25mm screws.
5. Install LED Display cover as instructed in the last two steps in [Section 21.2](#).
6. Attach left and right Front Trim Rings (5177586 & 5177586-2).
7. Install Front Bridge Cover (5328959) using provided fasteners.

22 Dock Installation

22.1 Removing Bridge Support Shipping Bracket

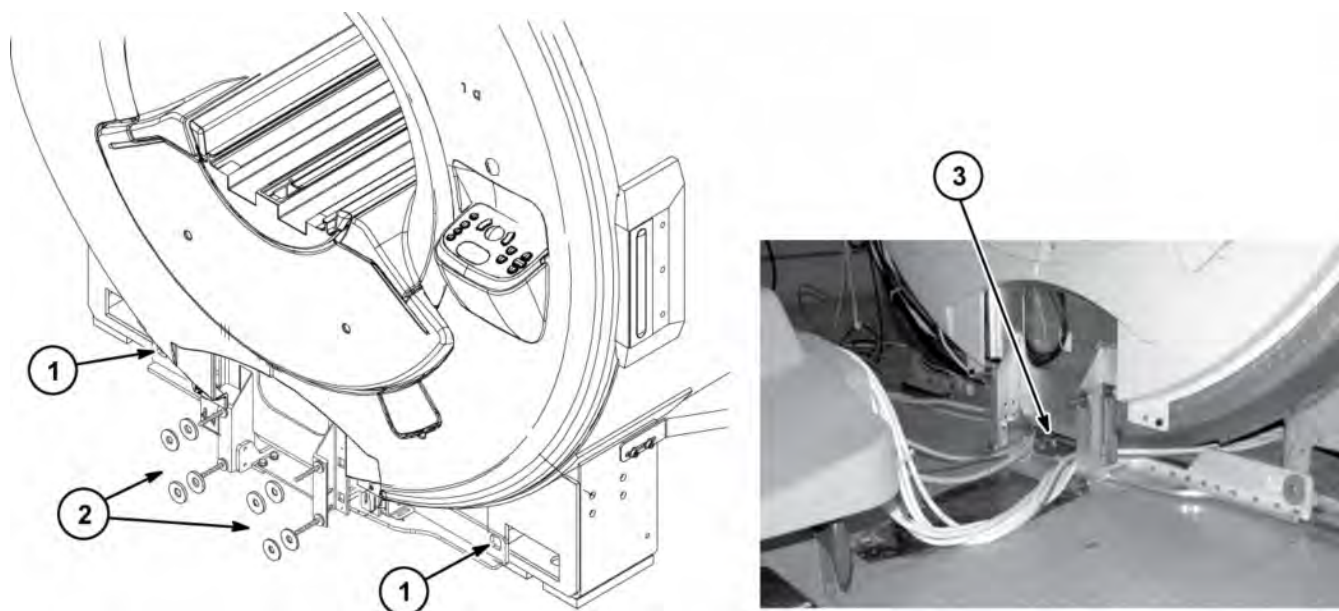
If still in place, remove yellow shipping bracket from front of magnet.

Illustration 4-58: Shipping Bracket Removal



22.2 Attach Dock to Magnet

Illustration 4-59: Dock Installation Preparation

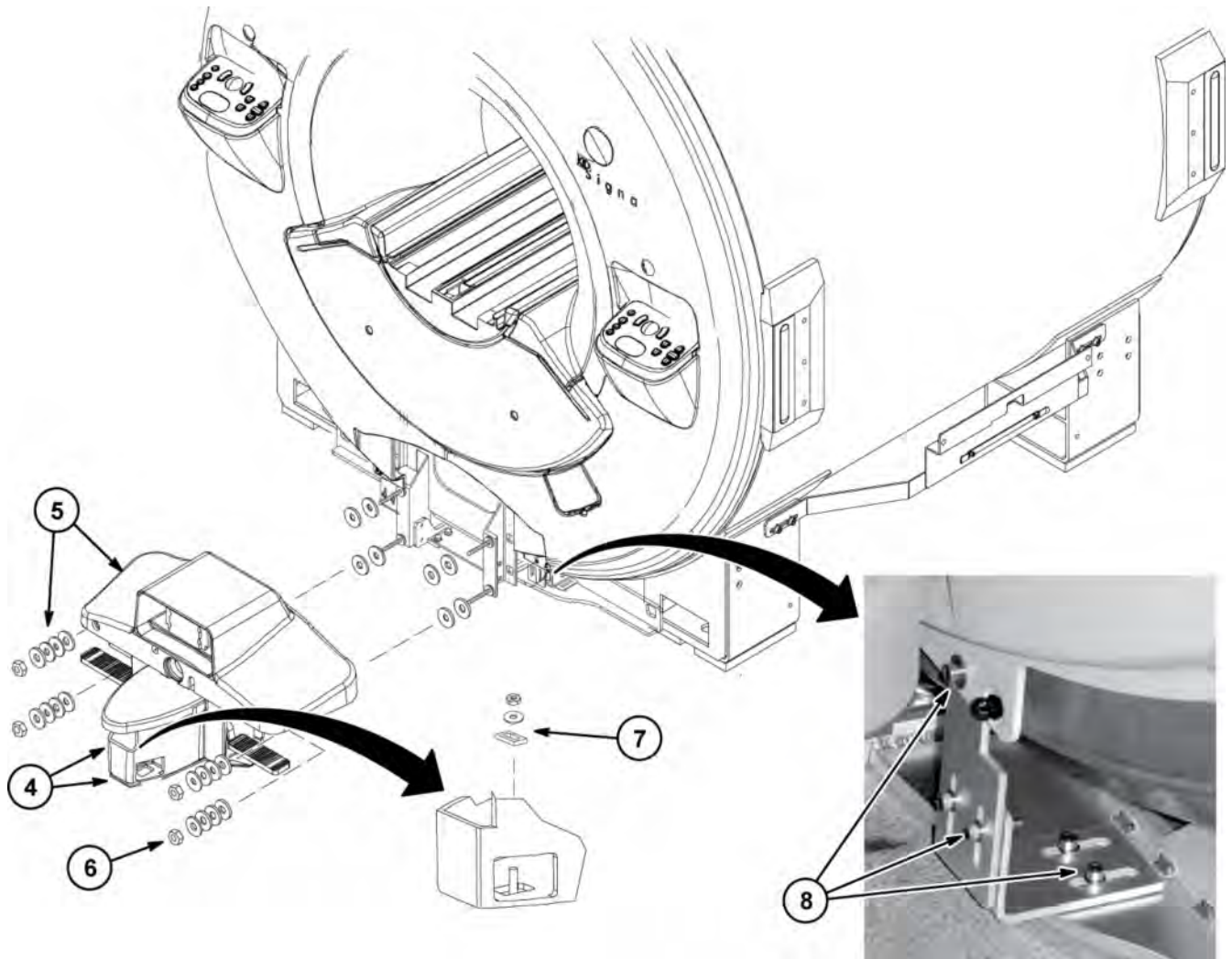


1. Make sure Dock Mounting Bracket is securely attached to the magnet legs.
2. Start with two (2) 11mm washers on each stud, both sides, between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is achieved.

NOTE: Make sure that the left side cable mounting bracket is detached from cables before routing the cables through the lower Dock Mounting Bracket.

3. Cables will be routed through lower Dock Mounting Bracket as Dock is moved into place.

Illustration 4-60: Install Dock



4. Position Dock in front of magnet and align Rubber Pad (2230001) under dock.
5. Move Dock into position on studs and place the remaining washers equally on each of the four mounting studs.
6. Fasten with provided nuts, one on each stud.
7. Fasten Dock to floor mounted stud with: (1) Dock Clamp Plate (2178874), (1) 11mm ID Washer (2109878-4), and (1) 10mm Hex Nut (2109875-4). Apply Loctite 242 as required.

NOTE: If adjustment is required for alignment of patient table, refer to calibration and setup procedures on Service Methods DVD until proper gap is achieved.

8. On each side, attach bottom of front cover to bracket on Dock Mounting Bracket. Adjust brackets as needed and tighten screws.

22.3 Dock Cable Connections - Left Side

NOTE: Care should be taken to not pinch cables between Dock assembly and magnet dock mounting brackets.

Illustration 4-61: Left Side Cables



1. Attach cable mounting bracket on magnet leg using two M10 x 25 screws (46-318508P20), M10 Washers (2109878-4) and M10 Nuts (2109875-4).
2. Route and connect cables from dock as shown. Cable connections to back side of bracket are coiled underneath left side of enclosure.
3. Insert and loop excess cable through hole at front of magnet leg.
4. Loosen bottom of cover and slide cables behind cover. This is required to install front Trim Covers.

22.4 Dock Cable Routing - Right Side

NOTE: Care should be taken to not pinch cables between Dock assembly and magnet dock mounting brackets.

Illustration 4-62: Right Side Cables



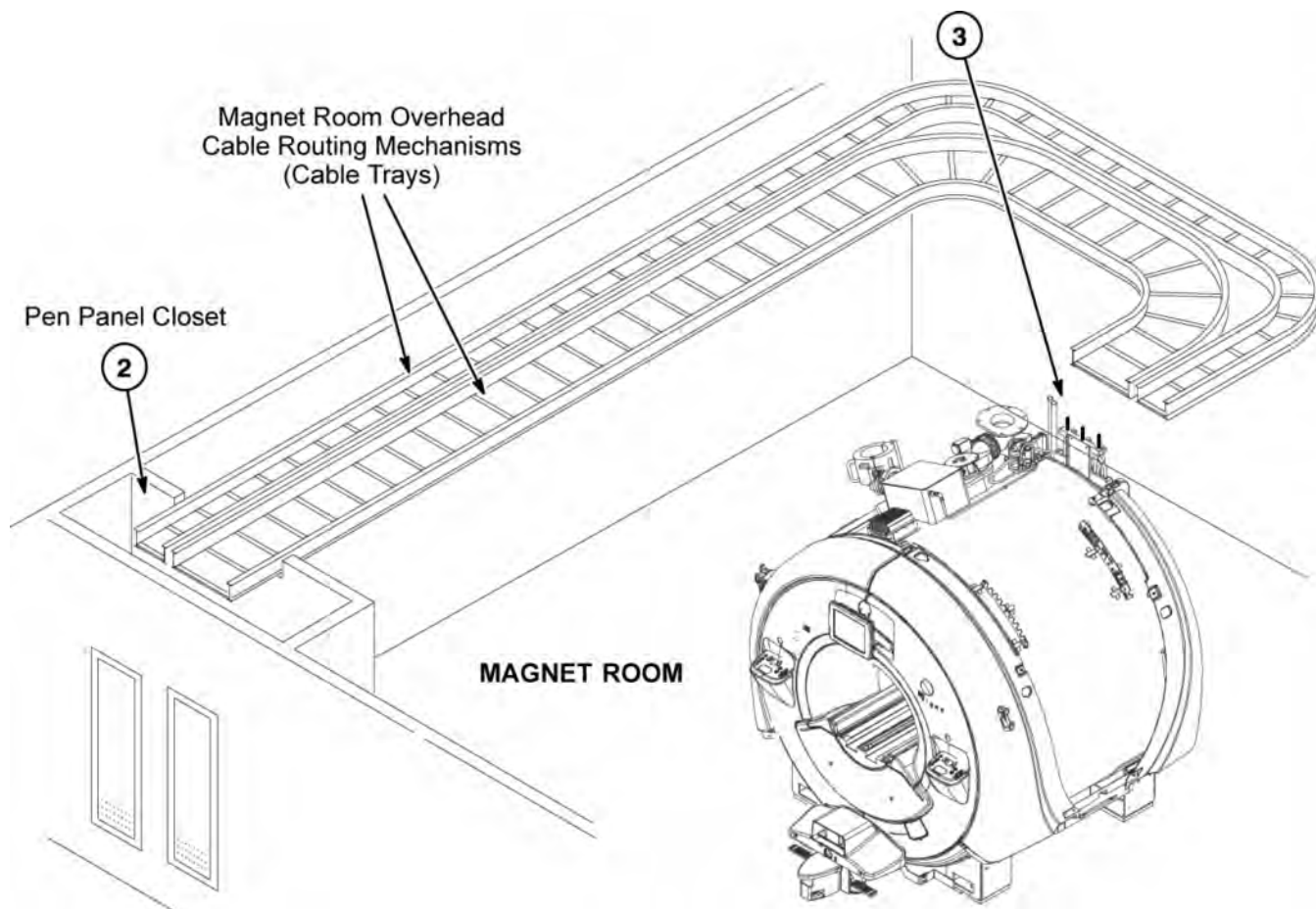
1. Route cables from dock through bottom opening of Bridge Support Magnet Mount Assembly as shown.
2. Continue routing cable under magnet towards the Rear Pedestal. These connections are described in the following section.

23 Route and Connect Cables in Magnet Room

23.1 Route Cables in Overhead Cable Routing Mechanisms

Refer to applicable magnet room cable map for location and routing of the remaining cables.

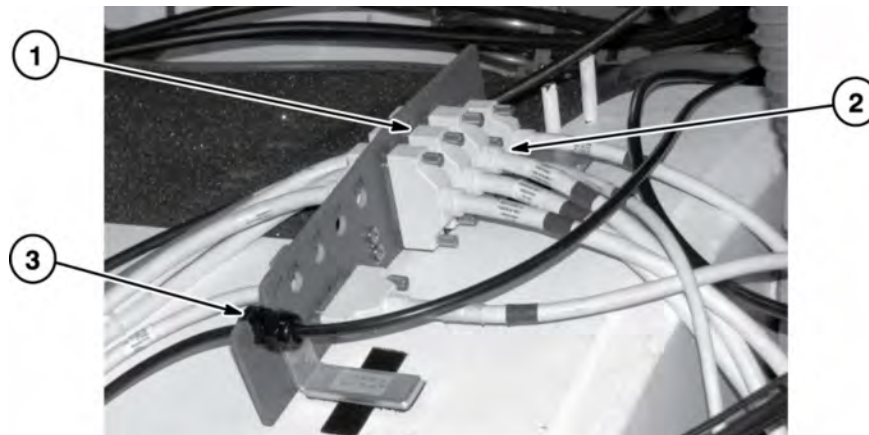
Illustration 4-63: Routing of cables in Overhead Trays



1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Route cables from Pen Panel Closet towards magnet. Excess cable length will be stored in Pen Panel Closet.
3. Refer to the following sections for connections and routing of cables around the magnet.

23.2 Connect Runs to Magnet I/O Panel

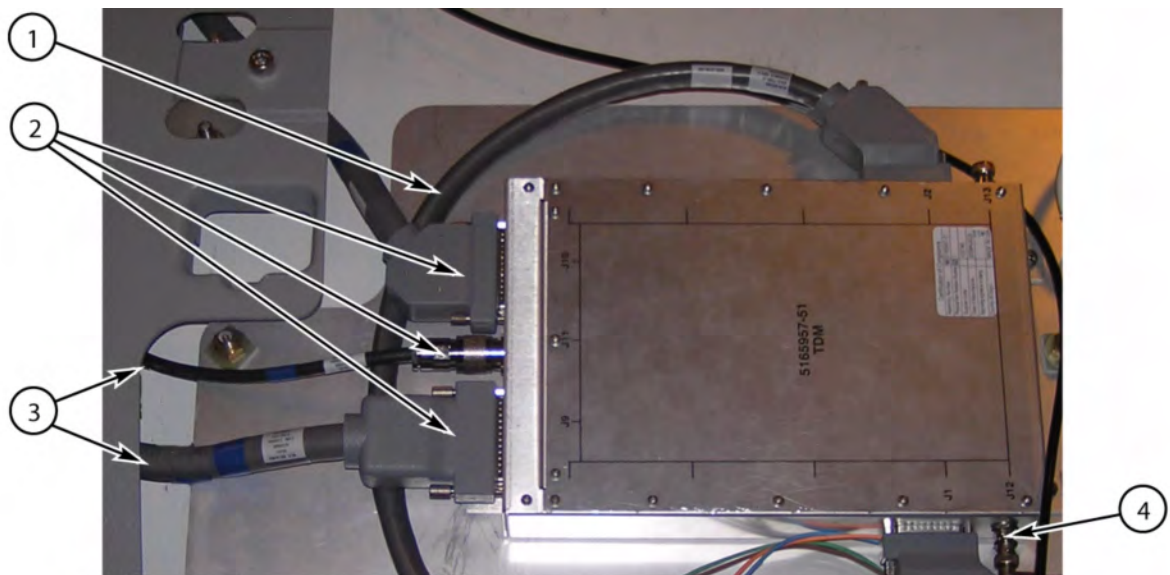
Illustration 4-64: Magnet I/O Cable Connections



1. Route and connect Runs M3031, M3302, M3303, M3312, and M3313 to I/O Panel as marked.
2. If IRD is installed, route and connect Run M3314 to J2.
3. If IRD is installed, fiber optic Run P2026 will be secured at J11 and connected to J28 at Fiber Optic I/O Panel.

23.3 Route and Connect Cables to MAG-TDM

Illustration 4-65: TDM Module Cable Connections

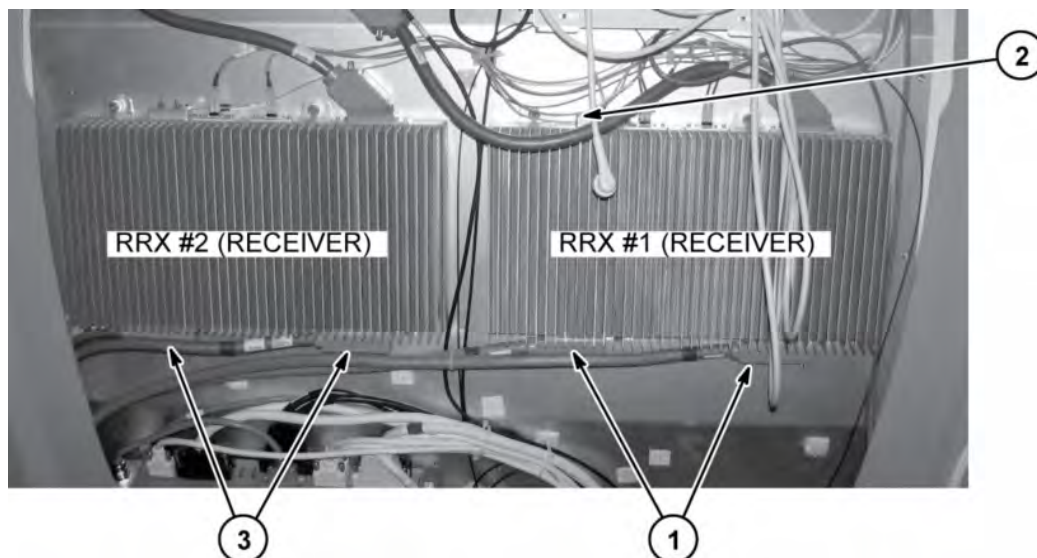


1. If Run M3368, TDM to Receiver 2, is installed, make sure it is routed under the three cables connected in the next step.
2. Route and connect Runs M1302, M3300, and M3301 from overhead cable trays to TDM module as marked.
3. Make sure cables are not pinched or subject to damage by sharp edges.

4. Route and connect Run M1315 to antenna in Rear Pedestal.

23.4 Route and Connect Cables to Receivers

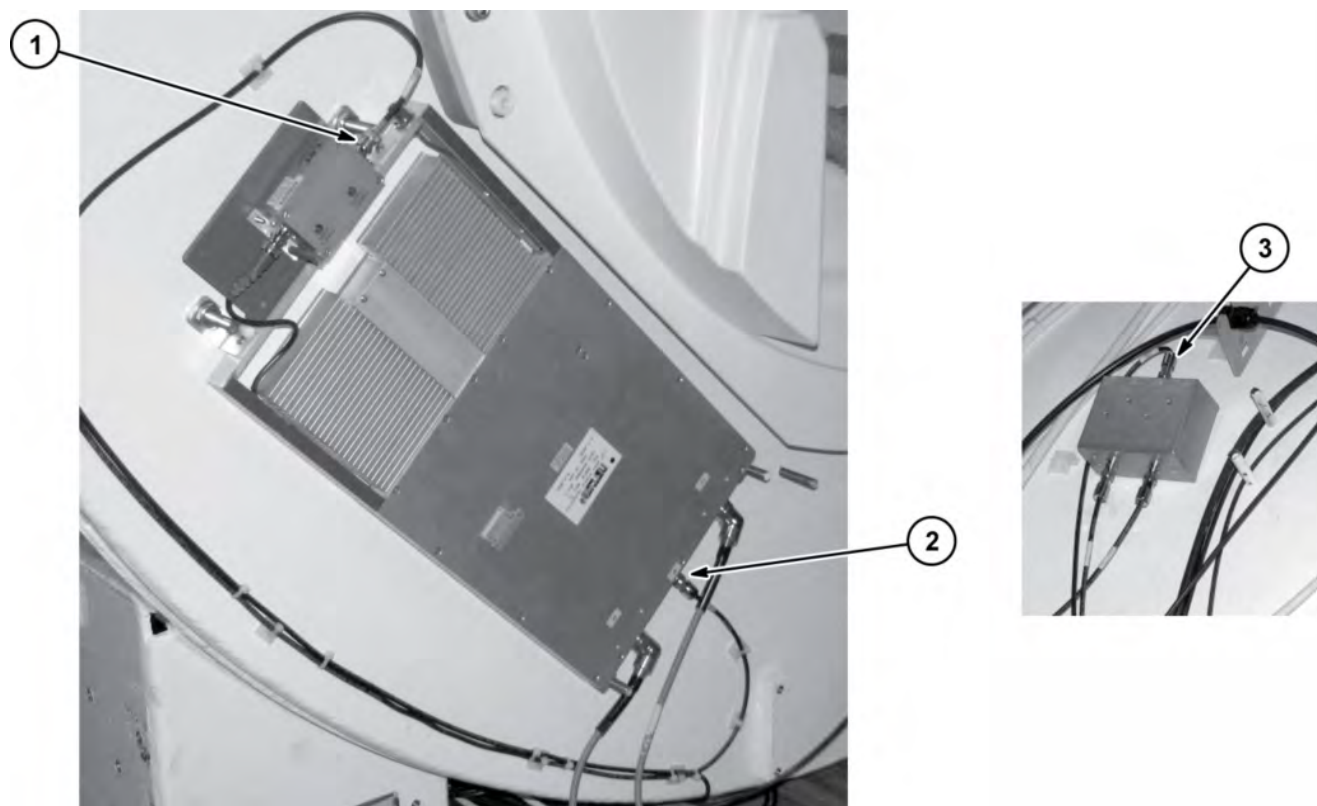
Illustration 4-66: RRX Cable Connections



1. Connect Runs M3320 and M3321 to RRX #1. Route other ends to Rear Pedestal PED-HUB.
2. Route Run M2300 to Rear Pedestal PED-HUB.
3. If system being installed is a 32 channel, connect Runs M3322 and M3323 to RRX#2. Route other ends to Rear pedestal PED-HUB.

23.5 Route and Connect Cables to MAG-HYB and MAG-DDSPLIT

Illustration 4-67: MAG-HYB and MAG-DDSPLIT Connections

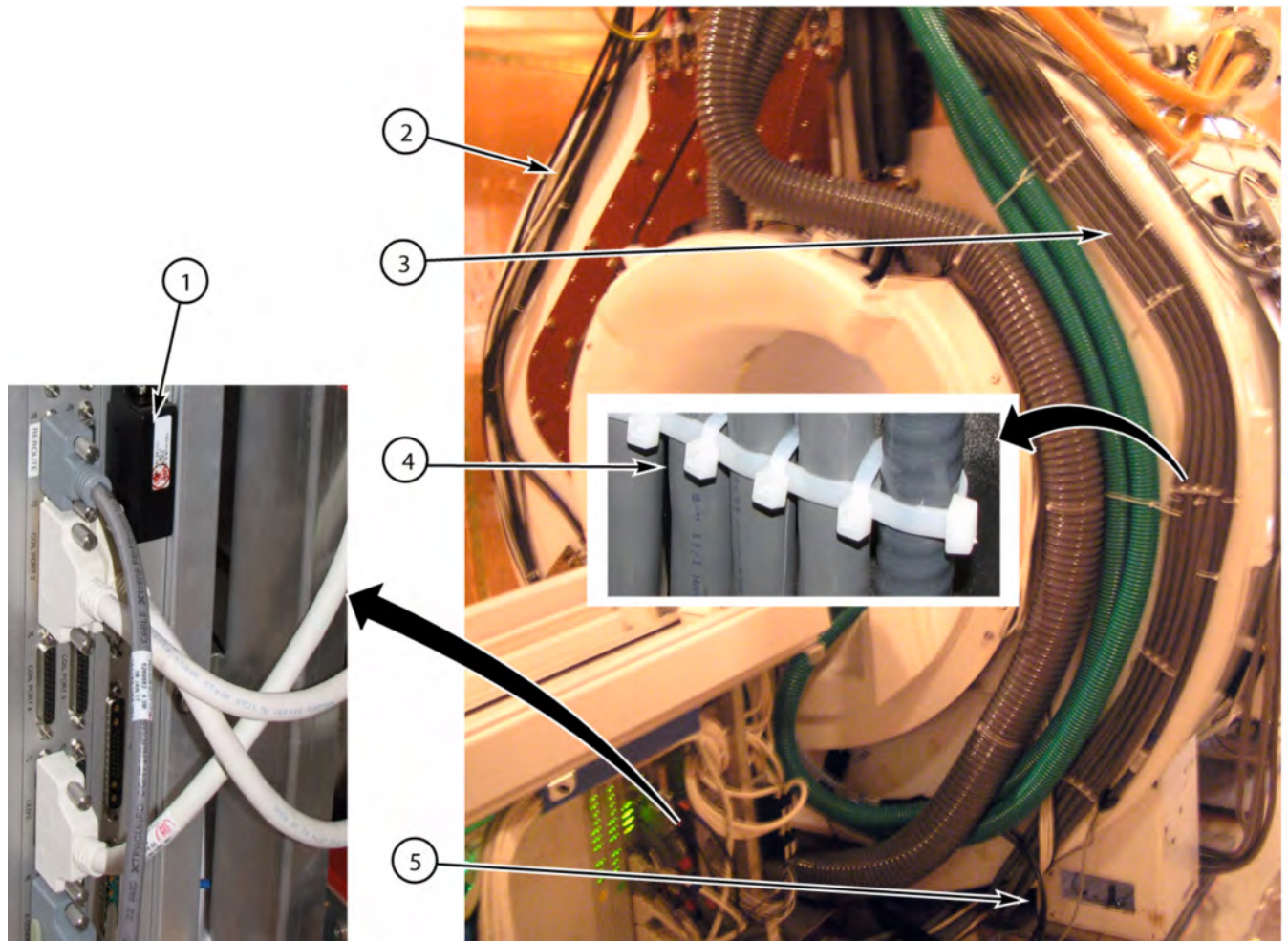


1. Route and connect Run M1335 from OUT on Pre-amp to either:
 - Slot 1 B1 on Rear Pedestal PED-HUB (for 16/0 or 32/0 systems)
 - J7 on Rear Pedestal PED-RBOX (for 32/32 systems)
2. Run M1312 was routed from PEN Panel J74. Connect to J6 on MAG-HYB.
3. Run M1320 was routed from PEN Panel J72. Connect to J9 on MAG-DDSPLIT module.

23.6 Route and Connect Cables to Rear Pedestal

NOTE: A minimum 18 inch (458mm) service loop must be provided between the magnet and the Rear Pedestal. This is required for service of the magnet.

Illustration 4-68: Routing of Cables at Rear of Magnet

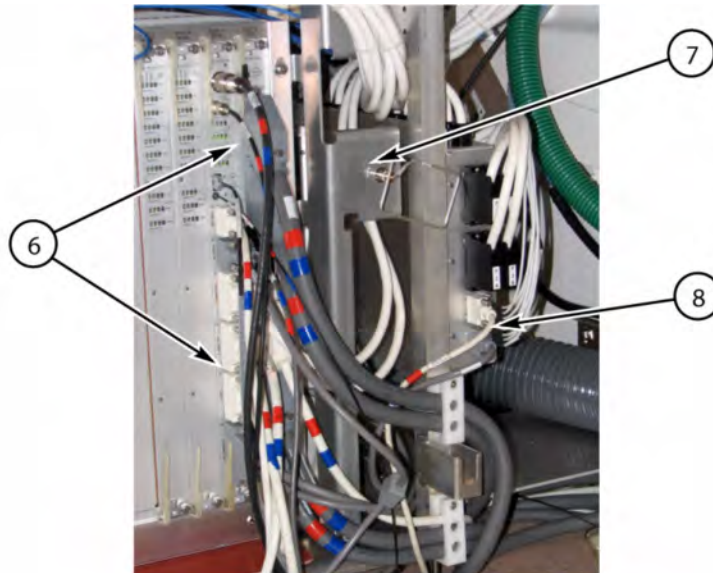


1. Install Shunt Connector (5183786) to Slot 12 J3 before connecting any cables to Rear Pedestal.
2. Route on right side of rear flange the following Runs: M1303, M1304, M3311, M3315, and M3316. Tie-wrap cables to pre-installed tie-wrap mounts located along outer edge of rear flange surface. Refer to cable map for connection designations in rear Pedestal.
3. Route on left side of rear flange the following Runs: M2025, M3304, M3306, M3307, M3308, and M3310
4. Tie-wrap cables together as shown. Attach to pre-installed tie-wrap mounts located along outer edge of rear flange surface.

**NOTICE**

Trim cable tie-wraps to prevent accidental cuts.

5. Make sure cables are routed between Trim Skirt mounting bracket and magnet.

Illustration 4-69: Rear Pedestal Left Side Cable Connections

6. Refer to cable map to route and connect the following cables to Slots 11 and 12: Runs M1303, M1304, M3304, M3306, M3307, M3308, M3310, and M3316. Secure cables with cable clamps on side of rear pedestal leg.
7. Route and connect Run M1315 to antenna.
8. If system being installed is a 32/32 channel, Route and connect Runs M3373 and M3364 to J9 and J10 on PED-RBOX.

Illustration 4-70: Rear Pedestal Right Side Cable Connections

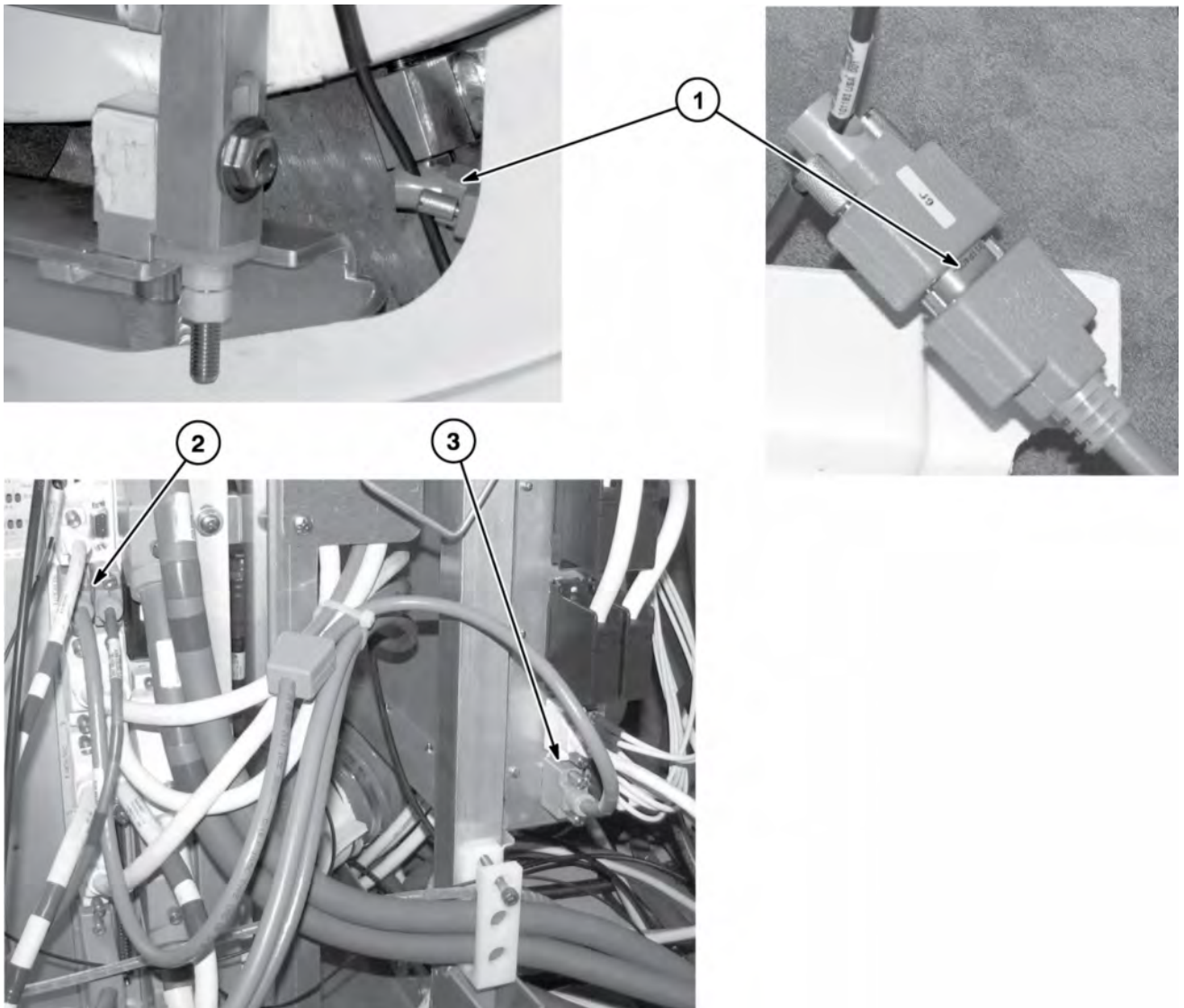
9. If system being installed is a 32/32 channel, Route and connect Runs M1330, M1331, M1332, M1322, and M1335 to PED-RBOX as marked.

NOTE: If system being installed is a 32/0 or a 16/0, Run M1335 is connected Slot 1 B1 on PEN-HUB.

10. Connect one end as marked of Run M3344 to Gradient Temperature cable. Connectors must be covered with Velcro wrap.
11. Route and connect Runs M1320 and M1321 to J101 and J102 on PED-HUB.
12. If system being installed is a 32/32 channel, Route and connect Runs M1322 and M1323 to J103 and J104 on PEN-HUB.

23.7 Route and Connect Run M3354

Illustration 4-71: Installing Run M1354

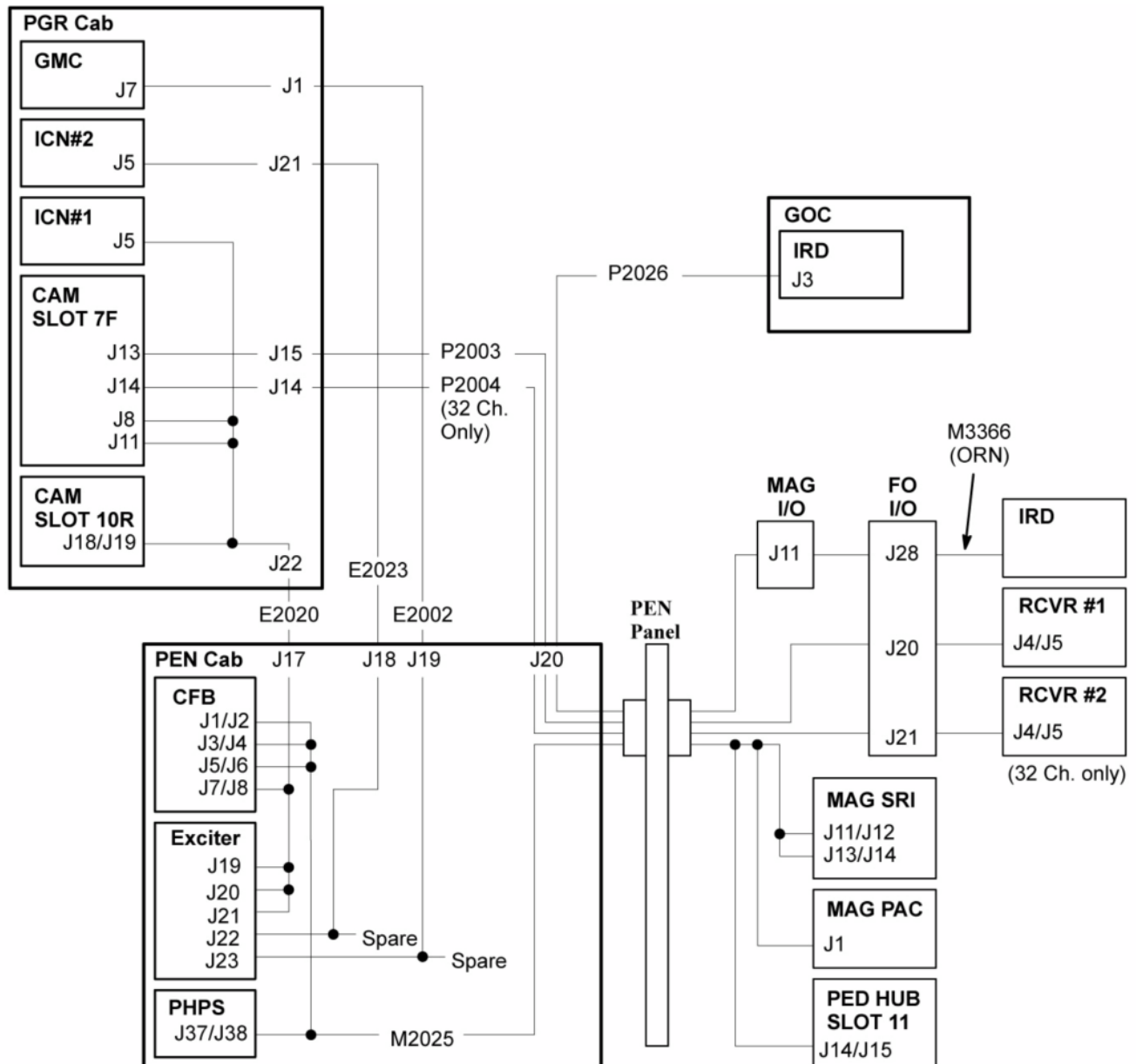


1. Connect to J9 located on RF Bias cable at front of magnet.
2. Route cable to rear pedestal and connect to Slot 11 J8 on PEN-HUB.
3. If 32/32 Channel system is installed, connect remaining connector to J10 on PED-RBOX.

24 Install Fiber Optic Cables

Refer to [Cable Locations in Equipment Room Overhead Cable Trays](#) for location of cables routed in Equipment Room and [Cable Locations in Magnet Room Overhead Cable Trays](#) for location of cables in Magnet Room in the overhead cable trays. Make sure cables are in their designated areas of the trays.

Illustration 4-72: Fiber Optic Cable Map



1. Route M2025 in overhead tray from top of magnet, thru fiber optic waveguide in PEN Panel to PEN cabinet.

2. Route P2003 from J20 and P2004 (32 Channel only) from J21 on Fiber Optic I/O panel on top of magnet, thru fiber optic waveguide in PEN Panel, thru J20 at top of PEN cabinet to PGR cabinet.
3. Route P2026 (if IRD is installed) from J28 on Fiber Optic I/O panel on top of magnet, thru fiber optic waveguide in PEN Panel, thru J20 at top of PEN cabinet to OW.
4. Route E2002, E2020, and E2023 from top of PEN cabinet to PGR cabinet

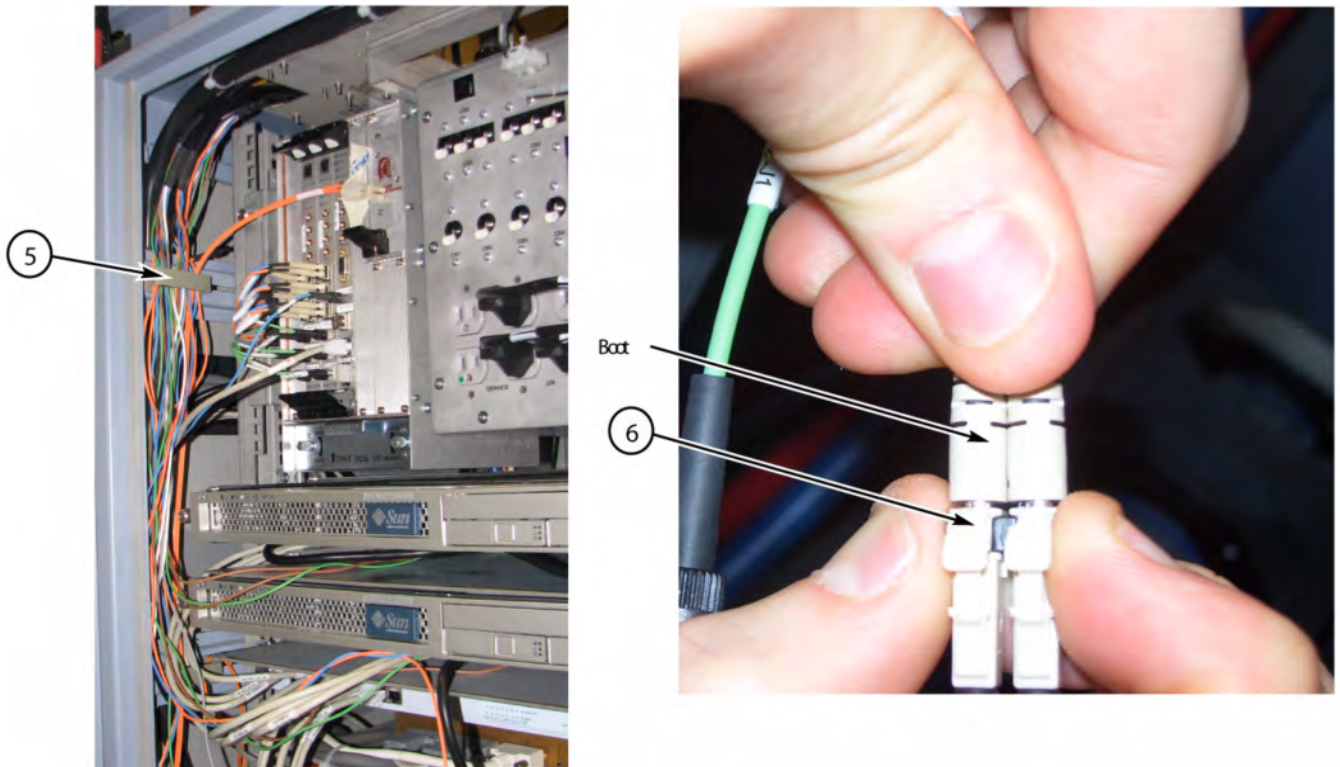
24.1 Route and Connect Fiber Optic Cables to PGR Cabinet

Illustration 4-73: Top of PGR Cabinet



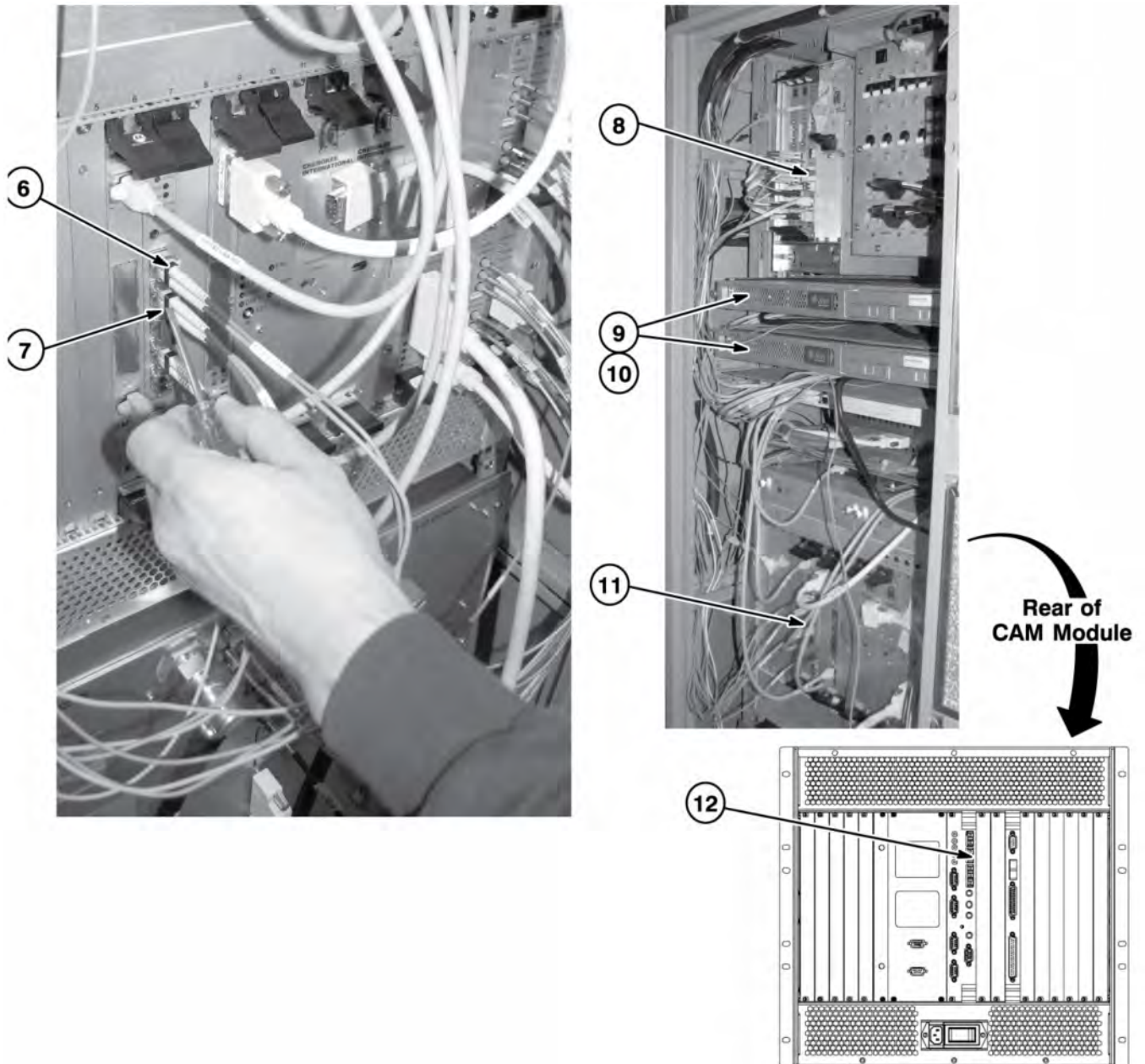
1. Starting at the PEN Cabinet, route fiber optic Runs P2003, P2004, E2002, E2020, and E2023 thru overhead cable trays to top of PGR Cabinet.
2. Connect cables as marked to fiber optic connection panel at top of PGR Cabinet.
3. Route fiber optic wires thru opening as shown.
4. Secure in place with clamps.

Illustration 4-74: Fiber Optic Connectors



5. Route fiber optic wires in PGR Cabinet behind guides.
6. These fiber optic connections are installed by pressing on each side of connector, not the boot, while inserting.

Illustration 4-75: Connect Fiber Optics at Front of Cabinet



7. For connections that do not provide space for fingers on the side, place a flat blade screwdriver between wire covers and insert using a small amount of force.

NOTE: EXERCISE CARE AS APPLYING EXCESSIVE FORCE CAN SEVER GLASS FIBERS PROTRUDING FROM CONNECT ENDS.

Refer to fiber optic cable map at front of this section for remainder of steps.

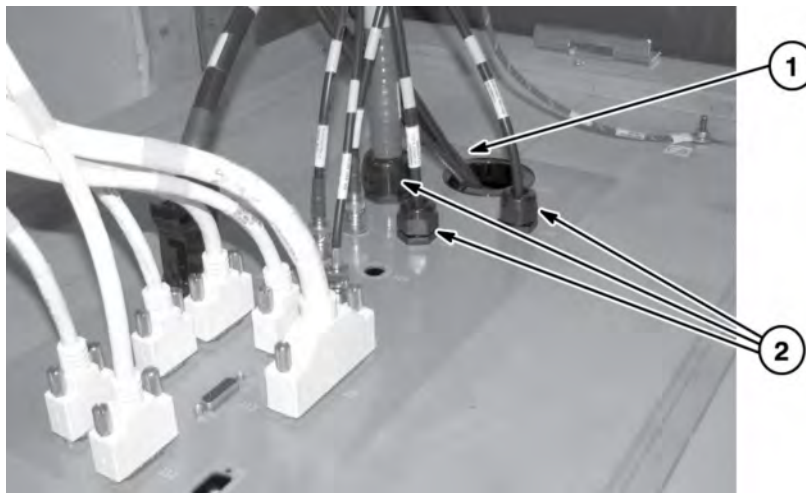
8. Connect to GMC J7 as marked.
9. Connections for ICN #1 (top) and #2 (lower) are at the back of the modules. Pull modules forward and connect applicable fiber optic to J5 on each module.

10. Connect double fiber optic cable to J1 on the back of each ICN #1 and #2 module.
11. Connect remaining fiber optics to J8, J11, J13, and J14 in CAM Slot 7F at front of CAM module.
12. Connection for CAM Slot 10R is at back of CAM module. Pull module forward and connect to J18/J19.

24.2 Route and Connect Fiber Optic Cables in PEN Cabinet

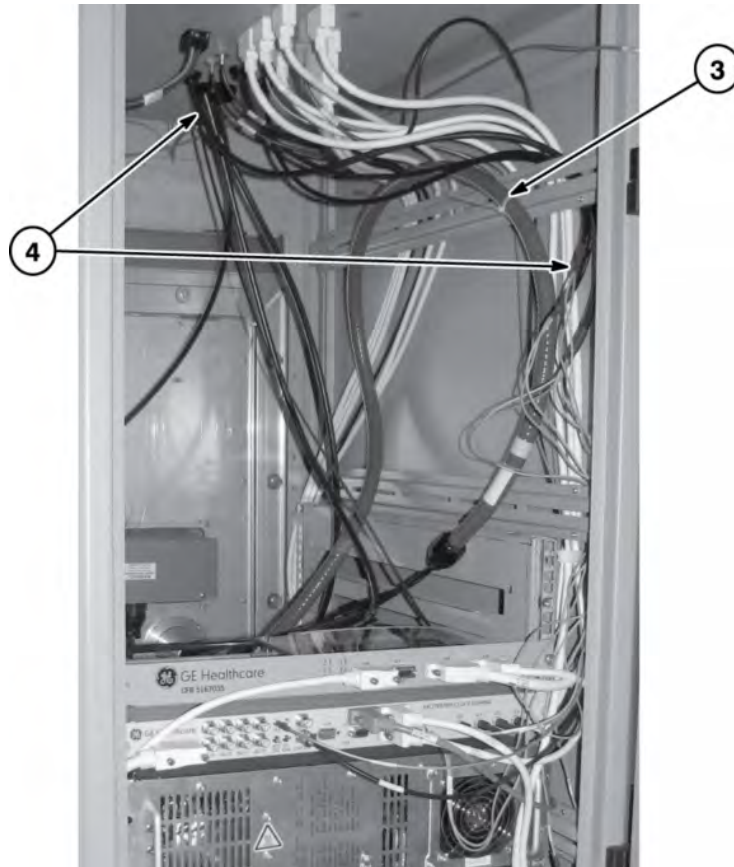
Route and connect fiber optic cables as shown below.

Illustration 4-76: Top of PEN Cabinet



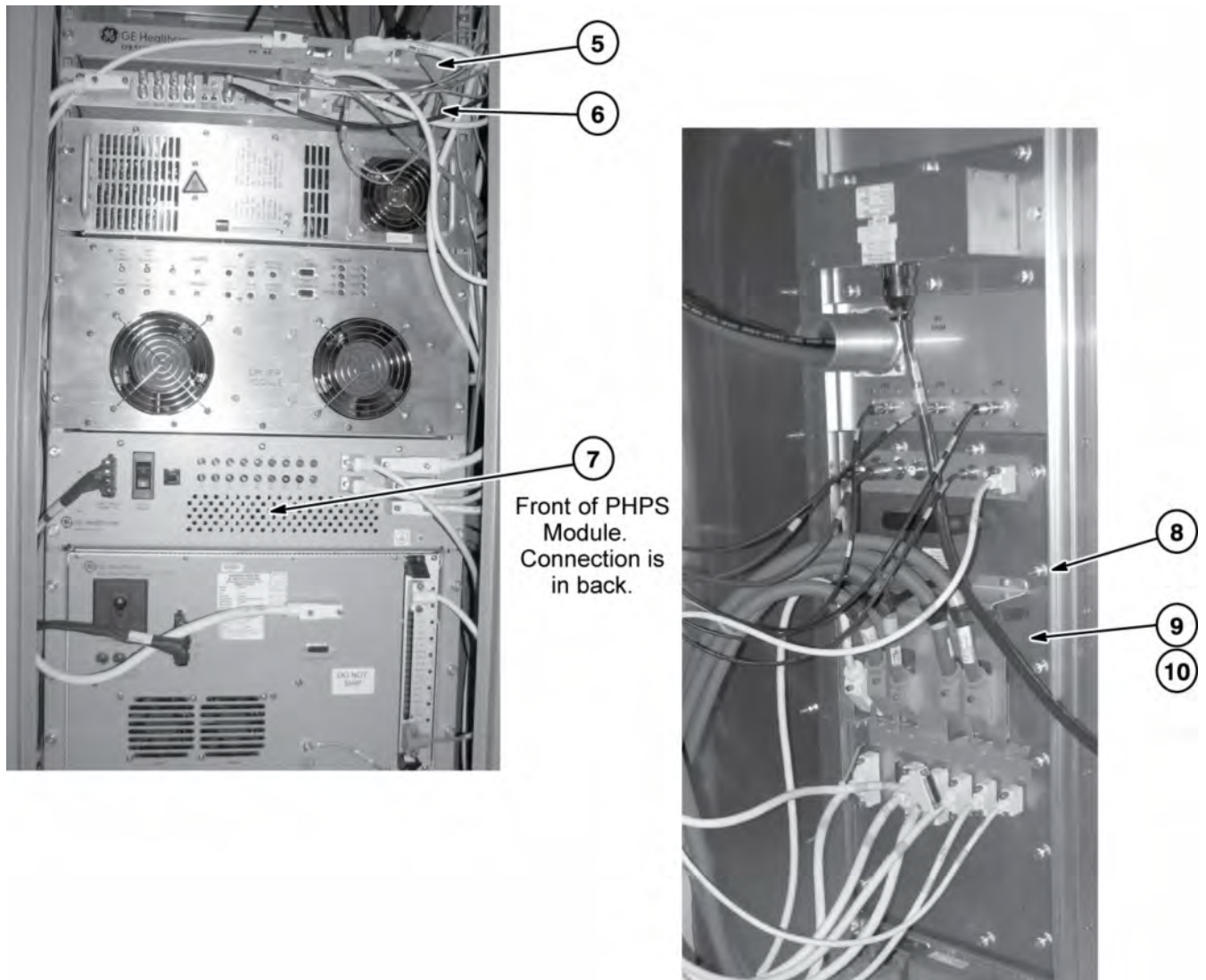
1. Runs P2003, P2004, and P2026 routed thru opening at top of cabinet.
2. Connect E2002, E2020, and E2023 to top of PEN cabinet as marked.

Illustration 4-77: Secure Fiber Optic Wires Inside PEN Cabinet



3. Route and secure with tie-wraps Run M2025 to side of cabinet
4. Route and secure with tie-wraps Runs E2002, E2020, and 2023 to side of cabinet

Illustration 4-78: Connect Fiber Optic Wires in PEN Cabinet

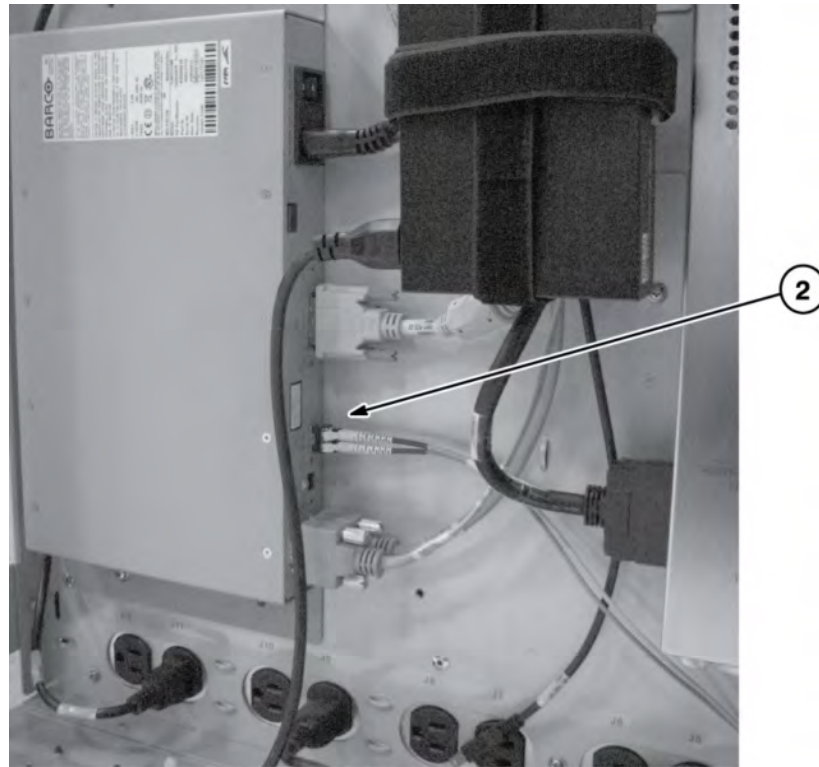


5. Connect fiber optic wires as marked to the CFB module.
6. Connect fiber optic wires as marked to the Exciter module.
7. J37/J38 is located on the back of the PHPS module. Refer to next steps for access and connection of wire.
8. On Magnet room side of PEN panel, unfasten the 16 fasteners attaching the access plate to the PEN Panel.
NOTE: Cables are attached on the interior side of the access plate.
9. Remove access plate enough to reach in and connect M2025 wire to J37/J38.
10. Reattach access plate to PEN Panel.

24.3 Route and Connect Fiber Optic Cable in Operator Workspace

NOTE: This cable is installed only if system has IRD (In Room Display) as part of installation.

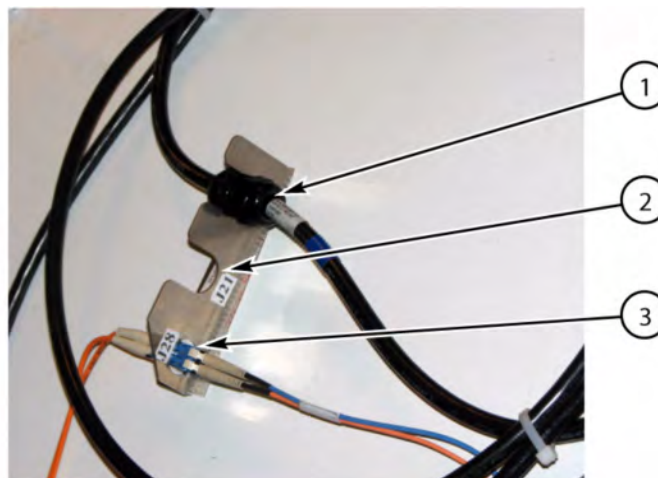
Illustration 4-79: Cable Connection in OW Cabinet



1. Route and attach fiber optic connector to bracket at rear of GOC cabinet.
2. Route and connect P2026 wire to J3 on IRD Module.

24.4 Route and Connect Fiber Optic Cables in Magnet Room

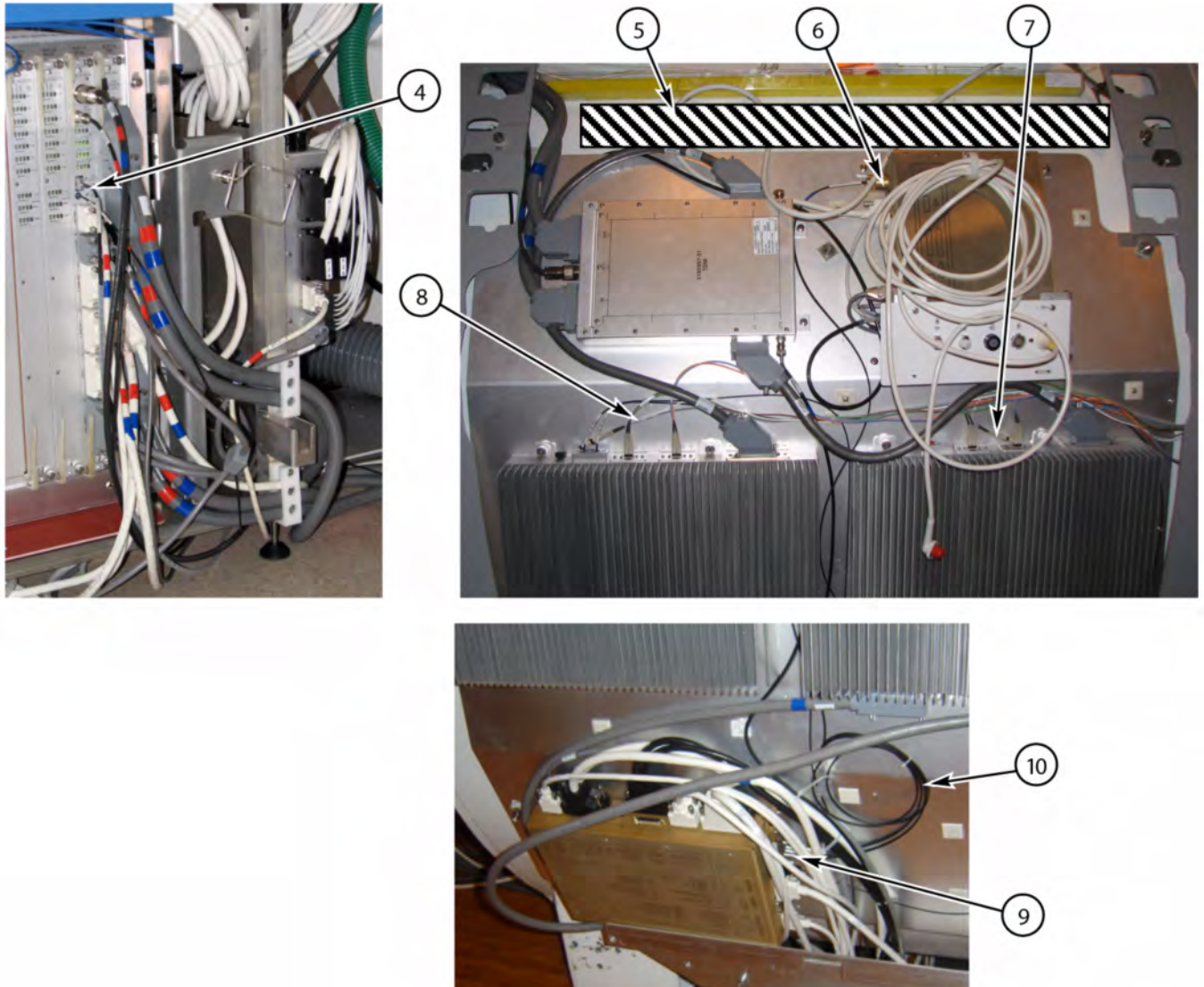
Illustration 4-80: Top of Magnet Fiber Optic Cables



1. Route and attach Run P2003 fiber optic connector to J20 on Fiber Optic I/O bracket on magnet.
2. If system has the second Receiver installed, RRX #2, route and attach Run P2004 fiber optic connector to J21 on Fiber Optic I/O bracket.

3. If IRD is installed, route and connect Run P2026 to fiber optic connector at J28 on I/O bracket. Run P2026 should already been routed through J11 on Magnet I/O bracket.

Illustration 4-81: Connect Fiber Optics to PAC and Rear Pedestal



4. Route M2025 at rear of magnet along side of cables routed on left side of rear magnet flange (as viewed from front of magnet) to Rear Pedestal. Make sure conduit is routed behind skirt cover bracket. Connect fiber optic wire M2025-D to PED HUB SLOT 11 - J14/J15 as marked.

NOTE: Secure in place with tie-wraps after all system cables have been connected and adjusted for cable service loop to Rear Pedestal

5. **IMPORTANT!** Do not route any cables on magnet in area indicated by diagonal shading. The service ladder side shield needs this area clear when it is used for servicing the magnet.
6. Route and connect wire M2025-C to J1 on PAC.
7. Route P2003 fiber optic wires over top of magnet and connect to J4/J5 on RCVR #1.

8. If system is 32 channel, route P2004 over top of magnet and connect to J4/J5 on RCVR #2.
9. Route and connect wires M2025-A and M2025-B marked J11/J12 and J13/J14 to SRI.
10. Coil and tie-wrap excess fiber optic wires. Make sure not to damage or crimp wires with too small of a bend radius.

25 Pneumatic Patient Alert Installation

The customer and users should be involved in the following decisions:

- Routing of Pneumatic Tubing
- Location of Control Box



NOTICE

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

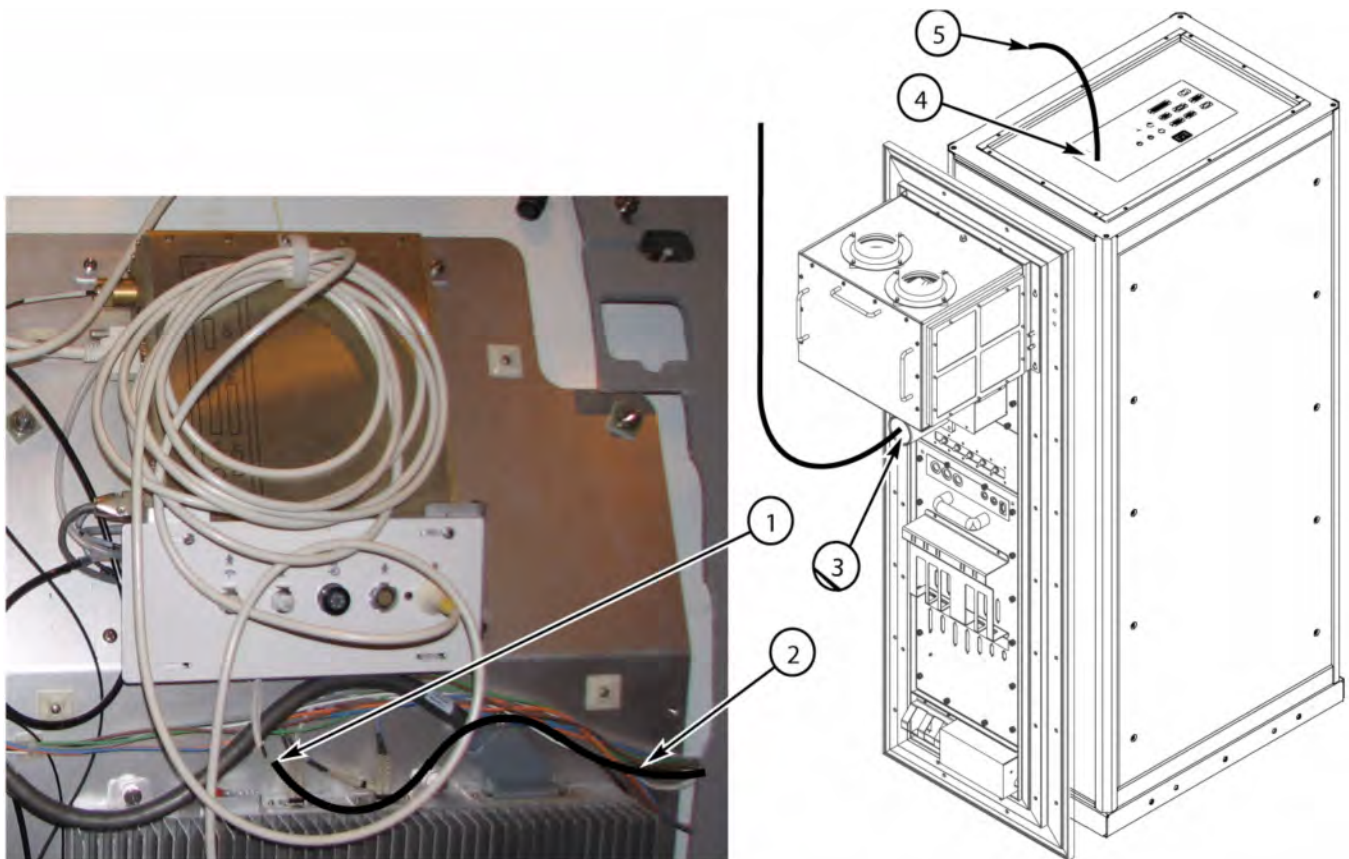
25.1 Magnet Room Routing of Pneumatic Tubing



NOTICE

Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched. Also, make sure that the pneumatic tubing can be routed from the PAC connector through the overhead cable tray, and to the Penetration Panel without being pinched along the route by other cables.

Illustration 4-82: Routing of Pneumatic Tube



1. Push end of pneumatic tubing onto PAC tubing connector located on left side of magnet.
2. Route tubing along path of fiber optic wires and secure in place with tie wraps. **Do not over-tighten tie-wraps. Tubing must not be deformed or kinked.**
NOTE: If the pneumatic tubing is to be pulled through fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.
3. Route Pneumatic Tubing from PAC, through the overhead cable tray to waveguide on Penetration Panel.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide, thru the fiber optic opening at the top of the PEN Cabinet to where the Control Box will be installed.
5. Refer to [Section 25.2](#) to determine if Extender Box is required. If Extender box is not required, route pneumatic tubing through overhead cable tray in Equipment room to Control Box location.

25.2 Determining Need of Extender Box

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46-317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:** The extender is not needed, proceed to [Section 25.5](#), *Control Box Installation*.
- **If the pneumatic tubing is not long enough to reach control box:** The extender is needed, proceed to [Section 25.3](#), *Installation of Extender Kit*.

25.3 Installation of Extender Kit, 46-317758P2 (If Required)



! DANGER

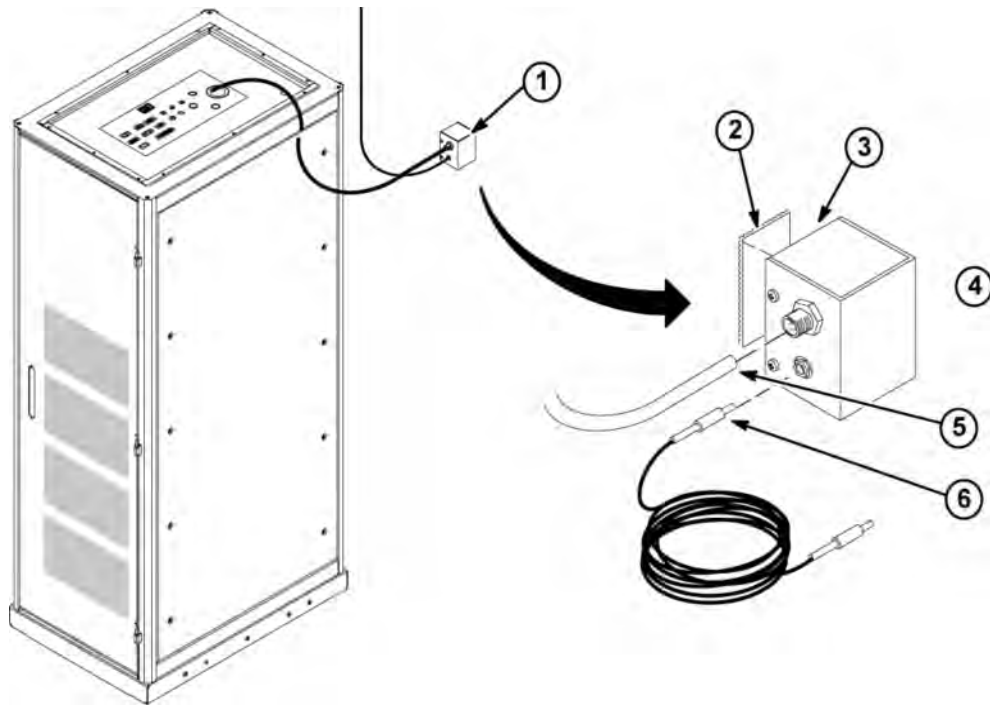
FERROUS MATERIAL HAZARD!

EXTENDER BOX CONTAINS FERROUS PARTS AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE.

DO NOT MOUNT EXTENDER BOX IN MAGNET ROOM.

25.3.1 Install Extender Box

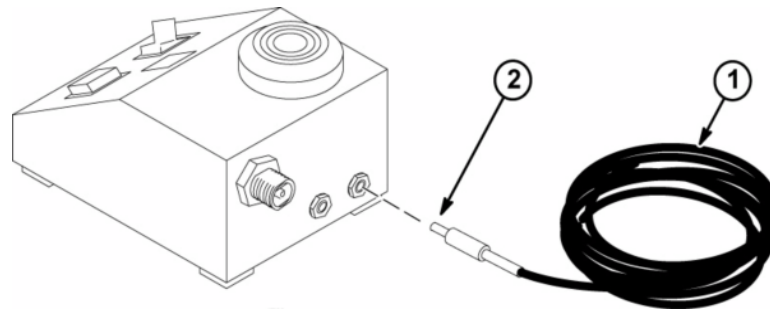
Illustration 4-83: Component Installation



1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach Velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed-thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.

25.3.2 Connection of Extender Wire to Control Box

Illustration 4-84: Extender Wire Connection



1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.

25.4 Control Box Installation Options

Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

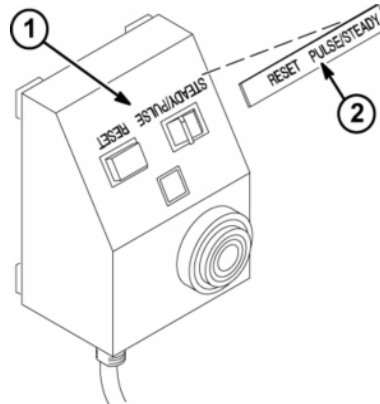
- Mount control box on a wall or other vertical surface per Procedures in [Section 25.5, Control Box Installation on Wall or Under Shelf](#).
- Mount control box under shelf per Procedures in [Section 25.5, Control Box Installation on Wall or Under Shelf](#).
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to [Section 25.9, Final Connections to Control Box](#).

NOTE: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

25.5 Control Box Installation on Wall or Under Shelf

1. Clean Control Box surface.
2. Remove adhesive backing from metal nameplate and apply over silk screened lettering as shown.

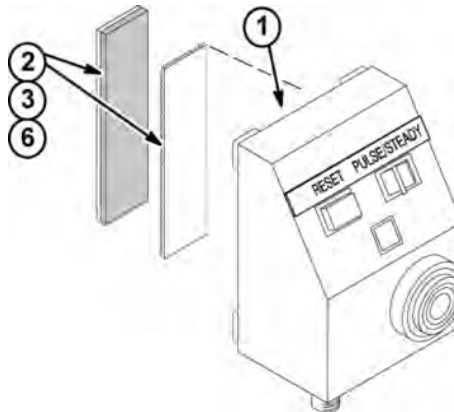
Illustration 4-85: Apply Nameplate



3. Select one of the following for installation of Control Box:
 - If control box will be mounted to WALL or SHELF with Velcro, go to [Section 25.6](#).
 - If control box will be mounted to WALL with screws, go to [Section 25.7](#)
 - If control box will be mounted to SHELF with screws, go to [Section 25.8](#)

25.6 Control Box Mounted with Velcro

Illustration 4-86: Apply Velcro



1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted **within sight of operator**.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.

7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to [Section 25.9, Final Connections to Control Box](#).

25.7 Installing Control Box with Mounting Screws on Vertical Surface

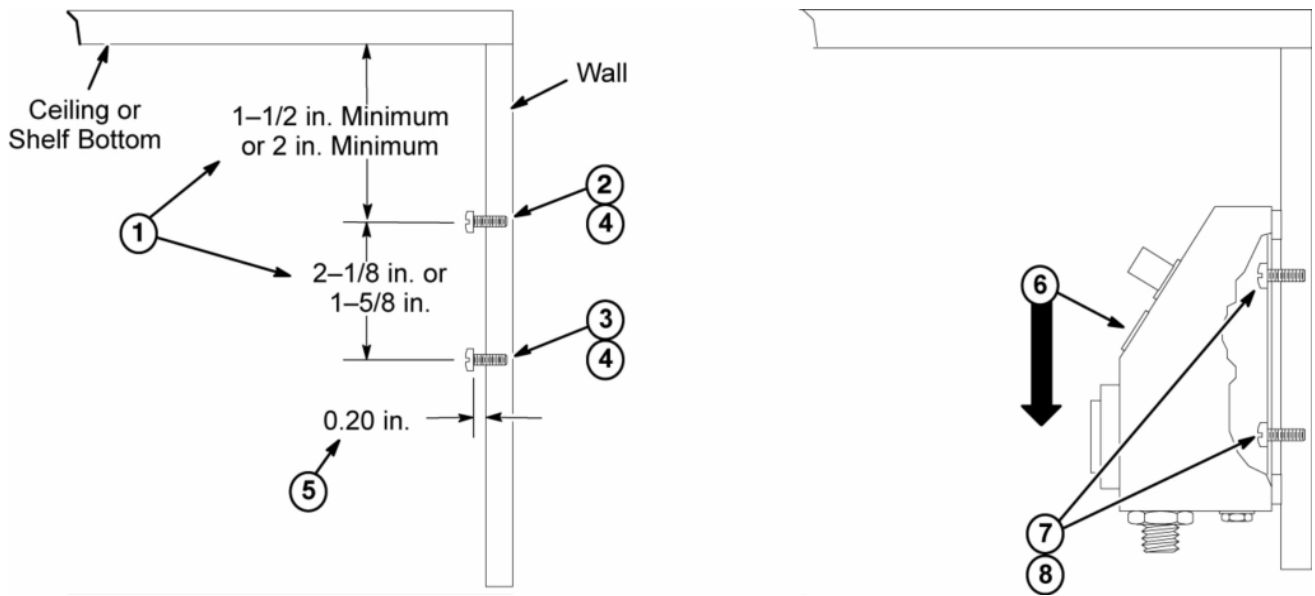
Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted **within sight of operator**.



NOTICE

If mounting on the wall, confirm with the site that no unforeseen hazards exist beneath the surface.

Illustration 4-87: Vertical Surface Installation

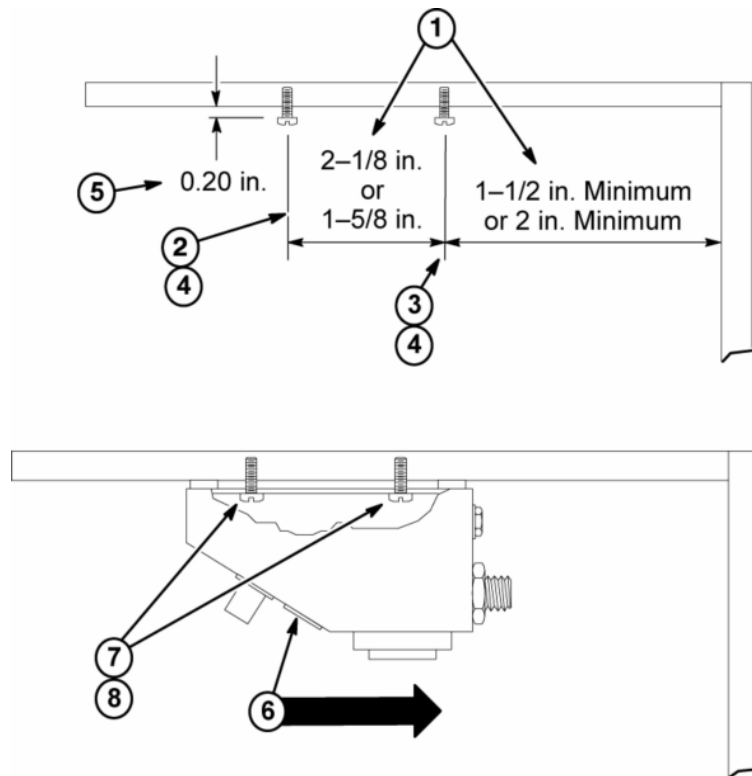


1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to [Section 25.9, Final Connections to Control Box](#).

25.8 Installing Control Box with Mounting Screws Under Shelf

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted **within sight of operator**.

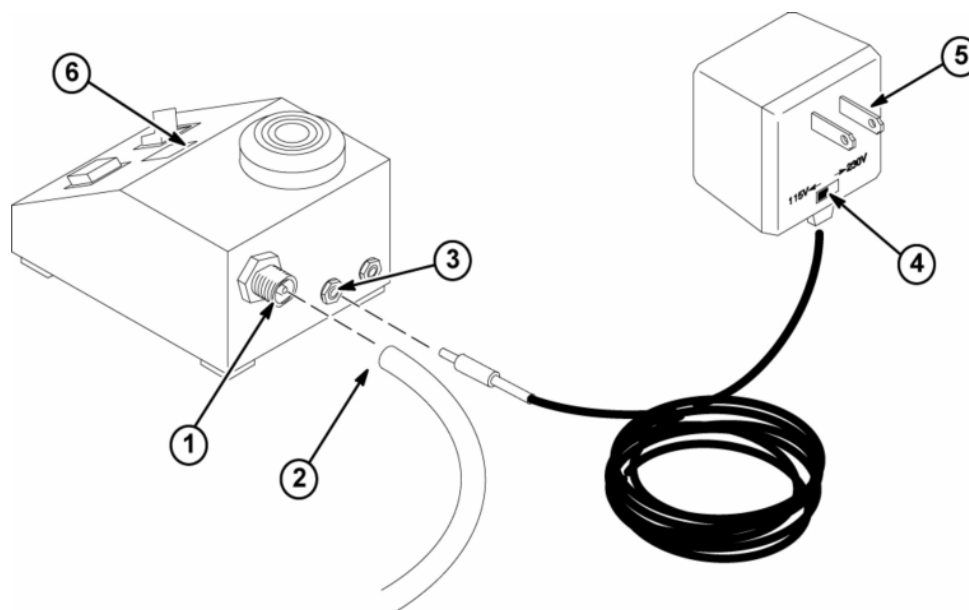
Illustration 4-88: Under Shelf Installation



1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to [Section 25.9](#), *Final Connections to Control Box*.

25.9 Final Connections to Control Box

Illustration 4-89: Control Box Connections

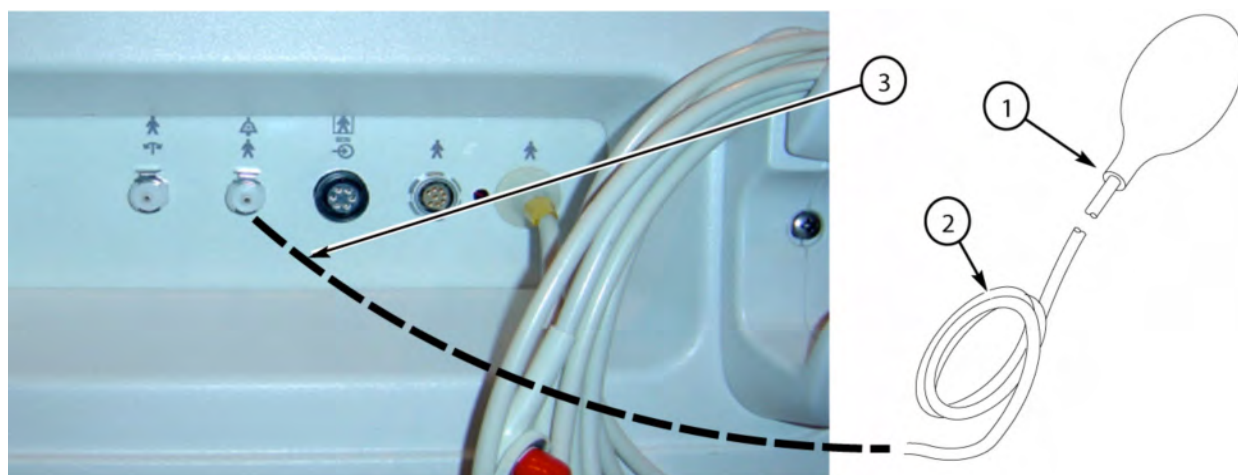


1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed-thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is lighted.

25.10 Squeeze Bulb Installation

The following steps should be completed after the side covers have been installed.

Illustration 4-90: Tube Connection to PAC Module



1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the PAC Interface without getting in the way of operator or patient during use.
3. Connect end of tubing to "CALL" connection on PAC Interface.

26 Bridge & Rear Pedestal Installation



WARNING

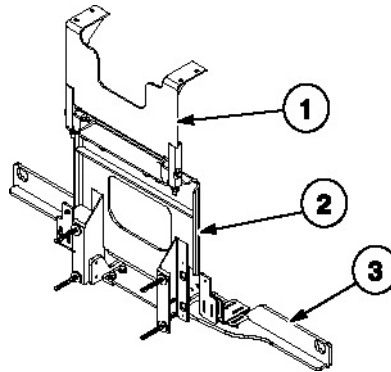
FERROUS MATERIAL HAZARD!
IF MAGNET IS ENERGIZED, FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

THE WHEEL ASSEMBLIES ATTACHED TO THE REAR PEDESTAL FOR SHIPPING AND DELIVERY MUST BE REMOVED BEFORE MOVING REAR PEDESTAL INTO MAGNET ROOM.

26.1 Bridge Front Support

The Bridge Front Support is an assembly of three brackets:

Illustration 4-91: Bridge Front Support Brackets



1. Bridge Support Upper Asm (5335143-2)
2. Bridge Support Lower Asm (5334546)
3. Bridge Support Magnet Mount Asm (5337932-2)



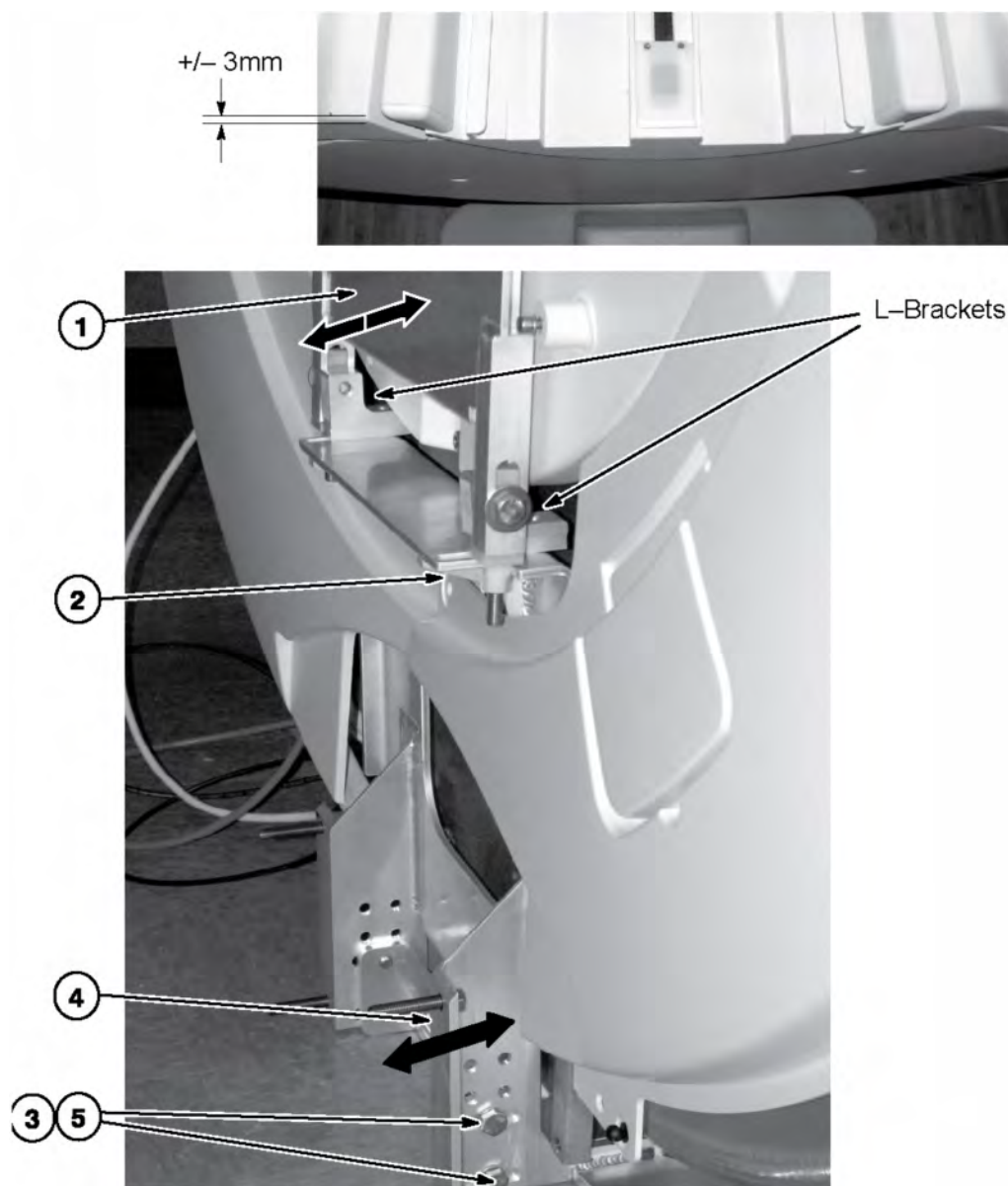
NOTICE

Make sure that after all final adjustments have been made and all fasteners tightened that no part of the Bridge Support Assembly is in contact with the magnet surface.

26.2 Z-Direction Bridge Alignment

Align the Bridge in the Z-direction (in/out of the magnet bore) until front surface of Bridge aligns with front surface of end bell to within +/- 3mm.

Illustration 4-92: Z-Direction Alignment



1. Alignment to front end bell is done by pushing or pulling Bridge with Support Upper attached which allows the L-Brackets to slide toward or away from the magnet.
2. When aligned, tighten the four M10 screws attaching L-Bracket to the Support Lower.
3. **NOTE:** If more adjustment is needed for the Z-axis adjustment, loosen the four bolts (two on each side) holding the Support Lower to the Bridge Support Magnet Mount Asm.
4. Move the Support Lower until the +/-3mm bridge-to-end bell alignment has been achieved.
5. Re-tighten bolts.

26.3 Attach Rear Pedestal to Magnet

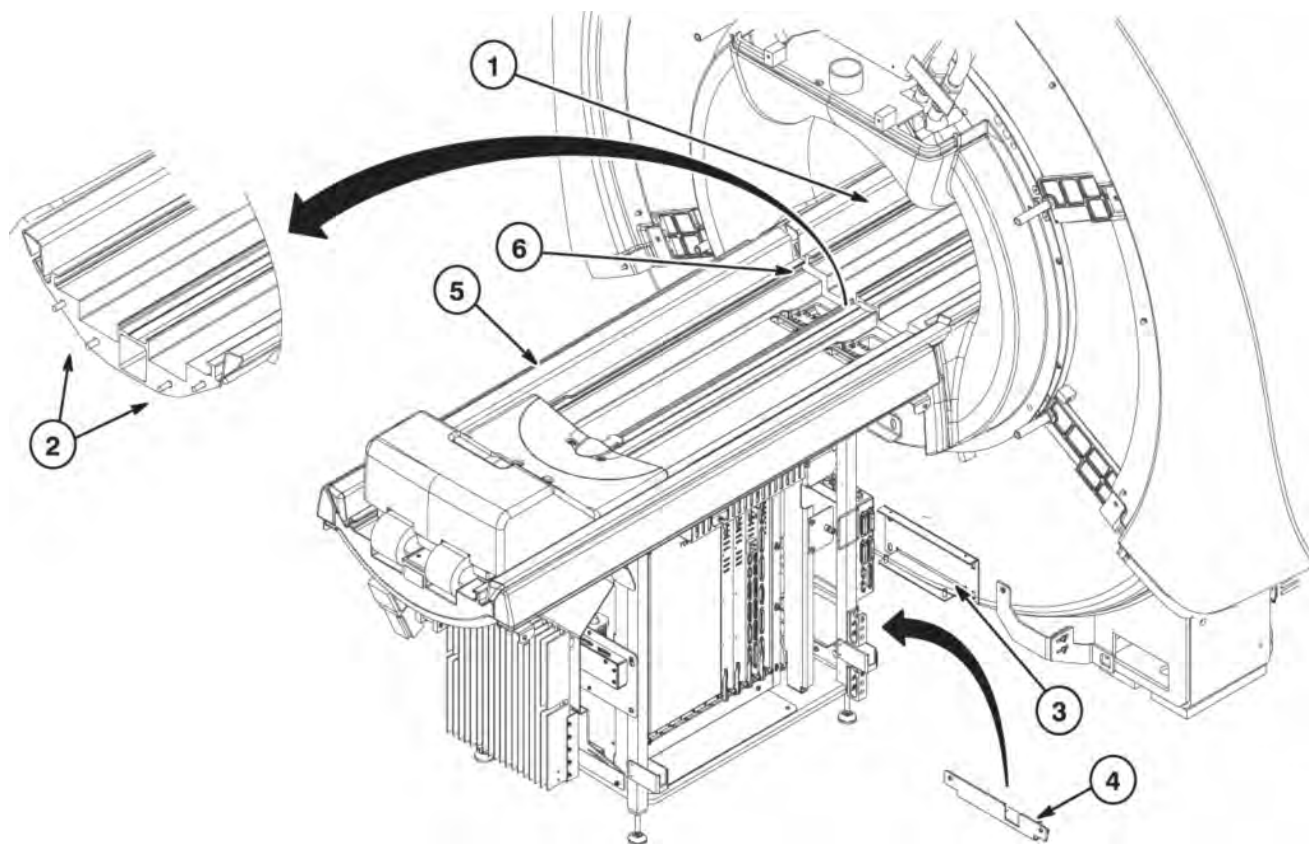


CAUTION

Potential Injury Hazard
Weight of Rear Pedestal

Rear Pedestal requires at least THREE people to move Rear Pedestal into position at rear of magnet.

Illustration 4-93: Attaching Front and Rear Split Bridge

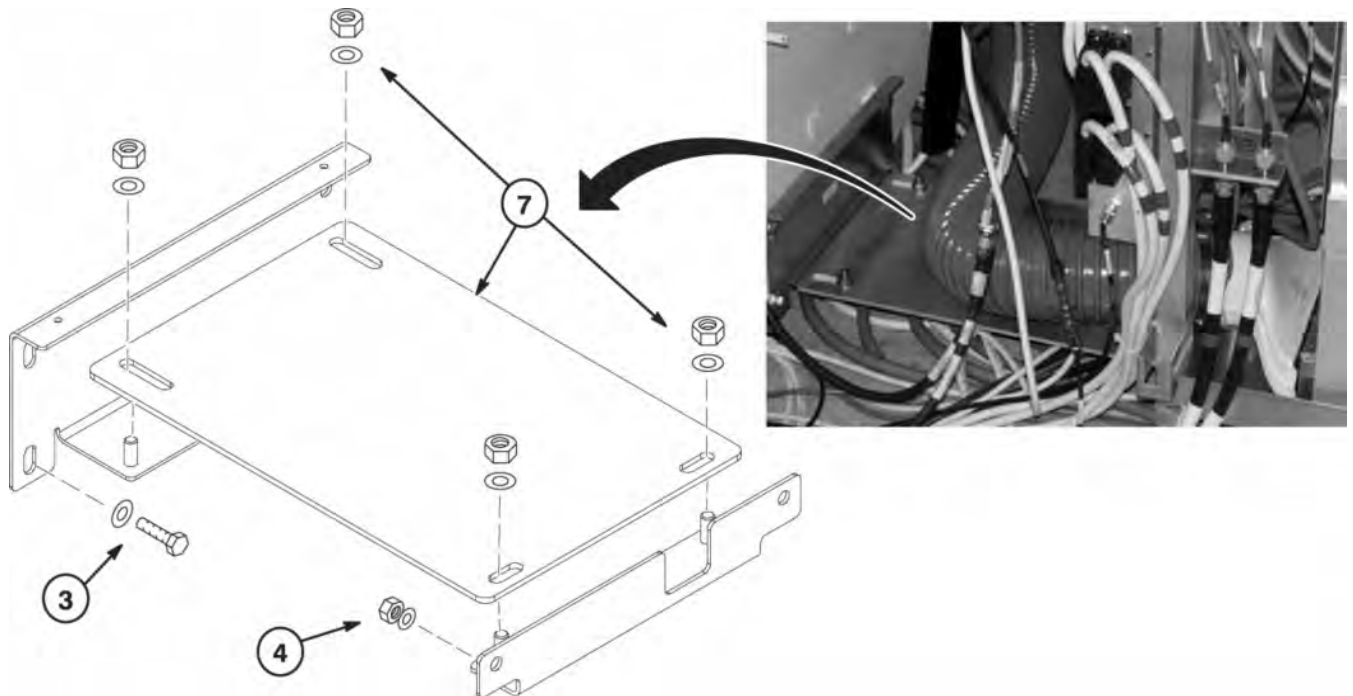


NOTICE

1. Before attaching Rear Pedestal, perform a preliminary check on the Y-direction distance at the front and rear of the bridge. Refer to [Section 26.4, Y-Direction Bridge Alignment](#), for instructions on setting required distance between Bridge and top of the patient bore.
2. Remove four M10 nuts and washers from studs on Front Split Bridge.
3. Attach 5343250 to rear magnet mounting blocks using the 4 nuts and 4 washers removed in previous step. **Do not tighten as adjustment may be needed.**

4. Attach 5343248 to two studs on rear pedestal using M10 washers and nuts.
5. Move Rear Pedestal closer to rear of magnet and adjust leveling feet to align top of rear split bridge with top of front split bridge in magnet bore.
6. Move Rear Pedestal forward and engage four studs on front split bridge. Secure with the four M10 nuts and washers previously removed. Remove foam under Bridge.

Illustration 4-94: Installing Lower Bracket



7. Using four (4) M10 washers and nuts, connect 5343250 and 5343248 with 5343249. **Do not tighten until bridge and table are completely adjusted.**

NOTE: Cables are routed below the bracket. Air hose is routed on top of bracket as shown in following illustration.

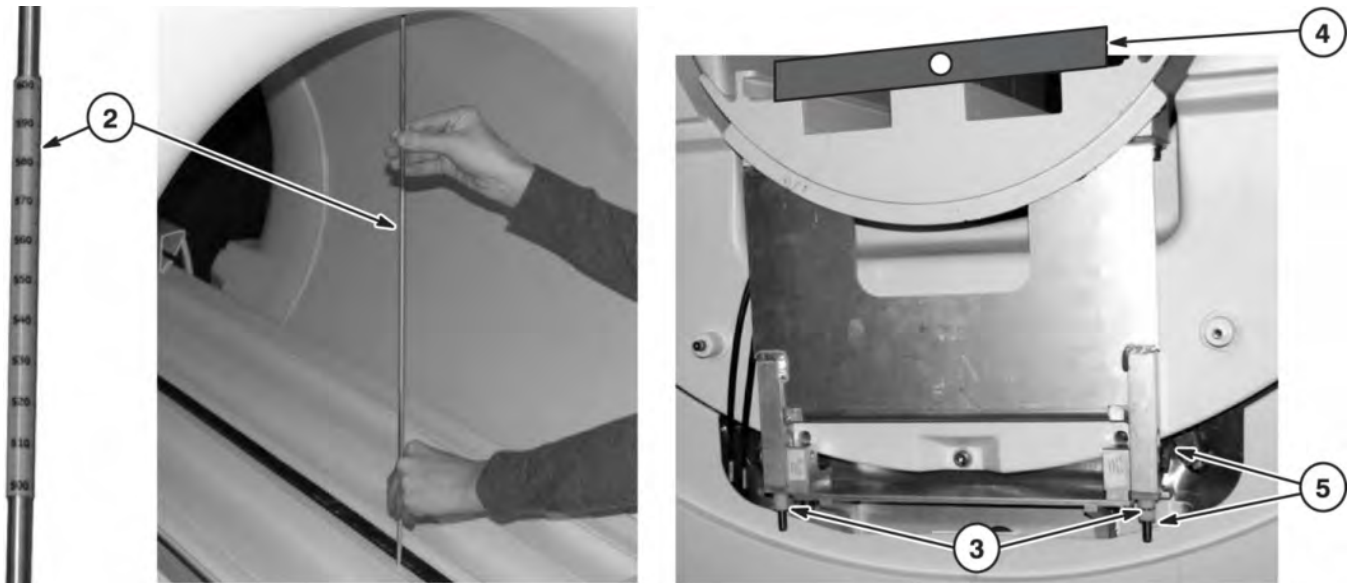
26.4 Y-Direction Bridge Alignment (Front of Magnet)

Illustration 4-95: Maintain Spacing from Floor



1. **IMPORTANT:** Make sure that the lower Bridge Support Magnet Mount Assembly surface, only by the magnet feet, does not come in contact with the magnet room floor (recommend a 2mm minimum clearance).

Illustration 4-96: Y-Direction Alignment



2. Align the Bridge in the Y-direction (up/down in the magnet bore) to within 582mm - 585mm using the Bridge height gauge (delivered to site packaged with Dock assembly) at the front of the magnet.
3. Adjust the M10 Nylon nuts at the bottom of the Bridge Support Upper bracket to raise and lower Bridge to attain the 582mm - 585mm distance in between the top of the patient bore and the surface under the belt on the bridge.
4. A level can be used to make sure bridge is level in the X-direction. Adjust M10 Nylon nuts as necessary.

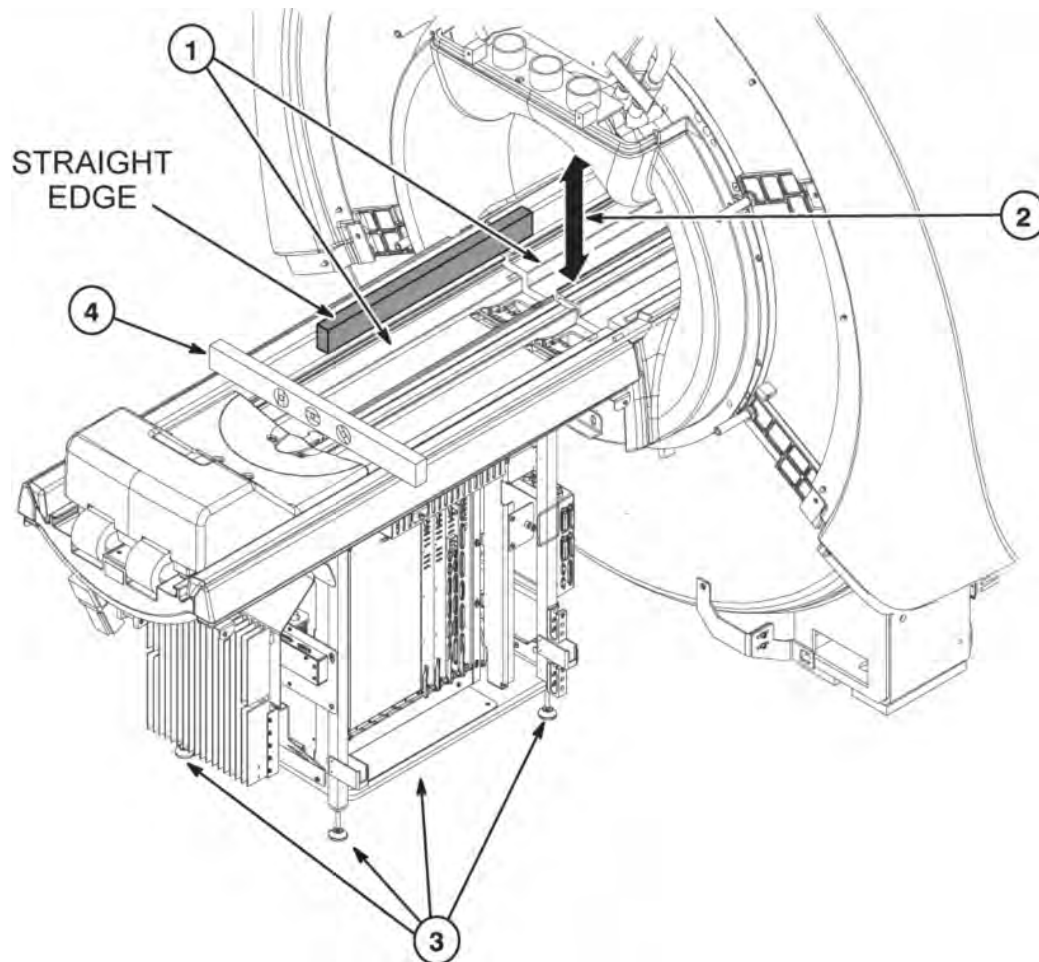
5. Tighten the two (2) M10 bolts and the M10 Nylon Hex Nuts on each side.

26.5 Rear Split Bridge Height Adjustment (Rear of Magnet)

Refer to **Y-Direction Bridge Alignment** section for distance between bridge and top of the RF Coil.

NOTE: The Bridge does not need to be “level” to Earth in the Z-direction in achieving the alignment of bridge surfaces. Therefore, you do NOT need a level at this point of the installation, you DO NEED a straight edge. (Though you can use a level as a straight edge)

Illustration 4-97: Rear Bridge Height Adjustment

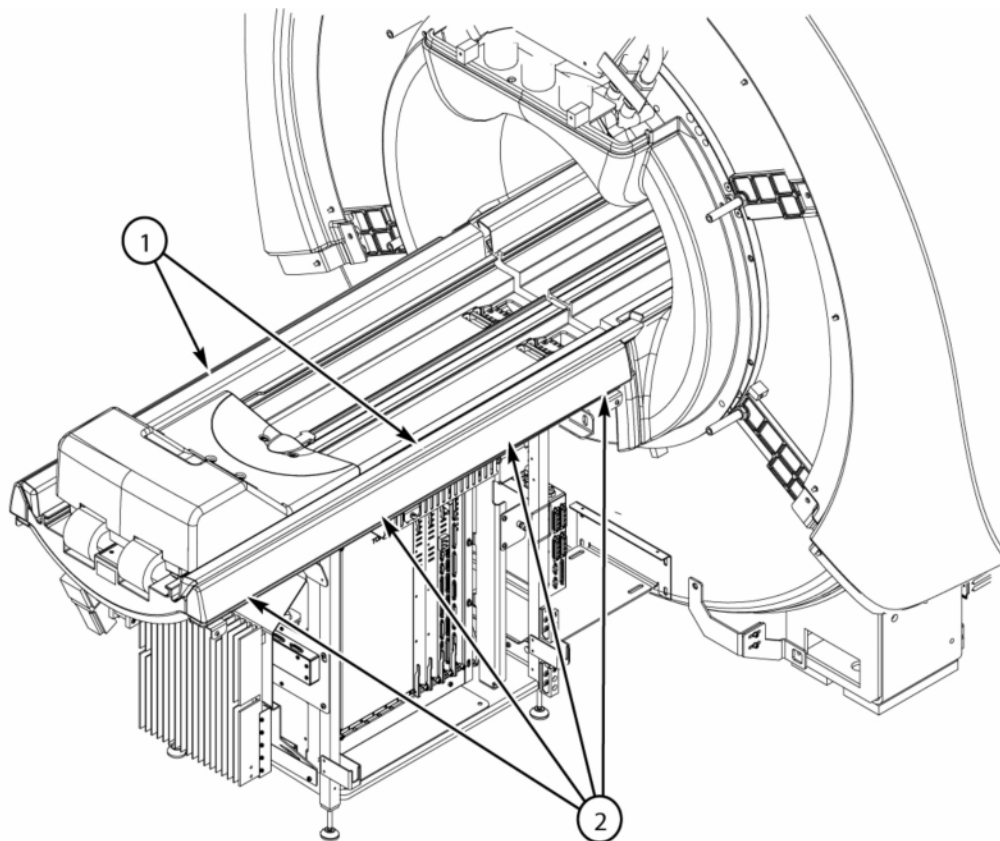


1. Use a Straight Edge to make sure that the front and rear Bridge surfaces, on both sides of Bridge where the LPCA wheels roll, are even. This will provide smooth travel for the Carriage wheels. If uneven, refer to next step.
2. Align the Bridge in the Y-direction (up/down in the magnet bore) to within 582mm - 585mm using the Bridge height gauge (delivered to site packaged with Dock assembly) at the rear of the magnet.
3. Adjust the rear pedestal feet (by turning) to raise and lower Bridge to attain the 582mm - 585mm distance in between the top of the patient bore and the surface under the belt on the bridge.

4. At this point in adjusting the Rear Pedestal, use a level to make sure the Bridge on the rear pedestal is level in the X-direction. Adjust Rear Pedestal feet as necessary.

26.6 Front and Rear Split Bridge Alignment Adjustment

Illustration 4-98: Bridge Alignment Adjustment



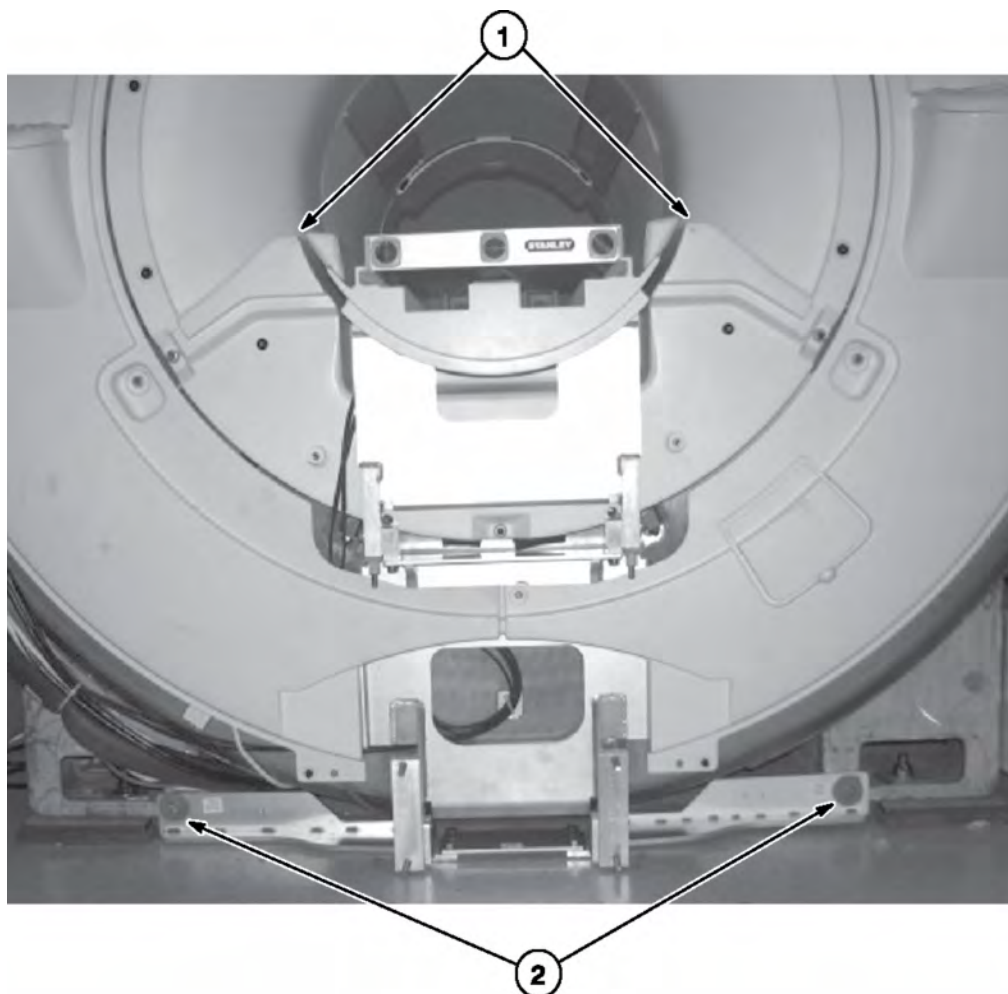
1. Install left and right Rear Pedestal Bumpers using M6 fasteners.

NOTE: Gap between Rear Split Bridge and left and right Rear Pedestal Bumpers should be uniform along entire length.

2. Adjustment for left and right Rear Pedestal Bumpers is made by loosening four fasteners underneath each bumper and moving bumper until uniform gap is made.

26.7 X-Direction Bridge Alignment

Illustration 4-99: X-Direction Alignment



1. Align the Bridge in the X-direction (left/right in the magnet bore) by shifting entire front Bridge Support Assembly until the distance between Bridge and RF Body Coil is equal on both sides of the Bridge. It may be necessary to loosen the two (2) M10 bolts that fasten the Bridge Support Assembly to the magnet feet.
IMPORTANT: Bridge must not be in contact with the RF Body Coil along its entire length.
2. Tighten the two (2) M10 bolts that fasten the Bridge Support Assembly to the magnet feet, once adjustment is complete.
3. Check to make sure the Bridge Support Upper Assembly is not in contact with the front end bell and the Support Lower is not touching the magnet. This is to ensure that there is not vibration “short circuit”.
4. Reattach front bridge cover.

26.8 Bridge Height Adjustment

Make sure the distance between the top of the patient bore and the surface under the belt on the bridge is between 582mm to 585mm at both ends of the magnet. Adjust bridge as necessary to meet distance.

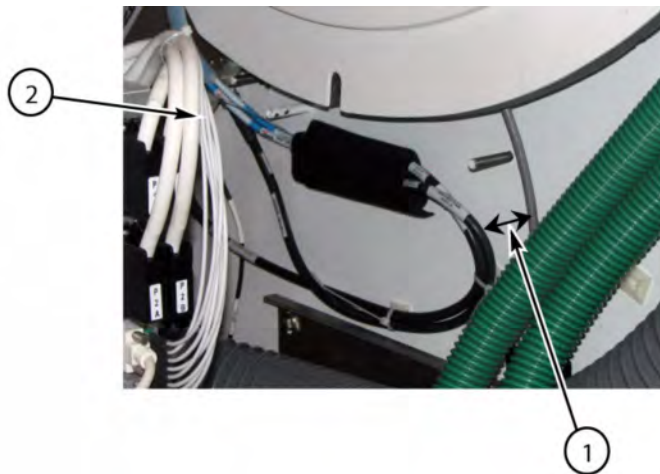
- Tighten all fastener connections on bridge and rear pedestal after adjustment is complete.
- Make sure the fasteners for the lower bracket between magnet and Rear Pedestal are tightened.

26.9 Cable Separation at Rear Pedestal Connection

For optimal performance of system, the “I” and “Q” cables must be spaced a minimum of 75mm (3 inches) from cables exiting the cable tracks from the LPCA and 30mm (1.2 inches) from the temperature Sensor cable routed from the gradient coil.

NOTE: For clarity, the illustration below shows I and Q cables without the foam insulation that will be pre-installed around cables on all installations.

Illustration 4-100: I and Q Cable Separation



1. Make sure the Temperature Sensor cable is separated at least 30mm from the “I” and “Q” cables.
2. Make sure the Cable Track cables are separated at least 75mm from the “I” and “Q” cables.

27 LPCA Installation

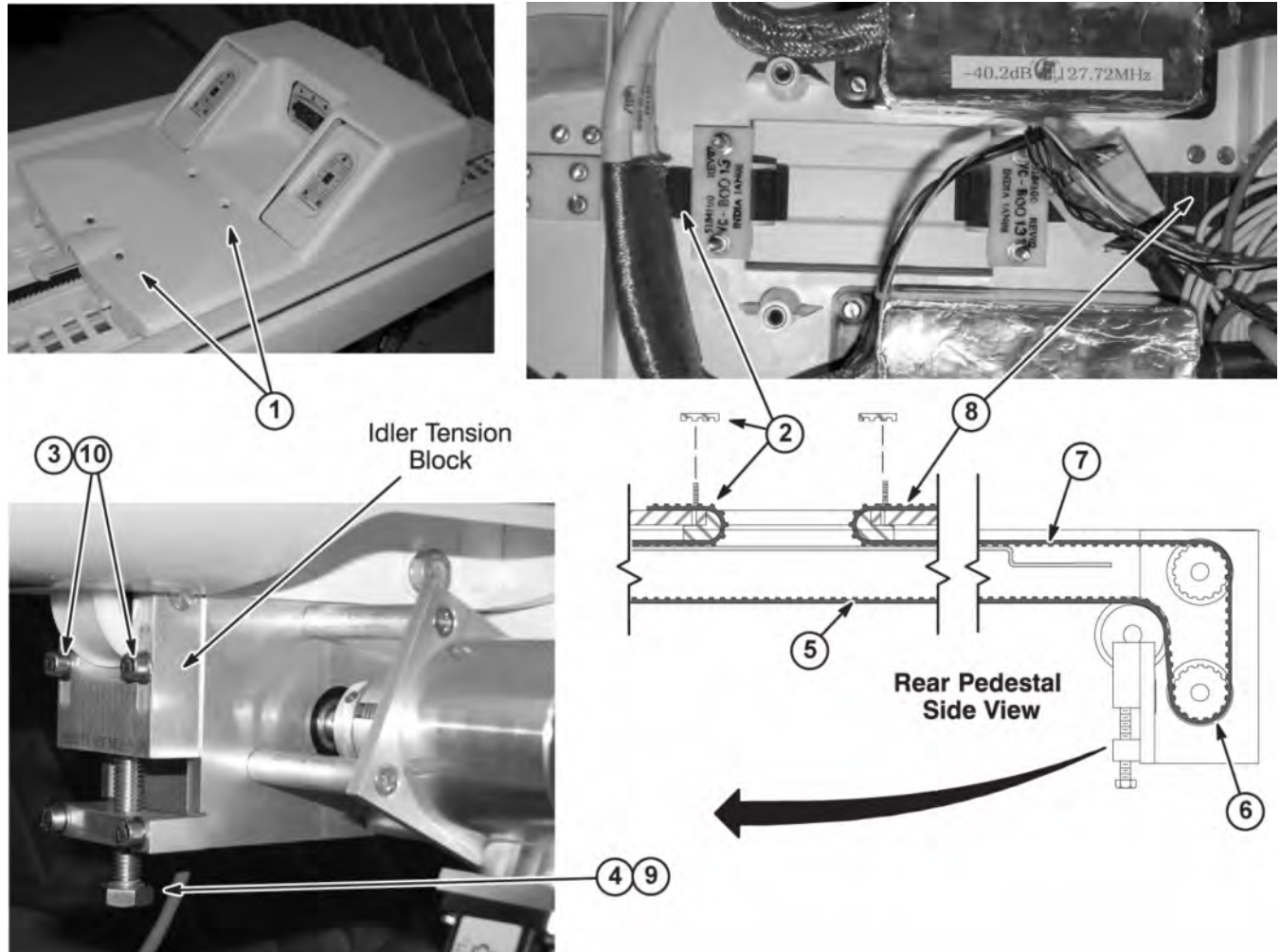
27.1 Route and Attach Drive Belt



NOTICE

Use care and avoid pinch points when performing drive belt installation.

Illustration 4-101: Drive Belt Installation



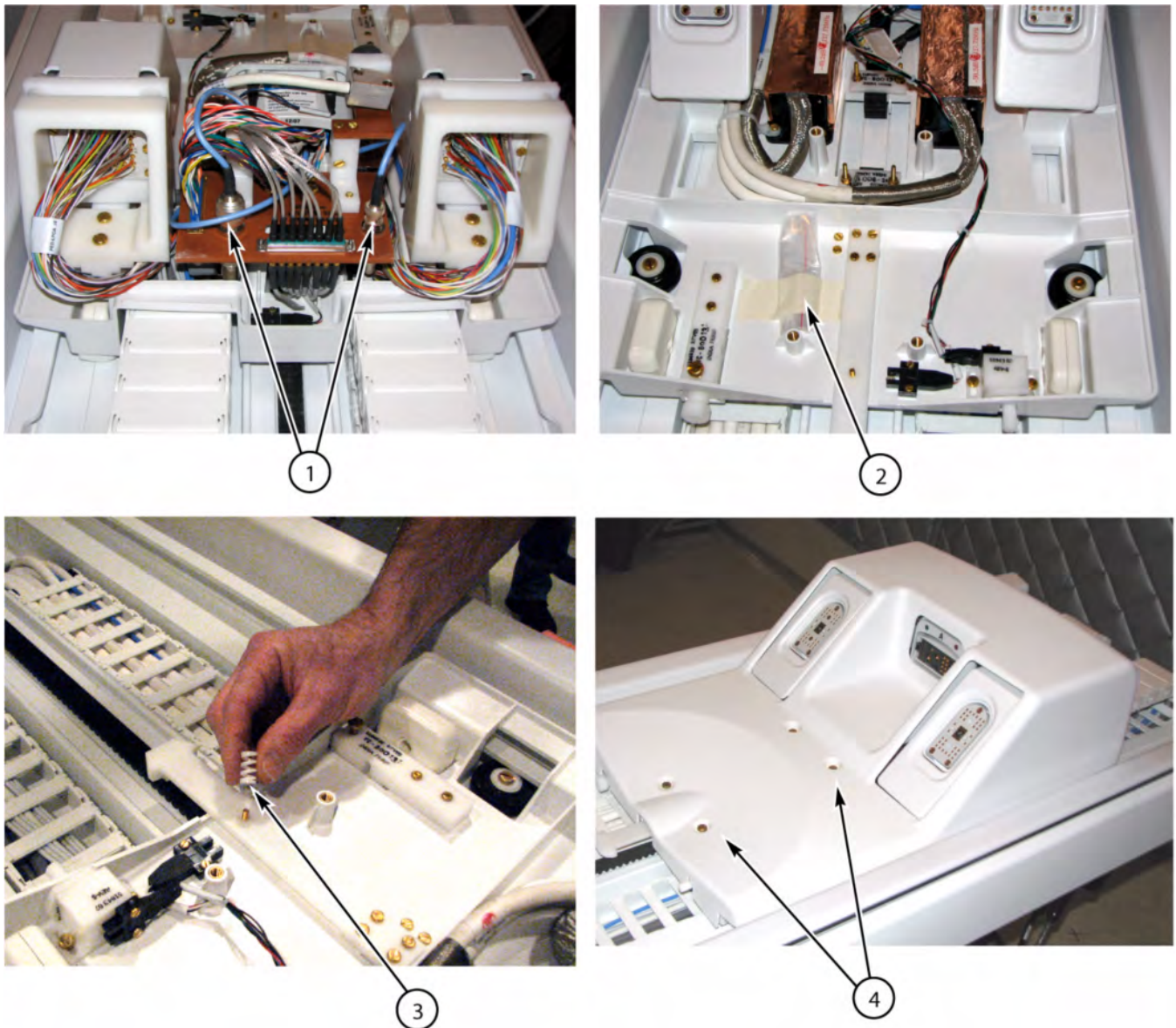
1. Unfasten four screws from top of LPCA and remove top cover.
2. Route one end of Drive Belt delivered with Front Split Bridge, through opening in LPCA Base Plate and attach with clamp. Make sure that 3 teeth are exposed after clamp is attached.
3. Under bridge, loosen the two socket head screws on Idler Tension Block.
4. Lower the Idler Tension Block with pulley to its lowest position by unscrewing hex head bolt.
5. Route other end of drive belt from front of Bridge towards Tension Pulley.
6. Feed back end of Belt underneath Bridge, around Tension Pulley and rear pulleys into Bridge.

NOTE: Check to verify that belt is not caught around spacers near rear pulley.

7. Route Belt under and through opening to top of Carriage.
8. Position Belt under Clamp Plate. There should be 15 teeth exposed past the clamp. Secure in place.
9. Tension the belt by raising the Idler Tension Bock. Turn the hex head bolt until the two socket head cap screws are at the bottom of the slot in the idler Tension Block.
10. Tighten the two socket head screws.

27.2 Install LPCA Cover

Illustration 4-102: LPCA Cover Installation



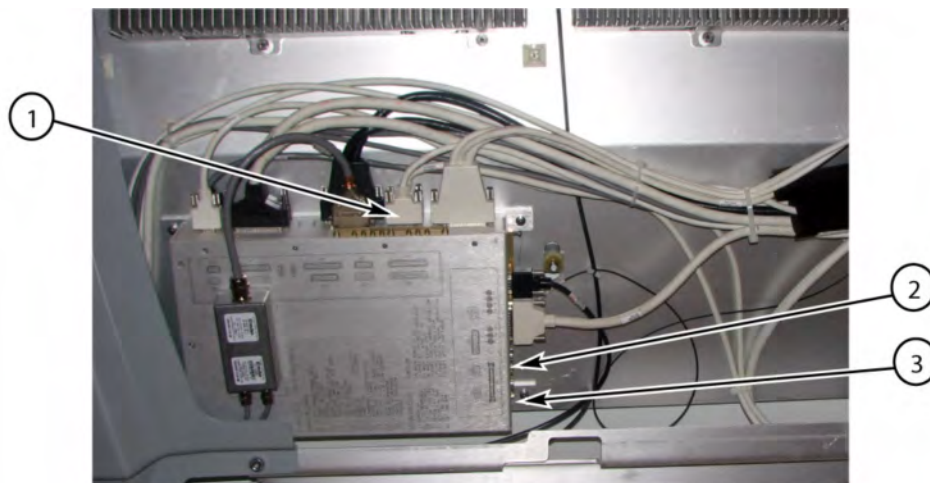
1. Check all connectors and make sure they are seated properly

2. Unfasten plastic bag from surface of LPCA and remove spring inside.
3. Set spring on Cradle Latch stud.
4. Carefully replace LPCA cover and fasten with four screws previously removed.

28 SRI Connections

Many of the cables connected to the SRI Module have been completed prior to delivery of system to site. Refer to applicable cable map for this site configuration and make sure that all cables, including those routed during this installation, have been connected to the SRI.

Illustration 4-103: SRI Cable Connections



1. Make sure Run M3342 is connected to J23.
2. Make sure Run M3345 is connected to J32.
3. Route Run M3346 from the cable track in the Rear Pedestal and connect to J31.

29 Emergency Off Connections

29.1 Overview

Two Emergency Off (E-off) pushbuttons are supplied with the system. One pushbutton is for installing in the magnet room. The second pushbutton is optional and can be installed in the equipment or operator room.

Refer to applicable site installation drawings for detailed wiring diagram.

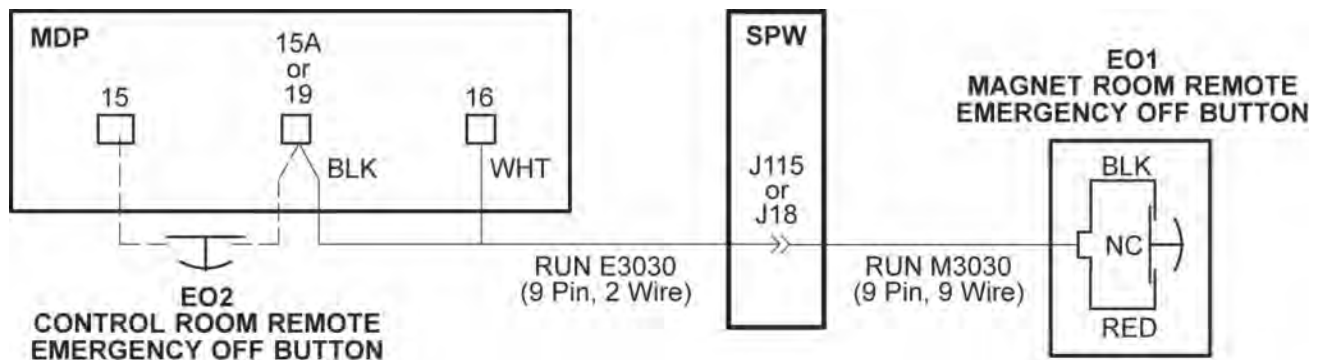
29.2 Magnet Room Pushbutton Installation

1. Connect Run M3030 to the SPW and route the cable to the remote E-off pushbutton location and trim cable to length. Locate Red and Black pair and prepare ends for applicable terminals.

NOTE: Run M3030 has nine wires. Make sure to use the black and red pair when connecting to the E-off pushbutton.

2. Connect Run E3030 to the Secondary Pen Wall (SPW) and route the cable to the Main Disconnect Panel (MDP) and trim cable to length. Locate white and black pair of wires and prepare ends for applicable terminals.
3. If the MDP power has been connected, have the proper personnel connect the E-off wiring. See illustration below for reference.

Illustration 4-104: Emergency Off Connections



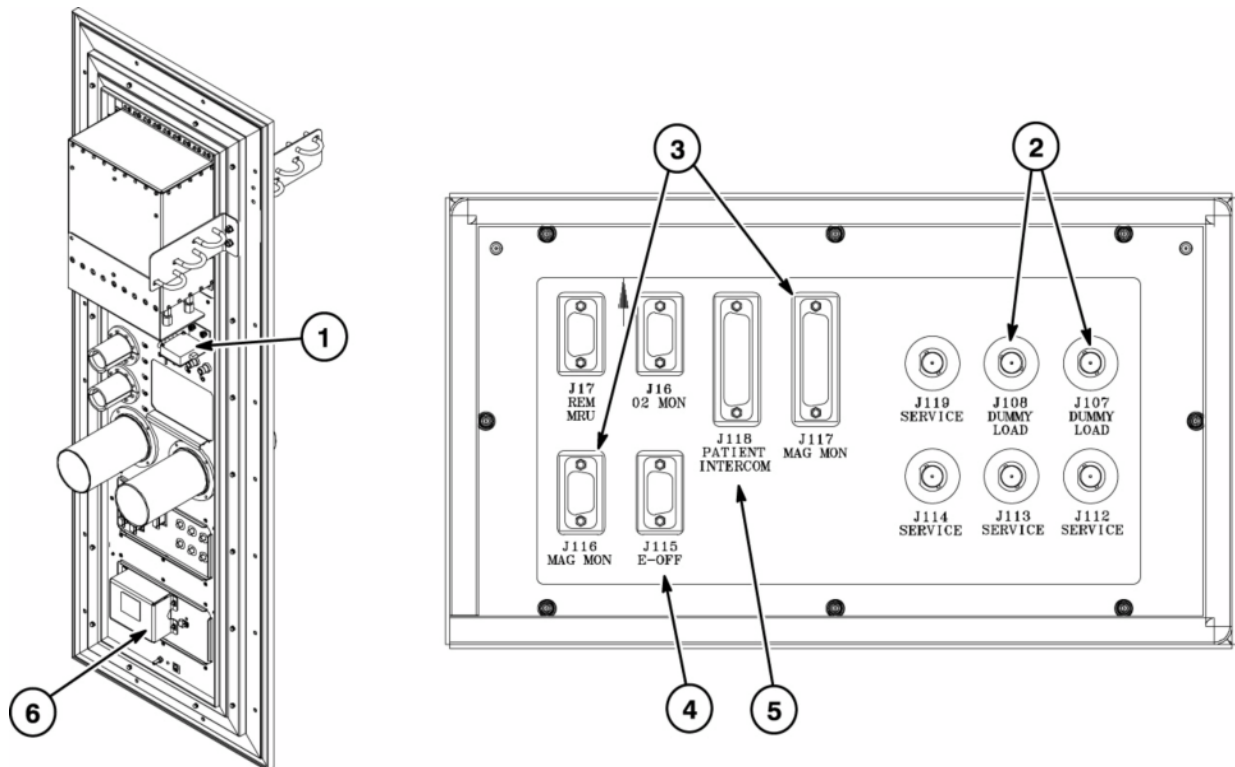
30 Route and Connect SPW Cables

The SPW (Secondary Pen Wall) was delivered to site and installed prior to magnet delivery.

- The gas lines, Runs 621 and 622, were routed thru the J110 and J111 waveguides when the magnet was installed.
- The gradient cables were routed and connected to the gradient filter in a previous procedure in the section.
- The coolant lines have been routed thru J100 and J101 waveguides.
- The 4 inch air hose from the PGR cabinet has been routed to J104 waveguide.

Refer to cable map and connect remaining cables to SPW after Grounding and Leakage test has been completed.

Illustration 4-105: SPW Cable Connections

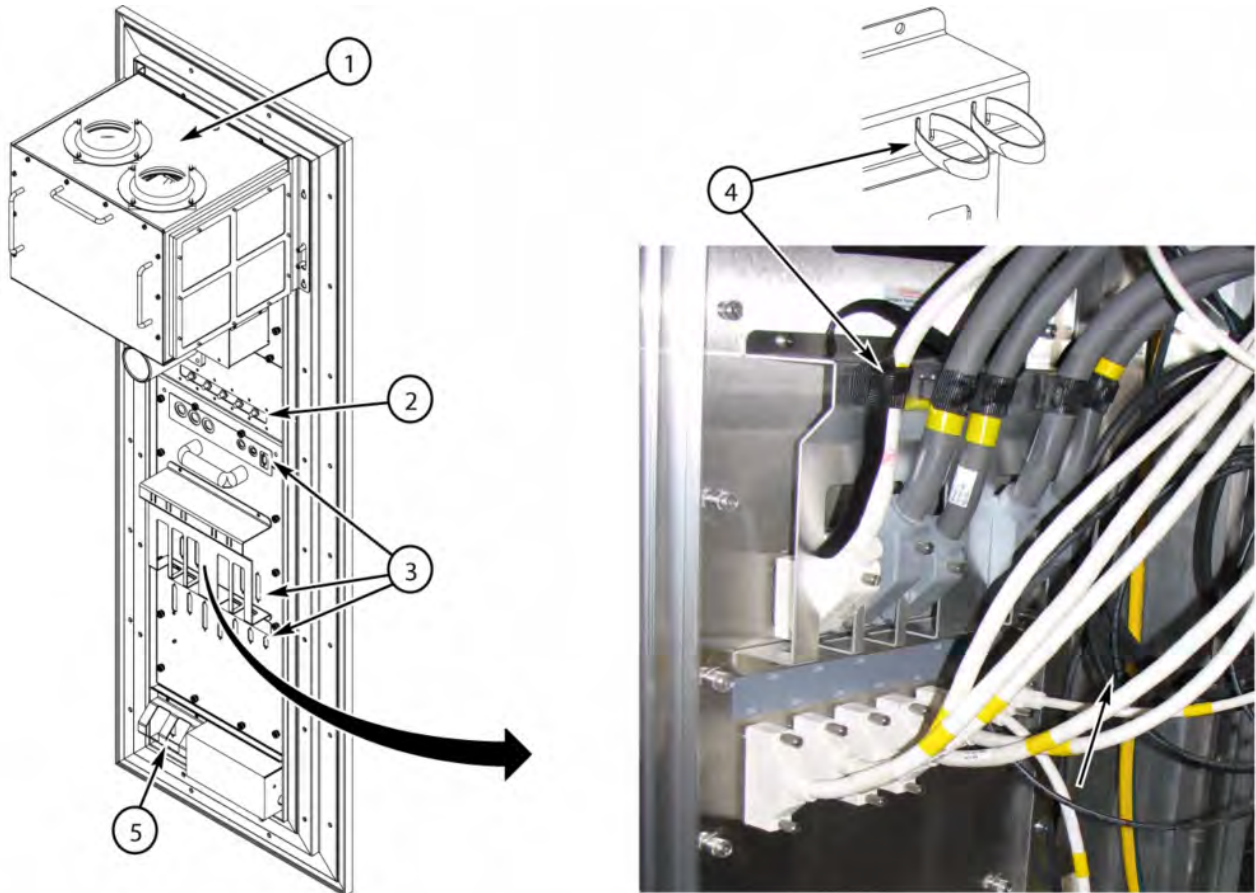


1. Route and connect Run M3309 to J123 on the Transmit Switch.
2. Not used. There are no connections to J107 and J108 at this time.
3. Route and connect Runs M3022 and M3023 from MagMon to J116 and J117.
4. Route and connect Run M3030 from E-Off Switch to J115.
5. Route and connect Run E3031 from GOC to J118.
6. Remove safety cover. Route and connect Run 623 to F1 and F3. Replace safety cover.

31 Route and Connect Penetration Panel Cables

1. The Blower Box hoses and cables should have been installed and connected in previous procedures.

Illustration 4-106: PEN Panel Cable Connections



2. Connect cables to TR/DD Filter: M3310 (J72), M1341 (J73), M3312 (J74), M1342 (J75), M3311 (J76)
3. Connect cables to access panel:
 - Top Row - M1302 (J29), M1304 (J30), M1303 (J32), M3316 (J34)
 - Middle Row - M3310 (J45), M3306 (J46), M3308 (J47), M3304 (J48).
 - Bottom Row - M3313 (J35), M3311 (J36), M3303 (J39), M3302 (J40), M3312 (J41), M3315 (J54), M3314 (J42) (option)
4. Make sure the middle row cables are routed in the correct orientation, connected, and secured in place with straps as shown.
5. Connect TDM power cables: M3300 (J51), M3301 (J52)

32 Installation of Magnet Enclosure Rear Covers

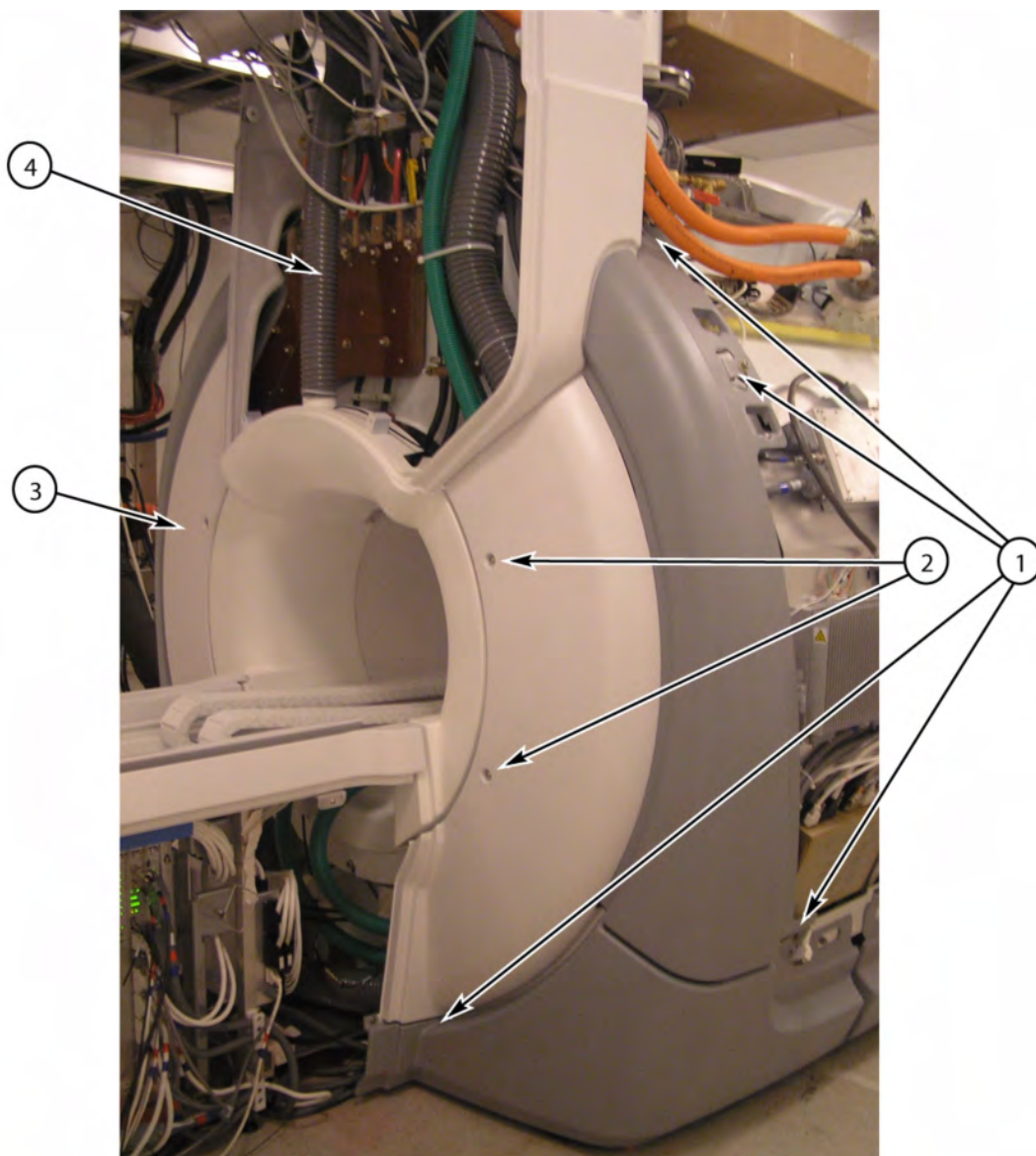


NOTICE

The rear covers are heavy. Avoid awkward body positioning. Use proper lifting techniques when handling covers.

NOTE: The lower Skirt Cover is shown in the view below, but is not installed at this time.

Illustration 4-107: Enclosure Rear Covers



1. Attach the Left Rear Cover using the four M10 x 25mm hex head cap screws (2109867-23) and four M10 nylon flat washers (46-294167P73) fasteners.

2. Install two M10 button head screws (2109872-12) and nylon flat washer (46-294167P73) at these two locations.
3. Attach the Right Rear Cover. Refer to steps 1 and 2.
4. Attach the Rear Cable Cover (not shown) using the two M6 x 16mm hex head cap screws (2109867-13) and M6 nylon washers (2229794).

33 Coil and Dress Cable in PEN Panel Closet

Inside the Pen Panel Closet, there are cable hooks for the purpose of storing excess cable lengths.

1. Separate, coil, and neatly dress any excess cable lengths and place them on available hooks.
2. Refer to [Storage of Excess Cable Length](#) for information on coiling cables.

Illustration 4-108: Storage of Excess Cable Lengths



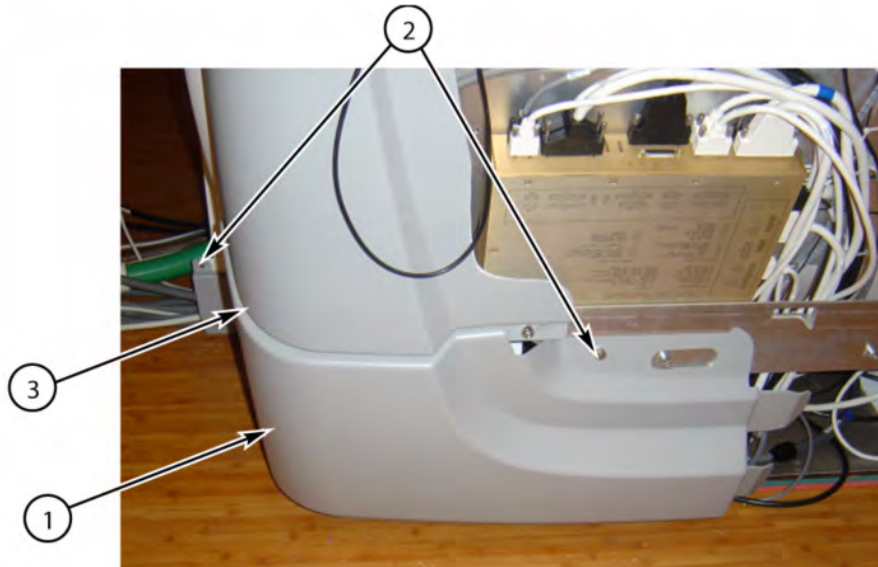
Chapter 5 Final Enclosure Installation

1 Install Enclosure Covers

1.1 Install Skirt Covers

Install the left and right side lower Skirt Covers as shown in the following steps.

Illustration 5-1: Installation of Skirt Cover



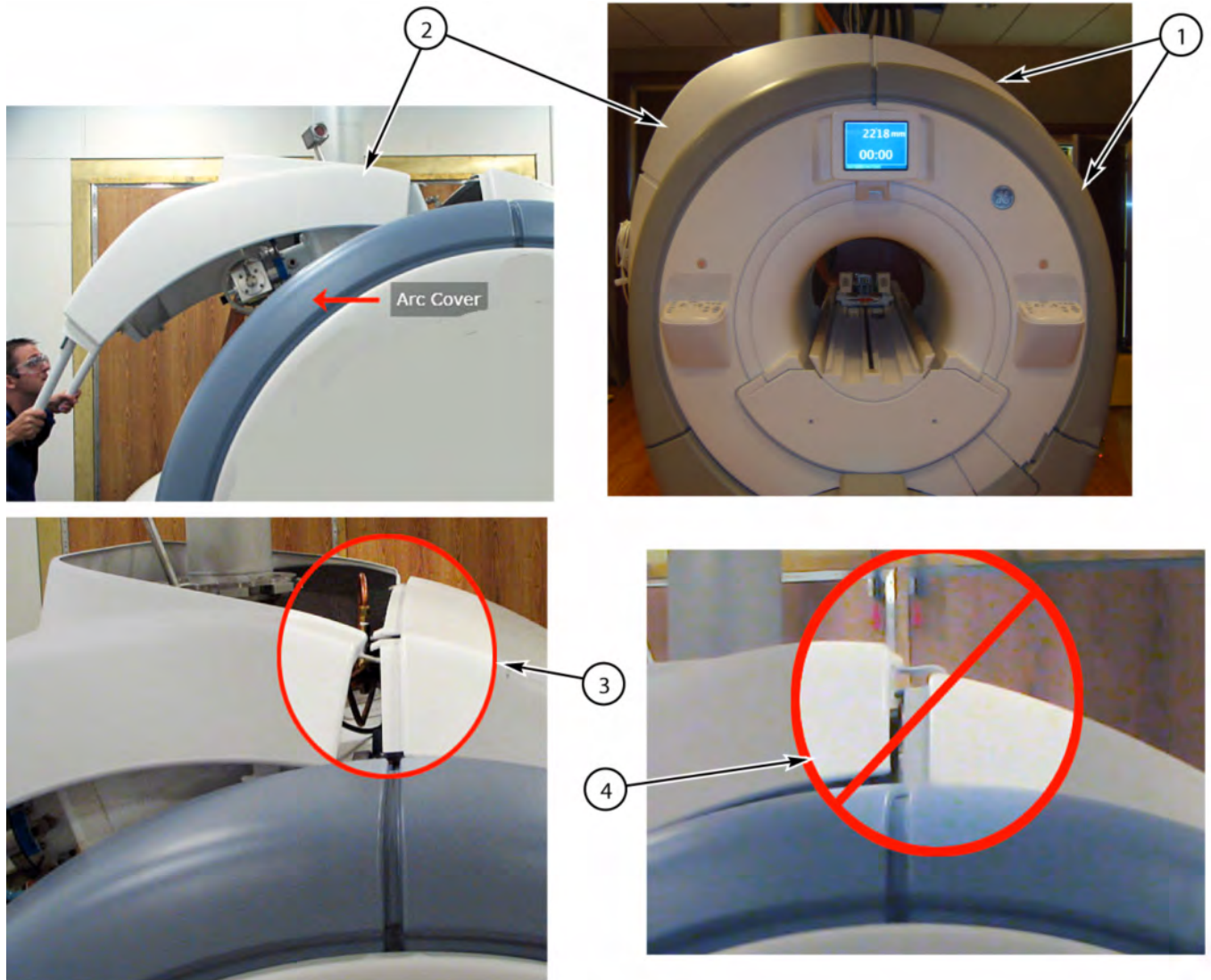
1. Position cover on magnet.
2. Attach using two socket head cap screws and washers.
3. Attach cover to popper.

NOTE: Rear trim covers have one popper, front skirt covers have two poppers.

4. Repeat above steps for remaining three covers.

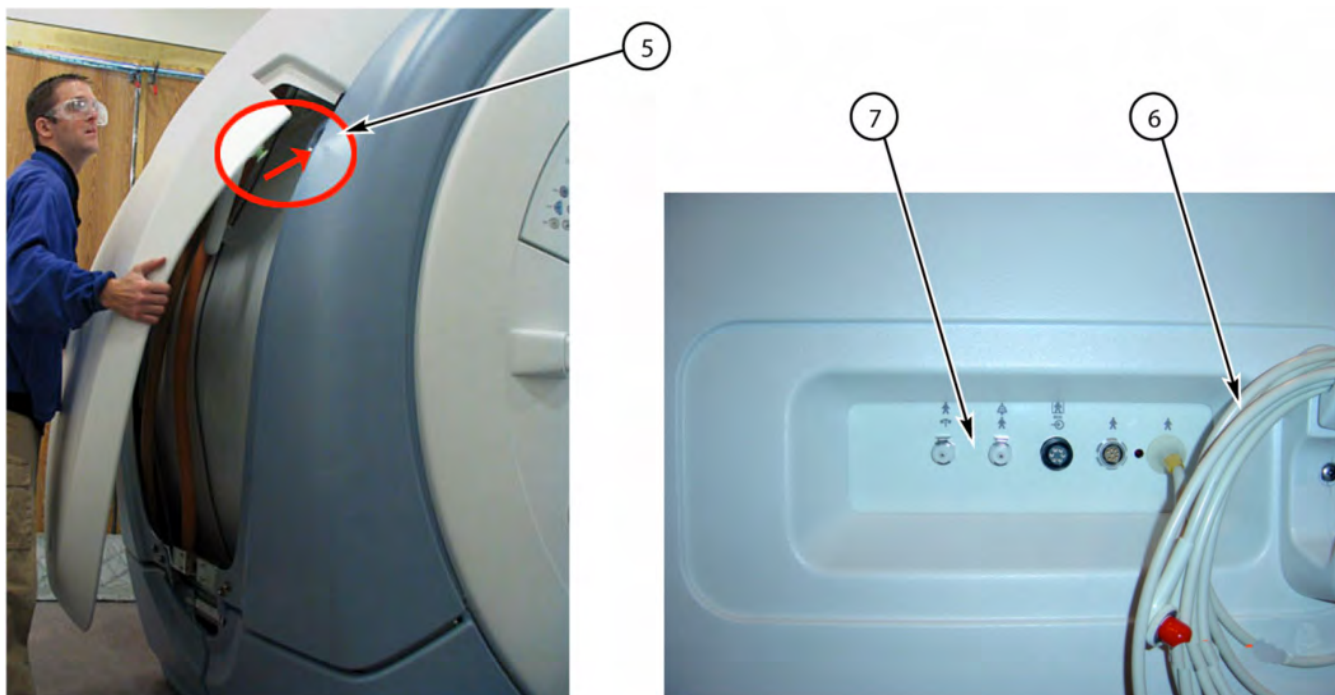
1.2 Install Side Covers

Illustration 5-2: Installation of Side Covers



1. Install the top right cover (5335653) and then the right side cover (5328957).
2. Hold the handles away from the enclosure when installing the top left side cover (5335654).
3. Make sure that the alignment tabs are properly slotted under the right side's slot.
4. Make sure the tab is properly placed.

Illustration 5-3: Installation of Left Side Cover

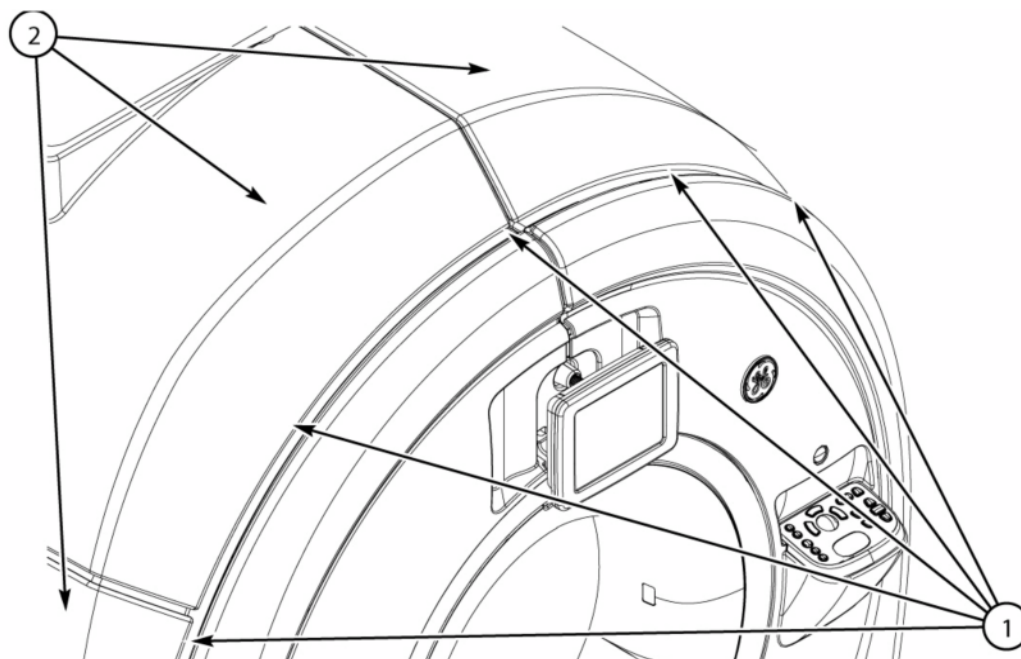


5. Hold the sides of the Lower Cover and align the round feet with the front and rear cover ovals. Route the PPG cable attached to PAC through the opening in the side cover. Snap in place.
6. Coil and place PPG cable on hook.
7. If connection points and symbols are not aligned properly, proceed to next sections for adjustment procedures.

1.3 Adjust Cover Fit

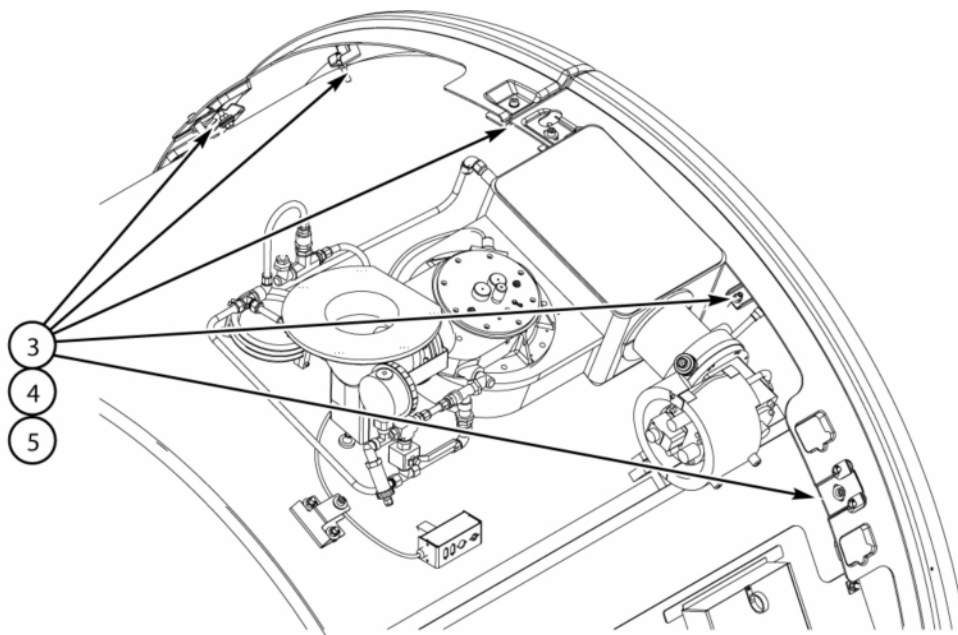
After installing the top and side covers, adjustment will most likely be required to adjust the gap spacing between the top & side covers and the front & rear covers.

Illustration 5-4: Check for Gaps



1. Observe and measure gaps at front and rear of enclosure between the covers. Only front view is shown in illustration.
2. Remove top and side covers.

Illustration 5-5: Adjustment Points



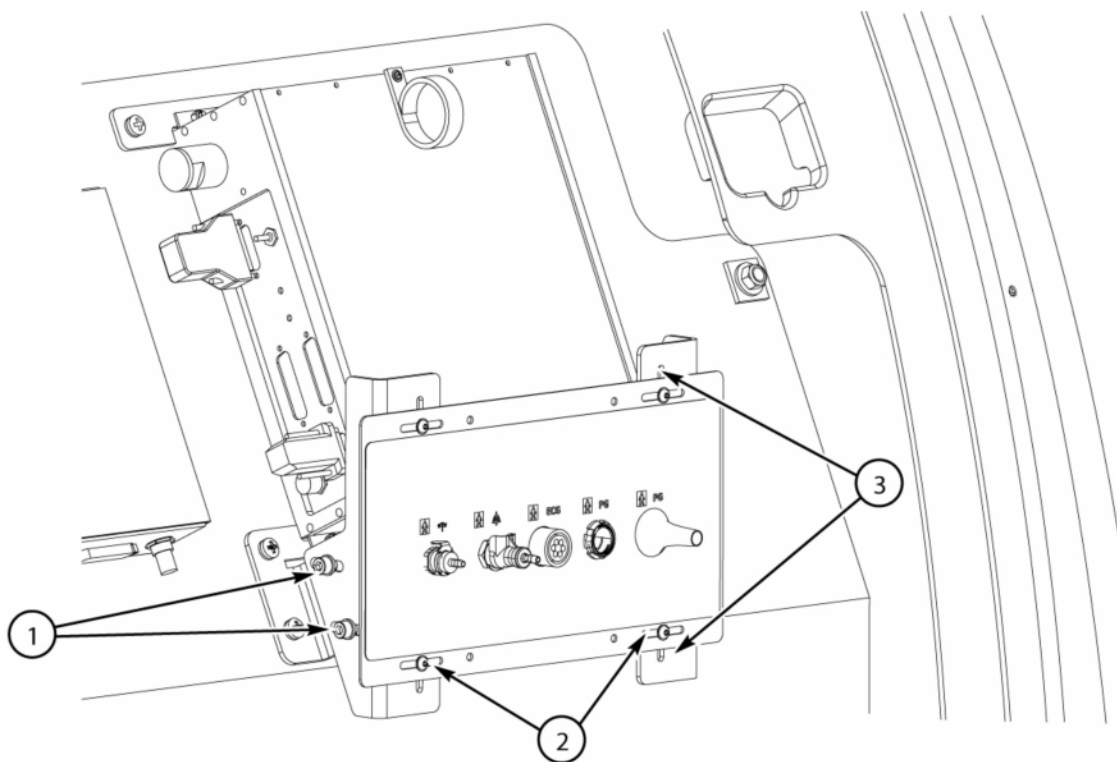
3. To adjust front and rear covers, turn the adjusting screws up or down to achieve proper height.
4. Install top and side covers again and check gap for uniformity.

5. Repeat above steps as necessary until uniform gap is achieved.

1.4 Adjust PAC Interface Panel

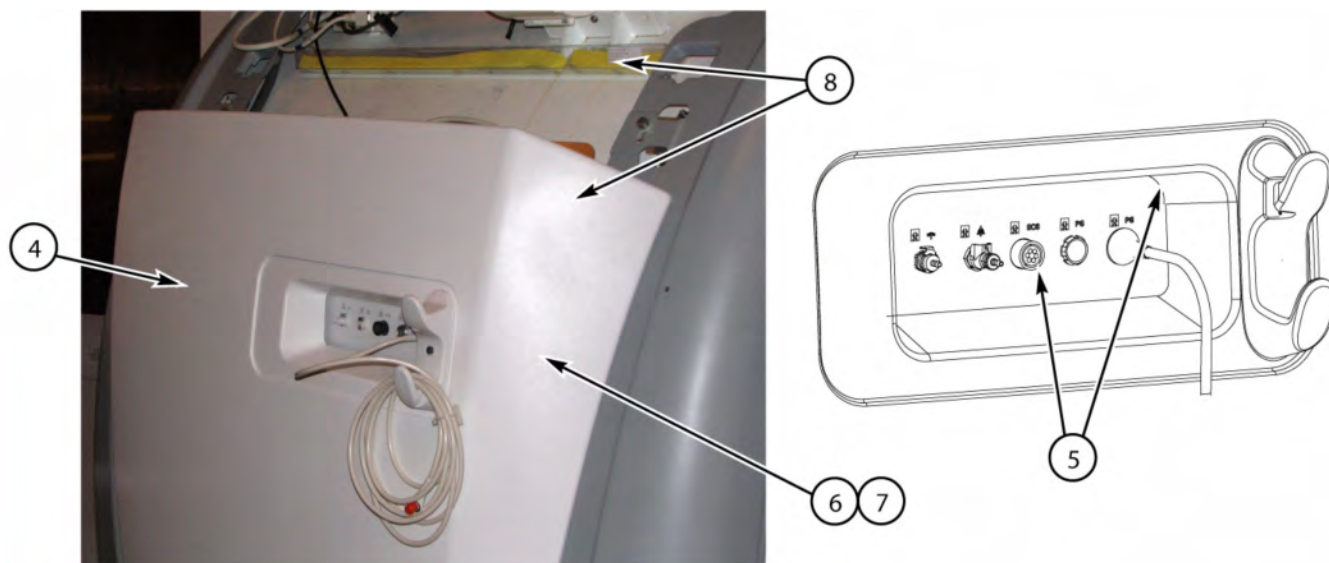
The PAC Interface panel can be adjusted in three directions. The five ports on the Interface Panel should be centered in the opening of the cover. It is suggested that the top cover be removed during this process. This will make the adjustments easier.

Illustration 5-6: Adjustment Locations



1. In/out direction
2. Side to side direction
3. Up/down direction

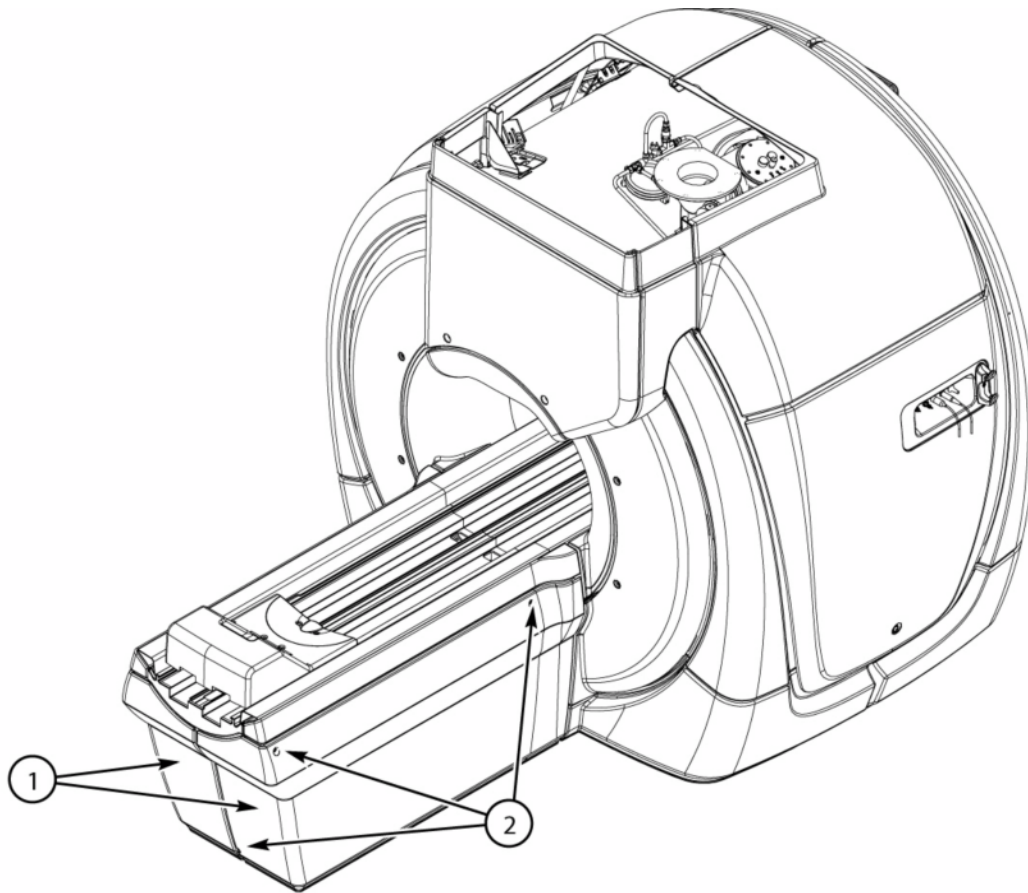
Illustration 5-7: Check Fit of Interface Panel



4. Remove top cover and reinstall side cover.
5. Check Interface panel for alignment gaps and center location in cover opening.
6. Remove side cover and adjust Interface Panel as necessary.
7. Repeat adjustment process until Interface Panel is centered and there are no gaps between side cover and Interface Panel.
8. Install top cover (not shown) and side cover.

1.5 Install Rear Pedestal Covers

Illustration 5-8: Installation of Rear Pedestal Covers

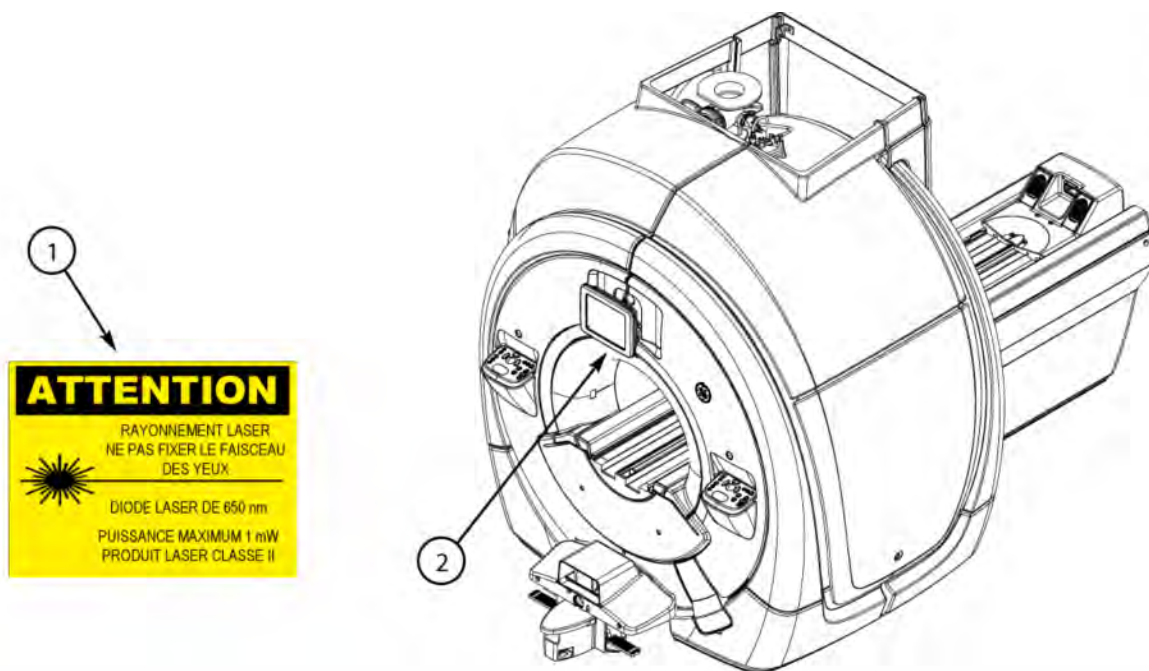


1. Move rear pedestal covers into place.
2. Attach each with three provided screws.

2 Install Laser Alignment Light Warning Label

The magnet enclosure is shipped with a label in English attached below the laser light.

Illustration 5-9: Warning Labels



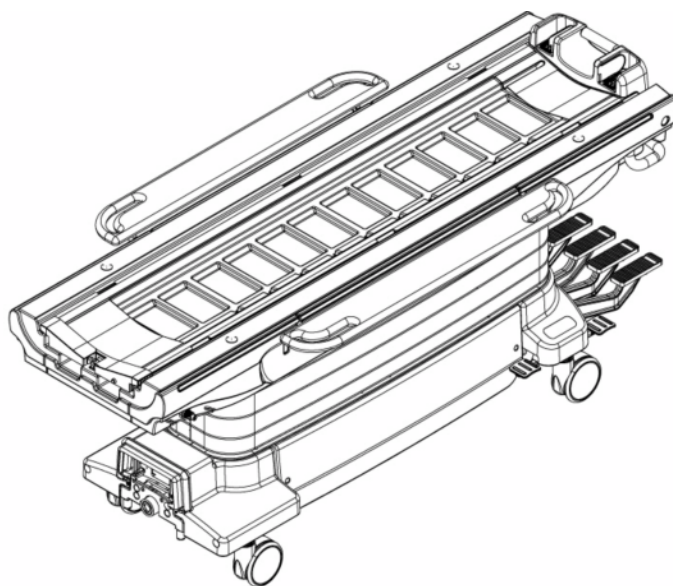
1. A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge. An example of one of the labels on the sheet is shown above.
2. For those sites which require a label other than English, peel off the appropriate label and affix over the English label at the laser light location.

3 Patient Transport Installation

Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

NOTE: If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

Illustration 5-10: Patient Transport

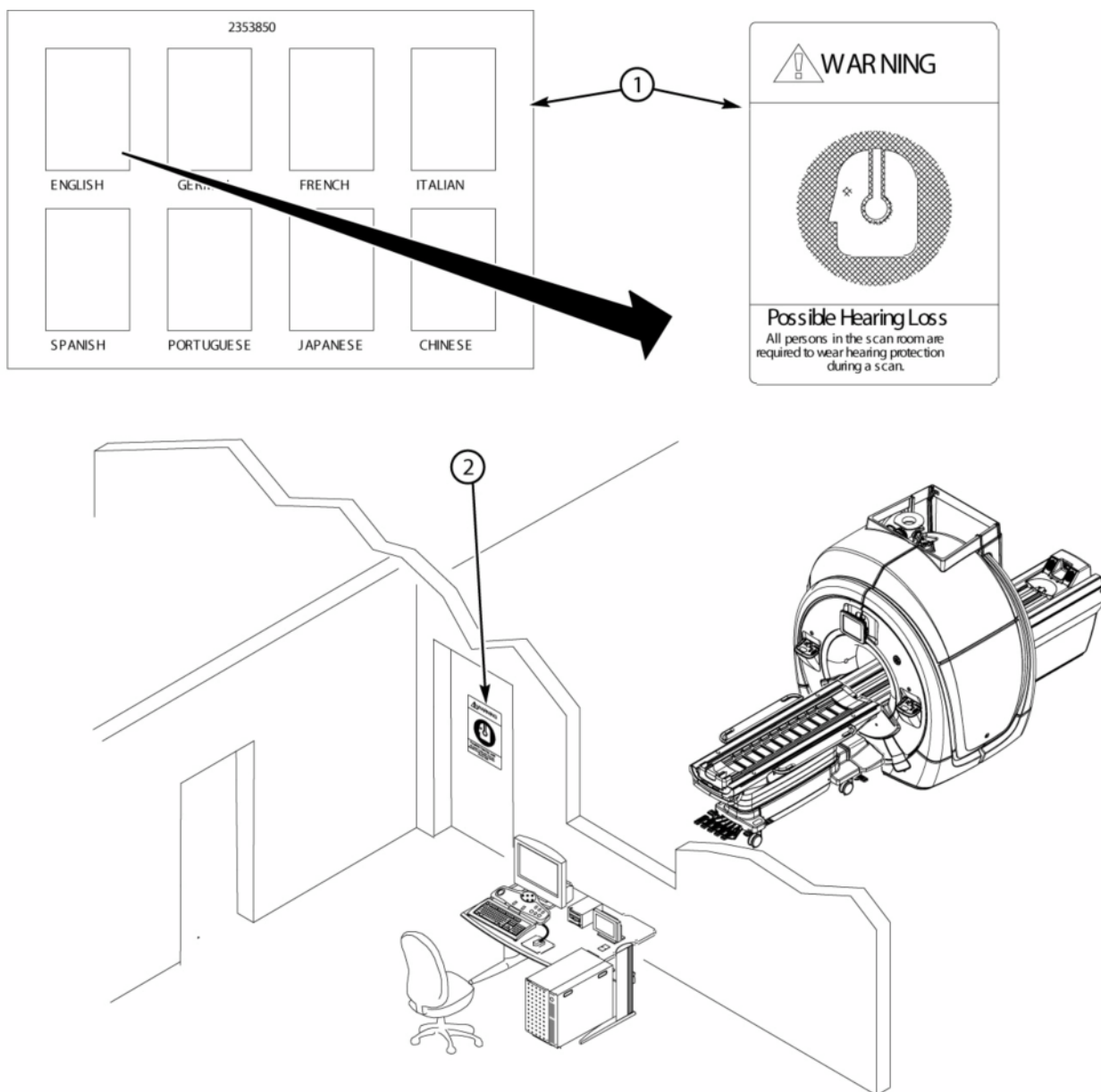


1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanical procedures in MR Service Methods DVD, Adjustments and Calibrations: Patient Table

4 Install Hearing Loss Warning Sign

1. Remove applicable language warning label from “HEARING PROTECTION REQUIRED” sheet of 8 labels (2353850).
2. Attach WARNING label to outside of magnet room door.

Illustration 5-11: Hearing Loss Warning Sign



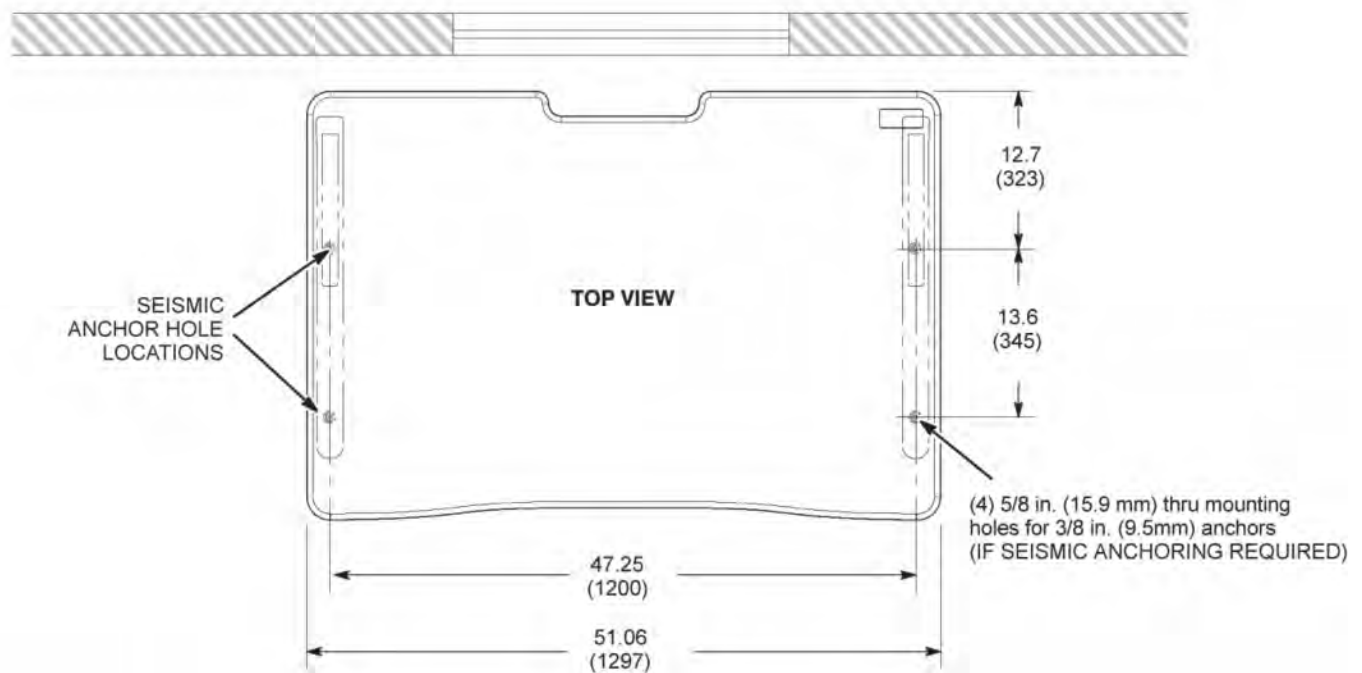
Chapter 6 Operator Workspace Installation

1 Operator Workspace

1.1 Operator Workspace Table Anchor Installation

The customer may have the Operator Workspace components installed on a customer designed countertop of other type of table. If the GE supplied Operator Workspace Table is to be installed, refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

Illustration 6-1: Position OW Table



1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions (5189137-50)* to assemble the model 5191031 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.



NOTICE

Two individuals are required to move and position Operator Workspace Table.

NOTE: The design of the new model 5191031 table does not require anti-tip anchors. This table meets the ANSA/BIFMA anti-tip standards.

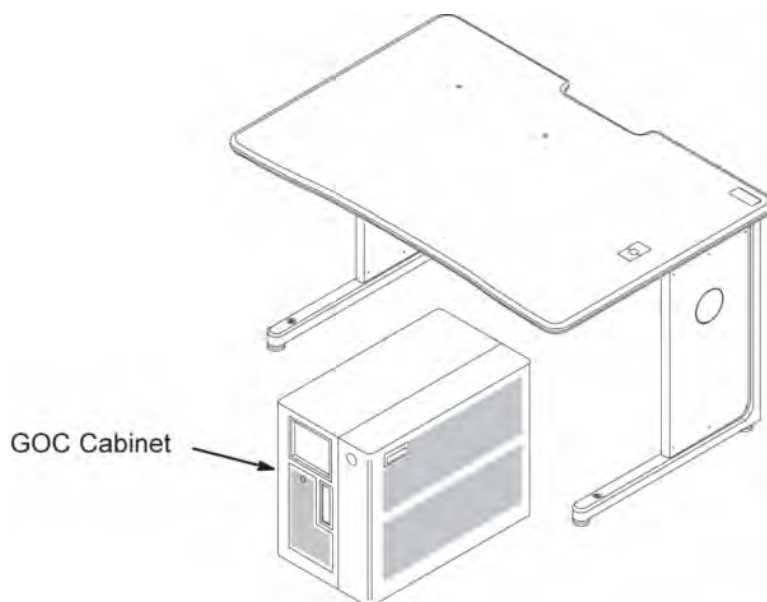
3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.

1.2 GOC Installation

1.2.1 Position GOC

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

Illustration 6-2: Position of GOC

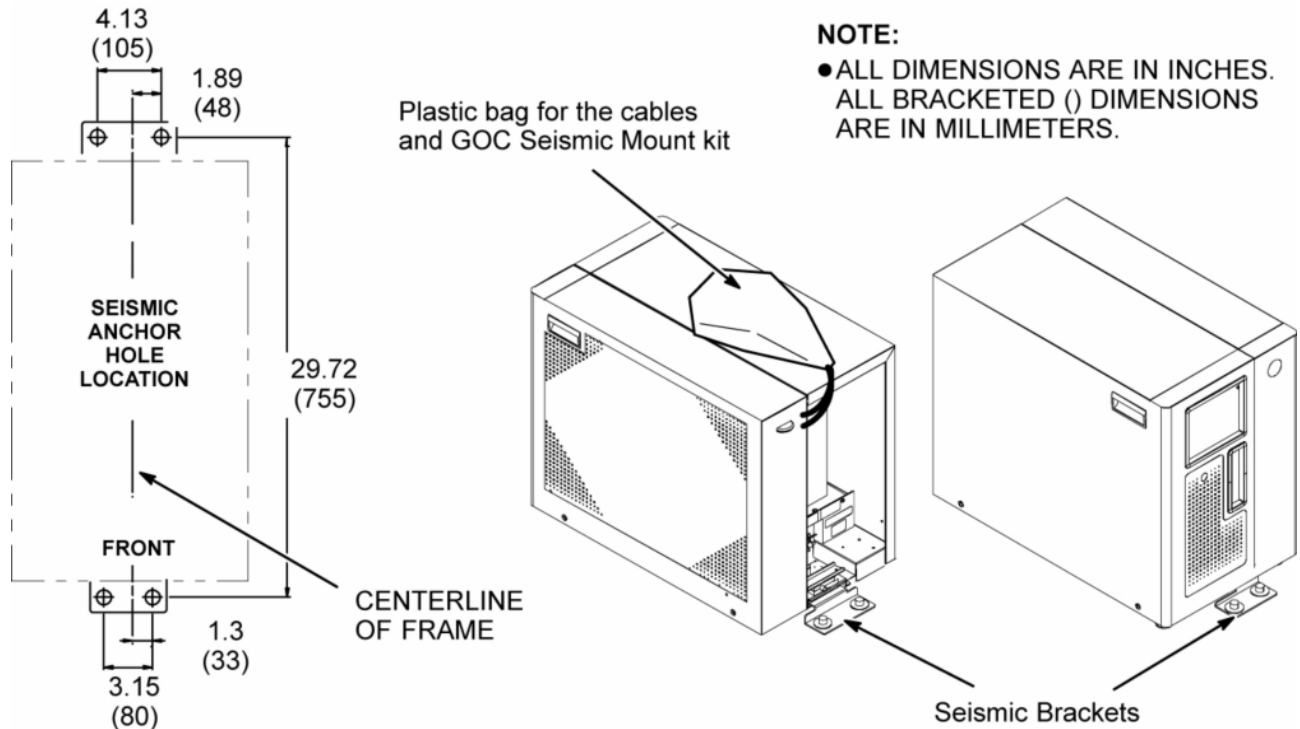


1.2.2 Seismic Anchoring (If Required)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Preinstallation manual. Illustration below indicates hole location for installing brackets.

Illustration 6-3: Install Seismic Brackets



1.3 Computer Cabinet Cable Routing and Connections

The GOC Cabinet is shipped with cables that will be routed and connected to components installed on the top of the OW Table.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (Option)

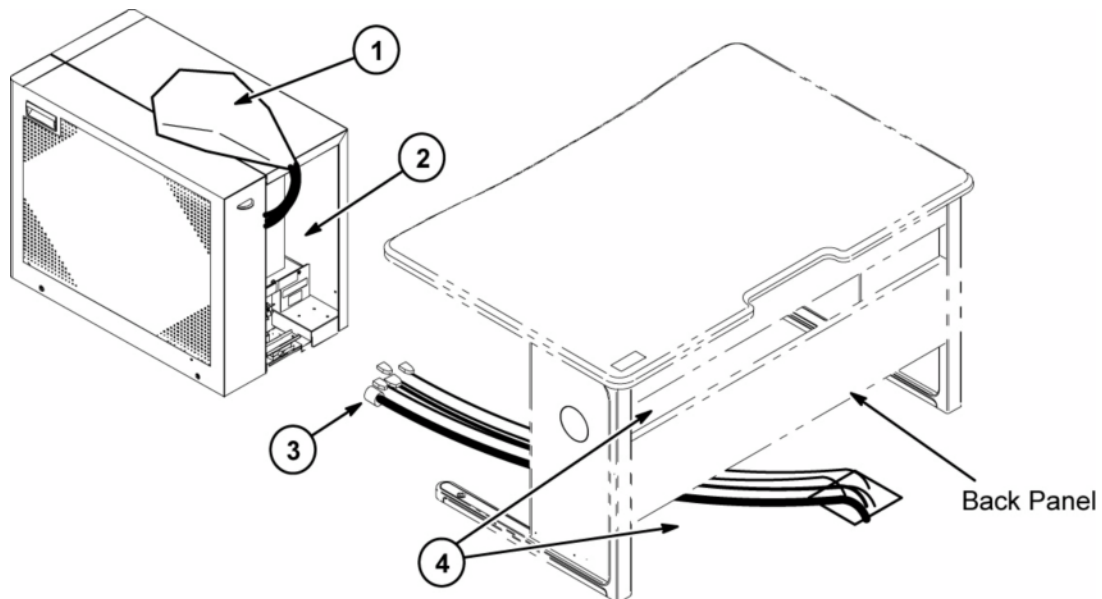
These cables are already connected to the computer as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will to be routed through the rear opening.



NOTICE

When routing cables at the customer site, confirm there are no unforeseen dangers where the cables are routed.

Illustration 6-4: Cable Connections to GOC



1. Runs E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening. (These cables are inside of plastic bag)
2. Runs E0003, E3011, E3015, E3017, E3018, the power cord for the Patient Alert system, and any other option cables will be routed through the rear opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.

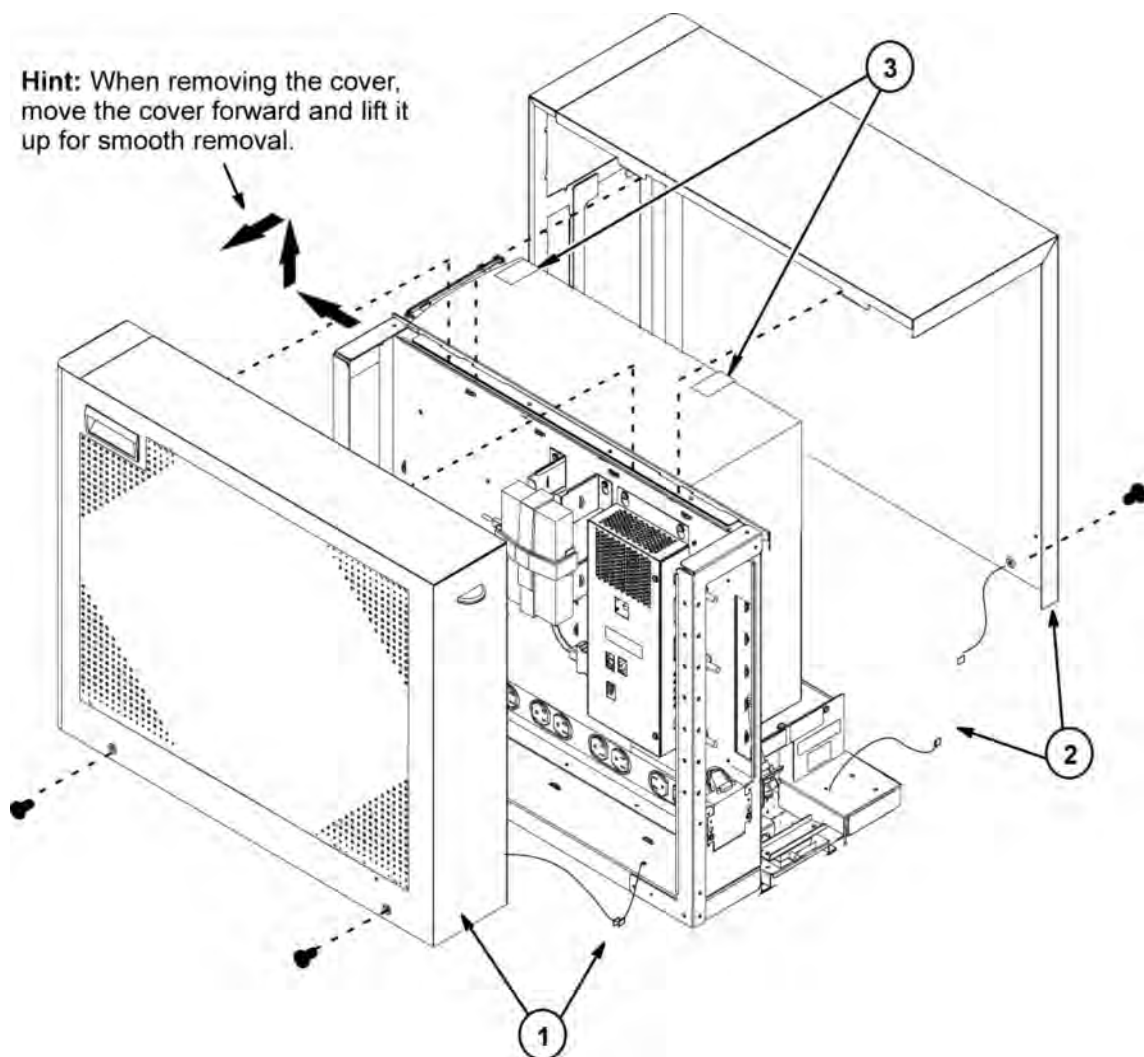
NOTE: Make sure a cable service loop of at least 4 feet (1300mm) is provided for future servicing

4. If GOC Cabinet is to be located under OW Table, the cable coming out of the opening at the rear of the cabinet will be routed below the Back Panel. Cables routed from components on top of table will be routed above back panel.
5. Instructions for connection of cables to GOC are provided in later procedures.

1.4 GOC Cabinet Cover Removal

The Computer Cabinet covers need to be removed to access the cable connection points.

Illustration 6-5: Remove GOC Covers



1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

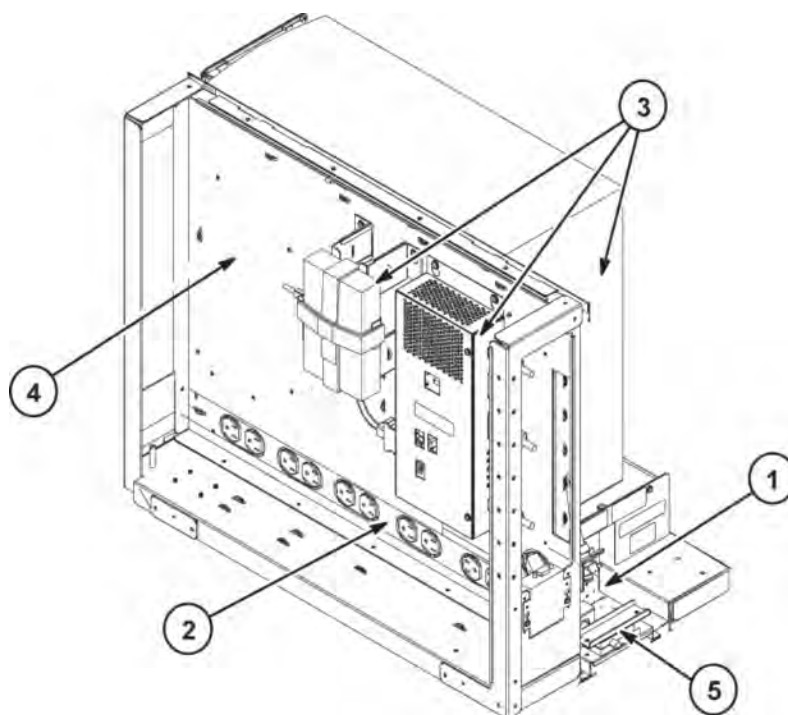
NOTE: A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

1.5 GOC Cable Connections

The connection points of cables routed to the GOC modules and the PC are described in the following sections. Refer to cable map for connection information.

Illustration 6-6: GOC Cable Connections

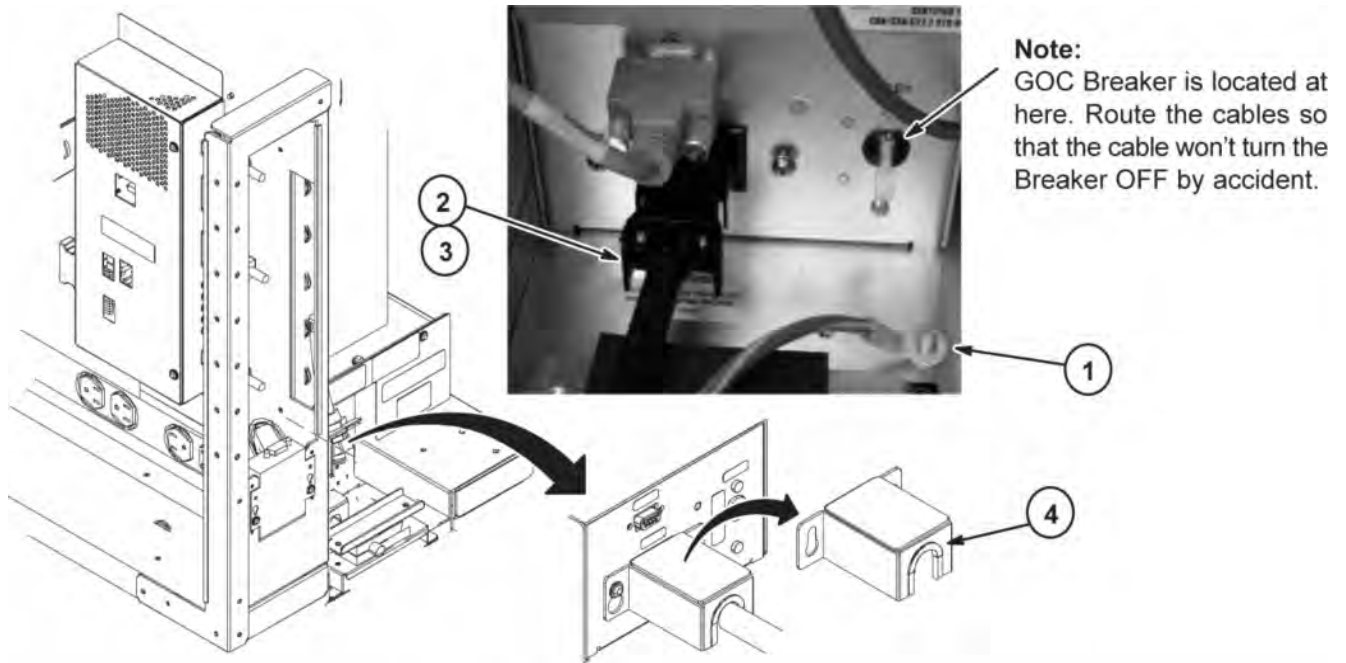


1. Refer to [Section 1.5.1](#) for system power and ground cable connections to GOC.
2. Refer to [Section 1.5.2](#) for OW component power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
3. Refer to [Section 1.5.3](#) for all cables connected to Linux PC computer and GOC modules..
4. If In Room Display (IRD) is installed for this site, refer to [Section 1.5.4](#) for IRD component installation procedures in GOC.
5. After all cables are installed, route cables through clamp. Tighten clamp to secure cables in place.

NOTE: GOC Breaker is located at here. Route the cables so that the cable won't turn the Breaker OFF by accident.

1.5.1 Power and Ground Connections

Illustration 6-7: GOC Power and Ground Cables



1. Connect Ground wire E4008 to GND stud.

NOTE: Make sure nut is firmly tightened to GND stud.

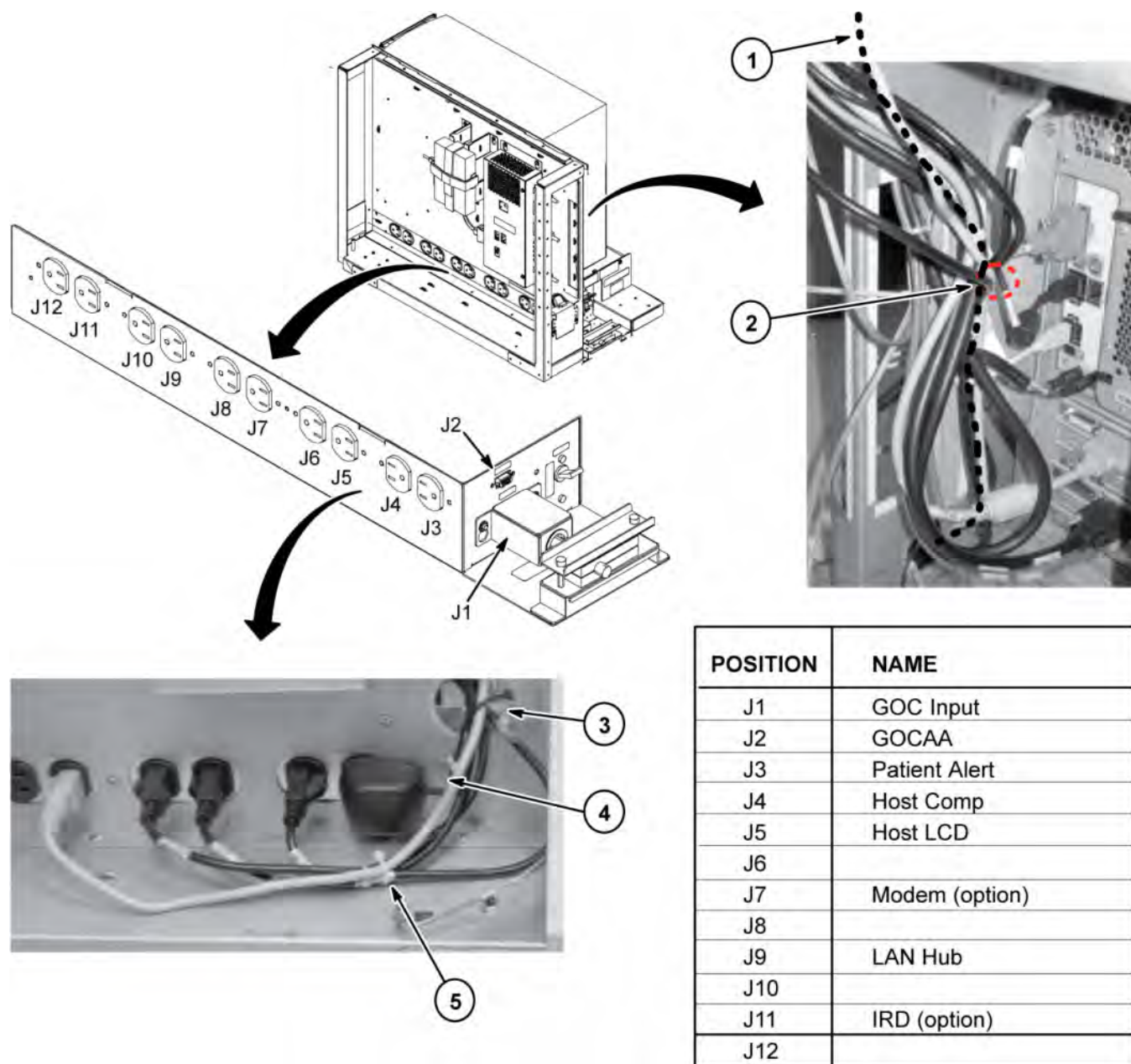
2. Before connecting GOC power cable E0003 to J1, loosen screws and remove cover.
3. Route and connect GOC power cable to J1 on PDU.
4. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover and tighten screws.

1.5.2 OW Component Power Cable Connections to PDM

The illustration shows the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable connections to OW1-PDM.

Illustration 6-8: Connect GOC Power Cables

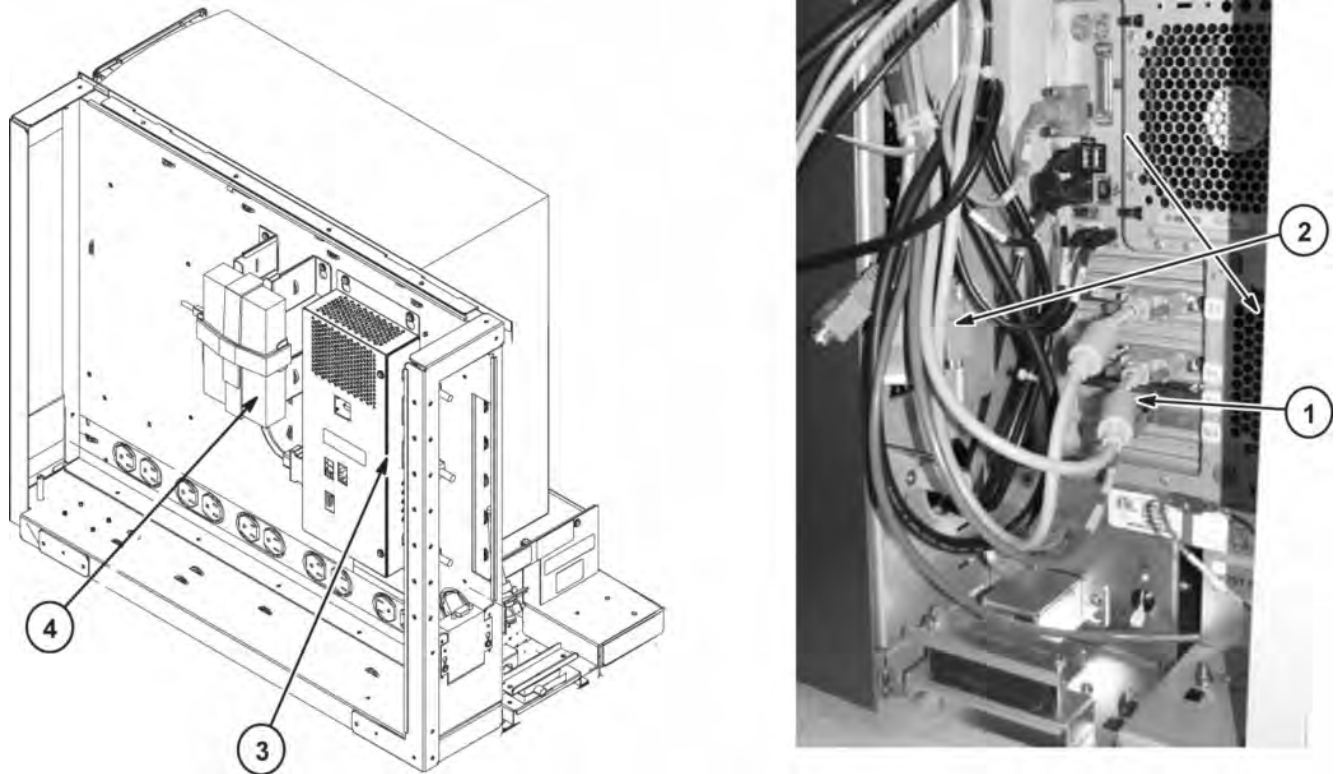


1. If Modem is installed, route power cable and connect to J7.
2. Tie-wrap point for Modem power cable.
3. Route power cables connected to J3 thru J7 through access hole.
4. Route and connect Patient Alert power cable to J3 and tie-wrap at this point.
5. Tie-wrap power cables J4 to J7 at this point.

1.5.3 Data Cable Connections

Many of the cables shown in illustration are pre-connected. Instructions for routing and connecting cables for USB mouse, keyboard, host display, and SCIM are provided in [Section 1.9](#).

Illustration 6-9: Cable Connections



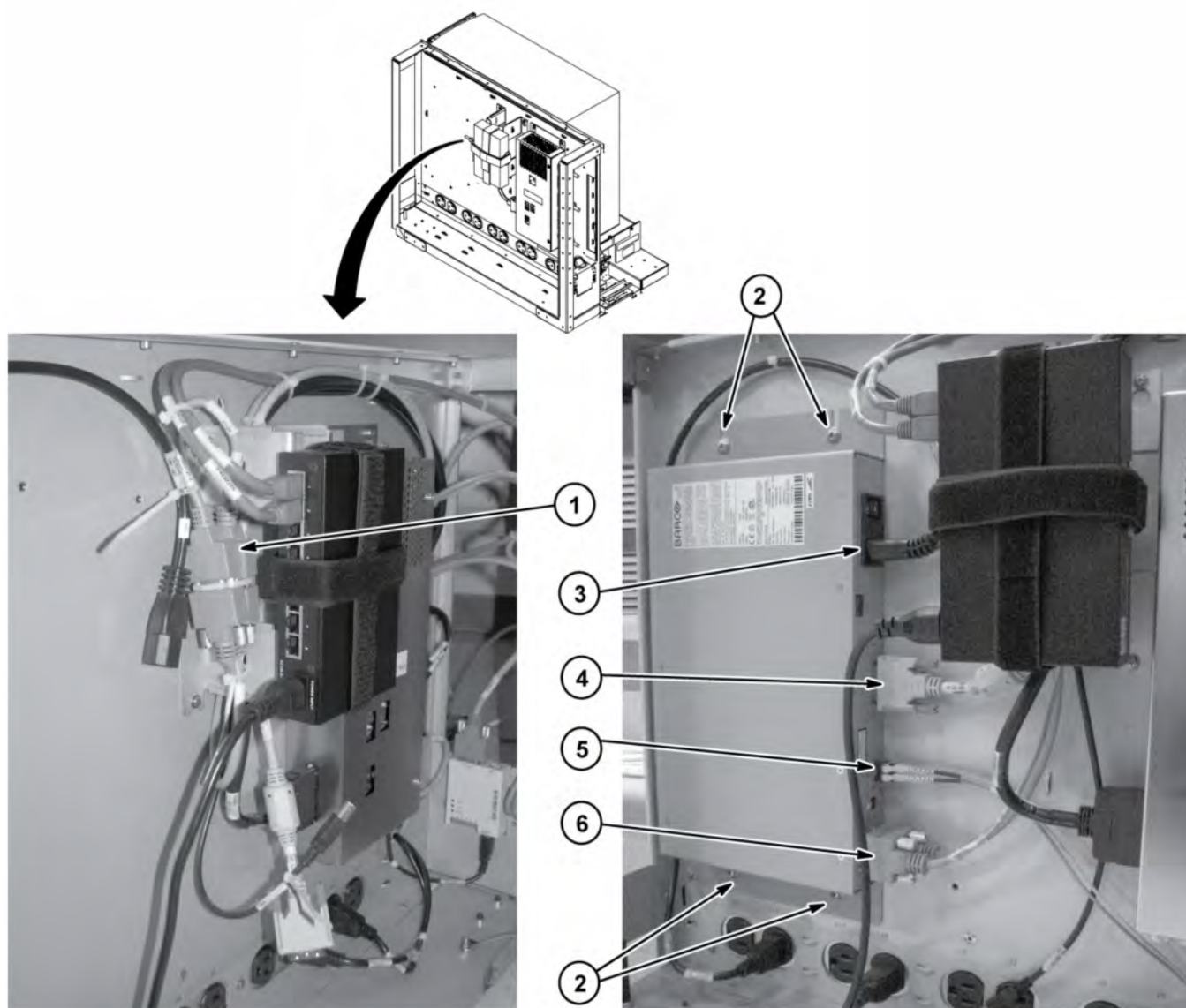
1. Connect Run E3011 to ethernet connection Port B
2. Connect Runs E3015, E3017, and E3018 to FJ6, FJ5, and FJ7 as marked.
3. Connect Run E3031 to J7 on GOC AA.
4. Use ports 3, 7 or 8 for connecting site ethernet cables such as AW.

NOTE: Do not use any other ports for site ethernet cables.

1.5.4 In Room Display (IRD) Installation (Option)

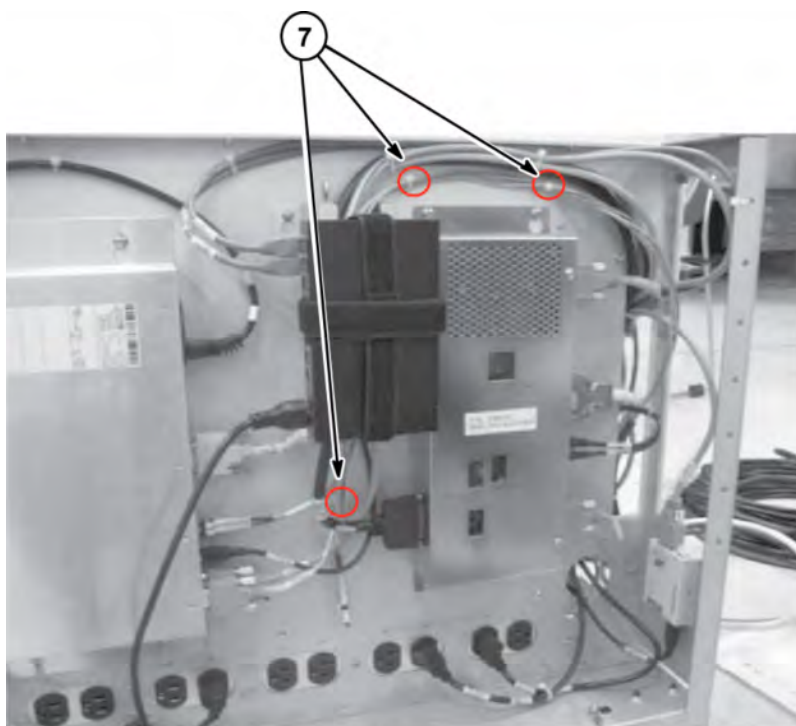
Complete this procedure if IRD is part of system installation. The GOC is delivered with cables pre-installed for connection to the Fiber Optic Converter Box.

Illustration 6-10: Install Fiber Optic Converter Box

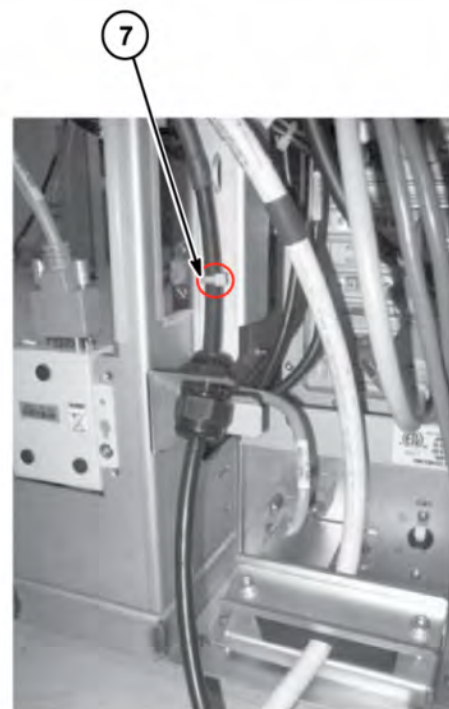


1. Disconnect and discard jumper cable 5267721-2 connected to E3019 and Sensor cable.
2. Install Fiber Optic Converter Box (5133768-2) using the provided four M5 screws
3. Connect the USB cable from the Host PC to J1.
4. Connect the DVI-D cable from the Host PC to J2.
5. Route fiber optic cable Run P2026 to GOC and connect to J3.
6. Connect Run E3019 to J4 and the Sensor cable to J5.

Illustration 6-11: Secure Fiber Optic Cable



GOC side view



GOC rear view

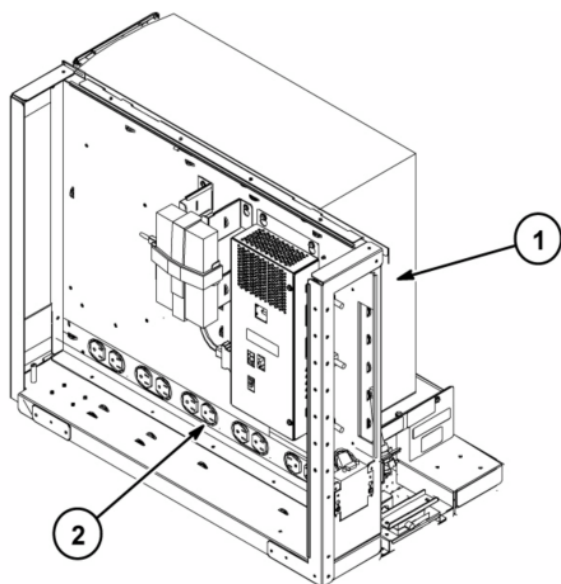
7. Secure the fiber optic cable with tie wraps at points indicated below. Tie the tie wrap loosely as to not damage the optical cable.

NOTE: Do NOT tighten the tie wraps firmly. Damage to the optical cable will occur if tie wrap is too tight.

1.6 Install Modem Cable to GOC Cabinet (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC

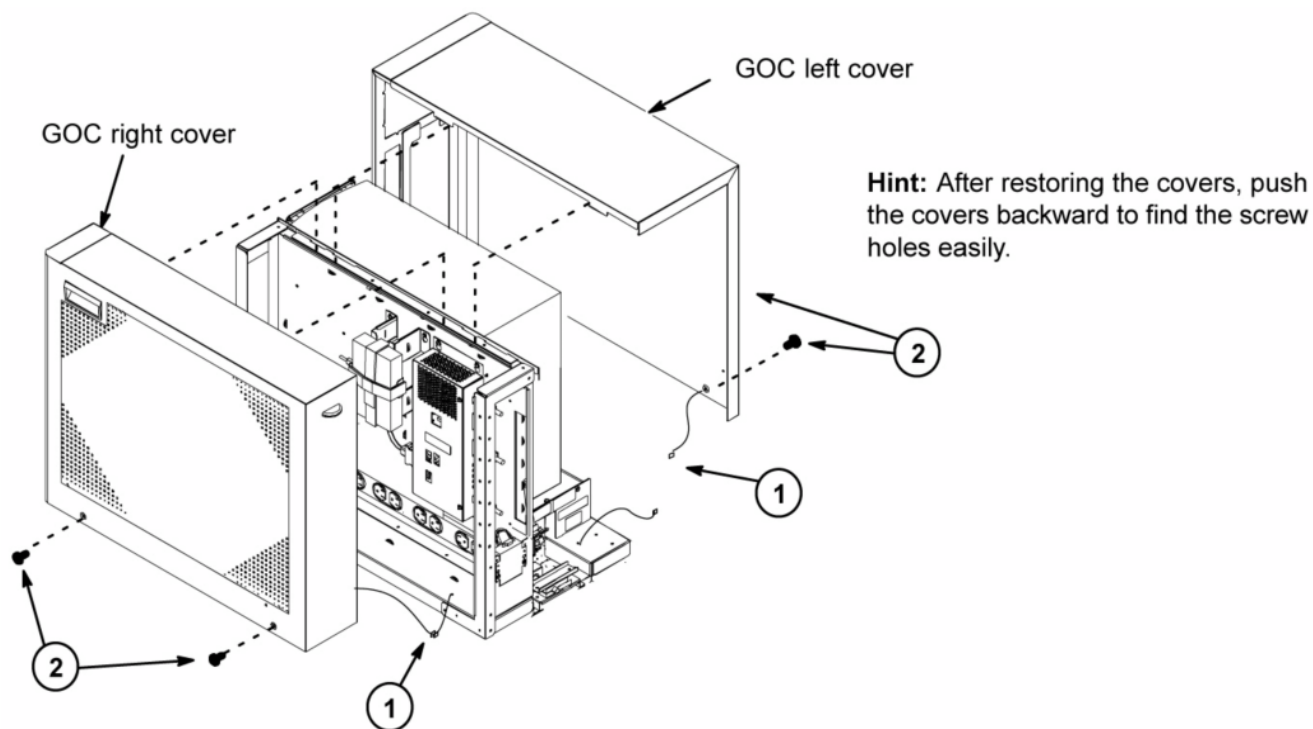
Illustration 6-12: Modem Installation



1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.

1.7 Restore GOC Covers

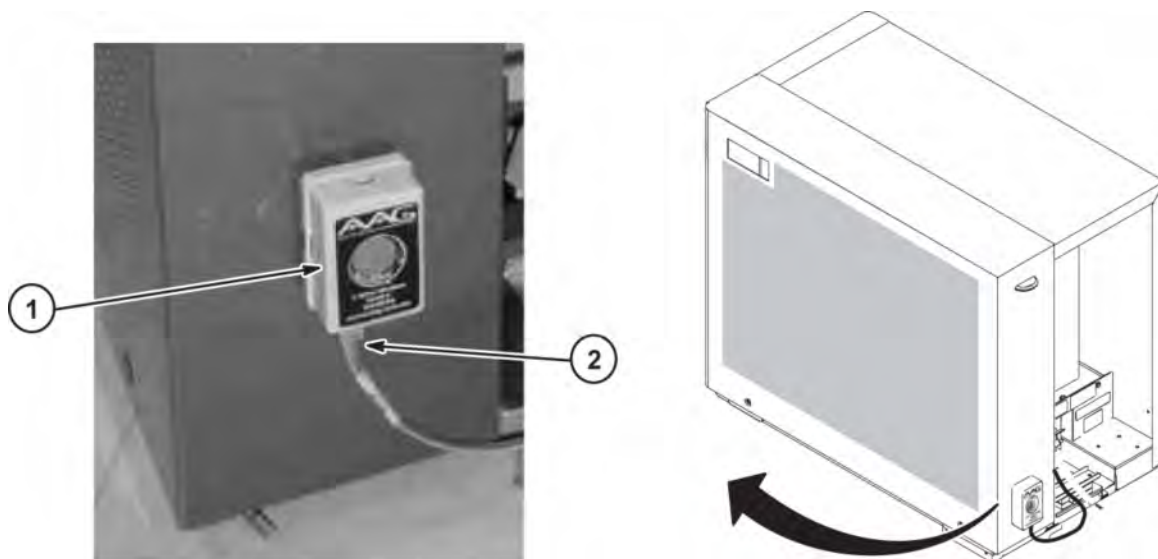
Illustration 6-13: Install GOC Covers



1. Connect Ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure.

1.8 Install Temperature & Humidity Sensor on GOC

Illustration 6-14: Install Sensor on GOC

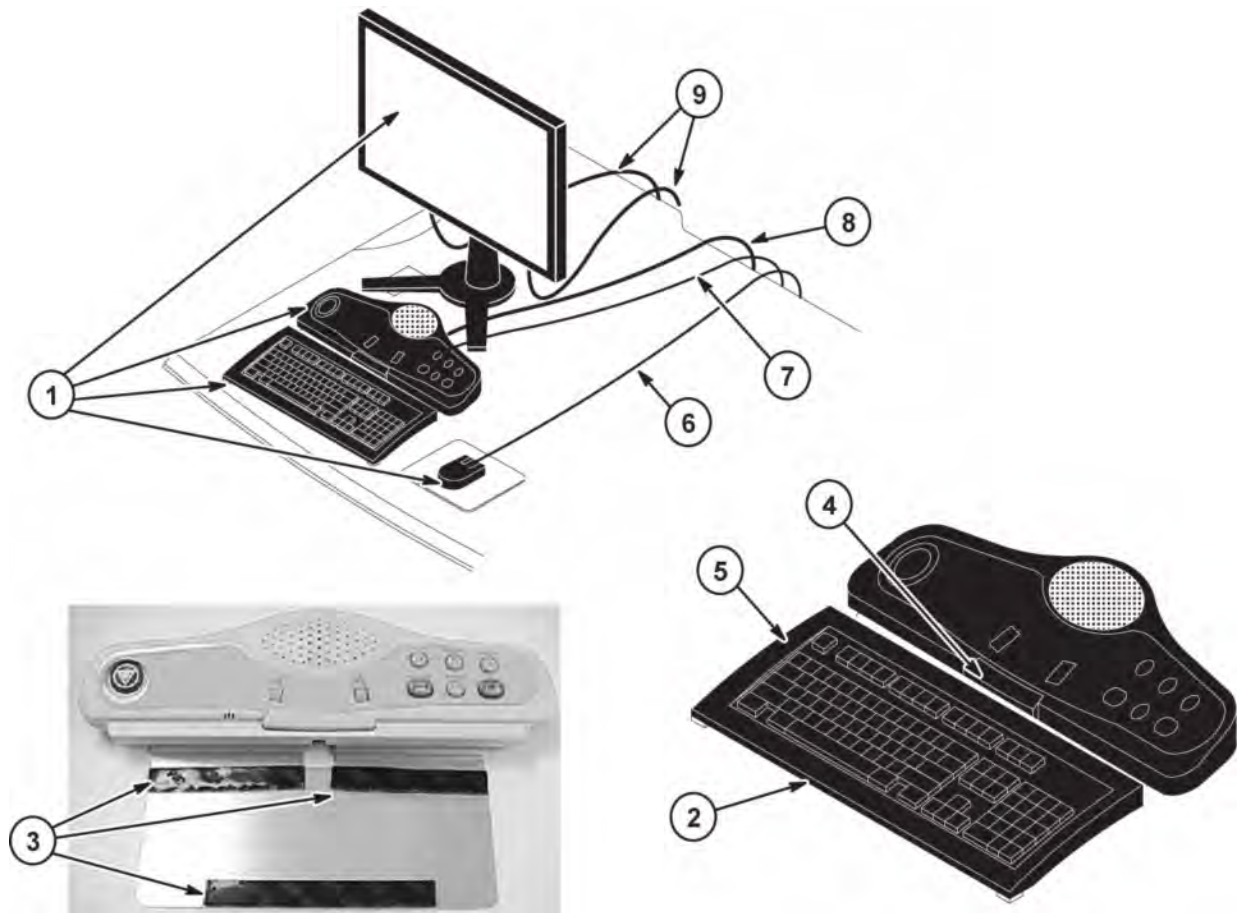


1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable (5267721) to Sensor.

1.9 Install Monitor and SCIM & Keyboard

Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

Illustration 6-15: Monitor, SCIM and Keyboard Installation



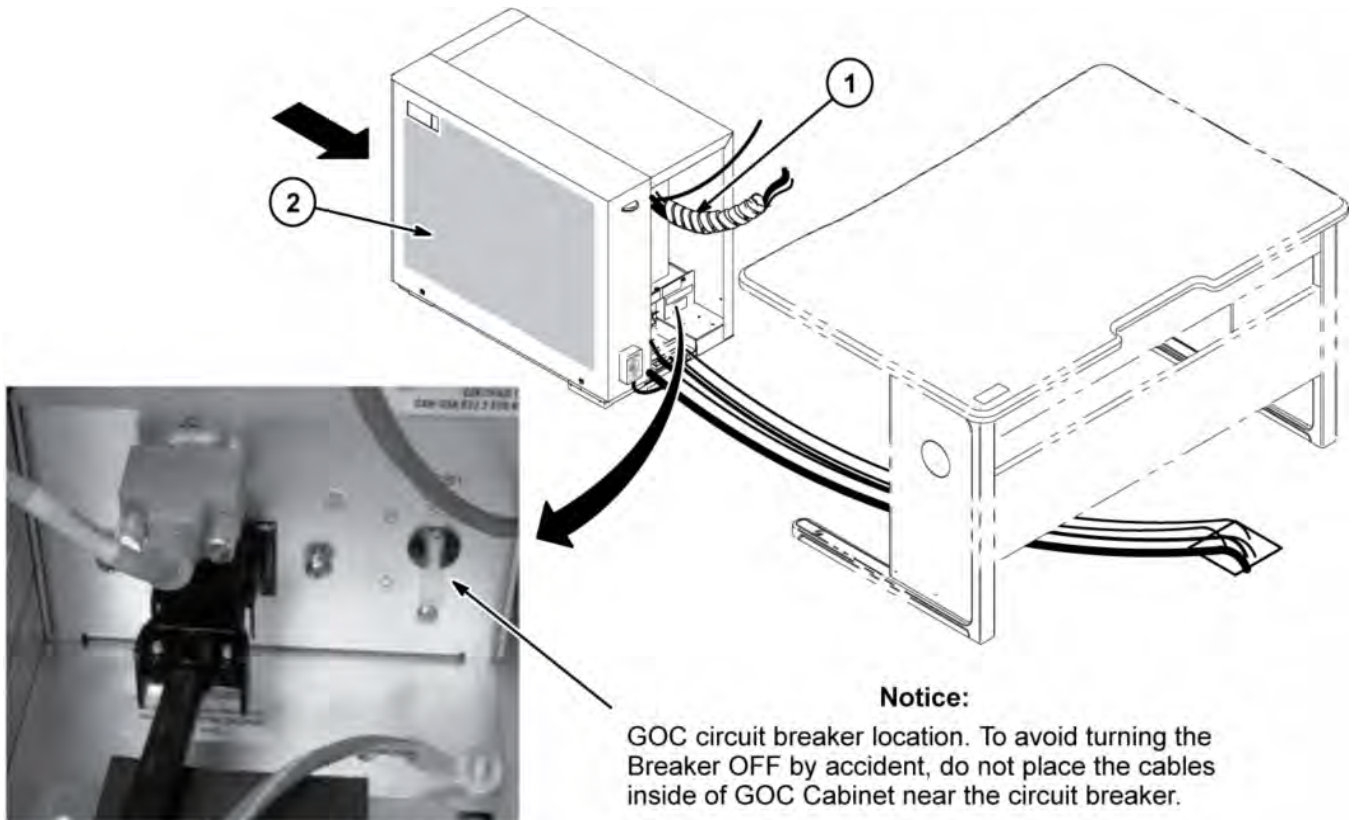
1. Position Host Display Monitor, SCIM, Mouse, and keyboard on table.
2. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
3. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).
4. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
5. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.
6. Connect Mouse cable to USB port at rear of computer in GOC.
7. Connect Run E3048 to USB Keyboard cable. Route and connect other end to computer in GOC.
8. Connect Run E3047 to SCIM. Route and connect other end to computer in GOC.
9. Route power cord Run E0015 and Run E3046 from GOC and connect to Display Monitor.

NOTE: The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

1.10 Final Positioning of GOC Cabinet

1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m) Spiral tube is included in the plastic bag which is shipped with GOC.

Illustration 6-16: Complete Cabling of GOC



- Two power cords
- One video cable
- Keyboard cable
- SCIM cable

2. Place cabinet in final position as determined by customer.

1.11 Power On Information

When powering up the GOC, make sure the Main breaker located in the rear of the GOC is ON.

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Chapter 7 Checklist

1 Completion Checklist

1.1 Mechanical Installation Completion Checklist

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel.

Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map
- ☐ Correct for wiring errors if necessary
- ☐ Recheck all coolant hose connections for tightness and that there are no leaks. Correct if necessary.
- ☐ Recheck all air hose connections for tightness and that there are no leaks. Correct if necessary.

1.2 Returning Shipping Material

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the "Return pile", Recycling Operation will route them to appropriate location.

1.3 Final Mechanical Installation Steps

1. Complete assembly of PPG cable and probe head.
2. Make sure all Cabinet, Magnet Enclosure, and Rear Pedestal covers have been installed.
3. Make sure all safety and warning labels have been installed.
4. Make sure Patient Transport is in position at front of Magnet.

2 GE Field Engineer Responsibilities

2.1 Phantoms and Service Kits

1. Locate Phantom Kit and SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.



NOTICE

Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

4. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
 - a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
 - b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
5. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.

2.2 Final GE Field Engineer Installation Steps

1. Record and enter applicable data into applicable site configuration files and records.
 2. Complete Product Locator information for all installed serialized components, new or updated, via either:
 - a. **(U.S. Only)** FE Site Verification Web Site
 - b. Process and return product locator installation cards for all serialized components to:
Product Locator File, P.O. Box 414, W-523, Milwaukee, WI 53201-0414
Refer to Product Locator section in “Getting Started” for details on submitting Product Locator information.
- NOTE:** Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.
3. Store the delivered site's set of service tools in service cabinet at site.
 4. Locate surface coil (“quick disconnect”) adapters and any optional surface coils and positioning accessories and place on Patient Transport.
 5. Locate Box “To be opened by Applications Specialist” and set aside for later visit by Applications Specialist.
 6. Leave the site's set of Manuals and CD-ROM at site. Install Service and Operator documentation on the host computer. Organize and set up reference cabinet for the Manuals.

7. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
8. Verify that coordination of application support for instruction of site users on the appropriate Software Release has been made before returning system to the users. Turn site over to applications who will instruct users.

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Chapter 8 Appendix

1 Cable Specification

Cables delivered to site for installation have specifications dependent on usage. The following tables contain information on cable runs for this system as it was originally installed.

Table provides definitions for the installed cables listed in Table

Table 8-1: Cable Table Definitions

Column Heading	Description	Special Notes
Description	Includes the Run number, destination labels of the cable, and length	Total length of cables delivered to site, function of cable specifies power from data
Number of Conductors	Wires in cable	
Certification	Type designation for the cable	
Voltage Rating	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given
Temperature Rating	Maximum temperature insulating material may be used without loss to basic properties	Specific to the core cable manufacturer given
Notes	Describe the cable's use in the MR system	

Table 8-2: MR450w Cables Installed on Site

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run 623, CRY to SPW, COLD HEAD CONTROL,LCC, 15000mm	4	16	CM, FT4	300	90°	
Run 624, SPW to MAG, COLD HEAD CONTROL,LCC, 15000mm	4	16	CM, FT4	300	90°	
Run E0003, PGR-J102 to OW-PDM-J1, Host Power, 30000mm	3	10	UL AWM 2587, FT4	600	90°	
Run E0004 PGR-J16 to Pen-J1, Pen Power, 17000mm	5	10	UL AWM 2587, FT4	600	90°	
Run E0004, PGR-J16 to PEN-J1, PEN Power, 10000mm	5	10	UL AWM 2587, FT4	600	90°	
Run E0009, HEC to CRY, INPUT POWER, SUMITOMO CSW-71D, 10000mm	4	12	LFNC	600	105°	Four 12AWG wires in conduit
Run E1001, PGR-J19 to SPW-J106, 1.5 RF Transmit, 16523mm ref. Aluminum braid shield.	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable Bend Radius -9 inch (229mm)
Run E1001, PGR-J19 to SPW-J106, 1.5T RF Transmit, 10327mm ref. Aluminum braid shield.	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable Bend Radius -9 inch (229mm)
Run E1002 SPW-J128 to PGR-J18, RF Transmit, 10218mm, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable Bend Radius -6 inch (152mm)
Run E1002 SPW-J128 to PGR-J18, RF Transmit, 16348mm, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable Bend Radius -6 inch (152mm)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E1003 PGR-J12 to PEN-J5, Body TR, 17000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable Bend Radius -2.5 inch (64mm)
Run E1003, PGR-J12 to PEN-J5, Body TR, 10000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable Bend Radius -2.5 inch (64mm)
Run E1004 PGR-J13 to Pen-J6, Head TR, 10000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable Bend Radius -2.5 inch (64mm)
Run E1004, PGR-J13 to PEN-J6, Head TR, 17000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable Bend Radius -2.5 inch (64mm)
Run E1005, PGR-J11 to PEN-J15, Tx Output, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable Bend Radius -2 inch (51mm)
Run E1005, PGR-J11 to PEN-J15, Tx Output, 17000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable Bend Radius -2 inch (51mm)
Run E2002 PGR-J1 to PEN-J19, 10000mm	2	N/A	OFNR	N/A	85°	optical fibers, 2 simplex connectors bundled on each end
Run E2002 PGR-J1 to PEN-J19, 17000mm	2	N/A	OFNR	N/A	85°	optical fibers, 2 simplex connectors bundled on each end
Run E2020 PGR-J22 to PEN-J17, 10000mm	6	N/A	LFNC	N/A	85°	optical fibers in conduit, 3 connectors bundled on each end
Run E2020 PGR-J22 to PEN-J17, 17000mm	6	N/A	LFNC	N/A	85°	optical fibers in conduit, 3 connectors bundled on each end
Run E2023 PGR-J21 to PEN-J18, 10000mm	2	N/A	OFNR	N/A	85°	optical fibers, 2 simplex connectors bundled on each end
Run E2023 PGR-J21 to PEN-J18, 17000mm	2	N/A	OFNR	N/A	85°	optical fibers, 2 simplex connectors bundled on each end
Run E3002 PGR-J103 to Pen-J11, 17000 mm	15	22	CL3, FT4	300	80°	
Run E3002, PGR-J103 to PEN-J11, E-Stop, 10000mm	15	22	CL3, FT4	300	80°	
Run E3006 PGR-J5 to Pen-J14, 17000 mm	9	22	CL3, FT4	300	90°	
Run E3006, PGR-J5 to PEN-J14, HART, 10000mm	9	22	CL3, FT4	300	90°	
Run E3008, PGR-J4 to PEN-J9, CAN, 10000mm	9	22	CL3, FT4	300	80°	
Run E3008, PGR-J9 to PEN-J9, CAN, 17000mm	9	22	CL3, FT4	300	80°	
Run E3009 PGR-J8 to Pen-J3, 10000 mm	37	22	CL3, FT4	300	80°	
Run E3009 PGR-J8 to Pen-J3, 17000 mm	37	22	CL3, FT4	300	80°	
Run E3011, PGR-J2 to OW1-Host, Ethernet, 30000mm	8	26	CMP	300	75°	
Run E3013, PGR-J6 to HEC-CAT 5, Cooling Control Ethernet, 10000mm	8	26	CMP	300	75°	
Run E3014, PGR-J20 to RF Door SW, E-Stop, 25000mm	2	22	CMG, FT4	300	105°	
Run E3015, PGR-J17 to OW-FJ6, E-stop, 30000mm	9	22	CL3, FT4	300	90°	
Run E3017, PEN-J10 to OW-FJ5, E-stop, 30000mm	9	22	CL3, FT4	300	90°	

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E3018, PEN-J12 to OW-FJ7, HART-RS422, 30000mm	15	22	CL3, FT4	300	80°	
Run E3020 Pen-J13 to Mon-J10, 17000mm	15	22	CL3, FT4	300	80°	
Run E3022, MON-J7 to SPW-J116, Instrumentation Box, 17000mm	6	22	CMG, FT4	300	80°	
Run E3023 SPW-J117 to MON-J8, 17000mm	25	22	CL3, FT4	300	80°	
Run E3025, CRY to MON-FJ4, Flow Monitor-Sumitomo, 10000mm	10	22	CMG, FT4	300	80°	
Run E3028, HEC-PLC-RS232 to PGR-J3, Cooling Control Communication, 10000mm	8	26	CM, FT4	300	75°	
Run E3030 E-Off Switch to SPW-J115, 17000mm	2	22	CMG, FT4	300	105°	
Run E3030, SPW-J115 to E-Off SW, E-off, 10000mm	2	22	CMG, FT4	300	105°	
Run E3031, SPW-J118 to OW-J7, Patient Intercom, 30000mm	25	22	CL3, FT4	300	80°	
Run E3034 MON-J5 to customer-supplied Network, 30000mm	8	26	CMP	300	75°	
Run E3317 Gradient Cable X, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E3317 Gradient Cable X, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E3318 Gradient Cable Y, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E3318 Gradient Cable Y, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E3319 Gradient Cable Z, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E3319 Gradient Cable Z, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run E4002, PEN-GND to PGR-GND, Ground Cable, 10000mm	1	10	CMG	300	80°	single conductor ground cable
Run E4002, PEN-GND to PGR-GND, Ground Cable, 17000mm	1	10	CMG	300	80°	single conductor ground cable
Run E4005 RF COM GND STUD to PGR-GND STUD, 10000mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run E4005 RF COM GND STUD to PGR-GND STUD, 17000mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run E4007, HEC-GND to PGR-GND, Ground Cable, 10000mm	1	10	CMG	300	80°	single conductor ground cable
Run E4008, OW-GOC-GND to PGR-GND, Ground Cable, 30000mm	1	10	CMG	300	80°	single conductor ground cable

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E4009 RF COM GND STUD to SPW-GND STUD, 2800mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run E4010 RF COM GND STUD to PEN-GND STUD, 2800mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run M1001, SPW-J106 to MAG-HYB-J1, 1.5T RF Body Transmit, 12394mm ref. Aluminum braid shield.	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable
Run M1001, SPW-J106 to MAG-HYB-J1, 1.5T RF Body Transmit, 16525mm ref. Aluminum braid shield.	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable
Run M1302, PEN-J29 to MAG-TDM-J11, Receiver Clock, Twinax, 11000mm	2	20	CL2, FT4	300	75°	twinaxial cable
Run M1302, PEN-J29 to MAG-TDM-J11, Receiver Clock, Twinax, 15000mm	2	20	CL2, FT4	300	75°	twinaxial cable
Run M1303, PEN-J32 to PED-HUB-SLOT11-J16, NB Exciter, 14500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M1303, PEN-J32 to PED-HUB-SLOT11-J16, NB Exciter, 18500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M1304, PEN-J30 to PED-HUB-SLOT11-J17, Rx LO - Exciter Reference Clock, Twinax, 13500mm	2	20	CL2, FT4	300	75°	twinaxial cable
Run M1304, PEN-J30 to PED-HUB-SLOT11-J17, Rx LO - Exciter Reference Clock, Twinax, 17500mm	2	20	CL2, FT4	300	75°	twinaxial cable
Run M1305, SPW-J122 to PED-IO-J1, P1 port RF Transmit, 14306mm reference, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable
Run M1305, SPW-J122 to PED-IO-J1, P1 port RF Transmit, 18393mm reference, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable
Run M1306, SPW-J120 to PED-IO-J2, A port RF Transmit, 13965mm reference, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable
Run M1306, SPW-J120 to PED-IO-J2, A port RF Transmit, 18052mm reference, Aluminum braid shield.	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable
Run M1310 PEN-J72 to MAG-J9, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M1310 PEN-J72 to MAG-J9, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M1312, DVw, Pen-J74 to MAG-HYB-J6, 13000 mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M1312, DVw, Pen-J74 to MAG-HYB-J6, 17000 mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable
Run M2025 MAG to PEN, CAN fiber optic, 12500mm	8	N/A	LFNC	N/A	85°	optical fibers in conduit, 4 duplex connectors bundled
Run M2025 MAG to PEN, CAN fiber optic, 16500mm	8	N/A	LFNC	N/A	85°	optical fibers in conduit, 4 duplex connectors bundled
Run M3022, SPW-J116 to MAG-P403, 14000 mm	9	22	CL3, FT4	300	90°	
Run M3023 SPW-J117 to Mag-FJ2, 14000mm	25	22	CL3, FT4	300	80°	

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3030, SPW-J115 to E-OFF, 30000mm	9	22	CL3, FT4	300	80°	
Run M3031, SPW-J118 to MAG-J1, Console Intercom, 10000mm	25	22	CL3, FT4	300	80°	
Run M3031, SPW-J118 to MAG-J1, Console Intercom, 14000mm	25	22	CL3, FT4	300	80°	
Run M3300,PEN-J51 to MAG-TDM-J9, TDM Power, 24W7, Low-Mag, 11200mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	
Run M3300,PEN-J51 to MAG-TDM-J9, TDM Power, 24W7, Low-Mag, 15200mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	
Run M3301, PEN-J52 to MAG-TDM-J10, TDM Power, 24W7, Low-Magnetic, 11200mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	
Run M3301,PEN-J52 to MAG-TDM-J10, TDM Power, 24W7, Low-Mag, 15200mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	
Run M3302 PEN-J40 to MAG-J4,SRI Power, 10000mm	37	22	CL3, FT4	300	80°	
Run M3302 PEN-J40 to MAG-J4,SRI Power, 14000mm	37	22	CL3, FT4	300	80°	
Run M3303 PEN-J39 to MAG-J3, MCPDB Interface, 10000mm	37	22	CL3, FT4	300	80°	
Run M3303 PEN-J39 to MAG-J3, MCPDB Interface, 14000mm	37	22	CL3, FT4	300	80°	
Run M3304 PEN-J48 to PED-HUB-SLOT12-J2, RFHub Power, 13W6, 17500mm	12	6/C 10AWG + 6/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3304,PEN-J48 to PED-HUB-SLOT12-J2, RFHub Power, 13W6, 13500mm	12	6/C 10AWG + 6/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3306, PEN-J46 to PED-HUB-SLOT12-J1, RFHub Power, 17C5, 13500mm	12	6/C 10AWG + 6/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3306, PEN-J46 to PED-HUB-SLOT12-J1, RFHub Power, 17C5, 17500mm	12	6/C 10AWG + 6/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3307, PEN-J50 to PED-HUB-SLOT12-J5, RFHub Power, 21WA4, 13500mm	8	4/C 10AWG + 4/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3307, PEN-J50 to PED-HUB-SLOT12-J5, RFHub Power, 21WA4, 17500mm	8	4/C 10AWG + 4/C 22AWG	UL AWM 2464, FT4	300	105°	

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3308, PEN-J47 to PED-HUB-SLOT12-J6, RFHub Power, 9C4, 13500mm	8	4/C 10AWG + 4/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3308, PEN-J47 to PED-HUB-SLOT12-J6, RFHub Power, 9W4, 17500mm	8	4/C 10AWG + 4/C 22AWG	UL AWM 2464, FT4	300	105°	
Run M3309, SPW-J123 to PED-HUB-SLOT_11-J9, Transmit switch control, 13500mm	7	22	CMG, FT4	300	80°	
Run M3309, SPW-J123 to PED-HUB-SLOT_11-J9, Transmit switch control, 17500mm	7	22	CMG, FT4	300	80°	
Run M3310 PEN-J45 to PED-HUB-SLOT12-J4, RF Power, 13500mm	25	22	CL3, FT4	300	80°	
Run M3310 PEN-J45 to PED-HUB-SLOT12-J4, RF Power, 17500mm	25	22	CL3, FT4	300	80°	
Run M3311 PEN-J36 to PED-MTR-J1, Long Drive Power, 15500mm	6	4/C 14AWG + 2/C 22AWG	CMG, FT4	300	80°	
Run M3311 PEN-J36 to PED-MTR-J1, Long Drive Power, 19500mm	6	4/C 14AWG + 2/C 22AWG	CMG, FT4	300	80°	
Run M3312 PEN-J41 to MAG-J6, PAC Power, 10000mm	25	22	CL3, FT4	300	80°	
Run M3312 PEN-J41 to MAG-J6, PAC Power, 14000mm	25	22	CL3, FT4	300	80°	
Run M3313 PEN-J35 to MAG-J5, Dock Power, 10000mm	37	22	CL3, FT4	300	80°	
Run M3313 PEN-J35 to MAG-J5, Dock Power, 14000mm	37	22	CL3, FT4	300	80°	
Run M3314 PEN-J42 to MAG-J2, LCD Power, 10000mm	15	22	CL3, FT4	300	80°	
Run M3314 Pen-J42 to Mag-J2, 14000 mm	15	22	CL3, FT4	300	80°	
Run M3315 Pen-J54 to PED-TIF-J4, 18500mm.	9	22	CL3, FT4	300	90°	
Run M3315, PEN-J54 to PED-TIF-J4 , Hall's Signal, 14500mm.	9	22	CL3, FT4	300	90°	
Run M3316 Pen-J34 to PED-HUB-SLOT11-J10, 17500mm.	9	22	CL3, FT4	300	90°	
Run M3316, PEN-J34 to PED-HUB-SLOT11-J10, TR_DD Unblank, 13500mm.	9	22	CL3, FT4	300	90°	
Run M3317 Gradient Cable X, Scan Room, Long, 14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC- ER, FT4	600	90°	
Run M3317 Gradient Cable X, Scan Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC- ER, FT4	600	90°	

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3318 Gradient Cable Y, Scan Room, Long,14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run M3318 Gradient Cable Y, Scan Room, Short,10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run M3319 Gradient Cable Z, Scan Room, Long,14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run M3319 Gradient Cable Z, Scan Room, Short,10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	UL Type TC-ER, FT4	600	90°	
Run M4005 RF COM GND STUD to MAG-GND STUD, 13500mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run M4005 RF COM GND STUD to MAG-GND STUD, 17500mm	1	1/0	UL AWM 1284, TEW	600	105°	single conductor ground cable
Run P2003 PGR-J15 to Mag-J20,31000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end
Run P2004 PGR-J14 to Mag-J21,20000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end
Run P2004 PGR-J14 to Mag-J21,31000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end
Run P2026 OW to Mag-J11,Scan Room Display,40000mm	4	N/A	OFNR	N/A	85°	optical fibers, 2 duplex connectors bundled on each end
Run P2026 OW to Mag-J11,Scan Room Display,44000mm	4	N/A	OFNR	N/A	85°	optical fibers, 2 duplex connectors bundled on each end

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